



DIFFERENTIAL GEOMETRY (MMTHCDG325)

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UNIT-I

Parametrized Curves, Regular curves, Arc length, Reparameterization by arc length, Arc length is independent of parameterization, plane and space curves, Frenet–Serret frame. Characterization of plane and space curves in terms of their curvature and torsion. Fundamental theorem of space curves. Helix, necessary and sufficient condition for a space curve to be helix. Involutes and evolutes of space curves. Global properties of curves: Simple closed curves, Isoperimetric Inequality, Four Vertex Theorem.

UNIT -II

Regular surfaces, Change of coordinates. The Tangent Plane, The First Fundamental Form (metric), Area element, and properties of metric, Orientable and Non-Orientable surfaces. Gauss map, differential of the Gauss map is self adjoint operator. Second Fundamental form. Normal curvature, Normal curvature in terms of second fundamental form. Principal curvatures and principle directions, Meusnier's theorem. Classification of points with prescribed principal curvatures, Euler's formula.

UNIT-III

Gaussian and mean curvature, Hilbert's theorem and its applications. Weingarten equation, Gaussian curvature in terms of area, line of curvature, Rodrigues' formula for line of curvature. Joachimsthal's Theorem, Dupin's Theorem. Ruled and Minimal surfaces. Isothermal coordinates. A surface is minimal if and only if its coordinate functions are harmonic. Isometry between surfaces, local isometry, and characterization of local isometry.

UNIT -IV

Christoffel symbols, Theorema Egregium. Gauss equations and Mainardi–Codazzi equations for surfaces, fundamental theorem for regular surface. (Statement only). Geodesics: geodesic curvature, geodesic curvature is intrinsic, equations of geodesic, geodesic on sphere and pseudo sphere, geodesic as distance minimizing curves. Gauss–Bonnet theorem (statement only), geodesic triangle on sphere, implication of Gauss–Bonnet theorem for geodesic triangle.

Chapter 1

Theory of Curves

Topics covered in Chapter 1

Parametrised curves, Regular curves, Arc length, Reparameterization by arc length, Arc length is independent of parameterization, plane and space curves, Frenet–Serret frame. Characterization of plane and space curves in terms of their curvature and torsion. Fundamental theorem of space curves. Helix, necessary and sufficient condition for a space curve to be helix. Involutives and evolutes of space curves. Global properties of curves: Simple closed curves, Isoperimetric Inequality, Four Vertex Theorem.

1.1 Differentiable Curves

In differential geometry, we treat a curve as a mapping from a real interval into Euclidean space.

Definition 1.1.1 (Differentiable Curve). A **differentiable curve** in $\mathbb{R}^n$ is a map $\alpha : I \rightarrow \mathbb{R}^n$ from an open interval $I \subseteq \mathbb{R}$ into $\mathbb{R}^n$ that is of class C^∞ (smooth). We write:

$$\alpha(t) = (x_1(t), x_2(t), \dots, x_n(t))$$

where each component function $x_i(t)$ is infinitely differentiable.

Notation 1.1.2. The variable $t \in I$ is called the **parameter** of the curve. The derivative vector

$$\alpha'(t) = \frac{d\alpha}{dt} = \left(\frac{dx_1}{dt}, \dots, \frac{dx_n}{dt} \right)$$

is called the **tangent vector** (or velocity vector) of the curve α at t .

1.2 Regular Points and Regular Curves

A regular curve should have a well-defined tangent line at every point. This leads to the concept of regularity.

Definition 1.2.1 (Regular Point). A point $t \in I$ is called a **regular point** of the curve α if:

$$\alpha'(t) \neq \mathbf{0}$$

where $\mathbf{0}$ is the zero vector in $\mathbb{R}^n$.

Definition 1.2.2 (Regular Curve). A curve $\alpha : I \rightarrow \mathbb{R}^n$ is said to be **regular** if $\alpha'(t) \neq \mathbf{0}$ for all $t \in I$.

Remark 1.2.3. Regularity ensures that the curve does not stop or form sharp cusps (singularities) where the direction of motion is undefined. Physically, if t represents time, a regular curve describes a particle that never comes to an instantaneous rest.

1.3 Parameterization and Reparameterization

The same geometric set of points (the trace of the curve) can be traversed in many different ways (different speeds or directions).

Definition 1.3.1 (Reparameterization). Let $\alpha : I \rightarrow \mathbb{R}^n$ be a differentiable curve. Let J be another interval and $\phi : J \rightarrow I$ be a C^∞ map such that $\phi'(u) \neq 0$ for all $u \in J$ (i.e., ϕ is a diffeomorphism). Then the curve

$$\beta = \alpha \circ \phi : J \rightarrow \mathbb{R}^n$$

is called a **reparameterization** of α .

Proposition 1.3.2. If α is a regular curve and β is a reparameterization of α , then β is also a regular curve.

Proof. By the chain rule,

$$\beta'(u) = \frac{d}{du}(\alpha(\phi(u))) = \alpha'(\phi(u)) \cdot \phi'(u)$$

Since α is regular, $\alpha'(\phi(u)) \neq 0$. Since ϕ is a diffeomorphism, $\phi'(u) \neq 0$. The product of two non-zero scalars/vectors is non-zero, hence $\beta'(u) \neq 0$. ■

It is often convenient to move along the curve at a constant speed of 1.

Definition 1.3.3 (Arc Length). The **arc length** of a regular curve $\alpha : [a, b] \rightarrow \mathbb{R}^n$ is defined as

$$s = \int_a^b \|\alpha'(t)\| dt$$

where $\|\cdot\|$ denotes the standard Euclidean norm.

Definition 1.3.4 (Unit-Speed Curve). The speed of a regular curve $\alpha : I \rightarrow \mathbb{R}$ is defined as

$$\|\alpha'(t)\| = \sqrt{x^2(t) + y^2(t) + z^2(t)}$$

A curve $\alpha : I \rightarrow \mathbb{R}^n$ is called a **unit-speed curve** (or parameterized by arc length) if:

$$\|\alpha'(s)\| = 1 \quad \text{for all } s \in I$$

Example 1.3.5 (Arc Length of a Circular Helix). (See Figure 1 in the Atlas.) Calculate the total length of one turn of the circular helix parameterized by:

$$\alpha(t) = (a \cos t, a \sin t, ct), \quad t \in [0, 2\pi]$$

where a and c are positive constants.

Solution. Differentiating, we obtain the velocity vector

$$\alpha'(t) = (-a \sin t, a \cos t, c)$$

Next compute the speed $\|\alpha'(t)\|$:

$$\|\alpha'(t)\| = \sqrt{(-a \sin t)^2 + (a \cos t)^2 + c^2}$$

$$\|\alpha'(t)\| = \sqrt{a^2(\sin^2 t + \cos^2 t) + c^2} = \sqrt{a^2 + c^2}$$

Since the speed is constant with respect to t , the arc length s is given by

$$s = \int_0^{2\pi} \|\alpha'(t)\| dt = \int_0^{2\pi} \sqrt{a^2 + c^2} dt$$

$$s = \left[t\sqrt{a^2 + c^2} \right]_0^{2\pi} = 2\pi\sqrt{a^2 + c^2}$$

■

Example 1.3.6 (Arc Length of a Semicubical Parabola). (See Figure 2 in the Atlas.) Find the length of the curve $\alpha(t) = (t^2, \frac{2}{3}t^3)$ from $t = 0$ to $t = 2$.

Solution. Differentiating, we obtain the tangent vector:

$$\alpha'(t) = \left(\frac{d}{dt}(t^2), \frac{d}{dt}\left(\frac{2}{3}t^3\right) \right) = (2t, 2t^2)$$

The speed of the curve is $\|\alpha'(t)\| = \sqrt{(2t)^2 + (2t^2)^2} = \sqrt{4t^2 + 4t^4}$. Factoring out $4t^2$ from the square root (noting $t \geq 0$):

$$\|\alpha'(t)\| = 2t\sqrt{1+t^2}$$

The arc length s is $s = \int_0^2 2t\sqrt{1+t^2} dt$. Let $u = 1+t^2$, then $du = 2t dt$. The limits of integration change from $t \in [0, 2]$ to $u \in [1, 5]$.

$$s = \int_1^5 \sqrt{u} du = \left[\frac{2}{3}u^{3/2} \right]_1^5$$

$$s = \frac{2}{3}(5\sqrt{5} - 1)$$

■

Example 1.3.7 (Arc Length of a Cycloid Arch). (See Figure 3 in the Atlas.) Calculate the arc length of one arch of the cycloid defined by:

$$\alpha(t) = (R(t - \sin t), R(1 - \cos t)), \quad t \in [0, 2\pi]$$

Solution. Differentiating with respect to t gives $x'(t) = R(1 - \cos t)$, $y'(t) = R \sin t$. The square of the speed is $\|\alpha'(t)\|^2 = R^2(1 - \cos t)^2 + R^2 \sin^2 t$.

$$\|\alpha'(t)\|^2 = R^2(1 - 2\cos t + \cos^2 t + \sin^2 t) = R^2(2 - 2\cos t)$$

Using the trigonometric identity $1 - \cos t = 2\sin^2(t/2)$, we obtain $\|\alpha'(t)\|^2 = R^2(2 \cdot 2\sin^2(t/2)) = 4R^2 \sin^2(t/2)$. Taking the square root, noting $\sin(t/2) \geq 0$ for $t \in [0, 2\pi]$:

$$\|\alpha'(t)\| = 2R \sin(t/2)$$

The arc length is

$$s = \int_0^{2\pi} 2R \sin(t/2) dt = [-4R \cos(t/2)]_0^{2\pi}$$

$$s = -4R \cos(\pi) - (-4R \cos(0)) = -4R(-1) + 4R(1) = 8R$$

■

Theorem 1.3.8 (Existence of Unit-Speed Reparameterization). *Let $\alpha : I \rightarrow \mathbb{R}^n$ be a regular differentiable curve. Then there exists a reparameterization of α that is a unit-speed curve.*

Proof. Let $\alpha(t)$ be a regular curve defined on an interval I . We define the arc length function $s : I \rightarrow \mathbb{R}$ starting from a fixed point $t_0 \in I$ as:

$$s(t) = \int_{t_0}^t \|\alpha'(u)\| du$$

By the fundamental theorem of calculus, the derivative of the arc length with respect to the parameter t is $\frac{ds}{dt} = \|\alpha'(t)\|$. Since α is a regular curve, $\|\alpha'(t)\| > 0$ for all $t \in I$. This implies that $s(t)$ is a strictly increasing and differentiable function. By the inverse function theorem, s has a differentiable inverse function $t(s)$ such that:

$$\frac{dt}{ds} = \frac{1}{ds/dt} = \frac{1}{\|\alpha'(t)\|}$$

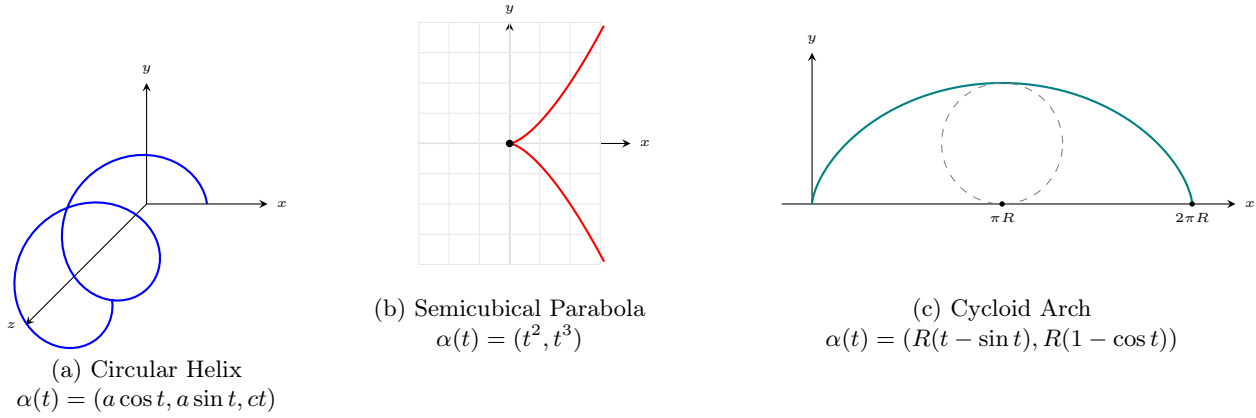


Figure 1.1: Figures for Examples 1.3.5, 1.3.6 and 1.3.7

Consider the reparameterized curve $\beta(s) = \alpha(t(s))$. To verify that β is unit-speed, we apply the Chain Rule to calculate its tangent vector:

$$\beta'(s) = \frac{d}{ds} \alpha(t(s)) = \alpha'(t(s)) \cdot \frac{dt}{ds}$$

Substituting the expression for $\frac{dt}{ds}$ into the equation above:

$$\beta'(s) = \alpha'(t) \cdot \frac{1}{\|\alpha'(t)\|} = \frac{\alpha'(t)}{\|\alpha'(t)\|}$$

Taking the Euclidean norm of both sides, we obtain

$$\|\beta'(s)\| = \left\| \frac{\alpha'(t)}{\|\alpha'(t)\|} \right\| = \frac{\|\alpha'(t)\|}{\|\alpha'(t)\|} = 1$$

Thus, $\|\beta'(s)\| = 1$ for all s in the domain of β , so β is a unit-speed curve. ■

A fundamental property of the arc length is that it is an intrinsic geometric property of the curve, independent of the choice of parameter.

Theorem 1.3.9 (Invariance of Arc Length). *The arc length of a curve is independent of its parameterization (up to orientation).*

Proof. Let $\alpha : [a, b] \rightarrow \mathbb{R}^n$ be a regular curve and let $\beta : [c, d] \rightarrow \mathbb{R}^n$ be a reparameterization given by $\beta = \alpha \circ \phi$, where $\phi : [c, d] \rightarrow [a, b]$ is a diffeomorphism. We consider the case where ϕ is orientation-preserving, meaning $\phi'(u) > 0$, $\phi(c) = a$, and $\phi(d) = b$.

The arc length of β is $L(\beta) = \int_c^d \|\beta'(u)\| du$. Applying the Chain Rule $\beta'(u) = \alpha'(\phi(u))\phi'(u)$:

$$L(\beta) = \int_c^d \|\alpha'(\phi(u))\phi'(u)\| du = \int_c^d \|\alpha'(\phi(u))\| |\phi'(u)| du$$

Since $\phi'(u) > 0$, we have $|\phi'(u)| = \phi'(u)$. Let $t = \phi(u)$, then $dt = \phi'(u)du$. As u goes from c to d , t goes from a to b . Substituting these expressions into the integral gives

$$L(\beta) = \int_a^b \|\alpha'(t)\| dt = L(\alpha)$$

If ϕ is orientation-reversing ($\phi'(u) < 0$), the limits of integration swap, but the absolute value $|\phi'(u)|$ introduces a negative sign that compensates, yielding the same result. ■

Example 1.3.10 (The Cusp). (See Figure 4 in the Atlas.) Consider the curve $\alpha : \mathbb{R} \rightarrow \mathbb{R}^2$ defined by:

$$\alpha(t) = (t^3, t^2)$$

Determine if the curve is regular and identify any singular points.

Solution. To test regularity, compute the tangent vector (velocity vector) $\alpha'(t)$:

$$\alpha'(t) = \frac{d}{dt}(t^3, t^2) = (3t^2, 2t)$$

The curve is regular precisely when $\alpha'(t) \neq (0, 0)$. We set the components to zero:

1. $3t^2 = 0 \implies t = 0$
2. $2t = 0 \implies t = 0$

Since both components vanish simultaneously at $t = 0$, we have $\alpha'(0) = (0, 0)$.

The curve is regular on the intervals $(-\infty, 0)$ and $(0, \infty)$. The point $t = 0$ is a **singular point**. Geometrically, this represents a "cusp" where the curve reverses its direction along the x-axis.

Example 1.3.11 (The Circular Helix). Calculate the arc length of the circular helix $\alpha : [0, 2\pi] \rightarrow \mathbb{R}^3$ defined by:

$$\alpha(t) = (a \cos t, a \sin t, bt)$$

where $a > 0$ and $b > 0$ are constants.

Solution. Differentiating, we obtain the velocity vector

$$\alpha'(t) = (-a \sin t, a \cos t, b)$$

Next compute the speed (the magnitude of the velocity vector):

$$\|\alpha'(t)\| = \sqrt{(-a \sin t)^2 + (a \cos t)^2 + b^2}$$

$$\|\alpha'(t)\| = \sqrt{a^2(\sin^2 t + \cos^2 t) + b^2} = \sqrt{a^2 + b^2}$$

Since the speed is constant, the arc length s from $t = 0$ to $t = 2\pi$ is

$$\begin{aligned} s &= \int_0^{2\pi} \|\alpha'(t)\| dt = \int_0^{2\pi} \sqrt{a^2 + b^2} dt \\ s &= \left[t\sqrt{a^2 + b^2} \right]_0^{2\pi} = 2\pi\sqrt{a^2 + b^2} \end{aligned}$$

Example 1.3.12 (Unit-Speed Reparameterization). Reparameterize the curve $\alpha(t) = (e^t \cos t, e^t \sin t)$ for $t \in \mathbb{R}$ by arc length starting from $t = 0$.

Solution. The speed of the curve is

$$\begin{aligned} \alpha'(t) &= (e^t \cos t - e^t \sin t, e^t \sin t + e^t \cos t) \\ \|\alpha'(t)\|^2 &= e^{2t}(\cos t - \sin t)^2 + e^{2t}(\sin t + \cos t)^2 \\ \|\alpha'(t)\|^2 &= e^{2t}(\cos^2 t - 2 \sin t \cos t + \sin^2 t + \sin^2 t + 2 \sin t \cos t + \cos^2 t) \\ \|\alpha'(t)\|^2 &= e^{2t}(1 + 1) = 2e^{2t} \implies \|\alpha'(t)\| = \sqrt{2}e^t \end{aligned}$$

The arc length function $s(t)$ is given by $s(t) = \int_0^t \sqrt{2}e^u du = \sqrt{2}(e^t - 1)$.

Therefore,

$$\frac{s}{\sqrt{2}} = e^t - 1 \implies e^t = \frac{s}{\sqrt{2}} + 1 \implies t(s) = \ln \left(\frac{s}{\sqrt{2}} + 1 \right)$$

Substituting $t(s)$ into $\alpha(t)$ to get the unit-speed curve $\beta(s)$:

$$\beta(s) = \alpha(t(s)) = \left(\left(\frac{s}{\sqrt{2}} + 1 \right) \cos \left(\ln \left(\frac{s}{\sqrt{2}} + 1 \right) \right), \left(\frac{s}{\sqrt{2}} + 1 \right) \sin \left(\ln \left(\frac{s}{\sqrt{2}} + 1 \right) \right) \right)$$

Example 1.3.13 (Different Parameterizations). Consider the line segment from $(0, 0)$ to $(1, 1)$ in $\mathbb{R}^2$. Show that the arc length is the same for the following two parameterizations:

1. $\alpha(t) = (t, t), \quad t \in [0, 1]$
2. $\beta(u) = (u^2, u^2), \quad u \in [0, 1]$

Solution. For parameterization α :

$$\alpha'(t) = (1, 1) \implies \|\alpha'(t)\| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$L(\alpha) = \int_0^1 \sqrt{2} dt = \sqrt{2}$$

For parameterization β :

$$\beta'(u) = (2u, 2u) \implies \|\beta'(u)\| = \sqrt{(2u)^2 + (2u)^2} = \sqrt{8u^2} = 2u\sqrt{2} \quad (\text{as } u \geq 0)$$

$$L(\beta) = \int_0^1 2u\sqrt{2} du = 2\sqrt{2} \left[\frac{u^2}{2} \right]_0^1 = 2\sqrt{2} \left(\frac{1}{2} \right) = \sqrt{2}$$

Thus, $L(\alpha) = L(\beta)$, demonstrating that arc length is independent of the parameterization chosen.

1.4 Curvature of Plane Curves

Curvature measures how fast the unit tangent vector $T(s)$ rotates as we move along the curve.

Definition 1.4.1 (Signed Curvature). Let $\alpha(s)$ be a unit-speed plane curve. The **signed curvature** $\kappa(s)$ is defined by the relation:

$$\kappa(s) = |T'(s)| = |\beta''(s)|.$$

Theorem 1.4.2. Let $\alpha(t)$ be a regular curve in $\mathbb{R}^3$. Then, its curvature is $\kappa = \frac{|\dot{\alpha} \times \ddot{\alpha}|}{|\dot{\alpha}|^3}$, where $\times$ denotes the cross product and a dot denotes differentiation with respect to t .

Proof. Let $\hat{\alpha}$ (with parameter s) be a unit-speed reparameterization of α , and let us denote d/ds by a dash. Then, by the chain rule,

$$\dot{\alpha} = \frac{d\alpha}{dt} = \frac{d\alpha}{ds} \frac{ds}{dt}, \quad \text{that is,} \quad \hat{\alpha}' \frac{ds}{dt} = \dot{\alpha},$$

so

$$\kappa = |\hat{\alpha}''| = \left| \frac{d}{ds} \left(\frac{\dot{\alpha}}{ds/dt} \right) \right| = \left| \frac{\frac{d}{dt} \left(\frac{\dot{\alpha}}{ds/dt} \right)}{ds/dt} \right| = \left| \frac{\ddot{\alpha} \frac{ds}{dt} - \dot{\alpha} \frac{d^2s}{dt^2}}{(ds/dt)^3} \right|.$$

Now,

$$\left(\frac{ds}{dt} \right)^2 = |\dot{\alpha}|^2 = \dot{\alpha} \cdot \dot{\alpha},$$

and differentiating with respect to t gives $\frac{ds}{dt} \frac{d^2s}{dt^2} = \dot{\alpha} \cdot \ddot{\alpha}$. Using this and the expression for κ , we obtain

$$\kappa = \left| \frac{\ddot{\alpha} \left(\frac{ds}{dt} \right)^2 - \dot{\alpha} \frac{d^2s}{dt^2} \frac{ds}{dt}}{(ds/dt)^4} \right| = \frac{|\ddot{\alpha}(\dot{\alpha} \cdot \dot{\alpha}) - \dot{\alpha}(\dot{\alpha} \cdot \ddot{\alpha})|}{|\dot{\alpha}|^4}.$$

Using the vector triple product identity

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$$

(where $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$), we obtain

$$\dot{\alpha} \times (\ddot{\alpha} \times \dot{\alpha}) = \ddot{\alpha}(\dot{\alpha} \cdot \dot{\alpha}) - \dot{\alpha}(\dot{\alpha} \cdot \ddot{\alpha}).$$

Further, $\dot{\alpha}$ and $\ddot{\alpha} \times \dot{\alpha}$ are perpendicular vectors, so

$$|\dot{\alpha} \times (\ddot{\alpha} \times \dot{\alpha})| = |\dot{\alpha}| |\ddot{\alpha} \times \dot{\alpha}|.$$

Hence,

$$\begin{aligned} \frac{|\ddot{\alpha}(\dot{\alpha} \cdot \dot{\alpha}) - \dot{\alpha}(\dot{\alpha} \cdot \ddot{\alpha})|}{|\dot{\alpha}|^4} &= \frac{|\dot{\alpha} \times (\ddot{\alpha} \times \dot{\alpha})|}{|\dot{\alpha}|^4} \\ &= \frac{|\dot{\alpha}| |\ddot{\alpha} \times \dot{\alpha}|}{|\dot{\alpha}|^4} \\ &= \frac{|\dot{\alpha} \times \ddot{\alpha}|}{|\dot{\alpha}|^3}. \end{aligned}$$

If α is a non-regular curve, its curvature is not defined in general. Note, however, that the formula shows that the curvature is defined at all regular points of the curve (where $\dot{\alpha}$ is non-zero). ■

Theorem 1.4.3 (General Curvature Formula). *Let $\alpha(t) = (x(t), y(t))$ be a regular plane curve. The signed curvature $\kappa(t)$ is given by*

$$\kappa(t) = \frac{x'(t)y''(t) - y'(t)x''(t)}{((x'(t))^2 + (y'(t))^2)^{3/2}}$$

Proof. Let $s(t)$ be the arc length and $\beta(s) = \alpha(t(s))$ be the reparameterization by arc length. The curvature is defined as $\kappa(s) = |\beta''(s)|$. Using the chain rule:

$$\beta'(s) = \alpha'(t) \frac{dt}{ds} \quad \text{and} \quad \beta''(s) = \alpha''(t) \left(\frac{dt}{ds} \right)^2 + \alpha'(t) \frac{d^2t}{ds^2}$$

Given that $\frac{ds}{dt} = \sqrt{x'^2 + y'^2}$, it follows that $\frac{dt}{ds} = (x'^2 + y'^2)^{-1/2}$. Squaring both sides gives,

$$\left(\frac{dt}{ds} \right)^2 = \frac{1}{x'^2 + y'^2}$$

Differentiating with respect to s gives $2 \frac{dt}{ds} \frac{d^2t}{ds^2} = -2 \frac{x'x'' + y'y''}{(x'^2 + y'^2)^2} \frac{dt}{ds}$, which simplifies to

$$\frac{d^2t}{ds^2} = -\frac{x'x'' + y'y''}{(x'^2 + y'^2)^2}$$

Substituting these expressions into $\beta''(s)$:

$$\begin{aligned} \beta''(s) &= \frac{(x'', y'')}{x'^2 + y'^2} - \frac{(x'x'' + y'y'')(x', y')}{(x'^2 + y'^2)^2} \\ &= \frac{1}{(x'^2 + y'^2)^2} ((x'', y'')(x'^2 + y'^2) - (x'x'' + y'y'')(x', y')) \\ &= \frac{1}{(x'^2 + y'^2)^2} ((x'^2x'' + y'^2x'', x'^2y'' + y'^2y'') - (x'^2x'' + x'y'y'', x'y'x'' + y'^2y'')) \\ &= \frac{1}{(x'^2 + y'^2)^2} (y'^2x'' - x'y'y'', x'^2y'' - x'y'x'') \\ &= \frac{1}{(x'^2 + y'^2)^2} (-y'(x'y'' - y'x''), x'(x'y'' - y'x'')) \\ &= \frac{x'y'' - y'x''}{(x'^2 + y'^2)^2} (-y', x') \end{aligned}$$

Taking norms gives

$$|\beta''(s)| = \frac{|x'y'' - y'x''|}{(x'^2 + y'^2)^2} \sqrt{x'^2 + y'^2} = \frac{|x'y'' - y'x''|}{(x'^2 + y'^2)^{3/2}}$$

■

2nd Proof. Let s be the arc length parameter and $v(t) = \frac{ds}{dt} = \|\alpha'(t)\|$. We have $\alpha'(t) = vT$. Differentiating with respect to t using the chain rule:

$$\alpha''(t) = \frac{dv}{dt}T + v\frac{dT}{dt} = v'T + v\left(\frac{dT}{ds}\frac{ds}{dt}\right) = v'T + v^2\kappa N$$

We use the determinant (scalar triple product in 2D) to isolate κ . For vectors $a = (a_1, a_2)$ and $b = (b_1, b_2)$, let $[a, b] = a_1b_2 - a_2b_1$. Then:

$$[\alpha'(t), \alpha''(t)] = [vT, v'T + v^2\kappa N] = vv'[T, T] + v^3\kappa[T, N]$$

Since $[T, T] = 0$ and $[T, N] = 1$ (by the positive orientation of the frame), we have

$$[\alpha'(t), \alpha''(t)] = v^3\kappa \implies \kappa = \frac{x'y'' - y'x''}{v^3}$$

Substituting $v = \sqrt{x'^2 + y'^2}$ completes the proof. ■

Example 1.4.4 (Curvature of a Parabola). Find the signed curvature $\kappa(t)$ of the parabola parameterized by $\alpha(t) = (t, t^2)$ for $t \in \mathbb{R}$. Determine the point where the curvature is maximal.

Solution. The curve is given by $x(t) = t$ and $y(t) = t^2$. We compute the first and second derivatives with respect to t :

$$\begin{aligned} x'(t) &= 1, & x''(t) &= 0 \\ y'(t) &= 2t, & y''(t) &= 2 \end{aligned}$$

The formula for the signed curvature of a plane curve is $\kappa(t) = \frac{x'y'' - y'x''}{((x')^2 + (y')^2)^{3/2}}$. Substituting these derivatives gives

$$\kappa(t) = \frac{(1)(2) - (2t)(0)}{(1^2 + (2t)^2)^{3/2}} = \frac{2}{(1 + 4t^2)^{3/2}}$$

To find where the curvature is maximal, we examine the denominator. Since the numerator is constant, $\kappa(t)$ is maximized when the denominator $(1 + 4t^2)^{3/2}$ is minimized. This occurs at $t = 0$. The maximum curvature is $\kappa(0) = \frac{2}{1^{3/2}} = 2$. This point $(0, 0)$ is the vertex of the parabola. ■

Example 1.4.5 (Curvature of the Logarithmic Spiral). (See Figure 5 in the Atlas.) Calculate the curvature of the logarithmic spiral $\alpha(t) = (e^t \cos t, e^t \sin t)$.

Solution. We have $x(t) = e^t \cos t$ and $y(t) = e^t \sin t$. Differentiating gives

$$\begin{aligned} x' &= e^t(\cos t - \sin t) \\ y' &= e^t(\sin t + \cos t) \\ x'' &= e^t(\cos t - \sin t) + e^t(-\sin t - \cos t) = -2e^t \sin t \\ y'' &= e^t(\sin t + \cos t) + e^t(\cos t - \sin t) = 2e^t \cos t \end{aligned}$$

First compute the speed squared $v^2 = (x')^2 + (y')^2$:

$$v^2 = e^{2t}(\cos^2 t - 2\cos t \sin t + \sin^2 t + \sin^2 t + 2\sin t \cos t + \cos^2 t) = 2e^{2t}$$

Thus, $v^3 = (2e^{2t})^{3/2} = 2\sqrt{2}e^{3t}$. Next compute the numerator $x'y'' - y'x''$:

$$\begin{aligned} x'y'' - y'x'' &= [e^t(\cos t - \sin t)][2e^t \cos t] - [e^t(\sin t + \cos t)][-2e^t \sin t] \\ &= 2e^{2t}(\cos^2 t - \sin t \cos t + \sin^2 t + \cos t \sin t) \\ &= 2e^{2t}(\cos^2 t + \sin^2 t) = 2e^{2t} \end{aligned}$$

Therefore, the curvature is

$$\kappa(t) = \frac{2e^{2t}}{2\sqrt{2}e^{3t}} = \frac{1}{\sqrt{2}e^t}$$
■

1.5 The Osculating Circle

The osculating circle is the circle that provides the best second-order approximation to a curve at a given point.

Definition 1.5.1 (Osculating Circle). Let $\alpha(s)$ be a unit-speed curve with $\kappa(s) \neq 0$. The **radius of curvature** is $\rho(s) = 1/|\kappa(s)|$. The **center of curvature** is $C(s) = \alpha(s) + \frac{1}{\kappa(s)}N(s)$. The **osculating circle** is the circle centered at $C(s)$ with radius $\rho(s)$.

Remark 1.5.2. The osculating circle lies on the concave side of the curve. If $\kappa > 0$, the center is in the direction of N ; if $\kappa < 0$, the center is in the direction of $-N$.

Example 1.5.3 (Osculating Circle of an Ellipse). (See Figure 7 in the Atlas.) Find the center and equation of the osculating circle to the ellipse $\alpha(t) = (2 \cos t, 3 \sin t)$ at the point $t = 0$.

Solution. At $t = 0$, the point on the curve is $\alpha(0) = (2, 0)$. At $t = 0$, the derivatives are

- $\alpha'(t) = (-2 \sin t, 3 \cos t) \implies \alpha'(0) = (0, 3)$
- $\alpha''(t) = (-2 \cos t, -3 \sin t) \implies \alpha''(0) = (-2, 0)$

The curvature at $t = 0$ is

$$\kappa(0) = \frac{x'(0)y''(0) - y'(0)x''(0)}{(x'(0)^2 + y'(0)^2)^{3/2}} = \frac{(0)(0) - (3)(-2)}{(0^2 + 3^2)^{3/2}} = \frac{6}{27} = \frac{2}{9}$$

The unit tangent is $T(0) = \frac{\alpha'(0)}{\|\alpha'(0)\|} = \frac{(0, 3)}{3} = (0, 1)$. Therefore, the oriented unit normal is $N(0) = (-y_T, x_T) = (-1, 0)$. Thus, the center of curvature $C(0)$ is

$$C(0) = \alpha(0) + \frac{1}{\kappa(0)}N(0) = (2, 0) + \frac{9}{2}(-1, 0) = (2 - 4.5, 0) = (-2.5, 0)$$

Therefore, the equation of the circle with center $(-2.5, 0)$ and radius $\rho = 1/|\kappa| = 9/2 = 4.5$ is

$$(x + 2.5)^2 + y^2 = (4.5)^2 \implies (x + \frac{5}{2})^2 + y^2 = \frac{81}{4}$$

■

Example 1.5.4 (The Twisted Cubic). (See Figure 9 in the Atlas.) Find the unit tangent vector T and the curvature κ for the twisted cubic $\alpha(t) = (t, t^2, t^3)$ at the origin $(0, 0, 0)$. Also, find the unit normal and unit binormal at $t = 0$.

Solution. We have $\alpha(t) = (t, t^2, t^3)$. Therefore, $\alpha'(t) = (1, 2t, 3t^2) \implies \alpha'(0) = (1, 0, 0)$. $\alpha''(t) = (0, 2, 6t) \implies \alpha''(0) = (0, 2, 0)$.

Hence, the unit tangent at $(0, 0, 0)$ is $T(0)$ given by

$$T(0) = \frac{(1, 0, 0)}{\|\alpha'(0)\|} = (1, 0, 0)$$

The curvature is given by $\kappa = \frac{\|\alpha' \times \alpha''\|}{\|\alpha'\|^3}$. At $t = 0$:

$$\alpha'(0) \times \alpha''(0) = (1, 0, 0) \times (0, 2, 0) = (0, 0, 2)$$

The magnitude is $\|\alpha' \times \alpha''\| = 2$. Since $\|\alpha'(0)\| = 1$:

$$\kappa(0) = \frac{2}{1^3} = 2$$

Also, we can see that $N(0) = (0, 1, 0)$ and $B(0) = (0, 0, 1)$ at the origin for this curve. ■

The osculating circle (from the Latin *osculum*, meaning "kiss") at a point P on a plane curve is the unique circle that shares the same tangent and the same curvature as the curve at that point.

Example 1.5.5 (Osculating Circle of a Parabola). (See Figure 6 in the Atlas.) Find the center and equation of the osculating circle to the parabola $y = x^2$ at the vertex $(0, 0)$.

Solution. We parameterize the parabola as $\alpha(x) = (x, x^2)$. At the vertex $x = 0$, the first and second derivatives are given by $\alpha'(0) = (1, 0)$ and $\alpha''(0) = (0, 2)$. Using the curvature formula for a plane curve, we find $\kappa(0) = \frac{x'y'' - y'x''}{(x'^2 + y'^2)^{3/2}} = \frac{(1)(2) - (0)(0)}{(1^2 + 0^2)^{3/2}} = 2$. The radius of curvature is thus $\rho = 1/|\kappa| = 1/2$. The unit tangent at the vertex is $T = (1, 0)$, which implies the oriented unit normal is $N = (0, 1)$. Consequently, the center of curvature is $C = \alpha(0) + \frac{1}{\kappa}N = (0, 0) + \frac{1}{2}(0, 1) = (0, 0.5)$. The equation of the osculating circle is therefore $x^2 + (y - 0.5)^2 = 0.25$, which simplifies to $x^2 + y^2 - y = 0$. ■

Example 1.5.6 (Osculating Circle of $y = e^x$). (See Figure 8 in the Atlas.) Find the radius and center of curvature for $y = e^x$ at the point $(0, 1)$.

Solution. Let the curve be parameterized as $\alpha(x) = (x, e^x)$. At $x = 0$, the point is $\alpha(0) = (0, 1)$ and the derivatives are $y' = 1$ and $y'' = 1$. The curvature is given by $\kappa = \frac{y''}{(1+y'^2)^{3/2}} = \frac{1}{(1+1^2)^{3/2}} = \frac{1}{2\sqrt{2}}$. This gives a radius of curvature $\rho = 1/\kappa = 2\sqrt{2}$. Since the velocity vector is $(1, 1)$, the unit tangent is $T = \frac{1}{\sqrt{2}}(1, 1)$ and the oriented unit normal is $N = \frac{1}{\sqrt{2}}(-1, 1)$. The center of curvature is computed as $C = (0, 1) + 2\sqrt{2} \left[\frac{1}{\sqrt{2}}(-1, 1) \right] = (0, 1) + (-2, 2) = (-2, 3)$. The resulting osculating circle has the equation $(x + 2)^2 + (y - 3)^2 = 8$. ■

1.6 The Tangent Line to a Plane Curve

Let $\alpha : I \rightarrow \mathbb{R}^2$ be a regular differentiable plane curve given by $\alpha(t) = (x(t), y(t))$. We seek the equation of the line tangent to the curve at a specific point corresponding to the parameter $t = t_0$.

Definition 1.6.1 (Tangent Vector). The **tangent vector** (or velocity vector) of the curve α at t_0 is the derivative:

$$\alpha'(t_0) = (x'(t_0), y'(t_0))$$

Geometrically, this vector is tangent to the curve at the point $P = \alpha(t_0)$.

Proposition 1.6.2 (Vector Equation of the Tangent Line). The tangent line L to the curve α at the point $\alpha(t_0)$ is the set of points $X = (X, Y)$ such that:

$$L(u) = \alpha(t_0) + u\alpha'(t_0), \quad u \in \mathbb{R}$$

where u is a scalar parameter for the line.

Derivation of the Cartesian Equation. Starting from the vector equation, we can write the components of the point (X, Y) on the tangent line:

$$\begin{aligned} X &= x(t_0) + ux'(t_0) \\ Y &= y(t_0) + uy'(t_0) \end{aligned}$$

To find the Cartesian (rectangular) equation, we eliminate the parameter u . From the first equation, assuming $x'(t_0) \neq 0$:

$$u = \frac{X - x(t_0)}{x'(t_0)}$$

Substituting this into the equation for Y :

$$Y = y(t_0) + \left(\frac{X - x(t_0)}{x'(t_0)} \right) y'(t_0)$$

Rearranging this gives the point-slope form of the line:

$$Y - y(t_0) = \frac{y'(t_0)}{x'(t_0)}(X - x(t_0))$$

Observe that $\frac{y'(t_0)}{x'(t_0)}$ is exactly the derivative $\frac{dy}{dx}$ at the point $(x(t_0), y(t_0))$ by the chain rule.

To obtain a form that is valid even if $x'(t_0) = 0$ (as long as the curve is regular), we cross-multiply:

$$y'(t_0)(X - x(t_0)) - x'(t_0)(Y - y(t_0)) = 0$$

This is the general linear equation of the tangent line to a plane curve. ■

Remark 1.6.3. In terms of the normal vector $N = (-y', x')$, the tangent line equation can be written compactly using the dot product:

$$\langle (X, Y) - \alpha(t_0), N(t_0) \rangle = 0$$

This shows that the tangent line consists of all points X such that the vector from $\alpha(t_0)$ to X is perpendicular to the normal vector of the curve.

Definition 1.6.4. The **normal line** (or simply the **normal**) to a regular plane curve $\alpha(t)$ at a point $\alpha(t_0) = (x(t_0), y(t_0))$ is the straight line in the plane $\mathbb{R}^2$ that passes through $\alpha(t_0)$ and is perpendicular to the tangent vector $\dot{\alpha}(t_0)$.

A vector in $\mathbb{R}^2$ perpendicular to the tangent vector $\dot{\alpha}(t_0) = (\dot{x}(t_0), \dot{y}(t_0))$ is given by $\mathbf{n}(t_0) = (-\dot{y}(t_0), \dot{x}(t_0))$. If we normalize this vector, we obtain the unit normal vector $\mathbf{N}(t_0)$:

$$\mathbf{N}(t_0) = \frac{(-\dot{y}(t_0), \dot{x}(t_0))}{\sqrt{\dot{x}(t_0)^2 + \dot{y}(t_0)^2}}$$

Equation of Normal

To derive the equation of the normal line at the point $(x_0, y_0) = (x(t_0), y(t_0))$, let (X, Y) represent an arbitrary point on the normal line.

Since the normal line is perpendicular to the tangent vector $\dot{\alpha}(t_0) = (\dot{x}(t_0), \dot{y}(t_0))$, the displacement vector from the point of tangency (x_0, y_0) to (X, Y) must be orthogonal to $\dot{\alpha}(t_0)$. Thus, their dot product must equal zero:

$$((X - x_0), (Y - y_0)) \cdot (\dot{x}(t_0), \dot{y}(t_0)) = 0$$

Expanding the dot product yields the Cartesian equation of the normal line:

$$\dot{x}(t_0)(X - x_0) + \dot{y}(t_0)(Y - y_0) = 0$$

Alternatively, if $\dot{y}(t_0) \neq 0$, we can express the equation of the normal line in slope-intercept form. Since the slope of the tangent line is $m_t = \frac{\dot{y}(t_0)}{\dot{x}(t_0)}$, the slope of the normal line is $m_n = -\frac{\dot{x}(t_0)}{\dot{y}(t_0)}$. The equation is therefore $Y - y_0 = -\frac{\dot{x}(t_0)}{\dot{y}(t_0)}(X - x_0)$.

In parametric form, the normal line can be expressed using a parameter $\lambda \in \mathbb{R}$ as:

$$\beta(\lambda) = \alpha(t_0) + \lambda \mathbf{N}(t_0)$$

Example 1.6.5. Consider the parabola parameterized by:

$$\alpha(t) = (t, t^2), \quad t \in \mathbb{R}$$

Find the equation of the normal line to this curve at the point where $t_0 = 1$.

Solution. First obtain the coordinates of the point on the curve at $t_0 = 1$:

$$x_0 = x(1) = 1, \quad y_0 = y(1) = 1^2 = 1$$

Thus, $(x_0, y_0) = (1, 1)$.

Next differentiate $\alpha(t)$ with respect to t to find the tangent vector:

$$\dot{\alpha}(t) = (1, 2t)$$

At $t_0 = 1$, the tangent vector is $\dot{\alpha}(1) = (\dot{x}(1), \dot{y}(1)) = (1, 2)$.

Using the Cartesian equation of the normal line:

$$\dot{x}(1)(X - x_0) + \dot{y}(1)(Y - y_0) = 0$$

Substituting the computed values into the equation:

$$1(X - 1) + 2(Y - 1) = 0$$

$$X - 1 + 2Y - 2 = 0$$

$$X + 2Y - 3 = 0$$

Hence the slope-intercept form of the normal line:

$$Y = -\frac{1}{2}X + \frac{3}{2}$$

■

1.7 The Frenet–Serret Formulas

The local geometry of a unit-speed curve $\alpha : I \rightarrow \mathbb{R}^3$ is entirely determined by its curvature $\kappa(s)$ and its torsion $\tau(s)$. The Frenet–Serret formulas describe how the moving orthonormal frame $\{T, N, B\}$ evolves along the curve.

In three-dimensional space, a regular curve $\alpha : I \rightarrow \mathbb{R}^3$ is described locally by three mutually orthogonal unit vectors known as the **Frenet–Serret Frame**. These vectors—the tangent, the normal, and the binormal—characterize the bending and twisting of the curve in space.

Definition 1.7.1 (The Unit Tangent Vector T). Let $\alpha(t)$ be a regular curve in $\mathbb{R}^3$. The **unit tangent vector** $T(t)$ is defined as the normalized velocity vector:

$$T(t) = \frac{\alpha'(t)}{\|\alpha'(t)\|}$$

If the curve is parameterized by arc length s , then $T(s) = \alpha'(s)$.

Definition 1.7.2 (The Principal Unit Normal Vector N). The **principal unit normal vector** $N(t)$ points in the direction in which the curve is bending. It is defined as

$$N(t) = \frac{T'(t)}{\|T'(t)\|}$$

The magnitude $\kappa(t) = \frac{\|T'(t)\|}{\|\alpha'(t)\|}$ is the **curvature** of the curve. N is only defined at points where $\kappa \neq 0$. Specifically, if $T(s) = (x(s), y(s))$, then $N(s) = (-y(s), x(s))$.

Proposition 1.7.3. Let $\mathbf{a}$ and $\mathbf{b}$ be two unit vectors in $\mathbb{R}^3$. Their cross product $\mathbf{a} \times \mathbf{b}$ is a unit vector if and only if $\mathbf{a} \perp \mathbf{b}$.

Proof. By the definition of the magnitude of a cross product, for any two vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^3$:

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta$$

where θ is the angle between the two vectors ($0 \leq \theta \leq \pi$). Since $\mathbf{a}$ and $\mathbf{b}$ are unit vectors, we have $\|\mathbf{a}\| = 1$ and $\|\mathbf{b}\| = 1$. Substituting these values into the equation:

$$\|\mathbf{a} \times \mathbf{b}\| = (1)(1) \sin \theta = \sin \theta$$

For the cross product $\mathbf{a} \times \mathbf{b}$ to be a unit vector, its magnitude must be equal to 1:

$$\sin \theta = 1$$

In the interval $[0, \pi]$, the sine function equals 1 only when $\theta = \frac{\pi}{2}$ (or 90°). Thus, the cross product of two unit vectors is a unit vector only if the vectors are orthogonal. ■

Definition 1.7.4 (The Binormal Vector B). The **binormal vector** $B(t)$ is defined as the cross product of the tangent and the normal vectors:

$$B(t) = T(t) \times N(t)$$

The triple $\{T, N, B\}$ forms a right-handed orthonormal basis at each point along the curve.

Proposition 1.7.5. For any regular differentiable curve $\alpha(s)$, the unit tangent vector $T(s)$ and the principal unit normal vector $N(s)$ are orthogonal; that is, $\langle T(s), N(s) \rangle = 0$.

Proof. By definition, $T(s)$ is a unit vector, which means its magnitude is constant:

$$\|T(s)\| = 1$$

Taking the square of the norm and expressing it as a dot product, we have $\langle T(s), T(s) \rangle = 1$. We differentiate both sides of this equation with respect to the parameter s . Using the product rule for dot products:

$$\begin{aligned} \frac{d}{ds} \langle T(s), T(s) \rangle &= \frac{d}{ds} (1) \\ \langle T'(s), T(s) \rangle + \langle T(s), T'(s) \rangle &= 0 \\ 2\langle T'(s), T(s) \rangle &= 0 \end{aligned}$$

Dividing by 2, we obtain $\langle T'(s), T(s) \rangle = 0$. Thus the derivative vector $T'(s)$ is always orthogonal to $T(s)$. Since the principal unit normal vector $N(s)$ is defined as the normalized derivative of the tangent vector:

$$N(s) = \frac{T'(s)}{\|T'(s)\|}$$

Since $N(s)$ is simply a scalar multiple of $T'(s)$, and $T'(s)$ is orthogonal to $T(s)$, it follows that

$$\langle T(s), N(s) \rangle = 0$$

Thus, the principal unit normal vector is perpendicular to the tangent vector at every point where it is defined. ■

Example 1.7.6 (The Circular Helix). Find the Frenet frame $\{T, N, B\}$ for the circular helix:

$$\alpha(t) = (a \cos t, a \sin t, bt)$$

where $a^2 + b^2 = c^2$.

Solution. First compute the velocity: $\alpha'(t) = (-a \sin t, a \cos t, b)$. The speed is

$$v = \|\alpha'(t)\| = \sqrt{a^2 \sin^2 t + a^2 \cos^2 t + b^2} = \sqrt{a^2 + b^2} = c.$$

Therefore,

$$T(t) = \frac{\alpha'(t)}{c} = \left(-\frac{a}{c} \sin t, \frac{a}{c} \cos t, \frac{b}{c} \right)$$

Differentiate $T(t)$: $T'(t) = \left(-\frac{a}{c} \cos t, -\frac{a}{c} \sin t, 0 \right)$. Its norm is

$$\|T'(t)\| = \sqrt{\frac{a^2}{c^2} (\cos^2 t + \sin^2 t)} = \frac{a}{c}.$$

Thus,

$$N(t) = \frac{T'(t)}{a/c} = (-\cos t, -\sin t, 0)$$

We compute the cross product $T \times N = B(t)$:

$$B(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\frac{a}{c} \sin t & \frac{a}{c} \cos t & \frac{b}{c} \\ -\cos t & -\sin t & 0 \end{vmatrix}$$

$$B(t) = \left(\frac{b}{c} \sin t, -\frac{b}{c} \cos t, \frac{a}{c} \right)$$

■

Definition 1.7.7 (Torsion). The **torsion** $\tau(s)$ of the curve α measures the departure of the curve from planarity and is defined by $B'(s) = -\tau(s)N(s)$.

Since $B(s)$ is a unit vector, $B'(s)$ is perpendicular to $B(s)$. Also, differentiating $\langle B, T \rangle = 0$ and using $T' = \kappa N$ shows that $B'(s)$ is perpendicular to $T(s)$. Hence $B'(s)$ must be parallel to $N(s)$.

The minus sign is a conventional choice that gives the Frenet–Serret formulas the standard form

$$T' = \kappa N, \quad N' = -\kappa T + \tau B, \quad B' = -\tau N.$$

Thus $|\tau(s)|$ measures the rate at which the binormal, and hence the osculating plane, rotates along the curve.

Example 1.7.8 (Calculating Torsion for a General Parameterization). Calculate the torsion of the twisted cubic $\alpha(t) = (t, t^2, t^3)$.

Solution. For a regular curve $\alpha(t)$ not necessarily parameterized by arc length, the torsion is

$$\tau(t) = \frac{(\alpha'(t) \times \alpha''(t)) \cdot \alpha'''(t)}{\|\alpha'(t) \times \alpha''(t)\|^2}.$$

We have

$$\begin{aligned} \alpha'(t) &= (1, 2t, 3t^2), \\ \alpha''(t) &= (0, 2, 6t), \\ \alpha'''(t) &= (0, 0, 6). \end{aligned}$$

The cross product is

$$\alpha'(t) \times \alpha''(t) = (6t^2, -6t, 2).$$

Therefore,

$$\|\alpha'(t) \times \alpha''(t)\|^2 = 36t^4 + 36t^2 + 4,$$

while $(\alpha'(t) \times \alpha''(t)) \cdot \alpha'''(t) = 12$.

Hence,

$$\tau(t) = \frac{12}{36t^4 + 36t^2 + 4} = \frac{3}{9t^4 + 9t^2 + 1}.$$

Since $\tau(t) \neq 0$ for all t , the twisted cubic is not planar. ■

Theorem 1.7.9 (Frenet–Serret Formulas). Let $\alpha(s)$ be a unit-speed curve in $\mathbb{R}^3$ with $\kappa(s) > 0$. The derivatives of the Frenet frame $\{T, N, B\}$ satisfy the following system:

$$\begin{aligned} T' &= \kappa N \\ N' &= -\kappa T + \tau B \\ B' &= -\tau N \end{aligned}$$

In matrix form, this is expressed as

$$\begin{pmatrix} T' \\ N' \\ B' \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}$$

Proof. Because $\{T, N, B\}$ forms an orthonormal basis for $\mathbb{R}^3$, the derivative of any member of the frame can be written as a linear combination of the basis vectors.

First, by definition, $T' = \kappa N$, where $\kappa = \|T'\|$ is the curvature.

Next, consider the binormal vector B . Since $\|B\| = 1$, we have $\langle B, B \rangle = 1$, and differentiation yields $\langle B', B \rangle = 0$, implying $B' \perp B$. Also, since $\langle B, T \rangle = 0$, differentiating yields $\langle B', T \rangle + \langle B, T' \rangle = 0$. Substituting $T' = \kappa N$ and noting $\langle B, N \rangle = 0$, we find $\langle B', T \rangle = 0$. Since B' is orthogonal to both B and T , it must be a multiple of N . We define the torsion τ by the relation $B' = -\tau N$.

Finally, we determine N' . Since $\{T, N, B\}$ is orthonormal, we have $N' = \langle N', T \rangle T + \langle N', N \rangle N + \langle N', B \rangle B$. From $\langle N, N \rangle = 1$, we have $\langle N', N \rangle = 0$. From $\langle N, T \rangle = 0$, we have $\langle N', T \rangle = -\langle N, T' \rangle = -\langle N, \kappa N \rangle = -\kappa$. From $\langle N, B \rangle = 0$, we have $\langle N', B \rangle = -\langle N, B' \rangle = -\langle N, -\tau N \rangle = \tau$. Substituting these values, we obtain the final relation:

$$N' = -\kappa T + \tau B$$

This proves the Frenet–Serret system. ■

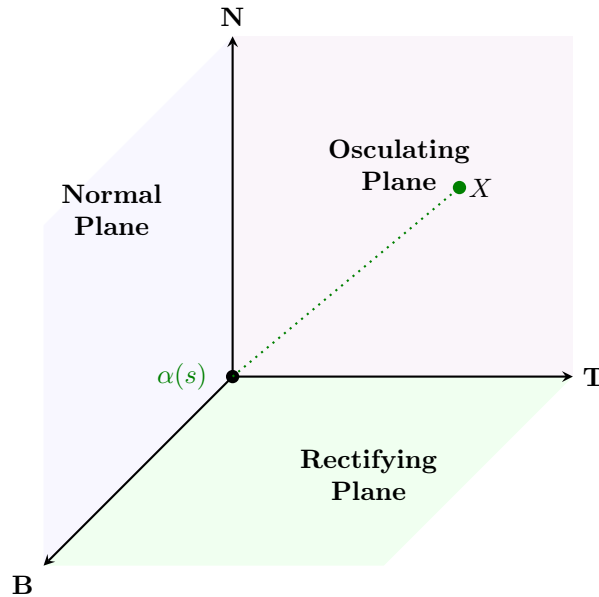


Figure 1.2: The normal, osculating, and rectifying planes of a space curve.

Remark 1.7.10. The skew-symmetry of the matrix in the Frenet–Serret formulas is a general property of the derivatives of a moving orthonormal frame. It reflects the fact that the frame undergoes an instantaneous rotation without changing the lengths of the basis vectors.

The orthonormal triad $\{T, N, B\}$ is referred to as the **Frenet–Serret frame** of the curve. These three vectors naturally determine three mutually perpendicular coordinate planes at each point along the curve, which describe its local spatial orientation. (See Figure 10 in the Atlas.)

Each pair of vectors in the Frenet–Serret frame spans a unique plane passing through the point $\alpha(s)$ defined below.

Definition 1.7.11. • The **normal plane** to the curve α at the point $\alpha(s)$ is the plane spanned by the principal normal $N(s)$ and the binormal $B(s)$. Since it is perpendicular to the tangent vector $T(s)$, its equation in terms of an arbitrary point $\mathbf{x} = (X, Y, Z)$ on the plane is given by $(\mathbf{x} - \alpha(s)) \cdot \mathbf{T}(s) = 0$.

- The **osculating plane** (literally, the “kissing plane”) is the plane spanned by the tangent vector $T(s)$ and the principal normal vector $N(s)$. This plane represents the flat surface that is locally closest to containing the curve. Its normal vector is the binormal $B(s)$, and its equation is $(\mathbf{x} - \alpha(s)) \cdot \mathbf{B}(s) = 0$.

- The **rectifying plane** is the plane spanned by the tangent vector $\mathbf{T}(s)$ and the binormal vector $\mathbf{B}(s)$. It is perpendicular to the principal normal vector $\mathbf{N}(s)$. Its equation is given by $(\mathbf{x} - \alpha(s)) \cdot \mathbf{N}(s) = 0$.

Remark 1.7.12. For a curve $\alpha(t)$ with a general parameter t , we can formulate these equations without converting to arc length. Letting $\dot{\alpha}$ and $\ddot{\alpha}$ denote derivatives with respect to t :

- The normal vector of the **normal plane** is parallel to $\dot{\alpha}(t)$.
- The normal vector of the **osculating plane** is parallel to $\dot{\alpha}(t) \times \ddot{\alpha}(t)$.
- The normal vector of the **rectifying plane** is parallel to $(\dot{\alpha}(t) \times \ddot{\alpha}(t)) \times \dot{\alpha}(t)$.

Example 1.7.13 (The Circular Helix). Let $\mathbf{r}(t) = (a \cos t, a \sin t, bt)$, with $a > 0$ and $b \neq 0$. Derive the Frenet frame $(\mathbf{T}, \mathbf{N}, \mathbf{B})$ and the equations of the three fundamental planes at an arbitrary point of the curve.

Solution. The derivative is $\mathbf{r}'(t) = (-a \sin t, a \cos t, b)$, and the speed is constant:

$$\|\mathbf{r}'(t)\| = \sqrt{a^2 \sin^2 t + a^2 \cos^2 t + b^2} = \sqrt{a^2 + b^2} =: c.$$

Hence the unit tangent vector is $\mathbf{T}(t) = \frac{1}{c}(-a \sin t, a \cos t, b)$.

Differentiating $\mathbf{T}$ gives $\mathbf{T}'(t) = \frac{1}{c}(-a \cos t, -a \sin t, 0)$, so

$$\|\mathbf{T}'(t)\| = \frac{a}{c}.$$

Therefore the principal normal is

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|} = (-\cos t, -\sin t, 0).$$

The binormal vector is obtained from the cross product:

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\frac{a}{c} \sin t & \frac{a}{c} \cos t & \frac{b}{c} \\ -\cos t & -\sin t & 0 \end{vmatrix}.$$

Expanding the determinant yields $\mathbf{B}(t) = \frac{1}{c}(b \sin t, -b \cos t, a)$.

We now determine the fundamental planes at the point

$$P = \mathbf{r}(t) = (a \cos t, a \sin t, bt).$$

Let $\mathbf{X} = (x, y, z)$.

Osculating plane. Its normal is $\mathbf{B}(t)$, hence

$$\mathbf{B}(t) \cdot (\mathbf{X} - \mathbf{r}(t)) = 0.$$

Multiplying by c and simplifying gives $b \sin t (x - a \cos t) - b \cos t (y - a \sin t) + a(z - bt) = 0$.

Normal plane. Its normal is $\mathbf{T}(t)$, so

$$\mathbf{T}(t) \cdot (\mathbf{X} - \mathbf{r}(t)) = 0,$$

which expands to

$$-a \sin t (x - a \cos t) + a \cos t (y - a \sin t) + b(z - bt) = 0.$$

Rectifying plane. Its normal is $\mathbf{N}(t)$, thus

$$\mathbf{N}(t) \cdot (\mathbf{X} - \mathbf{r}(t)) = 0.$$

Substituting $\mathbf{N}(t)$ and $\mathbf{r}(t)$ gives $-\cos t (x - a \cos t) - \sin t (y - a \sin t) = 0$, which simplifies to

$$\cos t x + \sin t y = a.$$

Thus the Frenet frame and all three fundamental planes are explicitly determined for every point of the helix. ■

Example 1.7.14 (The Twisted Cubic). Consider the twisted cubic $\mathbf{r}(t) = (t, t^2, t^3)$. Determine the Frenet frame and the fundamental planes at the point corresponding to $t = 0$.

Solution. The curve is regular since $\mathbf{r}'(t) = (1, 2t, 3t^2) \neq \mathbf{0}$ for all t . At $t = 0$, the point is $P = \mathbf{r}(0) = (0, 0, 0)$.

The derivative at $t = 0$ is $\mathbf{r}'(0) = (1, 0, 0)$, $\|\mathbf{r}'(0)\| = 1$, so the unit tangent vector is $\mathbf{T}(0) = (1, 0, 0)$.

To find the principal normal, we compute $\mathbf{T}'(0)$. Since the speed $\|\mathbf{r}'(t)\|$ equals 1 at $t = 0$, the derivative of the denominator in the definition of $\mathbf{T}$ vanishes at that point; hence $\mathbf{T}'(0) = \mathbf{r}''(0)$. Now

$$\mathbf{r}''(t) = (0, 2, 6t), \quad \mathbf{r}''(0) = (0, 2, 0),$$

so

$$\|\mathbf{T}'(0)\| = 2.$$

Therefore

$$\mathbf{N}(0) = \frac{\mathbf{T}'(0)}{\|\mathbf{T}'(0)\|} = \frac{(0, 2, 0)}{2} = (0, 1, 0).$$

The binormal vector is the cross product:

$$\mathbf{B}(0) = \mathbf{T}(0) \times \mathbf{N}(0) = (1, 0, 0) \times (0, 1, 0) = (0, 0, 1).$$

We now write the equations of the three fundamental planes at $P = (0, 0, 0)$. Let $\mathbf{X} = (x, y, z)$.

Osculating plane. Since the normal is $\mathbf{B}(0) = (0, 0, 1)$, we have

$$(0, 0, 1) \cdot (x, y, z) = 0 \implies z = 0.$$

Normal plane. The normal is $\mathbf{T}(0) = (1, 0, 0)$, so

$$(1, 0, 0) \cdot (x, y, z) = 0 \implies x = 0.$$

Rectifying plane. The normal is $\mathbf{N}(0) = (0, 1, 0)$, hence

$$(0, 1, 0) \cdot (x, y, z) = 0 \implies y = 0.$$

Thus, at the origin, the osculating plane is the xy -plane, the normal plane is the yz -plane, and the rectifying plane is the xz -plane. These are precisely the three coordinate planes. ■

The following theorem gives the curvature of space curves in any regular parametrization.

Theorem 1.7.15. Let $\alpha(t)$ be a regular curve in $\mathbb{R}^3$ with nowhere vanishing curvature. Then, denoting d/dt by a dot, its torsion is given by

$$\tau = \frac{(\dot{\alpha} \times \ddot{\alpha}) \cdot \ddot{\alpha}}{|\dot{\alpha} \times \ddot{\alpha}|^2}$$

Proof. **Case 1:** We first treat the case in which α is unit-speed. Using Frenet Frame,

$$\tau = -\mathbf{N} \cdot \dot{\mathbf{B}} = -\mathbf{N} \cdot (\mathbf{T} \times \mathbf{N})' = -\mathbf{N} \cdot (\dot{\mathbf{T}} \times \mathbf{N} + \mathbf{T} \times \dot{\mathbf{N}}) = -\mathbf{N} \cdot (\mathbf{T} \times \dot{\mathbf{N}}).$$

Now, $\mathbf{N} = \frac{1}{\kappa} \dot{\mathbf{T}} = \frac{1}{\kappa} \ddot{\alpha}$, so

$$\begin{aligned} \tau &= -\frac{1}{\kappa} \ddot{\alpha} \cdot \left(\dot{\alpha} \times \frac{d}{dt} \left(\frac{1}{\kappa} \ddot{\alpha} \right) \right) \\ &= -\frac{1}{\kappa} \ddot{\alpha} \cdot \left(\dot{\alpha} \times \left(\frac{1}{\kappa} \ddot{\alpha} - \frac{\dot{\kappa}}{\kappa^2} \ddot{\alpha} \right) \right) \\ &= \frac{1}{\kappa^2} \ddot{\alpha} \cdot (\dot{\alpha} \times \ddot{\alpha}), \end{aligned}$$

since $\ddot{\alpha} \cdot (\dot{\alpha} \times \ddot{\alpha}) = 0$ and $\ddot{\alpha} \cdot (\dot{\alpha} \times \ddot{\alpha}) = -\ddot{\alpha} \cdot (\dot{\alpha} \times \ddot{\alpha})$. Again since α is unit-speed, $\dot{\alpha}$ and $\ddot{\alpha}$ are perpendicular, so

$$|\dot{\alpha} \times \ddot{\alpha}| = |\dot{\alpha}||\ddot{\alpha}| = |\ddot{\alpha}| = \kappa.$$

Case 2: In the general case, let s be arc length along α and denote d/ds by a dash. Then,

$$\begin{aligned}\frac{d\alpha}{dt} &= \dot{\alpha} = \frac{d\alpha}{ds} \frac{ds}{dt} = \frac{ds}{dt} \alpha', \\ \ddot{\alpha} &= \left(\frac{ds}{dt}\right)^2 \alpha'' + \frac{d^2s}{dt^2} \alpha', \\ \ddot{\alpha} &= \left(\frac{ds}{dt}\right)^3 \alpha''' + 3 \frac{ds}{dt} \frac{d^2s}{dt^2} \alpha'' + \frac{d^3s}{dt^3} \alpha'.\end{aligned}$$

Hence,

$$\begin{aligned}\dot{\alpha} \times \ddot{\alpha} &= \left(\frac{ds}{dt}\right)^3 \alpha' \times \alpha'', \\ \ddot{\alpha} \cdot (\dot{\alpha} \times \ddot{\alpha}) &= \left(\frac{ds}{dt}\right)^6 \alpha''' \cdot (\alpha' \times \alpha''),\end{aligned}$$

and so

$$\frac{\ddot{\alpha} \cdot (\dot{\alpha} \times \ddot{\alpha})}{|\dot{\alpha} \times \ddot{\alpha}|^2} = \frac{\alpha''' \cdot (\alpha' \times \alpha'')}{|\alpha' \times \alpha''|^2}.$$

■

Theorem 1.7.16. A regular unit-speed curve $\alpha : I \rightarrow \mathbb{R}^3$ is a straight line if and only if its curvature $\kappa(s) = 0$ for all $s \in I$.

Proof. Let $\mathbf{T}(s) = \alpha'(s)$. By definition, $\kappa(s) = |\mathbf{T}'(s)| = |\alpha''(s)|$.

If α is a straight line, it can be parameterized as $\alpha(s) = \mathbf{a}s + \mathbf{b}$ for constant vectors $\mathbf{a}, \mathbf{b}$ with $|\mathbf{a}| = 1$. Then $\alpha''(s) = \mathbf{0}$, hence $\kappa(s) = |\mathbf{0}| = 0$.

If $\kappa(s) = 0$ for all $s \in I$, then $|\alpha''(s)| = 0 \implies \alpha''(s) = \mathbf{0}$. Integrating twice with respect to s yields $\alpha(s) = \mathbf{a}s + \mathbf{b}$ for some constant vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$, which is a straight line. ■

Theorem 1.7.17. A regular unit-speed curve $\alpha : I \rightarrow \mathbb{R}^3$ with $\kappa(s) > 0$ is a plane curve if and only if its torsion $\tau(s) = 0$ for all $s \in I$.

Proof. Suppose $\alpha(s)$ is a plane curve. Then there exists a constant unit vector $\mathbf{k}$ and a fixed point P_0 such that the curve satisfies the plane equation:

$$(\alpha(s) - P_0) \cdot \mathbf{k} = 0.$$

Differentiating with respect to s yields $\alpha'(s) \cdot \mathbf{k} = 0$, which implies:

$$\mathbf{T}(s) \cdot \mathbf{k} = 0$$

Differentiating a second time gives $\mathbf{T}'(s) \cdot \mathbf{k} = 0$. Applying the Frenet relation $\mathbf{T}' = \kappa \mathbf{N}$:

$$\kappa(s) \mathbf{N}(s) \cdot \mathbf{k} = 0$$

Given $\kappa > 0$, it follows that $\mathbf{N}(s) \cdot \mathbf{k} = 0$. Since $\mathbf{k}$ is a constant unit vector orthogonal to both $\mathbf{T}$ and $\mathbf{N}$, the binormal vector $\mathbf{B} = \mathbf{T} \times \mathbf{N}$ must be:

$$\mathbf{B}(s) = \pm \mathbf{k}$$

Because $\mathbf{k}$ is constant, its derivative is zero, meaning $\mathbf{B}'(s) = \mathbf{0}$. Finally, using the Frenet-Serret formula $\mathbf{B}' = -\tau \mathbf{N}$, we have

$$-\tau(s) \mathbf{N}(s) = \mathbf{0} \implies \tau(s) = 0 \text{ for all } s.$$

Suppose $\tau = 0$ for all s . From $\mathbf{B}' = -\tau\mathbf{N}$, we have $\mathbf{B}' = \mathbf{0}$, which implies $\mathbf{B}$ is a constant vector, $\mathbf{B}_0$. Consider the scalar function $f(s) = (\alpha(s) - \alpha(s_0)) \cdot \mathbf{B}_0$. Its derivative is

$$f'(s) = \alpha'(s) \cdot \mathbf{B}_0 = \mathbf{T}(s) \cdot \mathbf{B}_0 = 0$$

Thus $f(s)$ is a constant. Since $f(s_0) = 0$, we conclude that $(\alpha(s) - \alpha(s_0)) \cdot \mathbf{B}_0 = 0$ for all s , which is the equation of a plane. ■

2nd Proof. Suppose the image of α lies in a plane with constant unit normal $\mathbf{u}$, meaning $(\alpha(s) - \mathbf{x}_0) \cdot \mathbf{u} = 0$. Differentiating once yields $\mathbf{T}(s) \cdot \mathbf{u} = 0$. Differentiating again yields $\mathbf{T}'(s) \cdot \mathbf{u} = \kappa(s)\mathbf{N}(s) \cdot \mathbf{u} = 0$. Since $\kappa(s) > 0$, we have $\mathbf{N}(s) \cdot \mathbf{u} = 0$. Because $\mathbf{u}$ is orthogonal to both $\mathbf{T}$ and $\mathbf{N}$, the binormal vector $\mathbf{B} = \mathbf{T} \times \mathbf{N}$ must be parallel to $\mathbf{u}$. Thus, $\mathbf{B}(s) = \pm\mathbf{u}$ is constant, meaning $\mathbf{B}'(s) = \mathbf{0}$. By the Frenet–Serret formula $\mathbf{B}'(s) = -\tau(s)\mathbf{N}(s)$, we conclude $\tau(s) = 0$ since $\mathbf{N}(s) \neq \mathbf{0}$.

If $\tau(s) = 0$, then $\mathbf{B}'(s) = -\tau(s)\mathbf{N}(s) = \mathbf{0}$, so $\mathbf{B}(s)$ is a constant vector $\mathbf{b}$. Consider the derivative:

$$\frac{d}{ds} [(\alpha(s) - \alpha(s_0)) \cdot \mathbf{b}] = \alpha'(s) \cdot \mathbf{b} = \mathbf{T}(s) \cdot \mathbf{B}(s) = 0$$

Integrating this expression yields $(\alpha(s) - \alpha(s_0)) \cdot \mathbf{b} = C$ for some constant C . Evaluating at $s = s_0$ gives $C = 0$. Therefore, $(\alpha(s) - \alpha(s_0)) \cdot \mathbf{b} = 0$ for all $s \in I$, which is the equation of a plane. ■

1.8 Characterization of Special Space Curves

In the differential geometry of space curves, the intrinsic properties—curvature (κ) and torsion (τ)—completely determine the shape of a curve up to a rigid motion. Certain geometric constraints, such as lying on a cylinder or a sphere, manifest as specific algebraic or differential relationships between κ and τ .

1.8.1 General Helices

A **general helix** is a curve whose tangent vector makes a constant angle with a fixed direction in space. (See Figure 11 in the Atlas.)

Theorem 1.8.1 (Lancret’s Theorem). *A regular curve $\alpha(s)$ with $\kappa(s) > 0$ is a general helix if and only if the ratio of its torsion to its curvature is constant:*

$$\frac{\tau(s)}{\kappa(s)} = c, \quad \text{for some constant } c.$$

Proof. Suppose α is a general helix. By definition, there exists a constant unit vector $\mathbf{u}$ and a constant angle θ such that:

$$\langle \mathbf{T}, \mathbf{u} \rangle = \cos \theta$$

Differentiating with respect to arc length s and using the Frenet formula $\mathbf{T}' = \kappa\mathbf{N}$:

$$\langle \kappa\mathbf{N}, \mathbf{u} \rangle = 0$$

Since $\kappa > 0$, we must have $\langle \mathbf{N}, \mathbf{u} \rangle = 0$. This implies that the fixed vector $\mathbf{u}$ lies in the plane spanned by $\mathbf{T}$ and $\mathbf{B}$ (the rectifying plane). We can write:

$$\mathbf{u} = \cos \theta \mathbf{T} + \sin \theta \mathbf{B}$$

Differentiating this constant vector $\mathbf{u}$ again:

$$\mathbf{0} = \cos \theta \mathbf{T}' + \sin \theta \mathbf{B}' = \cos \theta (\kappa\mathbf{N}) + \sin \theta (-\tau\mathbf{N}) = (\kappa \cos \theta - \tau \sin \theta) \mathbf{N}$$

Since $\mathbf{N} \neq \mathbf{0}$, the coefficient must be zero:

$$\kappa \cos \theta = \tau \sin \theta \implies \frac{\tau}{\kappa} = \cot \theta = \text{constant}.$$

Conversely, suppose $\kappa/\tau = c$, where c is a constant. Then $\kappa = c\tau$. Multiplying by $\mathbf{N}$:

$$\kappa\mathbf{N} = c\tau\mathbf{N}$$

This gives

$$\mathbf{T}' = -c\mathbf{B}' \quad (\text{since } \mathbf{T}' = \kappa\mathbf{N} \text{ and } \mathbf{B}' = -\tau\mathbf{N})$$

or,

$$\mathbf{T}' + c\mathbf{B}' = 0.$$

This implies

$$\frac{d}{ds}(\mathbf{T} + c\mathbf{B}) = 0,$$

which further implies $\mathbf{T} + c\mathbf{B} = \mathbf{a}$, where $\mathbf{a}$ is a constant vector. Taking the dot product of both sides with $\mathbf{T}$:

$$\begin{aligned} \mathbf{T} \cdot (\mathbf{T} + c\mathbf{B}) &= \mathbf{a} \cdot \mathbf{T} \\ 1 + 0 &= |\mathbf{a}| |\mathbf{T}| \cos \theta \\ 1 &= |\mathbf{a}| \cos \theta \\ \implies \cos \theta &= \frac{1}{|\mathbf{a}|} = \text{constant} \end{aligned}$$

where θ is the angle between $\mathbf{T}$ and $\mathbf{a}$. Since the tangent vector $\mathbf{T}$ of the curve α makes a constant angle with the fixed vector $\mathbf{a}$, the curve α is a helix. ■

A curve is called **spherical** if it lies entirely on the surface of a sphere of radius R centered at some point $\mathbf{C}$.

Theorem 1.8.2 (Characterization of Spherical Curves). *(See Figure 12 in the Atlas.) A curve $\alpha(s)$ with $\kappa > 0$ and $\tau \neq 0$ lies on a sphere if and only if:*

$$\frac{\tau}{\kappa} + \frac{d}{ds} \left(\frac{1}{\tau} \frac{d}{ds} \left(\frac{1}{\kappa} \right) \right) = 0$$

Alternatively, in terms of the radius of curvature $\rho = 1/\kappa$ and the radius of torsion $\sigma = 1/\tau$, the condition is $\rho^2 + (\sigma\rho')^2 = R^2$ (a constant).

Proof. Suppose $\alpha(s)$ lies on a sphere of radius R centered at $\mathbf{C}$. Then $\|\alpha(s) - \mathbf{C}\|^2 = R^2$. Differentiating three times with respect to s :

1. $2\langle \alpha - \mathbf{C}, T \rangle = 0 \implies \alpha - \mathbf{C}$ is in the NB -plane.
2. $\langle T, T \rangle + \langle \alpha - \mathbf{C}, \kappa N \rangle = 0 \implies 1 + \kappa \langle \alpha - \mathbf{C}, N \rangle = 0 \implies \langle \alpha - \mathbf{C}, N \rangle = -1/\kappa$.
3. Differentiating $\langle \alpha - \mathbf{C}, N \rangle = -1/\kappa$:

$$\langle T, N \rangle + \langle \alpha - \mathbf{C}, -\kappa T + \tau B \rangle = (1/\kappa)'$$

$$0 + 0 + \tau \langle \alpha - \mathbf{C}, B \rangle = (1/\kappa)' \implies \langle \alpha - \mathbf{C}, B \rangle = \frac{1}{\tau} (1/\kappa)'$$

Thus, we can express the vector from the center to the curve as:

$$\alpha - \mathbf{C} = 0T - \frac{1}{\kappa}N + \frac{1}{\tau} \left(\frac{1}{\kappa} \right)' B$$

Since the distance $\|\alpha - \mathbf{C}\|$ is the constant radius R , we have

$$\left(\frac{1}{\kappa} \right)^2 + \left[\frac{1}{\tau} \left(\frac{1}{\kappa} \right)' \right]^2 = R^2$$

Differentiating this constant expression with respect to s yields the differential equation mentioned in the theorem. ■

Example 1.8.3 (The Circular Helix). Show that the circular helix $\alpha(t) = (a \cos t, a \sin t, bt)$ is a general helix using Lancret's Theorem.

Solution. From previous calculations for the circular helix, we found:

$$\kappa = \frac{a}{a^2 + b^2} \quad \text{and} \quad \tau = \frac{b}{a^2 + b^2}$$

Both κ and τ are constants. Therefore, their ratio is

$$\frac{\tau}{\kappa} = \frac{b/(a^2 + b^2)}{a/(a^2 + b^2)} = \frac{b}{a}$$

Since a and b are constants, the ratio τ/κ is constant. By Lancret's Theorem, the circular helix is a general helix. (Note: Because κ and τ are both constant, it is a very specific type of helix that lies on a circular cylinder). ■

Example 1.8.4 (A Non-Spherical Curve). Determine if a curve with constant curvature $\kappa = 1$ and constant torsion $\tau = 1$ can lie on a sphere.

Solution. Let us check the spherical condition: $\rho^2 + (\sigma\rho')^2 = R^2$. Given $\kappa = 1$, we have $\rho = 1/\kappa = 1$. The derivative is $\rho' = \frac{d}{ds}(1) = 0$. Substituting into the formula:

$$(1)^2 + (\sigma \cdot 0)^2 = 1$$

This results in $1 = R^2$, which is a constant. This suggests the curve *could* be spherical. However, we must also check the second derivative condition:

$$\frac{\tau}{\kappa} + \frac{d}{ds}(\sigma\rho') = 0$$

$$\frac{1}{1} + \frac{d}{ds}(1 \cdot 0) = 1 + 0 = 1$$

Since $1 \neq 0$, the differential equation for a spherical curve is not satisfied. A curve with constant κ and τ (which is a circular helix) does *not* lie on a sphere. It lies on a cylinder. ■

1.9 Involutives and Evolutes

The geometric relationship between a curve and its "unwrapped" counterpart is described through the concepts of involutes and evolutes. Originally developed by Christiaan Huygens for the design of isochronous pendulum clocks, these constructions remain a beautiful application of the Frenet–Serret frame.

1.9.1 The Involute of a Curve

Intuitively, an involute is the path traced by the end of a taut string as it is unwrapped from a curve.

Definition 1.9.1 (Involute). (See Figure 14 in the Atlas.) Let $\alpha : I \rightarrow \mathbb{R}^2$ be a unit-speed planar curve. The curve $\bar{\alpha}$ defined by

$$\bar{\alpha}(s) = \alpha(s) + (c - s)\mathbf{T}(s)$$

for a constant c , is called an **involute** of α .

The term $(c - s)$ represents the length of the "unwrapped" portion of the string at arc length s . The following theorem establishes the fundamental geometric link between the two curves.

Theorem 1.9.2. *The tangent line to the original curve α at any point s is the normal line to its involute $\bar{\alpha}$ at the corresponding point.*

Proof. Differentiating the expression for $\bar{\alpha}(s)$ with respect to s , we obtain

$$\bar{\alpha}'(s) = \alpha'(s) - \mathbf{T}(s) + (c - s)\mathbf{T}'(s)$$

Since α is unit-speed, $\alpha' = \mathbf{T}$. Applying the Frenet formula $\mathbf{T}' = \kappa\mathbf{N}$, we have

$$\bar{\alpha}'(s) = \mathbf{T}(s) - \mathbf{T}(s) + (c - s)\kappa\mathbf{N}(s) = (c - s)\kappa\mathbf{N}(s)$$

This shows that the velocity vector $\bar{\alpha}'(s)$ is parallel to the normal vector $\mathbf{N}(s)$ of the original curve. Consequently, the tangent of the involute is perpendicular to the tangent of the original curve: $\bar{\alpha}'(s) \cdot \mathbf{T}(s) = 0$. ■

1.9.2 The Evolute of a Curve

While the involute is formed by unwrapping a string, the evolute is the "source" curve from which such a string would be unwrapped.

Definition 1.9.3 (Evolute). (See Figure 13 in the Atlas.) Let $\alpha(s)$ be a regular planar curve with curvature $\kappa(s) \neq 0$. The **evolute** $\beta(s)$ of α is the locus of the centers of curvature of α . It is given by $\beta(s) = \alpha(s) + \rho(s)\mathbf{N}(s)$, where $\rho(s) = 1/\kappa(s)$ is the radius of curvature.

There is a reciprocal relationship between these two constructions: a curve β is the evolute of α if and only if α is an involute of β .

Theorem 1.9.4 (Tangents of the Evolute). *The tangent vector to the evolute $\beta(s)$ is always parallel to the principal normal vector $\mathbf{N}(s)$ of the original curve.*

Proof. Differentiating $\beta(s) = \alpha(s) + \rho(s)\mathbf{N}(s)$ with respect to arc length s :

$$\beta'(s) = \alpha'(s) + \rho'(s)\mathbf{N}(s) + \rho(s)\mathbf{N}'(s)$$

Substituting the Frenet–Serret relations $\alpha' = \mathbf{T}$ and $\mathbf{N}' = -\kappa\mathbf{T}$ (noting that $\tau = 0$ for planar curves):

$$\beta'(s) = \mathbf{T}(s) + \rho'(s)\mathbf{N}(s) + \rho(s)(-\kappa(s)\mathbf{T}(s))$$

$$\beta'(s) = \mathbf{T}(s) + \rho'(s)\mathbf{N}(s) - (\rho(s)\kappa(s))\mathbf{T}(s)$$

Using the property $\rho = 1/\kappa$, the tangent terms cancel:

$$\beta'(s) = \mathbf{T}(s) + \rho'(s)\mathbf{N}(s) - \mathbf{T}(s) = \rho'(s)\mathbf{N}(s)$$

Thus, the velocity of the evolute $\beta'(s)$ is a scalar multiple of the normal vector $\mathbf{N}(s)$. This confirms that the tangent to the evolute is precisely the normal to the original curve. ■

Summary

- To find an **Involute**, we move away from the curve along its **Tangent** vector.
- To find the **Evolute**, we move away from the curve along its **Normal** vector.

1.10 The Fundamental Theorem of Space Curves

In the derivation of the *Fundamental Theorem of Space Curves*, we encounter a system of linear differential equations of the form $\mathbf{F}'(s) = \mathbf{A}(s)\mathbf{F}(s)$. For such linear systems, if the coefficients (the curvature κ and torsion τ) are continuous on an interval $[0, L]$, the Lipschitz condition is automatically satisfied on any compact sub-interval. Consequently, the Picard–Lindelöf Theorem guarantees that once an initial orientation (frame) is chosen, the entire geometry of the curve is uniquely determined for all s . The Picard–Lindelöf Theorem provides the sufficient conditions under which a first-order initial value problem (IVP) is guaranteed to have a unique solution in a local neighborhood of the initial point.

Theorem: [Picard–Lindelöf] Let $D \subseteq \mathbb{R} \times \mathbb{R}^n$ be an open set, and consider the initial value problem:

$$\frac{dy}{dt} = \mathbf{f}(t, \mathbf{y}), \quad \mathbf{y}(t_0) = \mathbf{y}_0$$

Suppose that the function $\mathbf{f}(t, \mathbf{y})$ satisfies the following two conditions:

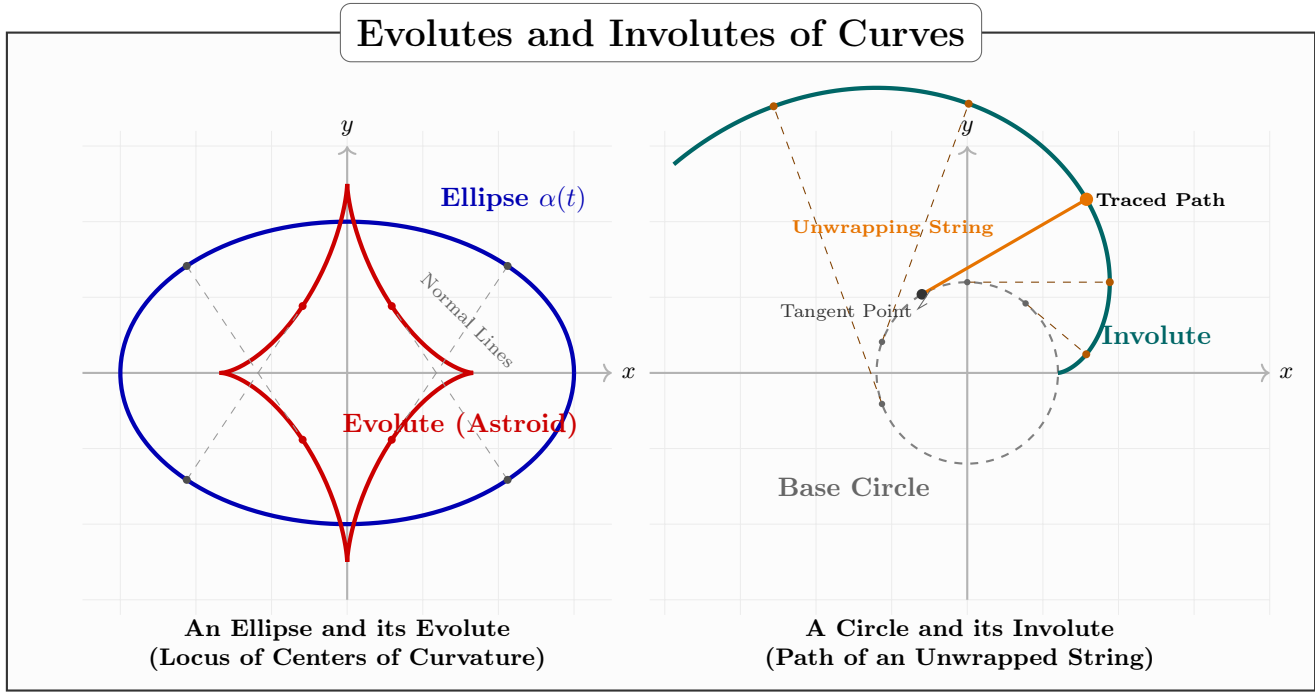


Figure 1.3: Caption

1. $\mathbf{f}$ is continuous in t and $\mathbf{y}$ on a closed rectangle $R \subset D$ centered at $(t_0, \mathbf{y}_0)$.
2. $\mathbf{f}$ is **Lipschitz continuous** in $\mathbf{y}$ on R . That is, there exists a constant $L \geq 0$ such that for all $(t, \mathbf{y}_1)$ and $(t, \mathbf{y}_2)$ in R :

$$\|\mathbf{f}(t, \mathbf{y}_1) - \mathbf{f}(t, \mathbf{y}_2)\| \leq L \|\mathbf{y}_1 - \mathbf{y}_2\|$$

Then, there exists some $\epsilon > 0$ such that the initial value problem has a **unique solution** $\mathbf{y}(t)$ on the interval $[t_0 - \epsilon, t_0 + \epsilon]$. The local geometry of a space curve is completely characterized by two scalar functions: its curvature $\kappa(s)$ and its torsion $\tau(s)$. The following theorem, often referred to as the existence and uniqueness theorem for space curves, asserts that these functions serve as a complete set of invariants.

Theorem 1.10.1 (Fundamental Theorem of Space Curves). *Let $\kappa(s)$ and $\tau(s)$ be continuous functions on an interval $I = [0, L]$, with $\kappa(s) > 0$ for all $s \in I$. There exists a unit-speed curve $\alpha : I \rightarrow \mathbb{R}^3$ having $\kappa(s)$ as its curvature and $\tau(s)$ as its torsion. Furthermore, any other curve $\tilde{\alpha}$ satisfying these conditions is congruent to α ; that is, they differ only by a rigid motion of $\mathbb{R}^3$.*

Proof. To establish the existence of such a curve, we first consider the Frenet-Serret equations as a system of first-order linear ordinary differential equations for three vector-valued functions $\mathbf{T}$, $\mathbf{N}$, and $\mathbf{B}$:

$$\frac{d}{ds} \begin{pmatrix} \mathbf{T} \\ \mathbf{N} \\ \mathbf{B} \end{pmatrix} = \begin{pmatrix} 0 & \kappa(s) & 0 \\ -\kappa(s) & 0 & \tau(s) \\ 0 & -\tau(s) & 0 \end{pmatrix} \begin{pmatrix} \mathbf{T} \\ \mathbf{N} \\ \mathbf{B} \end{pmatrix}$$

Let $A(s)$ be the skew-symmetric matrix in the system above. By the Picard-Lindelöf theorem, given any initial right-handed orthonormal basis $\{\mathbf{T}_0, \mathbf{N}_0, \mathbf{B}_0\}$ at $s = 0$, there exists a unique solution for the frame $\{\mathbf{T}(s), \mathbf{N}(s), \mathbf{B}(s)\}$ for all $s \in I$.

We must verify that this solution remains orthonormal for all s . Let $M(s)$ be the matrix whose rows are the vectors $\mathbf{T}$, $\mathbf{N}$, and $\mathbf{B}$. The derivative of the product MM^T is given by $(MM^T)' = M'M^T + M(M')^T$. Substituting $M' = AM$, and noting that A is skew-symmetric ($A^T = -A$), we find $(MM^T)' = AMM^T - MM^TA$. Since MM^T equals the identity matrix I at $s = 0$, and I is a stationary solution to this equation, the uniqueness of solutions to differential equations ensures that $MM^T = I$ for all s . Thus, the vectors remain unit-length and mutually orthogonal. We then define the required

curve by integrating the unit tangent vector, $\alpha(s) = \int_0^s \mathbf{T}(u)du + \mathbf{P}_0$, which by construction possesses the prescribed curvature and torsion.

To prove uniqueness, suppose there exist two curves α and $\tilde{\alpha}$ sharing the same curvature and torsion functions. By applying a rigid motion to $\tilde{\alpha}$, we may assume without loss of generality that $\alpha(0) = \tilde{\alpha}(0)$ and that their initial Frenet frames $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ and $\{\tilde{\mathbf{T}}, \tilde{\mathbf{N}}, \tilde{\mathbf{B}}\}$ coincide at $s = 0$. We define the scalar-valued alignment function:

$$f(s) = \langle \mathbf{T}, \tilde{\mathbf{T}} \rangle + \langle \mathbf{N}, \tilde{\mathbf{N}} \rangle + \langle \mathbf{B}, \tilde{\mathbf{B}} \rangle$$

Differentiating $f(s)$ and applying the Frenet–Serret formulas to both frames, we obtain

$$\begin{aligned} f'(s) &= \langle \kappa \mathbf{N}, \tilde{\mathbf{T}} \rangle + \langle \mathbf{T}, \kappa \tilde{\mathbf{N}} \rangle + \langle -\kappa \mathbf{T} + \tau \mathbf{B}, \tilde{\mathbf{N}} \rangle + \langle \mathbf{N}, -\kappa \tilde{\mathbf{T}} + \tau \tilde{\mathbf{B}} \rangle + \langle -\tau \mathbf{N}, \tilde{\mathbf{B}} \rangle + \langle \mathbf{B}, -\tau \tilde{\mathbf{N}} \rangle \\ &= \kappa \langle \mathbf{N}, \tilde{\mathbf{T}} \rangle + \kappa \langle \mathbf{T}, \tilde{\mathbf{N}} \rangle - \kappa \langle \mathbf{T}, \tilde{\mathbf{N}} \rangle + \tau \langle \mathbf{B}, \tilde{\mathbf{N}} \rangle - \kappa \langle \mathbf{N}, \tilde{\mathbf{T}} \rangle + \tau \langle \mathbf{N}, \tilde{\mathbf{B}} \rangle - \tau \langle \mathbf{N}, \tilde{\mathbf{B}} \rangle - \tau \langle \mathbf{B}, \tilde{\mathbf{N}} \rangle \end{aligned}$$

Observing that all terms cancel, we conclude $f'(s) = 0$, implying $f(s)$ is constant. Since the frames were identical at the origin, $f(0) = 1 + 1 + 1 = 3$. Because the inner product of two unit vectors is at most 1, the only way for their sum to reach 3 is if $\langle \mathbf{T}, \tilde{\mathbf{T}} \rangle = 1$, $\langle \mathbf{N}, \tilde{\mathbf{N}} \rangle = 1$, and $\langle \mathbf{B}, \tilde{\mathbf{B}} \rangle = 1$. This implies that $\mathbf{T}(s) = \tilde{\mathbf{T}}(s)$ for all s . Integrating this equality, and noting that the curves share the same initial point, it follows that $\alpha(s) = \tilde{\alpha}(s)$, completing the proof. ■

1.11 Global Properties of Curves

While local differential geometry focuses on the behavior of a curve in the neighborhood of a point (such as curvature and torsion), global differential geometry examines the curve as a whole. Here, we consider properties that depend on the entire trace of the curve and its topological nature.

A curve that returns to its starting point and does not intersect itself is a fundamental object in global geometry.

Definition 1.11.1 (Simple Closed Curve). A parametrized curve $\alpha : [a, b] \rightarrow \mathbb{R}^n$ is said to be **closed** if $\alpha(a) = \alpha(b)$ and all its derivatives match at the endpoints; that is, $\alpha^{(k)}(a) = \alpha^{(k)}(b)$ for all $k \geq 1$. A closed curve is **simple** if it is injective on the half-open interval $[a, b)$.

Remark 1.11.2 (The Jordan Curve Theorem). A famous result by Camille Jordan states that every simple closed curve in the plane divides $\mathbb{R}^2$ into exactly two disjoint regions: a bounded "interior" and an unbounded "exterior." Although intuitively obvious, the proof is famously non-trivial and forms a cornerstone of topology.

1.11.1 The Isoperimetric Inequality

Green's Theorem establishes a fundamental relationship between a line integral around a simple closed curve and a double integral over the plane region enclosed by the curve.

Theorem 1.11.3 (Green's Theorem). Let C be a positively oriented, piecewise smooth, simple closed curve in the plane, and let D be the region bounded by C . If $P(x, y)$ and $Q(x, y)$ have continuous partial derivatives on an open region containing D , then:

$$\oint_C (P dx + Q dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

In the context of the Isoperimetric Inequality, we are interested in the case where the double integral represents the area A of the region D . By choosing specific functions for P and Q such that $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$, we obtain several equivalent formulas for the area:

1. Let $P = 0$ and $Q = x$. Then $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 - 0 = 1$. Thus

$$A = \oint_C x dy$$

Proof. Consider Green's Theorem in its general form, which relates a line integral around a simple closed curve C to a double integral over the region D it encloses:

$$\oint_C (P dx + Q dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

To derive the specific area formula $A = \oint_C x dy$, we define the functions $P(x, y) = 0$ and $Q(x, y) = x$. We then evaluate the required partial derivatives:

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(x) = 1, \quad \frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(0) = 0$$

Substituting these derivatives into the right-hand side of the theorem yields $\iint_D (1 - 0) dA = \iint_D 1 dA = A$, where A represents the area of the region D . Substituting the definitions of P and Q into the left-hand side line integral gives $\oint_C (0 \cdot dx + x \cdot dy) = \oint_C x dy$. Equating the simplified left-hand and right-hand sides, we arrive at the area identity:

$$A = \oint_C x dy$$

This demonstrates that the area of a region D can be calculated by integrating the x -coordinate with respect to y along its boundary C . ■

2. Let $P = -y$ and $Q = 0$. Then $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0 - (-1) = 1$. Thus

$$A = - \oint_C y dx$$

3. Let $P = -\frac{1}{2}y$ and $Q = \frac{1}{2}x$. Then $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{1}{2} - (-\frac{1}{2}) = 1$. Thus

$$A = \frac{1}{2} \oint_C (x dy - y dx)$$

One of the oldest problems in geometry is determining which closed curve of a fixed length encloses the maximum area. The answer is provided by the Isoperimetric Inequality.

Theorem 1.11.4 (Isoperimetric Inequality). (See Figure 16 in the Atlas.) If L is the length of a simple closed plane curve α and A is the area of the region it encloses, then

$$L^2 \geq 4\pi A$$

Furthermore, equality holds if and only if the curve is a circle.

Proof. Let $\alpha(s) = (x(s), y(s))$ be a simple closed curve parameterized by arc length $s \in [0, L]$. We begin by placing the curve between two parallel lines, say $x = -r$ and $x = r$, such that the lines are tangent to the curve and the curve lies entirely within the strip they define. This implies that the maximum width of the curve in the x -direction is $2r$.

Consider a circle γ of radius r centered at the origin, which can be represented as $\gamma(s) = (x(s), \bar{y}(s))$. Observe that while $x(s)$ is the same coordinate function as in $\alpha(s)$, the function $\bar{y}(s)$ is defined such that $(x(s), \bar{y}(s))$ lies on the circle, meaning $x(s)^2 + \bar{y}(s)^2 = r^2$. Using Green's Theorem, the area A enclosed by α and the area of the circle can be expressed as

$$A = \int_0^L x(s)y'(s) ds \quad \text{and} \quad \pi r^2 = - \int_0^L \bar{y}(s)x'(s) ds$$

Adding these two integrals allows us to apply the Cauchy-Schwarz inequality to the integrand. We have

$$A + \pi r^2 = \int_0^L (xy' - \bar{y}x') ds = \int_0^L \langle (x, -\bar{y}), (y', x') \rangle ds$$

By the Cauchy-Schwarz inequality, $\langle (x, -\bar{y}), (y', x') \rangle \leq \sqrt{x^2 + \bar{y}^2} \sqrt{(y')^2 + (x')^2}$. Since $(x, \bar{y})$ lies on a circle of radius r , we know $\sqrt{x^2 + \bar{y}^2} = r$. Furthermore, since α is a unit-speed curve, $\sqrt{(x')^2 + (y')^2} = 1$. Substituting these values into the integral yields $A + \pi r^2 \leq \int_0^L r \cdot 1 \, ds = Lr$. Using the inequality between the arithmetic and geometric means, specifically that $2\sqrt{A \cdot \pi r^2} \leq A + \pi r^2$, we arrive at:

$$2\sqrt{A\pi r^2} \leq Lr$$

Squaring both sides gives $4\pi Ar^2 \leq L^2 r^2$, and dividing by r^2 (which is non-zero) results in the desired isoperimetric inequality:

$$4\pi A \leq L^2$$

The condition for equality $L^2 = 4\pi A$ requires that $A = \pi r^2$ and that the Cauchy-Schwarz inequality holds as an equality at every point. This occurs only if the vectors $(x, -\bar{y})$ and (y', x') are proportional. Specifically, for the magnitudes to match, we must have $x = ry'$ and $-\bar{y} = rx'$. Squaring and adding these relations, we find $x^2 + \bar{y}^2 = r^2((y')^2 + (x')^2)$, which simplifies to $r^2 = r^2(1)$. This system of differential equations characterizes a circle of radius r , thus proving that the circle is the unique minimizer of the perimeter for a fixed area. ■

1.11.2 The Four Vertex Theorem

To understand how the curvature of a closed curve must fluctuate, we define the concept of a vertex.

Definition 1.11.5 (Vertex). A point $\alpha(s)$ on a smooth curve is called a **vertex** if the derivative of its curvature vanishes at that point, i.e., $\kappa'(s) = 0$.

Geometrically, a vertex is a point where the curvature reaches a local maximum or minimum. For example, an ellipse (that is not a circle) has exactly four vertices: two at the ends of the major axis and two at the ends of the minor axis.

Theorem 1.11.6 (Four Vertex Theorem). (See Figure 15 in the Atlas.) Every simple closed convex plane curve has at least four vertices.

Proof. Let $\alpha(s) = (x(s), y(s))$ be a simple closed convex curve parameterized by arc length $s \in [0, L]$. A vertex is defined as a point where the derivative of the curvature vanishes, $\kappa'(s) = 0$. Since $\kappa(s)$ is a continuous periodic function on the compact interval $[0, L]$, the Extreme Value Theorem guarantees that $\kappa(s)$ must attain both a global maximum and a global minimum. These two points provide at least two distinct vertices on the curve.

We proceed by contradiction. Suppose that α possesses only two vertices, occurring at s_1 and s_2 . Let $L(x, y) = ax + by + c = 0$ be the equation of the line passing through the points $P = \alpha(s_1)$ and $Q = \alpha(s_2)$. Because the curve is convex and simple, this line divides the curve into two arcs such that the linear function $f(s) = ax(s) + by(s) + c$ changes sign only at s_1 and s_2 . Furthermore, since s_1 and s_2 are the only relative extrema of the curvature, the derivative $\kappa'(s)$ also changes sign only at these two points.

Consider the integral of the product of these functions over the entire curve:

$$I = \int_0^L \kappa'(s) f(s) \, ds = \int_0^L \kappa'(s) (ax(s) + by(s) + c) \, ds$$

Because both $\kappa'(s)$ and $f(s)$ change sign at exactly the same points, their product $\kappa'(s)f(s)$ is either non-negative or non-positive throughout the interval $[0, L]$. Since the curvature is not constant for a non-circular convex curve, the product is not identically zero, and thus the integral I must be non-zero.

However, we can evaluate this integral using integration by parts. First, observe that for any constant c :

$$\int_0^L \kappa'(s) c \, ds = c[\kappa(L) - \kappa(0)] = 0$$

Next examine the terms involving the coordinates. Using the Frenet–Serret relations, specifically that $x'(s) = \cos \theta(s)$ and $y'(s) = \sin \theta(s)$ with $\theta'(s) = \kappa(s)$, we have

$$\int_0^L \kappa(s)x'(s) ds = \int_0^L \theta'(s) \cos \theta(s) ds = [\sin \theta(s)]_0^L = 0$$

By integration by parts, $\int \kappa' x ds = \kappa x - \int \kappa x' ds$. For a closed curve, the boundary term vanishes, yielding:

$$\int_0^L \kappa'(s)x(s) ds = - \int_0^L \kappa(s)x'(s) ds = 0$$

Similarly, $\int_0^L \kappa'(s)y(s) ds = 0$. This implies that $I = 0$, which contradicts our earlier finding that I must be non-zero. Therefore, the assumption that there are only two vertices is false. Since the sign changes of the derivative of a periodic function must occur in even pairs, there must be at least four points where $\kappa'(s) = 0$, concluding the proof. ■

Remark 1.11.7. It is worth noting that the Four Vertex Theorem was originally proven for convex curves, but was later shown to hold for all simple closed curves. This theorem highlights a deep connection between the local analytic property of a derivative (κ') and the global topological property of being a closed loop.

The Circle as a Degenerate Case

For a circle of radius R , the curvature is constant at every point along its perimeter:

$$\kappa(s) = \frac{1}{R} \quad \text{for all } s$$

Consequently, the derivative of the curvature with respect to arc length is zero everywhere:

$$\frac{d\kappa}{ds} = 0 \quad \text{for all } s$$

Because the curvature is constant, every single point on a circle satisfies the definition of a local extremum (albeit a non-strict one). Thus, a circle possesses **infinitely many vertices**.

While this infinite nature makes the circle a degenerate case, it trivially satisfies the theorem's lower bound of “at least four” vertices.

Deforming the Circle (Geometric Intuition)

The physical and geometric significance of the theorem becomes clearer when the circle is deformed, breaking its perfect symmetry:

- **The Ellipse:** Stretches a circle into an ellipse, collapsing the infinite set of vertices into exactly four distinct, strict local extrema:
 - **Two local maxima** of curvature at the vertices of the major axis (where the curve curves most sharply).
 - **Two local minima** of curvature at the vertices of the minor axis (where the curve is flattest).
- **General Smooth Deformations:** If a circle is deformed in any arbitrary, smooth, closed way, the infinite vertices cannot simply vanish. Because the curve must close back up on itself, the curvature is forced to rise and fall in a balanced manner. This topological constraint guarantees that the curve will always be left with at least two local maxima and two local minima—resulting in a minimum of four vertices.

Chapter 2

Differential Geometry of Surfaces

2.1 Introduction

The study of surfaces in three-dimensional Euclidean space forms a central part of classical differential geometry. Intuitively, a surface is a two-dimensional object that is smooth enough to admit a tangent plane at each point. The precise mathematical formulation is the concept of a *regular surface*. This notion excludes objects with sharp edges, cusps, or self-intersections, and allows us to apply the tools of calculus.

Definition 2.1.1. A subset $S \subset \mathbb{R}^3$ is called a **regular surface** if for every point $p \in S$ there exist an open neighbourhood $V \subset \mathbb{R}^3$ containing p , an open set $U \subset \mathbb{R}^2$, and a map $\mathbf{x} : U \rightarrow S \cap V$ such that:

1. $\mathbf{x}$ is **smooth** (i.e., C^∞);
2. $\mathbf{x}$ is a **homeomorphism** onto its image (continuous, bijective, with continuous inverse);
3. The differential $d\mathbf{x}_q : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is **injective** for every $q \in U$. Equivalently, the partial derivatives $\mathbf{x}_u = \frac{\partial \mathbf{x}}{\partial u}$ and $\mathbf{x}_v = \frac{\partial \mathbf{x}}{\partial v}$ are linearly independent at each point.

The map $\mathbf{x}$ is called a **local parametrization** (or a **coordinate chart**) of S .

The third condition is the *regularity condition*. It guarantees that the image of $\mathbf{x}$ is not “folded” or “pinched”. In practice, we often write $\mathbf{x}(u, v) = (x(u, v), y(u, v), z(u, v))$, and the linear independence of $\mathbf{x}_u$ and $\mathbf{x}_v$ is equivalent to

$$\mathbf{x}_u \times \mathbf{x}_v \neq \mathbf{0}.$$

Remark 2.1.2. A regular surface is sometimes called a “smooth surface” or “embedded C^∞ surface” in $\mathbb{R}^3$. The definition ensures that each point has a neighbourhood that is diffeomorphic to an open set in $\mathbb{R}^2$.

We now examine several fundamental examples of regular surfaces.

Example 2.1.3 (The Plane). (See Figure 17 in the Atlas.) The simplest example is any plane in $\mathbb{R}^3$. For the xy -plane,

$$\mathbf{x}(u, v) = (u, v, 0), \quad (u, v) \in \mathbb{R}^2.$$

Here $\mathbf{x}_u = (1, 0, 0)$, $\mathbf{x}_v = (0, 1, 0)$, which are clearly linearly independent. Hence the whole plane is a regular surface.

Example 2.1.4 (Graph of a Smooth Function). (See Figure 18 in the Atlas.) Let $f : U \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ be a C^∞ function defined on an open set U . Then its graph

$$S = \{(x, y, f(x, y)) \mid (x, y) \in U\}$$

is a regular surface. A natural parametrization is $\mathbf{x}(x, y) = (x, y, f(x, y))$. Then $\mathbf{x}_x = (1, 0, f_x)$ and $\mathbf{x}_y = (0, 1, f_y)$. Their cross product is $(-f_x, -f_y, 1)$, which never vanishes. Therefore the graph is regular.

Example 2.1.5 (The Sphere). (See Figure 19 in the Atlas.) The sphere of radius $R > 0$,

$$S^2(R) = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = R^2\},$$

is a regular surface. It cannot be covered by a single parametrization (any such attempt would fail at least at one pole), but it can be covered by two charts. One standard parametrization uses spherical coordinates (with the poles omitted):

$$\mathbf{x}(\theta, \phi) = (R \sin \theta \cos \phi, R \sin \theta \sin \phi, R \cos \theta),$$

where $0 < \theta < \pi$ and $0 < \phi < 2\pi$. The partial derivatives are

$$\mathbf{x}_\theta = (R \cos \theta \cos \phi, R \cos \theta \sin \phi, -R \sin \theta),$$

$$\mathbf{x}_\phi = (-R \sin \theta \sin \phi, R \sin \theta \cos \phi, 0).$$

Their cross product is

$$\mathbf{x}_\theta \times \mathbf{x}_\phi = R^2 \sin \theta (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta),$$

which is non-zero because $\sin \theta \neq 0$ in the given range. To cover the north pole ($\theta = 0$) and south pole ($\theta = \pi$), we can use another chart, e.g., the stereographic projection. Hence the sphere is regular.

Example 2.1.6 (Circular Cylinder). (See Figure 20 in the Atlas.) The cylinder

$$C = \{(x, y, z) \mid x^2 + y^2 = R^2\}, \quad R > 0,$$

is a regular surface. A parametrization is

$$\mathbf{x}(\theta, z) = (R \cos \theta, R \sin \theta, z), \quad \theta \in (0, 2\pi), \quad z \in \mathbb{R}.$$

Then

$$\mathbf{x}_\theta = (-R \sin \theta, R \cos \theta, 0), \quad \mathbf{x}_z = (0, 0, 1).$$

Their cross product is $(R \cos \theta, R \sin \theta, 0)$, which is never zero. The whole cylinder is covered by two such charts (e.g., using $\theta \in (0, 2\pi)$ and $\theta \in (\pi, 3\pi)$) to avoid boundaries.

Example 2.1.7 (Torus). (See Figure 21 in the Atlas.) The torus T^2 (surface of a doughnut) is a regular surface. Its standard parametrization is

$$\mathbf{x}(\theta, \phi) = ((R + r \cos \theta) \cos \phi, (R + r \cos \theta) \sin \phi, r \sin \theta),$$

with $0 < r < R$ and $\theta, \phi \in (0, 2\pi)$. Compute

$$\mathbf{x}_\theta = (-r \sin \theta \cos \phi, -r \sin \theta \sin \phi, r \cos \theta),$$

$$\mathbf{x}_\phi = (-(R + r \cos \theta) \sin \phi, (R + r \cos \theta) \cos \phi, 0).$$

The cross product is

$$\mathbf{x}_\theta \times \mathbf{x}_\phi = r(R + r \cos \theta) (\cos \theta \cos \phi, \cos \theta \sin \phi, -\sin \theta).$$

Because $r > 0$ and $R + r \cos \theta > R - r > 0$, the cross product never vanishes. Thus the torus is regular.

Example 2.1.8 (Surfaces of Revolution). (See Figure 22 in the Atlas.) Let $\alpha(u) = (f(u), 0, g(u))$, $u \in I \subset \mathbb{R}$, be a regular curve in the xz -plane with $f(u) > 0$ (so the curve does not touch the z -axis). Rotating α around the z -axis yields a surface of revolution:

$$\mathbf{x}(u, \theta) = (f(u) \cos \theta, f(u) \sin \theta, g(u)), \quad u \in I, \quad \theta \in (0, 2\pi).$$

Then

$$\mathbf{x}_u = (f'(u) \cos \theta, f'(u) \sin \theta, g'(u)), \quad \mathbf{x}_\theta = (-f(u) \sin \theta, f(u) \cos \theta, 0).$$

Their cross product is

$$\mathbf{x}_u \times \mathbf{x}_\theta = f(u) (g'(u) \cos \theta, g'(u) \sin \theta, -f'(u)).$$

Since $f(u) > 0$ and $(f'(u), g'(u)) \neq (0, 0)$ (the curve is regular), the cross product never vanishes. Hence any surface of revolution generated by a regular profile curve with $f > 0$ is regular.

Example 2.1.9 (Enneper's Surface). (See Figure 23 in the Atlas.) Enneper's surface given by

$$\mathbf{x}(u, v) = \left(u - \frac{u^3}{3} + uv^2, v - \frac{v^3}{3} + u^2v, u^2 - v^2 \right).$$

One can compute

$$\mathbf{x}_u = (1 - u^2 + v^2, 2uv, 2u), \quad \mathbf{x}_v = (2uv, 1 - v^2 + u^2, -2v).$$

Their cross product becomes $\mathbf{x}_u \times \mathbf{x}_v = (-2u(1 + u^2 + v^2), -2v(1 + u^2 + v^2), 1 - (u^2 + v^2)^2)$. This vector is zero only when $u = v = 0$. Hence Enneper's surface is regular on $\mathbb{R}^2 \setminus \{(0, 0)\}$, and the origin corresponds to a self-intersection. This shows that regularity can fail at isolated points.

In the following, we assume $S \subset \mathbb{R}^3$ is a regular surface and that local parametrizations have the usual properties (smooth, bijective, with linearly independent partial derivatives).

Theorem 2.1.10 (Tangent plane). (See Figure 32 in the Atlas.) Let S be a regular surface and let $\mathbf{x} : U \subset \mathbb{R}^2 \rightarrow S$ be a local parametrization with $p = \mathbf{x}(u_0, v_0)$. Then the set of all tangent vectors to curves on S passing through p (denoted $T_p S$) is the two-dimensional subspace of $\mathbb{R}^3$ spanned by the partial derivatives $\mathbf{x}_u(u_0, v_0)$ and $\mathbf{x}_v(u_0, v_0)$:

$$T_p S = \{a \mathbf{x}_u(u_0, v_0) + b \mathbf{x}_v(u_0, v_0) \mid a, b \in \mathbb{R}\}.$$

Moreover, $T_p S$ is precisely the plane through p orthogonal to the normal vector

$$\mathbf{N}(p) = \frac{\mathbf{x}_u(u_0, v_0) \times \mathbf{x}_v(u_0, v_0)}{\|\mathbf{x}_u(u_0, v_0) \times \mathbf{x}_v(u_0, v_0)\|}.$$

Proof. We prove the two inclusions.

1. $\text{Span}(\mathbf{x}_u, \mathbf{x}_v) \subseteq T_p S$. For arbitrary $a, b \in \mathbb{R}$, define the curve $\alpha(t) = \mathbf{x}(u_0 + at, v_0 + bt)$ for small t . Then $\alpha(0) = p$ and

$$\alpha'(0) = a \mathbf{x}_u(u_0, v_0) + b \mathbf{x}_v(u_0, v_0),$$

which is therefore a tangent vector.

2. $T_p S \subseteq \text{Span}(\mathbf{x}_u, \mathbf{x}_v)$. Let $\beta : (-\varepsilon, \varepsilon) \rightarrow S$ be any smooth curve with $\beta(0) = p$ and $\beta'(0) = \mathbf{v} \in T_p S$. Because $\mathbf{x}$ is a local diffeomorphism onto its image, we can write $\beta(t) = \mathbf{x}(u(t), v(t))$ for sufficiently small t , with $u(0) = u_0$, $v(0) = v_0$, and $u(t), v(t)$ smooth. Differentiating using the chain rule gives

$$\mathbf{v} = \beta'(0) = \mathbf{x}_u(u_0, v_0) u'(0) + \mathbf{x}_v(u_0, v_0) v'(0),$$

which lies in the span of $\mathbf{x}_u, \mathbf{x}_v$.

The equivalence with the orthogonal plane follows because $\mathbf{x}_u \times \mathbf{x}_v$ is perpendicular to both $\mathbf{x}_u$ and $\mathbf{x}_v$, so the plane through p consisting of all vectors $\mathbf{w}$ satisfying $(\mathbf{w} - p) \cdot (\mathbf{x}_u \times \mathbf{x}_v) = 0$ is exactly $p + T_p S$. ■

Remark 2.1.11. For a surface given implicitly by $F(x, y, z) = c$ with $\nabla F \neq 0$ on S , the gradient $\nabla F(p)$ is orthogonal to the tangent plane. Hence $T_p S$ can be described as $\{\mathbf{v} \in \mathbb{R}^3 \mid \nabla F(p) \cdot \mathbf{v} = 0\}$.

Remark 2.1.12. The tangent plane theorem provides a concrete way to compute the linear approximation of a surface. The key steps are

- For a parametric surface, compute $\mathbf{x}_u$ and $\mathbf{x}_v$ and take their span.
- For a graph $z = f(x, y)$, the tangent plane at $(x_0, y_0, f(x_0, y_0))$ is $z - f(x_0, y_0) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$ (a special case of the parametric formula).
- For an implicit surface $F(x, y, z) = c$, use $\nabla F(p) \cdot (x - p) = 0$.

All methods are equivalent and yield the same geometric object.

Example 2.1.13 (Plane). Consider the surface S given by $\mathbf{x}(u, v) = (2u + v, u - 3v, 4u + 5v)$. Find the tangent plane at $p = \mathbf{x}(0, 0)$.

Solution. Compute the partial derivatives:

$$\mathbf{x}_u = (2, 1, 4), \quad \mathbf{x}_v = (1, -3, 5).$$

These are linearly independent (their cross product is $(2, 1, 4) \times (1, -3, 5) = (17, -6, -7) \neq \mathbf{0}$). At $p = \mathbf{x}(0, 0) = (0, 0, 0)$, the tangent plane is simply $\text{span}\{\mathbf{x}_u, \mathbf{x}_v\}$, i.e. the plane through the origin spanned by these two vectors. Its equation can be obtained using the normal vector $\mathbf{n} = \mathbf{x}_u \times \mathbf{x}_v = (17, -6, -7)$:

$$17x - 6y - 7z = 0.$$

Example 2.1.14 (Sphere). Let S be the sphere $x^2 + y^2 + z^2 = 25$. Find the tangent plane at the point $p = (3, 4, 0)$.

Solution. We use the implicit method. Define $F(x, y, z) = x^2 + y^2 + z^2$. Then $\nabla F = (2x, 2y, 2z)$. At $p = (3, 4, 0)$, $\nabla F(p) = (6, 8, 0)$. The tangent plane consists of points (x, y, z) such that

$$(6, 8, 0) \cdot (x - 3, y - 4, z - 0) = 0 \implies 6(x - 3) + 8(y - 4) = 0.$$

Simplify: $6x - 18 + 8y - 32 = 0$, i.e. $6x + 8y = 50$, or $3x + 4y = 25$.

Example 2.1.15 (Elliptic Paraboloid). (See Figure 24 in the Atlas.) Parametrize the surface $z = x^2 + y^2$ as $\mathbf{x}(x, y) = (x, y, x^2 + y^2)$. Find the tangent plane at $p = (1, 1, 2)$.

Solution. Compute $\mathbf{x}_x = (1, 0, 2x)$, $\mathbf{x}_y = (0, 1, 2y)$. At $(x, y) = (1, 1)$:

$$\mathbf{x}_x = (1, 0, 2), \quad \mathbf{x}_y = (0, 1, 2).$$

A normal vector is

$$\mathbf{n} = \mathbf{x}_x \times \mathbf{x}_y = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 2 \\ 0 & 1 & 2 \end{vmatrix} = (-2, -2, 1).$$

The tangent plane equation at p is

$$(-2, -2, 1) \cdot (x - 1, y - 1, z - 2) = 0 \implies -2(x - 1) - 2(y - 1) + (z - 2) = 0.$$

Simplify: $z - 2 = 2(x - 1) + 2(y - 1)$, so $z = 2x + 2y - 2$.

Example 2.1.16 (Cylinder). Consider the cylinder $x^2 + y^2 = 1$. Find the tangent plane at the point $p = (0, 1, 5)$.

Solution. We can parametrize by $\mathbf{x}(\theta, z) = (\cos \theta, \sin \theta, z)$. At $p = (0, 1, 5)$ we have $\theta = \pi/2$, $z = 5$. Then

$$\mathbf{x}_\theta = (-\sin \theta, \cos \theta, 0) = (-1, 0, 0), \quad \mathbf{x}_z = (0, 0, 1).$$

Thus $T_p S$ spans $(-1, 0, 0)$ and $(0, 0, 1)$; it is the xz -plane shifted to pass through p . The equation is $y = 1$. Alternatively, using the implicit form $F(x, y, z) = x^2 + y^2$, $\nabla F = (2x, 2y, 0)$; at $p = (0, 1, 5)$, $\nabla F = (0, 2, 0)$. Then $(0, 2, 0) \cdot (x - 0, y - 1, z - 5) = 0$ gives $2(y - 1) = 0$, i.e. $y = 1$.

Example 2.1.17 (Hyperbolic Paraboloid). (See Figure 25 in the Atlas.) Let S be the surface $z = xy$. Find the tangent plane at $(2, 3, 6)$.

Solution. Parametrize by $\mathbf{x}(x, y) = (x, y, xy)$. Then $\mathbf{x}_x = (1, 0, y)$, $\mathbf{x}_y = (0, 1, x)$. At $(x, y) = (2, 3)$:

$$\mathbf{x}_x = (1, 0, 3), \quad \mathbf{x}_y = (0, 1, 2).$$

Normal vector:

$$\mathbf{n} = (1, 0, 3) \times (0, 1, 2) = (-3, -2, 1).$$

The tangent plane equation at $(2, 3, 6)$ is

$$(-3, -2, 1) \cdot (x - 2, y - 3, z - 6) = 0 \implies -3(x - 2) - 2(y - 3) + (z - 6) = 0.$$

Simplify: $-3x + 6 - 2y + 6 + z - 6 = 0 \implies z = 3x + 2y - 6$.

What is NOT a Regular Surface?

Example 2.1.18 (Cone without vertex). (See Figure 26 in the Atlas.) The set $\{(x, y, z) \mid z = \sqrt{x^2 + y^2}, (x, y) \neq (0, 0)\}$ is not a regular surface because the function $\sqrt{x^2 + y^2}$ is not differentiable at $(0, 0)$. If we remove the apex, the remainder is actually a regular surface (it is a graph of a smooth function on $\mathbb{R}^2 \setminus \{(0, 0)\}$). The problem is that the apex itself cannot be included.

Example 2.1.19 (Self-intersecting surface). (See Figure 27 in the Atlas.) The Whitney umbrella given by $\mathbf{x}(u, v) = (uv, u, v^2)$ is not a regular surface because it is not a homeomorphism onto its image; points with $v = 0$ and $u \neq 0$ map to the same x coordinate? Actually it self-intersects along the x -axis, violating the homeomorphism condition.

Example 2.1.20 (Surface with a cusp). (See Figure 28 in the Atlas.) The set $z = (x^2 + y^2)^{1/3}$ is not C^1 at the origin, so it fails the smoothness condition.

Many surfaces arise as level sets $F(x, y, z) = c$ where $\nabla F \neq 0$ on the set. The implicit function theorem guarantees that such a level set is a regular surface.

Proposition 2.1.21. Let $F : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a smooth function and let $c \in \mathbb{R}$. If $\nabla F(p) \neq 0$ for every p with $F(p) = c$, then $S = F^{-1}(c)$ is a regular surface.

Example 2.1.22. The sphere $x^2 + y^2 + z^2 = R^2$ has $\nabla F = (2x, 2y, 2z)$ which vanishes only at $(0, 0, 0)$, not on the sphere. Hence the sphere is regular.

Example 2.1.23. The cylinder $x^2 + y^2 = R^2$ has $\nabla F = (2x, 2y, 0)$, never zero on the cylinder.

Example 2.1.24. The hyperboloid $x^2 + y^2 - z^2 = 1$ has $\nabla F = (2x, 2y, -2z)$; if this were zero then $x = y = z = 0$, but $(0, 0, 0)$ does not satisfy the equation. Therefore the hyperboloid is regular. (See Figure 29 in the Atlas.)

Table of Examples for Quick Reference

Surface	Parametrization	Regularity condition
Plane	$(u, v, 0)$	Always
Graph $z = f(x, y)$	$(x, y, f(x, y))$	Always
Sphere	spherical coordinates	$\sin \theta \neq 0$ (charts cover all)
Cylinder	$(R \cos \theta, R \sin \theta, z)$	$R > 0$
Torus	$((R + r \cos \theta) \cos \phi, \dots)$	$R > r > 0$
Surface of revolution	$(f(u) \cos \theta, f(u) \sin \theta, g(u))$	$f(u) > 0, (f', g') \neq (0, 0)$
Enneper's surface	$(u - u^3/3 + uv^2, v - v^3/3 + u^2v, u^2 - v^2)$	$(u, v) \neq (0, 0)$

Exercises

1. Show that the helicoid $\mathbf{x}(u, v) = (u \cos v, u \sin v, v)$ is a regular surface for $u > 0$.
2. Prove that the hyperboloid $x^2 + y^2 - z^2 = 1$ is a regular surface using the implicit function theorem.
3. Find all points where the parametrization $\mathbf{x}(u, v) = (u^3, v^3, uv)$ fails to be regular.
4. Give an example of a subset of $\mathbb{R}^3$ that is a regular surface but cannot be represented as the graph of a function of the form $z = f(x, y)$ over any open set.

2.1.1 Change of Coordinates

If $\mathbf{x} : U \rightarrow S$ and $\mathbf{y} : V \rightarrow S$ are two parametrizations with overlapping images, then the transition map $\mathbf{x}^{-1} \circ \mathbf{y} : \mathbf{y}^{-1}(\mathbf{x}(U) \cap \mathbf{y}(V)) \rightarrow \mathbf{x}^{-1}(\mathbf{x}(U) \cap \mathbf{y}(V))$ is a smooth diffeomorphism between open sets of $\mathbb{R}^2$. This ensures that geometric quantities (such as curvature) are independent of the chosen parametrization.

Theorem: [Inverse Function Theorem in $\mathbb{R}^3$] Let $F : \mathcal{U} \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a smooth map. If at a point $q = (u_0, v_0, t_0) \in \mathcal{U}$, the differential dF_q is invertible (i.e., $\det(dF_q) \neq 0$), then there exist open neighborhoods U of q and V of $F(q)$ such that $F : U \rightarrow V$ is a diffeomorphism.

Theorem 2.1.25 (Change of Coordinates). Let $S \subset \mathbb{R}^3$ be a regular surface. Let $\mathbf{x} : U \rightarrow S$ and $\mathbf{y} : V \rightarrow S$ be two coordinate patches such that $W = \mathbf{x}(U) \cap \mathbf{y}(V)$ is non-empty. Then the transition map

$$h = \mathbf{x}^{-1} \circ \mathbf{y} : \mathbf{y}^{-1}(W) \longrightarrow \mathbf{x}^{-1}(W)$$

is a diffeomorphism.

Proof. Let $p \in W$. Since $\mathbf{x}$ and $\mathbf{y}$ are coordinate patches, they are, by definition, homeomorphisms from open sets in $\mathbb{R}^2$ onto open sets in S . Consequently, $h = \mathbf{x}^{-1} \circ \mathbf{y}$ is a composition of homeomorphisms and is itself a homeomorphism. It remains to show that h is smooth (of class C^∞).

Let $q \in \mathbf{y}^{-1}(W)$ such that $\mathbf{y}(q) = p$. We represent the coordinates in U as (u, v) . Since $\mathbf{x}$ is a regular patch, the partial derivatives $\mathbf{x}_u$ and $\mathbf{x}_v$ are linearly independent at $\mathbf{x}^{-1}(p)$. We define the unit normal vector field $\mathbf{n}(u, v) = \frac{\mathbf{x}_u \times \mathbf{x}_v}{|\mathbf{x}_u \times \mathbf{x}_v|}$.

Consider the map $F : U \times \mathbb{R} \rightarrow \mathbb{R}^3$ defined by:

$$F(u, v, t) = \mathbf{x}(u, v) + t \mathbf{n}(u, v) \quad (2.1)$$

The map F is differentiable because $\mathbf{x}$ and $\mathbf{n}$ are smooth. The Jacobian of F at $(u, v, 0)$ is

$$\det(dF) = \det(\mathbf{x}_u, \mathbf{x}_v, \mathbf{n}) = \langle \mathbf{x}_u \times \mathbf{x}_v, \mathbf{n} \rangle = |\mathbf{x}_u \times \mathbf{x}_v|$$

Since $\mathbf{x}$ is regular, $|\mathbf{x}_u \times \mathbf{x}_v| \neq 0$. Thus, dF is non-singular. By the **Inverse Function Theorem**, there exists an open neighborhood $\mathcal{O}$ of p in $\mathbb{R}^3$ such that F^{-1} exists and is differentiable on $\mathcal{O}$.

Observe that for any point $p' \in S \cap \mathcal{O}$, the t -coordinate of $F^{-1}(p')$ is zero. Therefore, the restriction of F^{-1} to the surface S yields the first two components (u, v) , which is precisely the map $\mathbf{x}^{-1}$. Formally, if $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is the projection $\pi(u, v, t) = (u, v)$, then:

$$\mathbf{x}^{-1} = \pi \circ F^{-1}|_{S \cap \mathcal{O}}$$

Since F^{-1} and $\mathbf{y}$ are differentiable, the transition map h in a neighborhood of q is given by $h = (\pi \circ F^{-1}) \circ \mathbf{y}$. As a composition of differentiable maps, h is differentiable at q . By an identical argument, $h^{-1} = \mathbf{y}^{-1} \circ \mathbf{x}$ is also differentiable. Thus, h is a diffeomorphism. ■

Explanation of the Geometry Used in the Proof. The proof is structured into two main parts: verifying that the map is a bijection and then proving that it is smooth (differentiable) using an extension into 3D space.

Establishing Bijectivity

By the definition of a regular surface, a coordinate patch $\mathbf{x}$ is a *homeomorphism* onto its image. This means $\mathbf{x}$ is continuous and has a continuous inverse. Since $\mathbf{y}$ is a bijection from $\mathbf{y}^{-1}(W)$ to W , and $\mathbf{x}^{-1}$ is a bijection from W to $\mathbf{x}^{-1}(W)$, their composition $h = \mathbf{x}^{-1} \circ \mathbf{y}$ is necessarily a bijection. Geometrically, this means every point in one coordinate system corresponds uniquely to exactly one point in the other.

The Differentiability Problem

We know $\mathbf{y}$ is differentiable, but we cannot immediately say $\mathbf{x}^{-1}$ is differentiable because its domain is the surface $S \subset \mathbb{R}^3$, which is not an open set in $\mathbb{R}^3$. To use the standard rules of calculus, we must "extend" our coordinate patch from the 2D surface into a 3D neighborhood.

Extending the Patch to 3D Space

Let $q \in \mathbf{y}^{-1}(W)$ and $p = \mathbf{y}(q) \in W$. We define a new map $F : U \times \mathbb{R} \rightarrow \mathbb{R}^3$ by "thickening" the surface in the direction of the normal vector:

$$F(u, v, t) = \mathbf{x}(u, v) + t \mathbf{n}(u, v)$$

Here, $(u, v) \in U$ are the surface coordinates, $\mathbf{n}$ is the unit normal vector to the surface at $\mathbf{x}(u, v)$, and t represents the distance from the surface.

Applying the Inverse Function Theorem

F is a map between two open sets in $\mathbb{R}^3$. Its derivative at a point $(u, v, 0)$ is a 3×3 matrix with columns $(\mathbf{x}_u, \mathbf{x}_v, \mathbf{n})$. The determinant of this Jacobian is

$$\det(dF) = (\mathbf{x}_u \times \mathbf{x}_v) \cdot \mathbf{n} = |\mathbf{x}_u \times \mathbf{x}_v|$$

Because the surface is **regular**, the vectors $\mathbf{x}_u$ and $\mathbf{x}_v$ are linearly independent, so their cross product is non-zero. Thus, $\det(dF) \neq 0$. By the **Inverse Function Theorem**, F has a differentiable inverse F^{-1} in a small 3D neighborhood of the point p .

Linking F^{-1} back to $\mathbf{x}^{-1}$

Observe that for points lying exactly on the surface S , the t coordinate is 0. Therefore, $F(u, v, 0) = \mathbf{x}(u, v)$. This implies that the inverse map F^{-1} , when restricted to the surface S , behaves exactly like $\mathbf{x}^{-1}$ (returning the u and v coordinates).

We can now write the transition map h as a composition of smooth functions:

$$h = \pi \circ F^{-1} \circ \mathbf{y}$$

where $\pi(u, v, t) = (u, v)$ is the standard projection map. Since $\mathbf{y}$, F^{-1} , and π are all differentiable, their composition h is differentiable at q .

By swapping the roles of $\mathbf{x}$ and $\mathbf{y}$, we can show that $h^{-1} = \mathbf{y}^{-1} \circ \mathbf{x}$ is also differentiable. Thus, h is a diffeomorphism, proving that the change of coordinates is smooth. ■

A regular surface is locally smooth, meaning it can be approximated by a plane at any given point. This plane is known as the tangent plane. In differential geometry, we distinguish between the **Tangent Space** (a set of vectors) and the **Tangent Plane** (the geometric set of points in $\mathbb{R}^3$).

Definition 2.1.26 (Tangent Vector). Let S be a regular surface and $p \in S$. A vector $w \in \mathbb{R}^3$ is a **tangent vector** to S at p if there exists a differentiable parameterized curve $\alpha : (-\epsilon, \epsilon) \rightarrow S$ such that $\alpha(0) = p$ and $\alpha'(0) = w$. The **tangent space** to S at p , denoted by $T_p S$, is the set of all tangent vectors to S at p .

Proposition 2.1.27 (Basis of the Tangent Space). Let $\mathbf{x} : U \subset \mathbb{R}^2 \rightarrow S$ be a local parametrization of a regular surface S such that $\mathbf{x}(q) = p$ for some $q = (u_0, v_0) \in U$. The tangent space $T_p S$ is a two-dimensional vector subspace of $\mathbb{R}^3$ spanned by the partial derivative vectors:

$$\mathbf{x}_u(q) = \frac{\partial \mathbf{x}}{\partial u}(q) \quad \text{and} \quad \mathbf{x}_v(q) = \frac{\partial \mathbf{x}}{\partial v}(q)$$

Proof. Let $w \in T_p S$. By definition, $w = \alpha'(0)$ for some curve $\alpha(t)$ on S . We can write the curve in the coordinate neighborhood as $\alpha(t) = \mathbf{x}(u(t), v(t))$. By the chain rule,

$$w = \alpha'(0) = \frac{d}{dt} [\mathbf{x}(u(t), v(t))]_{t=0} = \mathbf{x}_u \frac{du}{dt}(0) + \mathbf{x}_v \frac{dv}{dt}(0)$$

This shows that w is a linear combination of $\mathbf{x}_u$ and $\mathbf{x}_v$. Since the surface is regular, $\mathbf{x}_u$ and $\mathbf{x}_v$ are linearly independent, thus they form a basis for $T_p S$. ■

Definition 2.1.28 (Normal Vector). Let $\mathbf{x}(u, v)$ be a parametrization of S at p . The **unit normal vector** to the surface at p is defined as

$$\mathbf{N}(p) = \frac{\mathbf{x}_u \times \mathbf{x}_v}{|\mathbf{x}_u \times \mathbf{x}_v|}$$

Theorem 2.1.29 (Equation of the Tangent Plane). *Let $S \subset \mathbb{R}^3$ be a regular surface and $\mathbf{x} : U \subset \mathbb{R}^2 \rightarrow S$ be a local parametrization of S such that $p = \mathbf{x}(u_0, v_0)$ for some $(u_0, v_0) \in U$. The tangent plane to S at $p = (x_0, y_0, z_0)$ is the set of points $P = (x, y, z) \in \mathbb{R}^3$ satisfying:*

$$\langle (P - p), \mathbf{N}(p) \rangle = 0$$

Furthermore, if the cross product of the partial derivatives is given by $\mathbf{x}_u \times \mathbf{x}_v = (A, B, C)$, then the Cartesian equation of the plane is $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$.

Proof. By the definition of a regular surface, the tangent space $T_p S$ is the 2D vector subspace of $\mathbb{R}^3$ spanned by the basis $\{\mathbf{x}_u, \mathbf{x}_v\}$.

Let $\mathbf{n} = \mathbf{x}_u \times \mathbf{x}_v$. By the properties of the cross product, $\mathbf{n}$ is orthogonal to $T_p S$. The regularity condition ($\mathbf{x}_u \times \mathbf{x}_v \neq \mathbf{0}$) ensures that $\mathbf{n}$ defines a unique normal direction.

A point P lies in the tangent plane at p if and only if the displacement vector $P - p$ is a tangent vector (i.e., $P - p \in T_p S$). This geometric condition is satisfied if and only if $P - p$ is orthogonal to the normal vector $\mathbf{n}$:

$$\langle P - p, \mathbf{n} \rangle = 0$$

Expanding this inner product with $P - p = (x - x_0, y - y_0, z - z_0)$ and $\mathbf{n} = (A, B, C)$ yields the Cartesian equation:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

■

Proposition 2.1.30 (Tangent Plane of a Level Set). *If a regular surface S is defined implicitly by $f(x, y, z) = a$, where a is a constant and $\nabla f(p) \neq 0$, then the tangent space $T_p S$ is the kernel of the linear map df_p . That is, a vector w is tangent to S at p if and only if:*

$$\langle \nabla f(p), w \rangle = 0$$

Example 2.1.31 (The Sphere - Implicit Method). Find the equation of the tangent plane to the sphere $x^2 + y^2 + z^2 = 14$ at the point $P(1, 2, 3)$.

Solution. The surface is given implicitly by $f(x, y, z) = x^2 + y^2 + z^2 - 14 = 0$. The gradient vector ∇f acts as the normal vector to the surface:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = (2x, 2y, 2z)$$

Evaluating the gradient at the point $P(1, 2, 3)$:

$$\mathbf{n} = \nabla f(1, 2, 3) = (2(1), 2(2), 2(3)) = (2, 4, 6)$$

The equation of the tangent plane at (x_0, y_0, z_0) is $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$:

$$2(x - 1) + 4(y - 2) + 6(z - 3) = 0$$

Dividing by 2 and simplifying:

$$(x - 1) + 2(y - 2) + 3(z - 3) = 0 \implies x + 2y + 3z = 14$$

■

Example 2.1.32 (The Hyperbolic Paraboloid - Parametric Method). Find the tangent plane to the surface $\mathbf{x}(u, v) = (u, v, u^2 - v^2)$ at the point where $u = 2, v = 1$.

Solution. First determine the point p on the surface:

$$p = \mathbf{x}(2, 1) = (2, 1, 2^2 - 1^2) = (2, 1, 3)$$

Next, compute the partial derivative vectors (the basis of the tangent space):

$$\mathbf{x}_u = (1, 0, 2u) \implies \mathbf{x}_u(2, 1) = (1, 0, 4)$$

$$\mathbf{x}_v = (0, 1, -2v) \implies \mathbf{x}_v(2, 1) = (0, 1, -2)$$

The normal vector $\mathbf{n}$ is the cross product of these basis vectors:

$$\mathbf{n} = \mathbf{x}_u \times \mathbf{x}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 4 \\ 0 & 1 & -2 \end{vmatrix} = (0 - 4)\mathbf{i} - (-2 - 0)\mathbf{j} + (1 - 0)\mathbf{k} = (-4, 2, 1)$$

The equation of the plane is

$$-4(x - 2) + 2(y - 1) + 1(z - 3) = 0 \implies -4x + 8 + 2y - 2 + z - 3 = 0$$

Simplifying, we obtain $4x - 2y - z = 3$. ■

Example 2.1.33 (The Ellipsoid - Implicit Method). (See Figure 30 in the Atlas.) Find the tangent plane to the ellipsoid $\frac{x^2}{4} + y^2 + \frac{z^2}{9} = 3$ at the point $P(-2, 1, 3)$.

Solution. Define the implicit function $f(x, y, z) = \frac{x^2}{4} + y^2 + \frac{z^2}{9} - 3 = 0$. The normal vector is the gradient:

$$\nabla f = \left(\frac{x}{2}, 2y, \frac{2z}{9} \right)$$

At the point $P(-2, 1, 3)$:

$$\mathbf{n} = \nabla f(-2, 1, 3) = \left(\frac{-2}{2}, 2(1), \frac{2(3)}{9} \right) = \left(-1, 2, \frac{2}{3} \right)$$

The tangent plane equation is $-1(x + 2) + 2(y - 1) + \frac{2}{3}(z - 3) = 0$. Multiply by 3 to clear the fraction:

$$-3(x + 2) + 6(y - 1) + 2(z - 3) = 0 \implies -3x - 6 + 6y - 6 + 2z - 6 = 0$$

Resulting in: $3x - 6y - 2z = -18$ or $3x - 6y - 2z + 18 = 0$. ■

Example 2.1.34 (The Helicoid - Parametric Method). (See Figure 31 in the Atlas.) Find the tangent plane to the helicoid $\mathbf{x}(u, v) = (u \cos v, u \sin v, v)$ at $(u, v) = (1, \pi/2)$.

Solution. First, determine the point coordinates:

$$p = \mathbf{x}(1, \pi/2) = (1 \cos(\pi/2), 1 \sin(\pi/2), \pi/2) = (0, 1, \pi/2)$$

Compute the partial derivatives:

$$\mathbf{x}_u = (\cos v, \sin v, 0) \implies \mathbf{x}_u(1, \pi/2) = (0, 1, 0)$$

$$\mathbf{x}_v = (-u \sin v, u \cos v, 1) \implies \mathbf{x}_v(1, \pi/2) = (-1, 0, 1)$$

Compute the normal vector via cross product:

$$\mathbf{n} = \mathbf{x}_u \times \mathbf{x}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = (1)\mathbf{i} - (0)\mathbf{j} + (1)\mathbf{k} = (1, 0, 1)$$

The equation of the tangent plane is

$$1(x - 0) + 0(y - 1) + 1(z - \pi/2) = 0 \implies x + z = \frac{\pi}{2}$$

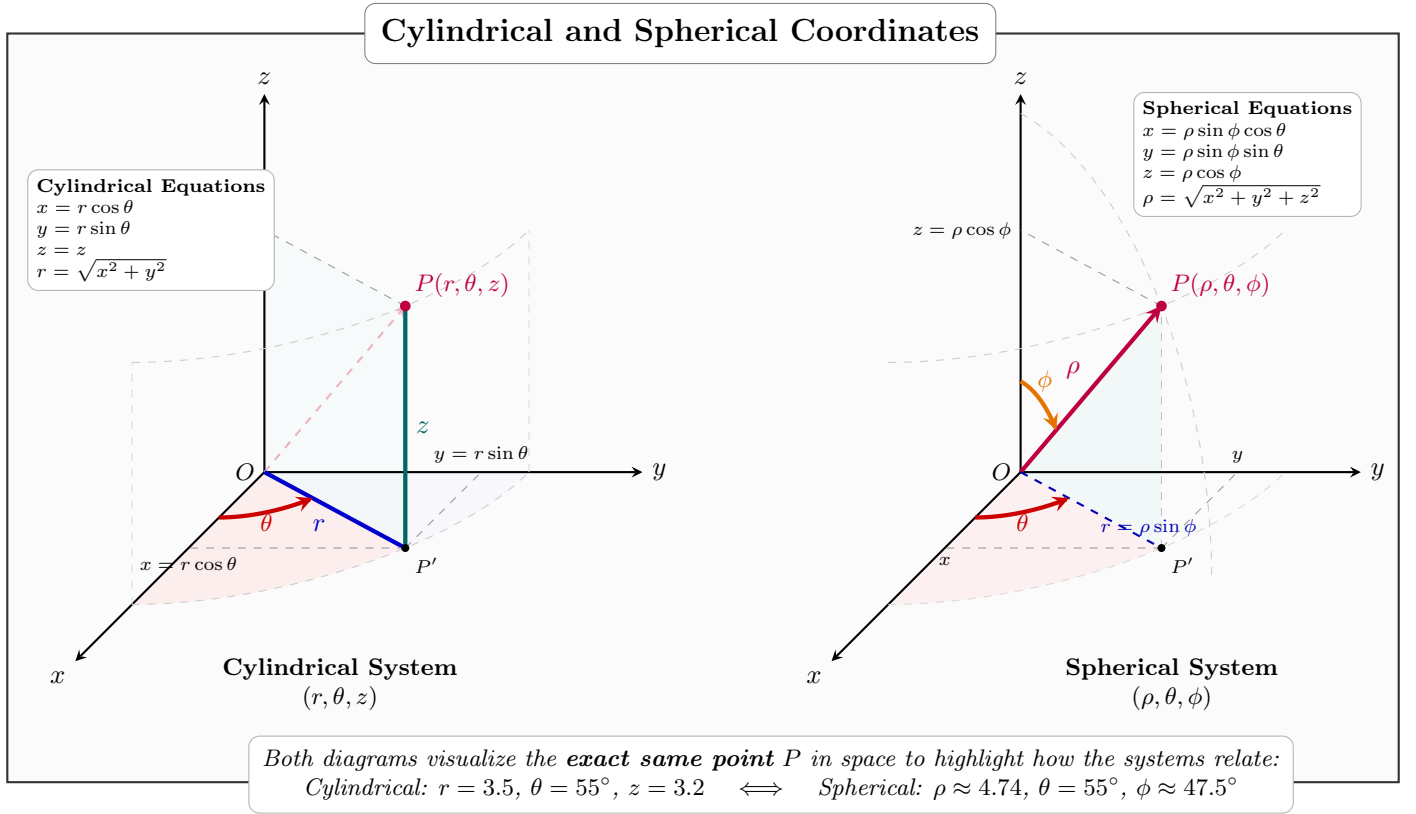


Figure 2.1: Caption

Remark 2.1.35 (Geometric Summary). The regularity of a surface ensures that the tangent plane is well-defined. If $\mathbf{x}_u \times \mathbf{x}_v = 0$, the "plane" would collapse into a line or a point, which happens at singularities (like the tip of a cone). Therefore, regularity is a requirement for the existence of a unique tangent plane.

A curvilinear coordinate system is described by a set of coordinates $(q_1, q_2, \dots, q_n)$. The local basis vectors at any point are tangent to the coordinate curves passing through that point. We can find these vectors by first defining the position vector $\mathbf{r}$ in terms of Cartesian coordinates:

$$\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}.$$

We then express $\mathbf{r}$ as a function of the curvilinear coordinates, $\mathbf{r}(q_1, q_2, q_3)$.

The (unnormalized) basis vectors $\mathbf{b}_i$ are obtained by taking the partial derivative of the position vector with respect to each coordinate:

$$\mathbf{b}_i = \frac{\partial \mathbf{r}}{\partial q_i}.$$

To obtain the orthonormal basis vectors $\hat{\mathbf{e}}_i$, we normalize these vectors:

$$\hat{\mathbf{e}}_i = \frac{\mathbf{b}_i}{\|\mathbf{b}_i\|}.$$

The magnitude $\|\mathbf{b}_i\|$ is known as the scale factor, often denoted h_i .

Example 2.1.36 (Unit Sphere (Spherical Coordinates)). A unit sphere is a 2-dimensional surface embedded in 3-dimensional space. Any point on its surface can be described using two coordinates: the polar angle θ (measured from the positive z -axis, $0 \leq \theta \leq \pi$) and the azimuthal angle ϕ (measured in the xy -plane from the positive x -axis, $0 \leq \phi < 2\pi$). The radius is fixed at $r = 1$.

The transformation from spherical to Cartesian coordinates is

$$\begin{aligned} x &= r \sin \theta \cos \phi, \\ y &= r \sin \theta \sin \phi, \\ z &= r \cos \theta. \end{aligned}$$

With $r = 1$, the position vector for a point on the unit sphere becomes

$$\mathbf{r}(\theta, \phi) = (\sin \theta \cos \phi) \hat{\mathbf{i}} + (\sin \theta \sin \phi) \hat{\mathbf{j}} + (\cos \theta) \hat{\mathbf{k}}.$$

Solution. Basis vector $\hat{\mathbf{e}}_\theta$: We take the partial derivative with respect to θ :

$$\frac{\partial \mathbf{r}}{\partial \theta} = (\cos \theta \cos \phi) \hat{\mathbf{i}} + (\cos \theta \sin \phi) \hat{\mathbf{j}} - (\sin \theta) \hat{\mathbf{k}}.$$

Its magnitude (the scale factor h_θ) is

$$\begin{aligned} \left\| \frac{\partial \mathbf{r}}{\partial \theta} \right\|^2 &= (\cos \theta \cos \phi)^2 + (\cos \theta \sin \phi)^2 + (-\sin \theta)^2 \\ &= \cos^2 \theta \cos^2 \phi + \cos^2 \theta \sin^2 \phi + \sin^2 \theta \\ &= \cos^2 \theta (\cos^2 \phi + \sin^2 \phi) + \sin^2 \theta \\ &= \cos^2 \theta (1) + \sin^2 \theta = 1. \end{aligned}$$

Since the magnitude is 1, the normalized basis vector is simply the partial derivative itself:

$$\hat{\mathbf{e}}_\theta = (\cos \theta \cos \phi) \hat{\mathbf{i}} + (\cos \theta \sin \phi) \hat{\mathbf{j}} - (\sin \theta) \hat{\mathbf{k}}.$$

Basis vector $\hat{\mathbf{e}}_\phi$: We take the partial derivative with respect to ϕ :

$$\frac{\partial \mathbf{r}}{\partial \phi} = (-\sin \theta \sin \phi) \hat{\mathbf{i}} + (\sin \theta \cos \phi) \hat{\mathbf{j}} + 0 \hat{\mathbf{k}}.$$

Its magnitude (the scale factor h_ϕ) is

$$\begin{aligned} \left\| \frac{\partial \mathbf{r}}{\partial \phi} \right\|^2 &= (-\sin \theta \sin \phi)^2 + (\sin \theta \cos \phi)^2 + 0^2 \\ &= \sin^2 \theta \sin^2 \phi + \sin^2 \theta \cos^2 \phi \\ &= \sin^2 \theta (\sin^2 \phi + \cos^2 \phi) = \sin^2 \theta (1) = \sin^2 \theta. \end{aligned}$$

Thus $\left\| \frac{\partial \mathbf{r}}{\partial \phi} \right\| = \sin \theta$. Normalizing gives

$$\hat{\mathbf{e}}_\phi = \frac{1}{\sin \theta} \left((-\sin \theta \sin \phi) \hat{\mathbf{i}} + (\sin \theta \cos \phi) \hat{\mathbf{j}} \right) = (-\sin \phi) \hat{\mathbf{i}} + (\cos \phi) \hat{\mathbf{j}}.$$

Result: The orthonormal basis for the tangent space of the unit sphere is $\{\hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi\}$, where:

$$\begin{aligned} \hat{\mathbf{e}}_\theta &= (\cos \theta \cos \phi) \hat{\mathbf{i}} + (\cos \theta \sin \phi) \hat{\mathbf{j}} - (\sin \theta) \hat{\mathbf{k}}, \\ \hat{\mathbf{e}}_\phi &= (-\sin \phi) \hat{\mathbf{i}} + (\cos \phi) \hat{\mathbf{j}}. \end{aligned}$$

Observe that a third basis vector, $\hat{\mathbf{e}}_r$, points radially outward and is normal to the surface; it is not part of the basis for the surface itself. ■

Example 2.1.37 (Cylindrical Coordinates). The cylindrical coordinate system is a three-dimensional coordinate system described by the coordinates (ρ, ϕ, z) , where ρ is the radial distance from the z -axis, ϕ is the azimuthal angle, and z is the height.

The transformation from cylindrical to Cartesian coordinates is given by

$$\begin{aligned} x &= \rho \cos \phi, \\ y &= \rho \sin \phi, \\ z &= z. \end{aligned}$$

The position vector of a point is represented as:

$$\mathbf{r}(\rho, \phi, z) = (\rho \cos \phi) \hat{\mathbf{i}} + (\rho \sin \phi) \hat{\mathbf{j}} + z \hat{\mathbf{k}}.$$

Solution. To find the orthonormal basis, we compute the partial derivative of the position vector with respect to each coordinate and normalize the resulting tangent vector.

1. Radial Direction (ρ): The tangent vector, its scale factor (magnitude), and the corresponding normalized unit vector are

$$\begin{aligned}\frac{\partial \mathbf{r}}{\partial \rho} &= (\cos \phi) \hat{\mathbf{i}} + (\sin \phi) \hat{\mathbf{j}}, \\ \left\| \frac{\partial \mathbf{r}}{\partial \rho} \right\| &= \sqrt{\cos^2 \phi + \sin^2 \phi} = 1, \\ \hat{\mathbf{e}}_\rho &= (\cos \phi) \hat{\mathbf{i}} + (\sin \phi) \hat{\mathbf{j}}.\end{aligned}$$

2. Azimuthal Direction (ϕ): Similarly, along the azimuthal direction, we compute:

$$\begin{aligned}\frac{\partial \mathbf{r}}{\partial \phi} &= (-\rho \sin \phi) \hat{\mathbf{i}} + (\rho \cos \phi) \hat{\mathbf{j}}, \\ \left\| \frac{\partial \mathbf{r}}{\partial \phi} \right\| &= \sqrt{(-\rho \sin \phi)^2 + (\rho \cos \phi)^2} = \rho, \\ \hat{\mathbf{e}}_\phi &= \frac{1}{\rho} \frac{\partial \mathbf{r}}{\partial \phi} = (-\sin \phi) \hat{\mathbf{i}} + (\cos \phi) \hat{\mathbf{j}}.\end{aligned}$$

3. Vertical Direction (z): Finally, along the vertical direction, the derivation yields

$$\begin{aligned}\frac{\partial \mathbf{r}}{\partial z} &= \hat{\mathbf{k}}, \\ \left\| \frac{\partial \mathbf{r}}{\partial z} \right\| &= 1, \\ \hat{\mathbf{e}}_z &= \hat{\mathbf{k}}.\end{aligned}$$

Result Summary: The final orthonormal basis $\{\hat{\mathbf{e}}_\rho, \hat{\mathbf{e}}_\phi, \hat{\mathbf{e}}_z\}$ for the cylindrical coordinate system is

$$\begin{aligned}\hat{\mathbf{e}}_\rho &= (\cos \phi) \hat{\mathbf{i}} + (\sin \phi) \hat{\mathbf{j}}, \\ \hat{\mathbf{e}}_\phi &= (-\sin \phi) \hat{\mathbf{i}} + (\cos \phi) \hat{\mathbf{j}}, \\ \hat{\mathbf{e}}_z &= \hat{\mathbf{k}}.\end{aligned}$$

■

2.2 The First Fundamental Form

Let a surface S in the Euclidean space $\mathbb{R}^3$ be given by a regular parametrization $\vec{r}: D \rightarrow \mathbb{R}^3$, where D is an open domain in the uv -plane.

$$\vec{r}(u, v) = (x(u, v), y(u, v), z(u, v))$$

At any point $P = \vec{r}(u_0, v_0)$ on the surface, the partial derivative vectors

$$\vec{r}_1(u, v) = \frac{\partial \vec{r}}{\partial u} \quad \text{and} \quad \vec{r}_2(u, v) = \frac{\partial \vec{r}}{\partial v}$$

are tangent to the coordinate curves passing through P . Since the parametrization is regular, the vectors $\vec{r}_1$ and $\vec{r}_2$ are linearly independent. They form a basis for the *tangent plane* to the surface at P , denoted $T_P(S)$. Any tangent vector $\vec{w} \in T_P(S)$ can be expressed as a unique linear combination $\vec{w} = \alpha \vec{r}_1 + \beta \vec{r}_2$.

Our goal is to measure geometric quantities like length and area on the surface S itself. We begin by considering the infinitesimal displacement vector $d\vec{r}$ along a curve on the surface. This vector can be expressed in terms of the basis vectors $\vec{r}_1, \vec{r}_2$:

$$d\vec{r} = \frac{\partial \vec{r}}{\partial u} du + \frac{\partial \vec{r}}{\partial v} dv = \vec{r}_1 du + \vec{r}_2 dv$$

The square of the length of this infinitesimal vector, denoted ds^2 , is found by taking its dot product with itself.

$$\begin{aligned} ds^2 &= d\vec{r} \cdot d\vec{r} \\ &= (\vec{r}_1 du + \vec{r}_2 dv) \cdot (\vec{r}_1 du + \vec{r}_2 dv) \\ &= (\vec{r}_1 \cdot \vec{r}_1) du^2 + 2(\vec{r}_1 \cdot \vec{r}_2) dudv + (\vec{r}_2 \cdot \vec{r}_2) dv^2 \end{aligned}$$

This quadratic form is fundamental to the study of surface geometry.

Definition 2.2.1 (Coefficients of the First Fundamental Form). The coefficients of the first fundamental form are three functions of (u, v) denoted by E, F, G :

$$\begin{aligned} E(u, v) &= \vec{r}_1 \cdot \vec{r}_1 = |\vec{r}_1|^2 \\ F(u, v) &= \vec{r}_1 \cdot \vec{r}_2 \\ G(u, v) &= \vec{r}_2 \cdot \vec{r}_2 = |\vec{r}_2|^2 \end{aligned}$$

Definition 2.2.2 (The First Fundamental Form). The **first fundamental form**, denoted by I , is the quadratic form on the tangent plane that gives the squared infinitesimal arc length ds^2 :

$$I = ds^2 = E(u, v) du^2 + 2F(u, v) dudv + G(u, v) dv^2$$

In matrix notation, this is written as

$$ds^2 = \begin{pmatrix} du & dv \end{pmatrix} \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} du \\ dv \end{pmatrix}$$

The symmetric matrix $g = (g_{ij}) = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$ is called the **metric tensor** of the surface.

Observe that since $\vec{r}_1$ and $\vec{r}_2$ are linearly independent, the determinant of the metric tensor, $\det(g) = EG - F^2$, is always positive.

We establish two fundamental properties of the first fundamental form. These properties confirm its role as a proper metric on the surface.

Theorem 2.2.3. *The first fundamental form $I = ds^2 = E du^2 + 2F du dv + G dv^2$ has the following properties:*

1. **Positive Definiteness:** *The form is positive definite. That is, $ds^2 \geq 0$, and $ds^2 = 0$ if and only if $du = 0$ and $dv = 0$.*
2. **Invariance:** *The value of the first fundamental form is invariant under a regular change of parameters (reparametrization).*

Proof. (1): By definition, the first fundamental form is the squared magnitude of the infinitesimal displacement vector $d\vec{r} = \vec{r}_1 du + \vec{r}_2 dv$.

$$ds^2 = d\vec{r} \cdot d\vec{r} = |\vec{r}_1 du + \vec{r}_2 dv|^2$$

The square of the magnitude of any real vector is always non-negative. Therefore, $ds^2 \geq 0$.

Now, we must show that $ds^2 = 0$ if and only if $du = 0$ and $dv = 0$. The equality $ds^2 = 0$ holds if and only if the vector $d\vec{r}$ is the zero vector:

$$\vec{r}_1 du + \vec{r}_2 dv = \vec{0}$$

The surface $\vec{r}(u, v)$ is assumed to be regular. By the definition of a regular surface, the tangent vectors $\vec{r}_1 = \partial\vec{r}/\partial u$ and $\vec{r}_2 = \partial\vec{r}/\partial v$ are linearly independent at every point.

The definition of linear independence states that the only linear combination of $\vec{r}_1$ and $\vec{r}_2$ that equals the zero vector is the trivial one, where all scalar coefficients are zero. In our case, the scalars are du and dv .

Therefore, the equation $\vec{r}_1 du + \vec{r}_2 dv = \vec{0}$ implies that $du = 0$ and $dv = 0$.

Conversely, if $du = 0$ and $dv = 0$, it is trivial to see that $ds^2 = E(0)^2 + 2F(0)(0) + G(0)^2 = 0$.

Thus, $ds^2 > 0$ for any non-zero pair of differentials (du, dv) , which proves that the first fundamental form is positive definite.

(2): Let the surface be described by two different regular parametrizations:

1. $\vec{r}(u, v)$, with first fundamental form $I = E du^2 + 2F du dv + G dv^2$.
2. $\vec{r}(u', v')$, with first fundamental form $I' = E' (du')^2 + 2F' du' dv' + G' (dv')^2$.

Let the change of parameters be given by a regular transformation:

$$u' = \phi(u, v) \quad \text{and} \quad v' = \psi(u, v)$$

This implies we can also write $u = u(u', v')$ and $v = v(u', v')$. The infinitesimal displacement vector on the surface, $d\vec{r}$, is a geometric entity. In the (u, v) coordinate system, it is expressed as

$$d\vec{r} = \frac{\partial\vec{r}}{\partial u} du + \frac{\partial\vec{r}}{\partial v} dv = \vec{r}_1 du + \vec{r}_2 dv$$

In the (u', v') coordinate system, the same vector $d\vec{r}$ is expressed as

$$d\vec{r} = \frac{\partial\vec{r}}{\partial u'} du' + \frac{\partial\vec{r}}{\partial v'} dv' = \vec{r}_{1'} du' + \vec{r}_{2'} dv'$$

Using the chain rule, we have the differential relations:

$$\begin{aligned} du &= \frac{\partial u}{\partial u'} du' + \frac{\partial u}{\partial v'} dv' \\ dv &= \frac{\partial v}{\partial u'} du' + \frac{\partial v}{\partial v'} dv' \end{aligned}$$

Substituting these into the first expression for $d\vec{r}$:

$$\begin{aligned} d\vec{r} &= \vec{r}_1 \left(\frac{\partial u}{\partial u'} du' + \frac{\partial u}{\partial v'} dv' \right) + \vec{r}_2 \left(\frac{\partial v}{\partial u'} du' + \frac{\partial v}{\partial v'} dv' \right) \\ &= \left(\vec{r}_1 \frac{\partial u}{\partial u'} + \vec{r}_2 \frac{\partial v}{\partial u'} \right) du' + \left(\vec{r}_1 \frac{\partial u}{\partial v'} + \vec{r}_2 \frac{\partial v}{\partial v'} \right) dv' \end{aligned}$$

By the chain rule for vector functions, the terms in the parentheses are precisely the definitions of the new tangent vectors:

$$\begin{aligned} \frac{\partial\vec{r}}{\partial u'} &= \frac{\partial\vec{r}}{\partial u} \frac{\partial u}{\partial u'} + \frac{\partial\vec{r}}{\partial v} \frac{\partial v}{\partial u'} = \vec{r}_1 \frac{\partial u}{\partial u'} + \vec{r}_2 \frac{\partial v}{\partial u'} = \vec{r}_{1'} \\ \frac{\partial\vec{r}}{\partial v'} &= \frac{\partial\vec{r}}{\partial u} \frac{\partial u}{\partial v'} + \frac{\partial\vec{r}}{\partial v} \frac{\partial v}{\partial v'} = \vec{r}_1 \frac{\partial u}{\partial v'} + \vec{r}_2 \frac{\partial v}{\partial v'} = \vec{r}_{2'} \end{aligned}$$

Substituting these back, we obtain $d\vec{r} = \vec{r}_{1'} du' + \vec{r}_{2'} dv'$. This confirms that our two expressions for $d\vec{r}$ represent the same vector in $\mathbb{R}^3$.

The first fundamental form is simply the squared magnitude of this vector, $ds^2 = |d\vec{r}|^2$. Since both coordinate systems produce the exact same vector $d\vec{r}$, their squared magnitudes must be identical.

$$I = |\vec{r}_1 du + \vec{r}_2 dv|^2 = |d\vec{r}|^2 = |\vec{r}_{1'} du' + \vec{r}_{2'} dv'|^2 = I'$$

Thus, the value of the first fundamental form is independent of the chosen parametrization, proving its invariance. ■

Example 2.2.4 (The Sphere). Consider a sphere of radius a with the standard spherical parametrization:

$$\vec{r}(u, v) = (a \sin u \cos v, a \sin u \sin v, a \cos u)$$

where $u \in [0, \pi]$ is the polar angle and $v \in [0, 2\pi)$ is the azimuthal angle. Find the first fundamental form.

Solution. First compute the partial derivatives $\vec{r}_1$ (w.r.t u) and $\vec{r}_2$ (w.r.t v):

$$\begin{aligned}\vec{r}_1 &= (a \cos u \cos v, a \cos u \sin v, -a \sin u) \\ \vec{r}_2 &= (-a \sin u \sin v, a \sin u \cos v, 0)\end{aligned}$$

Next compute the coefficients E, F, G :

$$\begin{aligned}E &= \vec{r}_1 \cdot \vec{r}_1 = a^2 \cos^2 u \cos^2 v + a^2 \cos^2 u \sin^2 v + a^2 \sin^2 u \\ &= a^2 \cos^2 u (\cos^2 v + \sin^2 v) + a^2 \sin^2 u \\ &= a^2 (\cos^2 u + \sin^2 u) = a^2 \\ F &= \vec{r}_1 \cdot \vec{r}_2 = -a^2 \cos u \sin u \cos v \sin v + a^2 \cos u \sin u \sin v \cos v + 0 = 0 \\ G &= \vec{r}_2 \cdot \vec{r}_2 = a^2 \sin^2 u \sin^2 v + a^2 \sin^2 u \cos^2 v + 0 \\ &= a^2 \sin^2 u (\sin^2 v + \cos^2 v) = a^2 \sin^2 u\end{aligned}$$

The first fundamental form for the sphere is $ds^2 = a^2 du^2 + a^2 \sin^2 u dv^2$. Since $F = 0$, the lines of constant latitude ($u = \text{const}$) and longitude ($v = \text{const}$) are orthogonal everywhere except at the poles where the parametrization is singular. The area element is $dA = \sqrt{EG - F^2} du dv = \sqrt{a^2 \cdot a^2 \sin^2 u} du dv = a^2 \sin u du dv$. ■

Example 2.2.5 (The General Plane). Let a plane in $\mathbb{R}^3$ be parametrized by $\vec{r}(u, v) = \vec{p}_0 + u\vec{a} + v\vec{b}$, where $\vec{p}_0$ is a point on the plane and $\{\vec{a}, \vec{b}\}$ is a basis of linearly independent (but not necessarily orthogonal) constant vectors. Find the first fundamental form.

Solution. Differentiating, we obtain the partial derivative vectors, which are constant for this parametrization.

$$\begin{aligned}\vec{r}_1 &= \frac{\partial}{\partial u}(\vec{p}_0 + u\vec{a} + v\vec{b}) = \vec{a} \\ \vec{r}_2 &= \frac{\partial}{\partial v}(\vec{p}_0 + u\vec{a} + v\vec{b}) = \vec{b}\end{aligned}$$

Now, we compute the coefficients E, F, G by taking the dot products of these basis vectors.

$$\begin{aligned}E &= \vec{r}_1 \cdot \vec{r}_1 = \vec{a} \cdot \vec{a} = |\vec{a}|^2 \\ F &= \vec{r}_1 \cdot \vec{r}_2 = \vec{a} \cdot \vec{b} \\ G &= \vec{r}_2 \cdot \vec{r}_2 = \vec{b} \cdot \vec{b} = |\vec{b}|^2\end{aligned}$$

Since the vectors $\vec{a}$ and $\vec{b}$ are constant, the coefficients E, F, G are constants for the entire surface. The first fundamental form is

$$ds^2 = |\vec{a}|^2 du^2 + 2(\vec{a} \cdot \vec{b}) du dv + |\vec{b}|^2 dv^2$$

Remark: If we choose an orthonormal basis, where $|\vec{a}| = 1$, $|\vec{b}| = 1$, and $\vec{a} \cdot \vec{b} = 0$, then $E = 1, F = 0, G = 1$. The metric becomes $ds^2 = du^2 + dv^2$, which is the familiar Pythagorean theorem in the Euclidean plane. This example shows how a non-zero F coefficient arises from using a non-orthogonal coordinate system. ■

Example 2.2.6 (The Right Circular Cone). Find the first fundamental form for a right circular cone parametrized by:

$$\vec{r}(u, v) = (u \cos v, u \sin v, ku), \quad u > 0, \quad k = \cot \alpha > 0$$

where α is the half-angle at the vertex.

Solution. First obtain the tangent vectors by differentiating with respect to u and v .

$$\begin{aligned}\vec{r}_1 &= \frac{\partial \vec{r}}{\partial u} = (\cos v, \sin v, k) \\ \vec{r}_2 &= \frac{\partial \vec{r}}{\partial v} = (-u \sin v, u \cos v, 0)\end{aligned}$$

Next compute the dot products to find E, F, G .

$$\begin{aligned}E &= \vec{r}_1 \cdot \vec{r}_1 = (\cos v)^2 + (\sin v)^2 + k^2 = \cos^2 v + \sin^2 v + k^2 = 1 + k^2 \\ F &= \vec{r}_1 \cdot \vec{r}_2 = (\cos v)(-u \sin v) + (\sin v)(u \cos v) + (k)(0) \\ &= -u \sin v \cos v + u \sin v \cos v = 0 \\ G &= \vec{r}_2 \cdot \vec{r}_2 = (-u \sin v)^2 + (u \cos v)^2 + 0^2 \\ &= u^2 \sin^2 v + u^2 \cos^2 v = u^2(\sin^2 v + \cos^2 v) = u^2\end{aligned}$$

The first fundamental form for the cone is $ds^2 = (1 + k^2)du^2 + u^2dv^2$. **Remark:** Since $F = 0$, the coordinate curves are orthogonal. The u -curves are straight lines along the generators of the cone, and the v -curves are circles of constant height. This metric shows that the cone is locally Euclidean (or "flat") because it can be unrolled into a sector of a plane without stretching, a property shared by all surfaces with zero Gaussian curvature. ■

Example 2.2.7 (The Helicoid). Compute the first fundamental form of the helicoid, given by the parametrization:

$$\vec{r}(u, v) = (u \cos v, u \sin v, cv), \quad u \in \mathbb{R}, \quad c \neq 0$$

Solution. We compute the partial derivatives of $\vec{r}(u, v)$.

$$\begin{aligned}\vec{r}_1 &= \frac{\partial \vec{r}}{\partial u} = (\cos v, \sin v, 0) \\ \vec{r}_2 &= \frac{\partial \vec{r}}{\partial v} = (-u \sin v, u \cos v, c)\end{aligned}$$

Now we calculate the coefficients E, F, G .

$$\begin{aligned}E &= \vec{r}_1 \cdot \vec{r}_1 = \cos^2 v + \sin^2 v + 0^2 = 1 \\ F &= \vec{r}_1 \cdot \vec{r}_2 = (\cos v)(-u \sin v) + (\sin v)(u \cos v) + (0)(c) = 0 \\ G &= \vec{r}_2 \cdot \vec{r}_2 = (-u \sin v)^2 + (u \cos v)^2 + c^2 \\ &= u^2 \sin^2 v + u^2 \cos^2 v + c^2 = u^2 + c^2\end{aligned}$$

The first fundamental form for the helicoid is $ds^2 = du^2 + (u^2 + c^2)dv^2$. **Remark:** The coordinate curves are again orthogonal because $F = 0$. The u -curves ($v = \text{const}$) are straight lines radiating horizontally from the z -axis. The v -curves ($u = \text{const}$) are helices of radius u . ■

Example 2.2.8 (Surface of Revolution). Let a regular plane curve (the profile curve) be given by $\vec{\alpha}(u) = (f(u), 0, g(u))$ with $f(u) > 0$. Find the first fundamental form of the surface generated by revolving this curve about the z -axis.

Solution. The parametrization of the surface of revolution is obtained by rotating the profile curve by an angle v around the z -axis.

$$\vec{r}(u, v) = (f(u) \cos v, f(u) \sin v, g(u))$$

We compute the tangent vectors $\vec{r}_1$ and $\vec{r}_2$. Let $f' = df/du$ and $g' = dg/du$.

$$\begin{aligned}\vec{r}_1 &= \frac{\partial \vec{r}}{\partial u} = (f'(u) \cos v, f'(u) \sin v, g'(u)) \\ \vec{r}_2 &= \frac{\partial \vec{r}}{\partial v} = (-f(u) \sin v, f(u) \cos v, 0)\end{aligned}$$

Next obtain the coefficients E, F, G .

$$\begin{aligned}
 E &= \vec{r}_1 \cdot \vec{r}_1 = (f'(u))^2 \cos^2 v + (f'(u))^2 \sin^2 v + (g'(u))^2 \\
 &= (f'(u))^2 (\cos^2 v + \sin^2 v) + (g'(u))^2 = (f'(u))^2 + (g'(u))^2 \\
 F &= \vec{r}_1 \cdot \vec{r}_2 = (f'(u) \cos v)(-f(u) \sin v) + (f'(u) \sin v)(f(u) \cos v) + (g'(u))(0) \\
 &= -f(u)f'(u) \sin v \cos v + f(u)f'(u) \sin v \cos v = 0 \\
 G &= \vec{r}_2 \cdot \vec{r}_2 = (-f(u) \sin v)^2 + (f(u) \cos v)^2 + 0^2 \\
 &= (f(u))^2 \sin^2 v + (f(u))^2 \cos^2 v = (f(u))^2
 \end{aligned}$$

The first fundamental form for a general surface of revolution is $ds^2 = ((f'(u))^2 + (g'(u))^2) du^2 + (f(u))^2 dv^2$. **Remark:** This is a powerful general result. For example, for a sphere of radius a , we can use the profile curve $f(u) = a \sin u, g(u) = a \cos u$. Then $f' = a \cos u, g' = -a \sin u$. $E = (a \cos u)^2 + (-a \sin u)^2 = a^2$ and $G = (a \sin u)^2 = a^2 \sin^2 u$. This yields $ds^2 = a^2 du^2 + a^2 \sin^2 u dv^2$, recovering the result from Example 2.2.4. ■

The first fundamental form allows us to compute all metric properties of the surface (those depending on measurements made along the surface) using only the coefficients E, F, G .

Theorem 2.2.9 (Length of a Curve). *Let C be a curve on the surface S given by $\vec{c}(t) = \vec{r}(u(t), v(t))$ for $t \in [a, b]$. The arc length L of C is given by*

$$L = \int_a^b \sqrt{E \left(\frac{du}{dt} \right)^2 + 2F \left(\frac{du}{dt} \right) \left(\frac{dv}{dt} \right) + G \left(\frac{dv}{dt} \right)^2} dt$$

Proof. The tangent vector to the curve is $\vec{c}'(t) = \frac{d\vec{r}}{dt} = \frac{\partial \vec{r}}{\partial u} \frac{du}{dt} + \frac{\partial \vec{r}}{\partial v} \frac{dv}{dt} = \vec{r}_1 u'(t) + \vec{r}_2 v'(t)$. The speed along the curve is $|\vec{c}'(t)|$. The square of the speed is

$$\begin{aligned}
 |\vec{c}'(t)|^2 &= \vec{c}'(t) \cdot \vec{c}'(t) \\
 &= (\vec{r}_1 u' + \vec{r}_2 v') \cdot (\vec{r}_1 u' + \vec{r}_2 v') \\
 &= (\vec{r}_1 \cdot \vec{r}_1)(u')^2 + 2(\vec{r}_1 \cdot \vec{r}_2)u'v' + (\vec{r}_2 \cdot \vec{r}_2)(v')^2 \\
 &= E(u')^2 + 2Fu'v' + G(v')^2
 \end{aligned}$$

The arc length is the integral of the speed, $L = \int_a^b |\vec{c}'(t)| dt$. Taking the square root of the expression above yields the desired formula. ■

Theorem 2.2.10 (Angle Between Tangent Vectors). *Let $\vec{w}_1 = a_1 \vec{r}_1 + b_1 \vec{r}_2$ and $\vec{w}_2 = a_2 \vec{r}_1 + b_2 \vec{r}_2$ be two tangent vectors at a point P on the surface. The angle θ between them is given by*

$$\cos \theta = \frac{\vec{w}_1 \cdot \vec{w}_2}{|\vec{w}_1| |\vec{w}_2|} = \frac{Ea_1a_2 + F(a_1b_2 + a_2b_1) + Gb_1b_2}{\sqrt{Ea_1^2 + 2Fa_1b_1 + Gb_1^2} \sqrt{Ea_2^2 + 2Fa_2b_2 + Gb_2^2}}$$

Proof. The proof is a direct computation using definition. ■

Corollary 2.2.11 (Orthogonality of Coordinate Curves). *The coordinate curves (u -curves and v -curves) are orthogonal at a point P if and only if $F(u, v) = 0$ at that point.*

Proof. The tangent vectors to the coordinate curves are $\vec{r}_1$ and $\vec{r}_2$. They are orthogonal if and only if their dot product is zero, which is precisely $F = \vec{r}_1 \cdot \vec{r}_2 = 0$. ■

Note: Since for any two vectors $\vec{a}, \vec{b} \in \mathbb{R}^3$, the magnitudes of their dot and cross products are related by the angle θ between them:

$$\begin{aligned}
 \vec{a} \cdot \vec{b} &= |\vec{a}| |\vec{b}| \cos \theta \\
 |\vec{a} \times \vec{b}| &= |\vec{a}| |\vec{b}| \sin \theta
 \end{aligned}$$

From these, we derive the celebrated **Lagrange's Identity**:

$$\begin{aligned} |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 &= |\vec{a}|^2 |\vec{b}|^2 - (|\vec{a}| |\vec{b}| \cos \theta)^2 \\ &= |\vec{a}|^2 |\vec{b}|^2 (1 - \cos^2 \theta) \\ &= |\vec{a}|^2 |\vec{b}|^2 \sin^2 \theta \\ &= (|\vec{a}| |\vec{b}| \sin \theta)^2 = |\vec{a} \times \vec{b}|^2 \end{aligned}$$

Thus, we have the identity $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$.

Theorem 2.2.12. Let S be a regular surface parametrized by $\vec{r}(u, v)$. Let $\vec{r}_1 = \partial \vec{r} / \partial u$ and $\vec{r}_2 = \partial \vec{r} / \partial v$ be the tangent vectors at a point on S . The squared magnitude of the cross product of these vectors is given by $|\vec{r}_1 \times \vec{r}_2|^2 = EG - F^2$, where E, F , and G are the coefficients of the first fundamental form.

Proof using Lagrange's Identity. We apply the Lagrange's identity to the tangent vectors $\vec{r}_1$ and $\vec{r}_2$.

$$|\vec{r}_1 \times \vec{r}_2|^2 = |\vec{r}_1|^2 |\vec{r}_2|^2 - (\vec{r}_1 \cdot \vec{r}_2)^2$$

By the definitions of the first fundamental coefficients, we have

- $E = \vec{r}_1 \cdot \vec{r}_1 = |\vec{r}_1|^2$
- $F = \vec{r}_1 \cdot \vec{r}_2$
- $G = \vec{r}_2 \cdot \vec{r}_2 = |\vec{r}_2|^2$

Substituting these definitions into the identity yields the desired result:

$$|\vec{r}_1 \times \vec{r}_2|^2 = EG - F^2$$

This proves the result. ■

Proof by Direct Computation. Let the surface be given in component form by

$$\vec{r}(u, v) = (x(u, v), y(u, v), z(u, v)).$$

The partial derivative vectors are

$$\vec{r}_1 = (x_u, y_u, z_u) \quad \text{and} \quad \vec{r}_2 = (x_v, y_v, z_v)$$

The cross product $\vec{r}_1 \times \vec{r}_2$ is given by

$$\vec{r}_1 \times \vec{r}_2 = (y_u z_v - z_u y_v, z_u x_v - x_u z_v, x_u y_v - y_u x_v)$$

The squared magnitude of this vector is the sum of the squares of its components:

$$|\vec{r}_1 \times \vec{r}_2|^2 = (y_u z_v - z_u y_v)^2 + (z_u x_v - x_u z_v)^2 + (x_u y_v - y_u x_v)^2 \quad (2.2)$$

Next compute the right-hand side, $EG - F^2$. The coefficients are

$$E = x_u^2 + y_u^2 + z_u^2$$

$$G = x_v^2 + y_v^2 + z_v^2$$

$$F = x_u x_v + y_u y_v + z_u z_v$$

Expanding the expression $EG - F^2$ gives

$$\begin{aligned} EG - F^2 &= (x_u^2 + y_u^2 + z_u^2)(x_v^2 + y_v^2 + z_v^2) - (x_u x_v + y_u y_v + z_u z_v)^2 \\ &= (x_u^2 y_v^2 - 2x_u y_v y_u x_v + y_u^2 x_v^2) + (x_u^2 z_v^2 - 2x_u z_v z_u x_v + z_u^2 x_v^2) \\ &\quad + (y_u^2 z_v^2 - 2y_u z_v z_u y_v + z_u^2 y_v^2) \\ &= (x_u y_v - y_u x_v)^2 + (x_u z_v - z_u x_v)^2 + (y_u z_v - z_u y_v)^2 \end{aligned}$$

Upon rearranging the terms, this becomes

$$EG - F^2 = (y_u z_v - z_u y_v)^2 + (z_u x_v - x_u z_v)^2 + (x_u y_v - y_u x_v)^2 \quad (2.3)$$

A direct comparison of equations (2.4) and (2.5) shows they are identical. ■

Theorem 2.2.13 (Surface Area). *Let $R \subset S$ be a region on the surface corresponding to a domain D in the uv -plane. The area of R , denoted $A(R)$, is given by $A(R) = \iint_D \sqrt{EG - F^2} du dv$.*

Proof. An infinitesimal parallelogram on the surface is spanned by the vectors $d\vec{u} = \vec{r}_1 du$ and $d\vec{v} = \vec{r}_2 dv$. The area of this parallelogram, dA , is the magnitude of their cross product:

$$dA = |d\vec{u} \times d\vec{v}| = |\vec{r}_1 du \times \vec{r}_2 dv| = |\vec{r}_1 \times \vec{r}_2| du dv$$

By Lagrange's identity, $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$. Applying this, we obtain

$$|\vec{r}_1 \times \vec{r}_2|^2 = |\vec{r}_1|^2 |\vec{r}_2|^2 - (\vec{r}_1 \cdot \vec{r}_2)^2 = EG - F^2$$

Thus, the area element is $dA = \sqrt{EG - F^2} du dv$. Integrating this over the domain D gives the total area of the region R . ■

Theorem 2.2.14. *The area of a regular surface is a geometric invariant, independent of its parametrization.*

Proof. Let R be a region on a regular surface S . Suppose this region can be described by two different regular parametrizations: $\vec{r}(u, v)$ over a domain D in the uv -plane and $\vec{s}(u', v')$ over a domain D' in the $u'v'$ -plane. The area of R can be computed using either parametrization:

$$A_1 = \iint_D |\vec{r}_u \times \vec{r}_v| du dv \quad \text{and} \quad A_2 = \iint_{D'} |\vec{s}_{u'} \times \vec{s}_{v'}| du' dv'$$

Our goal is to prove that $A_1 = A_2$. We will achieve this by transforming the first integral into the second using the change of variables formula. The heart of the proof lies in understanding how the area element transforms between the two coordinate systems.

Let the coordinate transformation be given by $u = u(u', v')$ and $v = v(u', v')$. By the chain rule, the tangent vectors of the two systems are related as follows:

$$\begin{aligned} \vec{s}_{u'} &= \frac{\partial \vec{r}}{\partial u'} = \frac{\partial \vec{r}}{\partial u} \frac{\partial u}{\partial u'} + \frac{\partial \vec{r}}{\partial v} \frac{\partial v}{\partial u'} = \vec{r}_u \frac{\partial u}{\partial u'} + \vec{r}_v \frac{\partial v}{\partial u'} \\ \vec{s}_{v'} &= \frac{\partial \vec{r}}{\partial v'} = \frac{\partial \vec{r}}{\partial u} \frac{\partial u}{\partial v'} + \frac{\partial \vec{r}}{\partial v} \frac{\partial v}{\partial v'} = \vec{r}_u \frac{\partial u}{\partial v'} + \vec{r}_v \frac{\partial v}{\partial v'} \end{aligned}$$

The key step is to compute the cross product $\vec{s}_{u'} \times \vec{s}_{v'}$. Using the bilinearity of the cross product and the fact that $\vec{v} \times \vec{w} = -\vec{w} \times \vec{v}$, we obtain

$$\begin{aligned} \vec{s}_{u'} \times \vec{s}_{v'} &= \left(\vec{r}_u \frac{\partial u}{\partial u'} + \vec{r}_v \frac{\partial v}{\partial u'} \right) \times \left(\vec{r}_u \frac{\partial u}{\partial v'} + \vec{r}_v \frac{\partial v}{\partial v'} \right) \\ &= (\vec{r}_u \times \vec{r}_v) \frac{\partial u}{\partial u'} \frac{\partial v}{\partial v'} + (\vec{r}_v \times \vec{r}_u) \frac{\partial v}{\partial u'} \frac{\partial u}{\partial v'} \\ &= (\vec{r}_u \times \vec{r}_v) \left(\frac{\partial u}{\partial u'} \frac{\partial v}{\partial v'} - \frac{\partial u}{\partial v'} \frac{\partial v}{\partial u'} \right) \end{aligned}$$

We recognize the scalar term in the parentheses as the Jacobian determinant of the transformation, denoted $\frac{\partial(u, v)}{\partial(u', v')}$. This gives the fundamental relationship between the normal vectors of the two parametrizations:

$$\vec{s}_{u'} \times \vec{s}_{v'} = (\vec{r}_u \times \vec{r}_v) \frac{\partial(u, v)}{\partial(u', v')}$$

Taking the magnitude of both sides yields the transformation rule for the area distortion factor:

$$|\vec{s}_{u'} \times \vec{s}_{v'}| = |\vec{r}_u \times \vec{r}_v| \left| \frac{\partial(u, v)}{\partial(u', v')} \right|$$

Now we apply the standard change of variables formula to the first area integral:

$$A_1 = \iint_D |\vec{r}_u \times \vec{r}_v| du dv = \iint_{D'} |\vec{r}_u \times \vec{r}_v| \left| \frac{\partial(u, v)}{\partial(u', v')} \right| du' dv'$$

Substituting the transformation rule we just derived, we obtain $A_1 = \iint_{D'} |\vec{s}_u \times \vec{s}_v| du' dv' = A_2$. Since the two expressions yield the same value, we conclude that the surface area is an intrinsic property of the surface, independent of the particular parametrization chosen to describe it. ■

Example 2.2.15. Use the first fundamental form to compute the surface area of a sphere of radius a .

Solution. We begin by choosing the standard spherical coordinate parametrization for a sphere of radius a :

$$\vec{r}(u, v) = (a \sin u \cos v, a \sin u \sin v, a \cos u)$$

The domain of the parameters that covers the entire sphere is $D = [0, \pi] \times [0, 2\pi]$, where u is the polar angle and v is the azimuthal angle. The area A is given by the integral of the area element over this domain:

$$A = \iint_D \sqrt{EG - F^2} du dv$$

Our first task is to compute the coefficients E, F , and G for this parametrization. First obtain the partial derivatives of $\vec{r}(u, v)$ with respect to u and v :

$$\begin{aligned}\vec{r}_1 &= \frac{\partial \vec{r}}{\partial u} = (a \cos u \cos v, a \cos u \sin v, -a \sin u) \\ \vec{r}_2 &= \frac{\partial \vec{r}}{\partial v} = (-a \sin u \sin v, a \sin u \cos v, 0)\end{aligned}$$

From these, we compute the coefficients of the first fundamental form by taking their dot products:

$$\begin{aligned}E &= \vec{r}_1 \cdot \vec{r}_1 = (a \cos u \cos v)^2 + (a \cos u \sin v)^2 + (-a \sin u)^2 \\ &= a^2 \cos^2 u (\cos^2 v + \sin^2 v) + a^2 \sin^2 u = a^2 (\cos^2 u + \sin^2 u) = \mathbf{a^2}\end{aligned}$$

$$F = \vec{r}_1 \cdot \vec{r}_2 = -a^2 \sin u \cos u \sin v \cos v + a^2 \sin u \cos u \sin v \cos v = \mathbf{0}$$

$$\begin{aligned}G &= \vec{r}_2 \cdot \vec{r}_2 = (-a \sin u \sin v)^2 + (a \sin u \cos v)^2 + (0)^2 \\ &= a^2 \sin^2 u (\sin^2 v + \cos^2 v) = \mathbf{a^2 \sin^2 u}\end{aligned}$$

With the coefficients determined, we obtain the term under the square root:

$$EG - F^2 = (a^2)(a^2 \sin^2 u) - (0)^2 = a^4 \sin^2 u$$

The area element is therefore

$$dA = \sqrt{EG - F^2} du dv = \sqrt{a^4 \sin^2 u} du dv = a^2 |\sin u| du dv$$

Since $u \in [0, \pi]$, $\sin u \geq 0$, so we may drop the absolute value. We now integrate the area element over the domain $D = [0, \pi] \times [0, 2\pi]$:

$$\begin{aligned}A &= \iint_D a^2 \sin u du dv \\ &= \int_{v=0}^{2\pi} \left(\int_{u=0}^{\pi} a^2 \sin u du \right) dv\end{aligned}$$

The inner integral with respect to u evaluates to:

$$\int_0^{\pi} a^2 \sin u du = a^2 [-\cos u]_0^{\pi} = a^2 (-(-1) - (-1)) = 2a^2$$

Substituting this constant result into the outer integral, we obtain the total area:

$$A = \int_0^{2\pi} 2a^2 dv = 2a^2 [v]_0^{2\pi} = 2a^2(2\pi) = 4\pi a^2$$

Example 2.2.16 (Area of a Cylinder). Find the lateral surface area of a right circular cylinder with radius a and height h .

Solution. We can parametrize the cylinder as follows:

$$\vec{r}(u, v) = (a \cos u, a \sin u, v)$$

The domain for the lateral surface is $D = [0, 2\pi] \times [0, h]$. First compute the tangent vectors $\vec{r}_1 = \partial \vec{r} / \partial u$ and $\vec{r}_2 = \partial \vec{r} / \partial v$:

$$\begin{aligned}\vec{r}_1 &= (-a \sin u, a \cos u, 0) \\ \vec{r}_2 &= (0, 0, 1)\end{aligned}$$

Next determine the coefficients of the first fundamental form:

$$\begin{aligned}E &= \vec{r}_1 \cdot \vec{r}_1 = (-a \sin u)^2 + (a \cos u)^2 + 0^2 = a^2 \\ F &= \vec{r}_1 \cdot \vec{r}_2 = 0 \\ G &= \vec{r}_2 \cdot \vec{r}_2 = 0^2 + 0^2 + 1^2 = 1\end{aligned}$$

The area element is therefore $\sqrt{EG - F^2} = \sqrt{(a^2)(1) - 0^2} = \sqrt{a^2} = a$. The surface area is found by integrating this constant value over the domain D :

$$\begin{aligned}A &= \iint_D a \, du \, dv = \int_{v=0}^h \int_{u=0}^{2\pi} a \, du \, dv \\ &= a \left(\int_0^{2\pi} du \right) \left(\int_0^h dv \right) \\ &= a(2\pi)(h) = 2\pi ah\end{aligned}$$

This is the well-known formula for the lateral area of a cylinder. ■

Example 2.2.17 (Area of a Cone). Find the lateral surface area of a right circular cone with base radius R and height H .

Solution. A cone can be parametrized by letting u be the distance from the apex and v be the angle of revolution. The relationship between height and radius is linear, so we set $z = (H/R)u$.

$$\vec{r}(u, v) = \left(u \cos v, u \sin v, \frac{H}{R}u \right)$$

The domain is $D = [0, R] \times [0, 2\pi]$, where u represents the radius of a circular cross-section. The tangent vectors are

$$\begin{aligned}\vec{r}_1 &= \left(\cos v, \sin v, \frac{H}{R} \right) \\ \vec{r}_2 &= (-u \sin v, u \cos v, 0)\end{aligned}$$

The first fundamental coefficients are

$$\begin{aligned}E &= \vec{r}_1 \cdot \vec{r}_1 = \cos^2 v + \sin^2 v + \left(\frac{H}{R} \right)^2 = 1 + \frac{H^2}{R^2} = \frac{R^2 + H^2}{R^2} \\ F &= \vec{r}_1 \cdot \vec{r}_2 = 0 \\ G &= \vec{r}_2 \cdot \vec{r}_2 = (-u \sin v)^2 + (u \cos v)^2 = u^2\end{aligned}$$

The area element integrand is

$$\sqrt{EG - F^2} = \sqrt{\left(\frac{R^2 + H^2}{R^2} \right) u^2 - 0} = \sqrt{\frac{L^2}{R^2} u^2} = \frac{L}{R} u$$

where $L = \sqrt{R^2 + H^2}$ is the slant height of the cone. Now we integrate over the domain:

$$\begin{aligned} A &= \iint_D \frac{L}{R} u \, du \, dv = \int_{v=0}^{2\pi} \int_{u=0}^R \frac{L}{R} u \, du \, dv \\ &= \frac{L}{R} \int_0^{2\pi} \left[\frac{u^2}{2} \right]_0^R dv \\ &= \frac{L}{R} \int_0^{2\pi} \frac{R^2}{2} dv = \frac{LR}{2} \int_0^{2\pi} dv \\ &= \frac{LR}{2} (2\pi) = \pi RL \end{aligned}$$

This gives the formula for the lateral surface area of a cone. ■

Example 2.2.18 (Area of a Torus). Find the surface area of a torus with major radius R and minor radius r ($R > r$).

Solution. A torus is generated by revolving a circle of radius r around an axis at a distance R from its center. A standard parametrization is $\vec{r}(u, v) = ((R + r \cos u) \cos v, (R + r \cos u) \sin v, r \sin u)$. The domain that covers the entire surface is $D = [0, 2\pi] \times [0, 2\pi]$. We compute the tangent vectors:

$$\begin{aligned} \vec{r}_1 &= (-r \sin u \cos v, -r \sin u \sin v, r \cos u) \\ \vec{r}_2 &= (-(R + r \cos u) \sin v, (R + r \cos u) \cos v, 0) \end{aligned}$$

Next calculate the fundamental coefficients:

$$\begin{aligned} E &= \vec{r}_1 \cdot \vec{r}_1 = (-r \sin u)^2 (\cos^2 v + \sin^2 v) + (r \cos u)^2 = r^2 \sin^2 u + r^2 \cos^2 u = r^2 \\ F &= \vec{r}_1 \cdot \vec{r}_2 = r \sin u \cos v (R + r \cos u) \sin v - r \sin u \sin v (R + r \cos u) \cos v + 0 = 0 \\ G &= \vec{r}_2 \cdot \vec{r}_2 = (R + r \cos u)^2 (\sin^2 v + \cos^2 v) = (R + r \cos u)^2 \end{aligned}$$

The area element is given by $\sqrt{EG - F^2} = \sqrt{r^2(R + r \cos u)^2 - 0} = r(R + r \cos u)$. (The absolute value is not needed since $R > r$ ensures $R + r \cos u > 0$). We integrate this over the domain:

$$\begin{aligned} A &= \iint_D r(R + r \cos u) \, du \, dv = \int_{v=0}^{2\pi} \int_{u=0}^{2\pi} (rR + r^2 \cos u) \, du \, dv \\ &= \int_0^{2\pi} [rRu + r^2 \sin u]_0^{2\pi} dv \\ &= \int_0^{2\pi} (rR(2\pi) + r^2 \sin(2\pi) - 0) dv \\ &= \int_0^{2\pi} 2\pi rR \, dv = 2\pi rR [v]_0^{2\pi} \\ &= 2\pi rR(2\pi) = 4\pi^2 Rr \end{aligned}$$

The surface area of the torus is $4\pi^2 Rr$. This result is consistent with Pappus's second centroid theorem: Area = (arc length of generating curve) $\times$ (distance travelled by centroid) = $(2\pi r)(2\pi R)$. ■

2.3 Orientability of Surfaces

An essential topological property of a surface is its orientability. Intuitively, a surface is orientable if it is "two-sided." For instance, a sphere has a distinct inside and outside; we can paint the outside blue and the inside red without the colors ever meeting. Similarly, a plane or a torus has two distinct sides.

In contrast, a surface like the Möbius strip is "one-sided." If one were to start painting its surface, one would eventually cover the entire strip without ever crossing an edge. This intuitive distinction is made precise in differential geometry using the concept of a continuously varying normal vector. An orientable surface is one on which we can define a consistent notion of "up" or "outward" at every point, whereas on a non-orientable surface, any such choice will inevitably lead to a contradiction.

2.3.1 The Differentiable Normal Field

Let S be a regular surface parametrized by $\vec{r}(u, v)$. At any point $P \in S$, the vectors $\vec{r}_1 = \partial\vec{r}/\partial u$ and $\vec{r}_2 = \partial\vec{r}/\partial v$ form a basis for the tangent plane $T_P(S)$. The cross product $\vec{r}_1 \times \vec{r}_2$ is a non-zero vector normal to the surface. We can define a unit normal vector $\vec{N}$ at P by

$$\vec{N}(u, v) = \frac{\vec{r}_1 \times \vec{r}_2}{|\vec{r}_1 \times \vec{r}_2|}$$

At each point, there are two choices for the unit normal vector, $\vec{N}$ and $-\vec{N}$. Orientability is the property that a choice can be made globally in a smooth, consistent manner.

Definition 2.3.1 (Orientability via a Normal Field). A regular surface $S \subset \mathbb{R}^3$ is said to be **orientable** if there exists a differentiable field of unit normal vectors $\vec{N} : S \rightarrow \mathbb{R}^3$. Such a choice of $\vec{N}$ is called an **orientation** of S . A surface that is not orientable is called **non-orientable**.

Remark 2.3.2. A vector field $\vec{N}$ on S is **differentiable** if, for any parametrization $\vec{r}(u, v)$ of S , the composite function $\vec{N}(\vec{r}(u, v))$ is a differentiable function of (u, v) . This formalizes the idea of the normal vector varying smoothly across the surface.

A direct consequence of the definition is a simple criterion for surfaces that are graphs of functions.

Theorem 2.3.3. *Every regular surface that can be represented globally as the graph of a differentiable function, $z = f(x, y)$, is orientable.*

Proof. Let the surface S be the graph of $z = f(x, y)$. We can parametrize S by $\vec{r}(x, y) = (x, y, f(x, y))$. The tangent vectors are

$$\begin{aligned}\vec{r}_1 &= \frac{\partial\vec{r}}{\partial x} = (1, 0, f_x) \\ \vec{r}_2 &= \frac{\partial\vec{r}}{\partial y} = (0, 1, f_y)\end{aligned}$$

Their cross product is a normal vector:

$$\vec{n}(x, y) = \vec{r}_1 \times \vec{r}_2 = (-f_x, -f_y, 1)$$

Since the z -component of $\vec{n}$ is always 1, this vector is never zero. As f is differentiable, its partial derivatives f_x and f_y are continuous, making $\vec{n}(x, y)$ a continuous (and thus differentiable) vector field on S . We can define a differentiable field of unit normal vectors by

$$\vec{N}(x, y) = \frac{\vec{n}(x, y)}{|\vec{n}(x, y)|} = \frac{(-f_x, -f_y, 1)}{\sqrt{f_x^2 + f_y^2 + 1}}$$

The existence of this field proves that S is orientable. The chosen orientation is the one with an "upward-pointing" normal. ■

Another powerful criterion involves level surfaces. In analogy to level curves (or contour lines) for a function of two variables, a level surface is the three-dimensional counterpart for a function of three variables. Imagine a scalar field in space, such as the temperature or electric potential at every point. A level surface is the set of all points in space where the value of this field is a constant. For instance, an isothermal surface is a level surface of the temperature function.

Definition 2.3.4 (Level Surface). Let U be an open subset of $\mathbb{R}^3$ and let $F : U \rightarrow \mathbb{R}$ be a real-valued, differentiable function of three variables, $F(x, y, z)$.

For any constant c in the range of F , the **level surface** (or **level set**) corresponding to c is the set of all points (x, y, z) in the domain U for which the function F takes the value c . This set, denoted S_c , is defined as

$$S_c = \{(x, y, z) \in U \mid F(x, y, z) = c\}$$

Remark 2.3.5 (Regular Surfaces). In differential geometry, we are particularly interested in level surfaces that are smooth, or "regular." A level surface S_c is guaranteed to be a **regular surface** at a point $P \in S_c$ if the gradient of F is non-zero at that point, i.e., $\nabla F(P) \neq \vec{0}$.

A constant c is called a **regular value** of F if $\nabla F(P) \neq \vec{0}$ for all points P such that $F(P) = c$. If c is a regular value, the entire level surface S_c is a regular surface. The gradient vector ∇F at any point on the surface is normal to the tangent plane of the surface at that point.

Example 2.3.6 (The Sphere). Consider the function $F(x, y, z) = x^2 + y^2 + z^2$. The level surfaces of this function are concentric spheres centered at the origin.

Let's choose a constant $c = a^2$ for some $a > 0$. The corresponding level surface is $S_{a^2} = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = a^2\}$. This is a sphere of radius a .

To check if this is a regular surface, we compute the gradient of F :

$$\nabla F(x, y, z) = \left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \right) = (2x, 2y, 2z)$$

The gradient ∇F is the zero vector only at the origin $(0, 0, 0)$. Since $a > 0$, the origin is not on the surface S_{a^2} . Therefore, $\nabla F \neq \vec{0}$ for all points on the sphere, which means $c = a^2$ is a regular value. Consequently, the sphere is a regular level surface.

Example 2.3.7 (A Plane). Let $F(x, y, z) = Ax + By + Cz$. The level surface for a constant D is the set of points where $Ax + By + Cz = D$, which is the equation of a plane. The gradient is $\nabla F = (A, B, C)$, which is a constant non-zero vector (assuming not all of A, B, C are zero). Therefore, any plane is a regular level surface.

Theorem 2.3.8. Let $F : U \subset \mathbb{R}^3 \rightarrow \mathbb{R}$ be a differentiable function. If c is a regular value of F (i.e., $\nabla F \neq \vec{0}$ for all points on the level set), then the level surface $S = F^{-1}(c)$ is orientable.

Proof. A fundamental property of the gradient is that ∇F is normal to the level surfaces of F . Since c is a regular value, $\nabla F(P) \neq \vec{0}$ for any point $P \in S$. Because F is differentiable, its gradient ∇F is a continuous vector field on U , and thus on S . Therefore, we can define a vector field on S by

$$\vec{N}(P) = \frac{\nabla F(P)}{|\nabla F(P)|}$$

This field consists of unit normal vectors and is differentiable. By definition, the existence of such a field implies that S is orientable. ■

Corollary 2.3.9. The sphere $x^2 + y^2 + z^2 = a^2$ is orientable.

Proof. Let $F(x, y, z) = x^2 + y^2 + z^2$. The sphere is the level surface $F = a^2$. The gradient is $\nabla F = (2x, 2y, 2z)$, which is zero only at the origin $(0, 0, 0)$. Since the origin is not on the sphere (for $a > 0$), ∇F is non-zero everywhere on the surface. Thus, the sphere is orientable. The normal field $\vec{N} = (x/a, y/a, z/a)$ defines the "outward" orientation. ■

2.3.2 The Oriented Atlas Criterion

An equivalent, more intrinsic definition of orientability involves the transition maps between coordinate charts of an atlas for the surface.

Definition 2.3.10 (Orientability via an Atlas). A regular surface S is **orientable** if it possesses an atlas $\{\vec{r}_\alpha : D_\alpha \rightarrow S\}$ such that for any two overlapping charts $\vec{r}_\alpha$ and $\vec{r}_\beta$, the Jacobian determinant of the transition map $\vec{r}_\beta^{-1} \circ \vec{r}_\alpha$ is positive on the domain of its definition. Such an atlas is called an **oriented atlas**.

Theorem 2.3.11. The two definitions of orientability are equivalent.

Proof. ($\Rightarrow$) Assume S is orientable and let $\vec{N}$ be a differentiable unit normal field on S . Let $\{\vec{s}_\alpha(u_\alpha, v_\alpha)\}$ be any atlas for S . For each chart $\vec{s}_\alpha$, we define its local unit normal

$$\vec{n}_\alpha = \frac{\vec{s}_{\alpha,1} \times \vec{s}_{\alpha,2}}{|\vec{s}_{\alpha,1} \times \vec{s}_{\alpha,2}|}$$

At each point, $\vec{n}_\alpha$ is either equal to $\vec{N}$ or $-\vec{N}$. If $\vec{n}_\alpha \cdot \vec{N} < 0$ on the chart's domain, we define a new chart $\vec{r}_\alpha(u_\alpha, v_\alpha) = \vec{s}_\alpha(v_\alpha, u_\alpha)$ (by swapping the parameters). This new chart has local normal $\vec{r}_{\alpha,1} \times \vec{r}_{\alpha,2} = \vec{s}_{\alpha,2} \times \vec{s}_{\alpha,1} = -(\vec{s}_{\alpha,1} \times \vec{s}_{\alpha,2})$, so its normal agrees with $\vec{N}$. The collection $\{\vec{r}_\alpha\}$ is a new atlas for S where every chart's local normal field agrees with the global orientation $\vec{N}$.

Now consider two overlapping charts $\vec{r}_\alpha(u_\alpha, v_\alpha)$ and $\vec{r}_\beta(u_\beta, v_\beta)$ from this new atlas. By the chain rule, their tangent vectors are related by the Jacobian matrix of the transition map:

$$(\vec{r}_{\beta,1}, \vec{r}_{\beta,2}) = (\vec{r}_{\alpha,1}, \vec{r}_{\alpha,2})J$$

where J is the Jacobian matrix of $\vec{r}_\alpha^{-1} \circ \vec{r}_\beta$. The normal vectors are related by

$$\vec{r}_{\beta,1} \times \vec{r}_{\beta,2} = (\det J)(\vec{r}_{\alpha,1} \times \vec{r}_{\alpha,2})$$

Since both normal vectors agree with the global field $\vec{N}$, they must point in the same direction. Therefore, $\det J$ must be positive. This confirms the atlas is oriented.

($\Leftarrow$) Assume S has an oriented atlas $\{\vec{r}_\alpha\}$. For each chart, define a local normal $\vec{n}_\alpha = (\vec{r}_{\alpha,1} \times \vec{r}_{\alpha,2})/|\dots|$. In any overlap, the relation $\vec{n}_\beta = \frac{\det J}{|\det J|} \vec{n}_\alpha$ holds. Since $\det J > 0$ for an oriented atlas, we have $\vec{n}_\beta = \vec{n}_\alpha$ on the overlap. This means the local normal fields agree on all intersections, allowing them to be "patched together" to form a single, globally defined, differentiable unit normal field $\vec{N}$. Therefore, S is orientable. ■

Theorem 2.3.12. *The Möbius strip is a non-orientable surface.*

Proof. We prove by contradiction. Assume the Möbius strip S is orientable, which implies the existence of a continuous unit normal field $\vec{N}$ on S . We will show that this assumption leads to a contradiction by examining a specific closed path on the surface.

Consider the standard parametrization of the Möbius strip of radius R :

$$\vec{r}(u, v) = \left(\left(R + v \cos \frac{u}{2} \right) \cos u, \left(R + v \cos \frac{u}{2} \right) \sin u, v \sin \frac{u}{2} \right)$$

for $(u, v) \in [0, 2\pi] \times (-1, 1)$.

Let us analyze the normal vector along the central circle of the strip, $\vec{\gamma}(u) = \vec{r}(u, 0)$, which is a closed loop since $\vec{r}(0, 0) = \vec{r}(2\pi, 0)$. Along this path (where $v = 0$), the tangent vectors are

$$\begin{aligned} \vec{r}_u(u, 0) &= (-R \sin u, R \cos u, 0) \\ \vec{r}_v(u, 0) &= \left(\cos \frac{u}{2} \cos u, \cos \frac{u}{2} \sin u, \sin \frac{u}{2} \right) \end{aligned}$$

Their cross product, a local normal vector $\vec{n}(u, 0)$, is found to be:

$$\vec{n}(u, 0) = \left(R \cos u \sin \frac{u}{2}, R \sin u \sin \frac{u}{2}, -R \cos \frac{u}{2} \right)$$

Now we evaluate this normal vector at the start and end of the parametric interval, which correspond to the same point $P = (R, 0, 0)$ on the strip.

At the start ($u = 0$):

$$\vec{n}(0, 0) = (0, 0, -R)$$

At the end ($u = 2\pi$):

$$\vec{n}(2\pi, 0) = (0, 0, R)$$

Thus, we have $\vec{n}(2\pi, 0) = -\vec{n}(0, 0)$.

This presents a contradiction. Any continuous normal field $\vec{N}$ on S must be continuously proportional to the local normal field $\vec{n}$ along the path. However, our calculation shows that after traversing a closed loop on the surface, the local normal vector reverses its direction. A continuous field cannot have two distinct, opposing values at the same point.

Therefore, our initial assumption must be false, and the Möbius strip is non-orientable. ■

2.3.3 The Gauss Map

Let $S \subset \mathbb{R}^3$ be a regular, orientable surface. The property of orientability guarantees the existence of a continuous, differentiable field of unit normal vectors $\vec{N} : S \rightarrow \mathbb{R}^3$. This normal field allows us to study the curvature of S by analyzing how the normal vector changes from point to point. The Gauss map provides the fundamental tool for this analysis.

Definition 2.3.13 (The Gauss Map). (See Figure 47 in the Atlas.) Let S be an oriented regular surface in $\mathbb{R}^3$ with a chosen orientation given by the unit normal field $\vec{N}$. The **Gauss map**, denoted $N : S \rightarrow S^2$, is the function that maps each point $P \in S$ to its corresponding unit normal vector $\vec{N}(P)$, viewed as a point on the unit sphere $S^2 \subset \mathbb{R}^3$.

$$N : S \rightarrow S^2, \quad P \mapsto \vec{N}(P)$$

Remark 2.3.14. The Gauss map essentially translates the surface S to the unit sphere. A flat region on S (where the normal vector is constant) will be mapped to a single point on S^2 . A highly curved region on S (where the normal vector changes rapidly) will be mapped to a large area on S^2 . The Gauss map thus encodes the extrinsic curvature of the surface.

2.3.4 The Differential of the Gauss Map (The Shape Operator)

Since the Gauss map N is a differentiable map between two surfaces, we can consider its differential. Let $P \in S$ and let $\vec{v} \in T_P(S)$ be a tangent vector at P . The differential of N at P , denoted dN_P , is a linear map from the tangent plane of S at P to the tangent plane of S^2 at the point $N(P)$.

$$dN_P : T_P(S) \rightarrow T_{N(P)}(S^2)$$

The action of dN_P on $\vec{v}$ is defined by the directional derivative of the vector field $\vec{N}$ in the direction of $\vec{v}$. Let $\vec{\alpha}(t)$ be a curve on S such that $\vec{\alpha}(0) = P$ and $\vec{\alpha}'(0) = \vec{v}$. Then

$$dN_P(\vec{v}) = \left. \frac{d}{dt} \right|_{t=0} N(\vec{\alpha}(t))$$

Crucially, the unit normal vector $\vec{N}(P)$ is, by its definition, orthogonal to the tangent plane $T_{N(P)}(S^2)$. Furthermore, the tangent plane $T_P(S)$ is also orthogonal to $\vec{N}(P)$. This means that the two tangent planes, $T_P(S)$ and $T_{N(P)}(S^2)$, are parallel. This parallelism allows us to identify them and view the differential dN_P as a linear operator on a single vector space, $T_P(S)$.

Definition 2.3.15 (The Shape Operator or Weingarten Map). The **shape operator** (or **Weingarten map**) at a point $P \in S$ is the linear operator $S_P : T_P(S) \rightarrow T_P(S)$ defined by

$$S_P(\vec{v}) = -dN_P(\vec{v})$$

for all $\vec{v} \in T_P(S)$.

Remark 2.3.16. The negative sign is a convention. With this sign, the eigenvalues of the shape operator (the principal curvatures) are positive for a surface that curves away from its tangent plane in the standard way (like a sphere oriented outwards). The operator $S_P(\vec{v})$ measures the rate of change of the normal vector as one moves along the surface in the direction $\vec{v}$, and is therefore a fundamental measure of the surface's curvature.

One of the most important properties of the shape operator is its symmetry with respect to the inner product (the first fundamental form) on the tangent plane.

Theorem 2.3.17. *The shape operator $S_P : T_P(S) \rightarrow T_P(S)$ is a self-adjoint (or symmetric) linear operator. That is, for any two tangent vectors $\vec{v}, \vec{w} \in T_P(S)$,*

$$\langle S_P(\vec{v}), \vec{w} \rangle = \langle \vec{v}, S_P(\vec{w}) \rangle$$

where $\langle \cdot, \cdot \rangle$ denotes the standard dot product in $\mathbb{R}^3$, which serves as the metric on the tangent plane.

Proof. Let $\vec{r}(u, v)$ be a local parametrization of the surface S around the point P . The vectors $\vec{r}_1 = \partial \vec{r} / \partial u$ and $\vec{r}_2 = \partial \vec{r} / \partial v$ form a basis for the tangent plane $T_P(S)$. To prove the theorem, it is sufficient to show that the identity holds for the basis vectors:

$$\langle S_P(\vec{r}_1), \vec{r}_2 \rangle = \langle \vec{r}_1, S_P(\vec{r}_2) \rangle$$

By definition, $S_P(\vec{r}_1) = -dN_P(\vec{r}_1) = -\frac{\partial \vec{N}}{\partial u} = -\vec{N}_1$. Similarly, $S_P(\vec{r}_2) = -\vec{N}_2$. The identity we need to prove becomes

$$\langle -\vec{N}_1, \vec{r}_2 \rangle = \langle \vec{r}_1, -\vec{N}_2 \rangle \quad \text{or} \quad \langle \vec{N}_1, \vec{r}_2 \rangle = \langle \vec{N}_2, \vec{r}_1 \rangle$$

To establish this, we start with the fundamental property that the normal vector $\vec{N}$ is orthogonal to any tangent vector. For the basis vector $\vec{r}_2$, this means:

$$\langle \vec{N}, \vec{r}_2 \rangle = 0$$

This identity holds over the entire coordinate patch. We can therefore differentiate it with respect to the parameter u :

$$\frac{\partial}{\partial u} \langle \vec{N}, \vec{r}_2 \rangle = 0$$

Applying the product rule for differentiation of an inner product, we obtain

$$\left\langle \frac{\partial \vec{N}}{\partial u}, \vec{r}_2 \right\rangle + \langle \vec{N}, \frac{\partial \vec{r}_2}{\partial u} \rangle = 0 \implies \langle \vec{N}_1, \vec{r}_2 \rangle + \langle \vec{N}, \vec{r}_{21} \rangle = 0$$

This gives us an expression for the left-hand side of our goal:

$$\langle \vec{N}_1, \vec{r}_2 \rangle = -\langle \vec{N}, \vec{r}_{21} \rangle \quad (2.4)$$

Similarly, we start with the orthogonality condition for $\vec{r}_1$:

$$\langle \vec{N}, \vec{r}_1 \rangle = 0$$

Differentiating this with respect to the parameter v yields

$$\frac{\partial}{\partial v} \langle \vec{N}, \vec{r}_1 \rangle = 0 \implies \left\langle \frac{\partial \vec{N}}{\partial v}, \vec{r}_1 \right\rangle + \langle \vec{N}, \frac{\partial \vec{r}_1}{\partial v} \rangle = 0 \implies \langle \vec{N}_2, \vec{r}_1 \rangle + \langle \vec{N}, \vec{r}_{12} \rangle = 0$$

This gives an expression for the right-hand side of our goal:

$$\langle \vec{N}_2, \vec{r}_1 \rangle = -\langle \vec{N}, \vec{r}_{12} \rangle \quad (2.5)$$

Since the parametrization $\vec{r}(u, v)$ is regular, its component functions are at least C^2 . By Clairaut's Theorem on the equality of mixed partial derivatives, we have $\vec{r}_{12} = \frac{\partial^2 \vec{r}}{\partial v \partial u} = \frac{\partial^2 \vec{r}}{\partial u \partial v} = \vec{r}_{21}$. Therefore, the right-hand sides of equations (2.4) and (2.5) are equal, implying that

$$\langle \vec{N}_1, \vec{r}_2 \rangle = \langle \vec{N}_2, \vec{r}_1 \rangle$$

This proves the result. ■

Corollary 2.3.18 (Existence of Principal Curvatures and Directions). *A direct and profound consequence of the self-adjointness of the shape operator S_P is that, at every point $P \in S$, there exists an orthonormal basis of $T_P(S)$ consisting of eigenvectors of S_P .*

- The eigenvalues of S_P , denoted k_1 and k_2 , are real numbers called the **principal curvatures**.
- The corresponding eigenvectors, $\vec{e}_1$ and $\vec{e}_2$, are called the **principal directions** of curvature.

This property is fundamental to the study of surface curvature, as it allows us to define the Gaussian curvature ($K = k_1 k_2$) and mean curvature ($H = \frac{1}{2}(k_1 + k_2)$) of the surface.

Proof. Proof shall be taken at a later stage in this chapter. ■

2.4 The Second Fundamental Form

Let $S \subset \mathbb{R}^3$ be a regular surface. For a point $p \in S$, choose a local parametrization

$$\mathbf{r}(u, v) : U \subset \mathbb{R}^2 \rightarrow S, \quad (u, v) \in U,$$

with $\mathbf{r}_u = \frac{\partial \mathbf{r}}{\partial u}$ and $\mathbf{r}_v = \frac{\partial \mathbf{r}}{\partial v}$ linearly independent. The **first fundamental form** is $I = E du^2 + 2F du dv + G dv^2$, where $E = \mathbf{r}_u \cdot \mathbf{r}_u$, $F = \mathbf{r}_u \cdot \mathbf{r}_v$, $G = \mathbf{r}_v \cdot \mathbf{r}_v$. The unit normal vector field (compatible with the orientation) is $\mathbf{N} = \frac{\mathbf{r}_u \times \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|}$.

Definition 2.4.1 (Shape operator). For $p \in S$ and $\mathbf{v} \in T_p S$, let $\alpha(t)$ be a curve on S with $\alpha(0) = p$ and $\alpha'(0) = \mathbf{v}$. The **shape operator** (or Weingarten map) $S_p : T_p S \rightarrow T_p S$ is

$$S_p(\mathbf{v}) = -d\mathbf{N}_p(\mathbf{v}) = - \left. \frac{d}{dt} \right|_{t=0} \mathbf{N}(\alpha(t)).$$

It is a linear, self-adjoint operator with respect to the first fundamental form.

Definition 2.4.2 (Second fundamental form). The **second fundamental form** $II_p : T_p S \times T_p S \rightarrow \mathbb{R}$ is the symmetric bilinear form

$$II_p(\mathbf{v}, \mathbf{w}) = \langle S_p(\mathbf{v}), \mathbf{w} \rangle = -\langle d\mathbf{N}_p(\mathbf{v}), \mathbf{w} \rangle,$$

where $\langle \cdot, \cdot \rangle$ is the Euclidean dot product.

Local coordinate expression

As $\mathbf{r}_u, \mathbf{r}_v$ spans $T_p S$. Take $\mathbf{v} = a\mathbf{r}_u + b\mathbf{r}_v$ and $\mathbf{w} = c\mathbf{r}_u + d\mathbf{r}_v$ in $T_p S$. Then

$$\begin{aligned} II_p(\mathbf{v}, \mathbf{w}) &= \langle -dN_P(a\vec{r}_u + b\vec{r}_v), c\vec{r}_u + d\vec{r}_v \rangle \\ &= -\langle a\mathbf{N}_u + b\mathbf{N}_v, c\mathbf{r}_u + d\mathbf{r}_v \rangle \\ &= -ac\langle \mathbf{N}_u, \mathbf{r}_u \rangle - ad\langle \mathbf{N}_u, \mathbf{r}_v \rangle - bc\langle \mathbf{N}_v, \mathbf{r}_u \rangle - bd\langle \mathbf{N}_v, \mathbf{r}_v \rangle. \end{aligned}$$

Because $\mathbf{N} \cdot \mathbf{r}_u = 0$ and $\mathbf{N} \cdot \mathbf{r}_v = 0$, differentiation gives

$$\langle \mathbf{N}_u, \mathbf{r}_u \rangle = -\mathbf{N} \cdot \mathbf{r}_{uu}, \quad \langle \mathbf{N}_u, \mathbf{r}_v \rangle = -\mathbf{N} \cdot \mathbf{r}_{uv}, \quad \langle \mathbf{N}_v, \mathbf{r}_u \rangle = -\mathbf{N} \cdot \mathbf{r}_{uv}, \quad \langle \mathbf{N}_v, \mathbf{r}_v \rangle = -\mathbf{N} \cdot \mathbf{r}_{vv}.$$

Define the coefficients

$$L = \mathbf{N} \cdot \mathbf{r}_{uu}, \quad M = \mathbf{N} \cdot \mathbf{r}_{uv}, \quad N = \mathbf{N} \cdot \mathbf{r}_{vv}.$$

Then

$$II_p(\mathbf{v}, \mathbf{w}) = acL + (ad + bc)M + bdN.$$

Identifying $du(\mathbf{v}) = a$, $dv(\mathbf{v}) = b$, $du(\mathbf{w}) = c$, $dv(\mathbf{w}) = d$, we obtain the quadratic form

$$II = L du^2 + 2M du dv + N dv^2.$$

Definition 2.4.3 (Coefficients of the Second Fundamental Form). The coefficients L, M, N of the second fundamental form are given by:

$$\begin{aligned} L(u, v) &= II(\vec{r}_1) = \langle S_P(\vec{r}_1), \vec{r}_1 \rangle \\ M(u, v) &= II(\vec{r}_1, \vec{r}_2) = \langle S_P(\vec{r}_1), \vec{r}_2 \rangle = \langle S_P(\vec{r}_2), \vec{r}_1 \rangle \\ N(u, v) &= II(\vec{r}_2) = \langle S_P(\vec{r}_2), \vec{r}_2 \rangle \end{aligned}$$

Explicit formulas for L, M, N via cross products

Using $\mathbf{N} = \frac{\mathbf{r}_u \times \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|}$,

$$L = \frac{(\mathbf{r}_u \times \mathbf{r}_v) \cdot \mathbf{r}_{uu}}{\|\mathbf{r}_u \times \mathbf{r}_v\|} = \frac{(\mathbf{r}_{uu} \times \mathbf{r}_u) \cdot \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|} \quad (\text{cyclic permutation}).$$

The alternative formula (with $\mathbf{N}$ taken as $\frac{\mathbf{r}_v \times \mathbf{r}_u}{\|\mathbf{r}_u \times \mathbf{r}_v\|}$) gives

$$L = \frac{(\mathbf{r}_{uu} \times \mathbf{r}_v) \cdot \mathbf{r}_u}{\|\mathbf{r}_u \times \mathbf{r}_v\|}, \quad M = \frac{(\mathbf{r}_{uv} \times \mathbf{r}_v) \cdot \mathbf{r}_u}{\|\mathbf{r}_u \times \mathbf{r}_v\|}, \quad N = \frac{(\mathbf{r}_{vv} \times \mathbf{r}_v) \cdot \mathbf{r}_u}{\|\mathbf{r}_u \times \mathbf{r}_v\|}.$$

Either orientation is acceptable; only the overall sign of II changes.

2.5 Weingarten Equations (Shape Operator in Coordinates)

The partial derivatives of $\mathbf{N}$ can be expressed in the basis $\{\mathbf{r}_u, \mathbf{r}_v\}$.

Theorem 2.5.1 (Weingarten Equations). *Let $S \subset \mathbb{R}^3$ be a regular surface parametrized by $\mathbf{r}(u, v)$ with unit normal $\mathbf{N}$. Denote the coefficients of the first and second fundamental forms by E, F, G and L, M, N respectively. Then the partial derivatives of $\mathbf{N}$ are given by*

$$\begin{aligned} \mathbf{N}_u &= \frac{FM - GL}{EG - F^2} \mathbf{r}_u + \frac{FL - EM}{EG - F^2} \mathbf{r}_v, \\ \mathbf{N}_v &= \frac{FN - GM}{EG - F^2} \mathbf{r}_u + \frac{FM - EN}{EG - F^2} \mathbf{r}_v. \end{aligned}$$

Proof. Since $\mathbf{N}$ is a unit normal, we have $\mathbf{N} \cdot \mathbf{r}_u = 0$ and $\mathbf{N} \cdot \mathbf{r}_v = 0$. Differentiating the first equality with respect to u and v yields

$$\begin{aligned} \mathbf{N}_u \cdot \mathbf{r}_u + \mathbf{N} \cdot \mathbf{r}_{uu} &= 0 \implies \mathbf{N}_u \cdot \mathbf{r}_u = -\mathbf{N} \cdot \mathbf{r}_{uu} = -L, \\ \mathbf{N}_v \cdot \mathbf{r}_u + \mathbf{N} \cdot \mathbf{r}_{uv} &= 0 \implies \mathbf{N}_v \cdot \mathbf{r}_u = -\mathbf{N} \cdot \mathbf{r}_{uv} = -M. \end{aligned}$$

Similarly, from $\mathbf{N} \cdot \mathbf{r}_v = 0$ we obtain

$$\mathbf{N}_u \cdot \mathbf{r}_v = -\mathbf{N} \cdot \mathbf{r}_{uv} = -M, \quad \mathbf{N}_v \cdot \mathbf{r}_v = -\mathbf{N} \cdot \mathbf{r}_{vv} = -N.$$

Now, because $\{\mathbf{r}_u, \mathbf{r}_v\}$ is a basis of the tangent plane, we can write

$$\mathbf{N}_u = \alpha \mathbf{r}_u + \beta \mathbf{r}_v,$$

for some functions α, β on the surface. Taking the dot product of this expression with $\mathbf{r}_u$ and $\mathbf{r}_v$ gives the linear system

$$\begin{cases} \alpha E + \beta F = \mathbf{N}_u \cdot \mathbf{r}_u = -L, \\ \alpha F + \beta G = \mathbf{N}_u \cdot \mathbf{r}_v = -M. \end{cases}$$

We solve for α and β . The determinant of the coefficient matrix is $EG - F^2 \neq 0$ (regularity). Using Cramer's rule:

$$\begin{aligned} \alpha &= \frac{\det \begin{pmatrix} -L & F \\ -M & G \end{pmatrix}}{\det \begin{pmatrix} E & F \\ F & G \end{pmatrix}} = \frac{(-L)G - F(-M)}{EG - F^2} = \frac{-LG + FM}{EG - F^2} = \frac{FM - GL}{EG - F^2}, \\ \beta &= \frac{\det \begin{pmatrix} E & -L \\ F & -M \end{pmatrix}}{EG - F^2} = \frac{E(-M) - (-L)F}{EG - F^2} = \frac{-EM + LF}{EG - F^2} = \frac{FL - EM}{EG - F^2}. \end{aligned}$$

Thus the first Weingarten equation is proved.

For $\mathbf{N}_v$, write $\mathbf{N}_v = \gamma \mathbf{r}_u + \delta \mathbf{r}_v$. Dotting with $\mathbf{r}_u$ and $\mathbf{r}_v$ yields

$$\gamma E + \delta F = \mathbf{N}_v \cdot \mathbf{r}_u = -M, \quad \gamma F + \delta G = \mathbf{N}_v \cdot \mathbf{r}_v = -N.$$

Solving similarly,

$$\begin{aligned} \gamma &= \frac{(-M)G - F(-N)}{EG - F^2} = \frac{-MG + FN}{EG - F^2} = \frac{FN - GM}{EG - F^2}, \\ \delta &= \frac{E(-N) - (-M)F}{EG - F^2} = \frac{-EN + MF}{EG - F^2} = \frac{FM - EN}{EG - F^2}. \end{aligned}$$

This proves the result. ■

Let $S \subset \mathbb{R}^3$ be a regular surface with local parametrization $\mathbf{r}(u, v)$. The shape operator $S_p : T_p S \rightarrow T_p S$ is defined by $S_p(\mathbf{v}) = -d\mathbf{N}_p(\mathbf{v})$, where $\mathbf{N}$ is a unit normal field. In the basis $\{\mathbf{r}_u, \mathbf{r}_v\}$ of the tangent plane, the matrix of S_p is obtained from the Weingarten equations:

$$\begin{aligned} \mathbf{N}_u &= \frac{FM - GL}{EG - F^2} \mathbf{r}_u + \frac{FL - EM}{EG - F^2} \mathbf{r}_v, \\ \mathbf{N}_v &= \frac{FN - GM}{EG - F^2} \mathbf{r}_u + \frac{FM - EN}{EG - F^2} \mathbf{r}_v. \end{aligned}$$

Since $S(\mathbf{r}_u) = -\mathbf{N}_u$ and $S(\mathbf{r}_v) = -\mathbf{N}_v$, the matrix of S with respect to $\{\mathbf{r}_u, \mathbf{r}_v\}$ is

$$[S] = \frac{1}{EG - F^2} \begin{pmatrix} GL - FM & GM - FN \\ EM - FL & EN - FM \end{pmatrix}.$$

Note: Let $M_1 = \begin{bmatrix} E & F \\ F & G \end{bmatrix}$ and $M_2 = \begin{bmatrix} L & M \\ M & N \end{bmatrix}$ be the matrices of the First and the Second fundamental forms, then the Weingarten equations can be written as

$$\begin{bmatrix} N_u \\ N_v \end{bmatrix} = M_1^{-1} M_2 = \frac{1}{EG - F^2} \begin{bmatrix} G & -F \\ -F & E \end{bmatrix} \begin{bmatrix} L & M \\ M & N \end{bmatrix} \begin{bmatrix} r_u \\ r_v \end{bmatrix}$$

Theorem 2.5.2 (Principal Curvatures). *At any point $p \in S$, the eigenvalues of the shape operator S_p are the **principal curvatures** k_1 and k_2 . They are given explicitly by*

$$k_{1,2} = H \pm \sqrt{H^2 - K},$$

where

$$H = \frac{EN - 2FM + GL}{2(EG - F^2)}, \quad K = \frac{LN - M^2}{EG - F^2}$$

are the mean curvature and Gaussian curvature, respectively.

Proof. The characteristic polynomial of $[S]$ is $\det([S] - kI) = 0$. The trace and determinant of $[S]$ are

$$\begin{aligned} \text{tr}(S) &= \frac{GL - FM + EN - FM}{EG - F^2} = \frac{EN + GL - 2FM}{EG - F^2} = 2H, \\ \det(S) &= \frac{(GL - FM)(EN - FM) - (GM - FN)(EM - FL)}{(EG - F^2)^2}. \end{aligned}$$

Expanding the numerator:

$$\begin{aligned} &(GL - FM)(EN - FM) - (GM - FN)(EM - FL) \\ &= (GL)(EN) - (GL)(FM) - (FM)(EN) + (FM)^2 \\ &\quad - [(GM)(EM) - (GM)(FL) - (FN)(EM) + (FN)(FL)] \\ &= GL EN - GL FM - FM EN + F^2 M^2 \\ &\quad - GM EM + GM FL + FN EM - FN FL. \end{aligned}$$

Using $EG - F^2 = \|\mathbf{r}_u \times \mathbf{r}_v\|^2 > 0$, one can simplify to

$$\det([S]) = \frac{(LN - M^2)(EG - F^2)}{(EG - F^2)^2} = \frac{LN - M^2}{EG - F^2} = K.$$

Thus the characteristic equation is $k^2 - (\text{tr } S)k + \det S = 0$, i.e.

$$k^2 - 2Hk + K = 0.$$

Solving the quadratic:

$$k = \frac{2H \pm \sqrt{4H^2 - 4K}}{2} = H \pm \sqrt{H^2 - K}.$$

Hence, the eigenvalues (principal curvatures) are as stated. ■

Example 2.5.3. Show that:

- (i) $L = \frac{1}{H}[\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_{11}]$
- (ii) $M = \frac{1}{H}[\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_{12}]$
- (iii) $N = \frac{1}{H}[\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_{22}]$

where $H = |\mathbf{r}_1 \times \mathbf{r}_2|$ and $[a, b, c]$ denotes the scalar triple product.

Solution. The coefficients of the second fundamental form are defined by the projection of the second-order partial derivatives onto the unit normal vector $\mathbf{n}$:

$$L = \mathbf{r}_{11} \cdot \mathbf{n}, \quad M = \mathbf{r}_{12} \cdot \mathbf{n}, \quad N = \mathbf{r}_{22} \cdot \mathbf{n}$$

The unit normal vector $\mathbf{n}$ to the surface $\mathbf{r}(u, v)$ is defined as

$$\mathbf{n} = \frac{\mathbf{r}_1 \times \mathbf{r}_2}{|\mathbf{r}_1 \times \mathbf{r}_2|} = \frac{\mathbf{r}_1 \times \mathbf{r}_2}{H}$$

Substituting this expression for $\mathbf{n}$ into the definitions:

1. For L :

$$L = \mathbf{r}_{11} \cdot \left(\frac{\mathbf{r}_1 \times \mathbf{r}_2}{H} \right) = \frac{1}{H}(\mathbf{r}_{11} \cdot (\mathbf{r}_1 \times \mathbf{r}_2)) = \frac{1}{H}[\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_{11}]$$

2. For M :

$$M = \mathbf{r}_{12} \cdot \left(\frac{\mathbf{r}_1 \times \mathbf{r}_2}{H} \right) = \frac{1}{H}(\mathbf{r}_{12} \cdot (\mathbf{r}_1 \times \mathbf{r}_2)) = \frac{1}{H}[\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_{12}]$$

3. For N :

$$N = \mathbf{r}_{22} \cdot \left(\frac{\mathbf{r}_1 \times \mathbf{r}_2}{H} \right) = \frac{1}{H}(\mathbf{r}_{22} \cdot (\mathbf{r}_1 \times \mathbf{r}_2)) = \frac{1}{H}[\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_{22}]$$

This confirms the required identities. ■

Definition 2.5.4 (Gaussian and Mean Curvature). Let k_1, k_2 be the principal curvatures at a point P .

- The **Gaussian curvature** is $K = k_1 k_2 = \det(S_P)$.
- The **Mean curvature** is $H = \frac{1}{2}(k_1 + k_2) = \frac{1}{2}\text{tr}(S_P)$.

These can be computed directly from the coefficients of the two fundamental forms:

$$K = \frac{LN - M^2}{EG - F^2} \quad \text{and} \quad H = \frac{EN - 2FM + GL}{2(EG - F^2)}$$

Corollary 2.5.5. *The principal curvatures are real because $H^2 - K \geq 0$ for any surface (the discriminant of the characteristic polynomial is non-negative for a self-adjoint operator). Moreover, the principal directions are the corresponding eigenvectors of $[S]$, obtained by solving $(S - k_i I)\mathbf{v} = 0$.*

Example 2.5.6 (The Sphere). Find the second fundamental form, Gaussian curvature, and mean curvature of a sphere of radius a .

Solution. Parametrize the sphere by $\vec{r}(u, v) = (a \sin u \cos v, a \sin u \sin v, a \cos u)$. We have previously found:

- $\vec{r}_1 = (a \cos u \cos v, a \cos u \sin v, -a \sin u)$
- $\vec{r}_2 = (-a \sin u \sin v, a \sin u \cos v, 0)$
- The outward unit normal is $\vec{N} = (\sin u \cos v, \sin u \sin v, \cos u)$.
- $E = a^2, F = 0, G = a^2 \sin^2 u$.

We compute the second partial derivatives:

$$\begin{aligned}\vec{r}_{11} &= (-a \sin u \cos v, -a \sin u \sin v, -a \cos u) \\ \vec{r}_{12} &= (-a \cos u \sin v, a \cos u \cos v, 0) \\ \vec{r}_{22} &= (-a \sin u \cos v, -a \sin u \sin v, 0)\end{aligned}$$

Now we compute L, M, N by taking dot products with $\vec{N}$:

$$\begin{aligned}L &= \vec{r}_{11} \cdot \vec{N} = -a \sin^2 u \cos^2 v - a \sin^2 u \sin^2 v - a \cos^2 u = -a(\sin^2 u + \cos^2 u) = -a \\ M &= \vec{r}_{12} \cdot \vec{N} = -a \cos u \sin u \sin v \cos v + a \cos u \sin u \cos v \sin v = 0 \\ N &= \vec{r}_{22} \cdot \vec{N} = -a \sin^2 u \cos^2 v - a \sin^2 u \sin^2 v = -a \sin^2 u(\cos^2 v + \sin^2 v) = -a \sin^2 u\end{aligned}$$

The second fundamental form is $II = -a du^2 - a \sin^2 u dv^2$. The curvatures are

$$\begin{aligned}K &= \frac{LN - M^2}{EG - F^2} = \frac{(-a)(-a \sin^2 u) - 0^2}{(a^2)(a^2 \sin^2 u) - 0^2} = \frac{a^2 \sin^2 u}{a^4 \sin^2 u} = \frac{1}{a^2} \\ H &= \frac{EN - 2FM + GL}{2(EG - F^2)} = \frac{a^2(-a \sin^2 u) - 0 + a^2 \sin^2 u(-a)}{2(a^4 \sin^2 u)} = \frac{-2a^3 \sin^2 u}{2a^4 \sin^2 u} = -\frac{1}{a}\end{aligned}$$

The negative sign for H depends on the choice of the outward normal. The magnitudes are constant, as expected. Principal curvatures are $k_1 = k_2 = -1/a$. ■

Example 2.5.7 (Circular Cylinder of Radius a). (See Figure 33 in the Atlas.) Compute the curvatures of the cylinder $x^2 + y^2 = a^2$.

Solution. We parametrize the cylinder by $\vec{r}(u, v) = (a \cos u, a \sin u, v)$.

First Fundamental Form: The tangent vectors are $\vec{r}_1 = (-a \sin u, a \cos u, 0)$ and $\vec{r}_2 = (0, 0, 1)$.

$$E = a^2, \quad F = 0, \quad G = 1$$

Normal Vector and Second Derivatives: For a more intuitive result where curvature is positive, we choose the inward-pointing normal:

$$\vec{N} = (-\cos u, -\sin u, 0)$$

The second derivatives are

$$\begin{aligned}\vec{r}_{11} &= (-a \cos u, -a \sin u, 0) \\ \vec{r}_{12} &= (0, 0, 0) \\ \vec{r}_{22} &= (0, 0, 0)\end{aligned}$$

Second Fundamental Form: The coefficients are

$$\begin{aligned} L &= \vec{r}_{11} \cdot \vec{N} = (-a \cos u)(-\cos u) + (-a \sin u)(-\sin u) = a \\ M &= \vec{r}_{12} \cdot \vec{N} = 0 \\ N &= \vec{r}_{22} \cdot \vec{N} = 0 \end{aligned}$$

The second fundamental form is $II = a du^2$.

Curvatures:

$$\begin{aligned} K &= \frac{LN - M^2}{EG - F^2} = \frac{a(0) - 0^2}{a^2(1) - 0^2} = 0 \\ H &= \frac{EN - 2FM + GL}{2(EG - F^2)} = \frac{a^2(0) - 0 + 1(a)}{2(a^2)} = \frac{1}{2a} \end{aligned}$$

The zero Gaussian curvature confirms that the cylinder is a developable surface. The principal curvatures are $k_1 = 1/a$ (the curvature of the circular cross-section) and $k_2 = 0$ (the curvature of the straight rulings). ■

Example 2.5.8 (Elliptic Paraboloid $z = x^2 + y^2$). Compute the curvatures of the elliptic paraboloid.

Solution. We use the graph parametrization $\vec{r}(x, y) = (x, y, x^2 + y^2)$.

First Fundamental Form: The tangent vectors are $\vec{r}_1 = (1, 0, 2x)$ and $\vec{r}_2 = (0, 1, 2y)$.

$$E = 1 + 4x^2, \quad F = 4xy, \quad G = 1 + 4y^2$$

Normal Vector and Second Derivatives: The upward-pointing normal is $\vec{r}_1 \times \vec{r}_2 = (-2x, -2y, 1)$. Normalizing gives $\vec{N} = \frac{(-2x, -2y, 1)}{\sqrt{1+4x^2+4y^2}}$. The second derivatives are constant:

$$\vec{r}_{11} = (0, 0, 2), \quad \vec{r}_{12} = (0, 0, 0), \quad \vec{r}_{22} = (0, 0, 2)$$

Second Fundamental Form: The coefficients are

$$\begin{aligned} L &= \vec{r}_{11} \cdot \vec{N} = \frac{2}{\sqrt{1+4x^2+4y^2}} \\ M &= \vec{r}_{12} \cdot \vec{N} = 0 \\ N &= \vec{r}_{22} \cdot \vec{N} = \frac{2}{\sqrt{1+4x^2+4y^2}} \end{aligned}$$

Curvatures: We first compute the denominators and numerators. Let $r^2 = x^2 + y^2$.

$$\begin{aligned} EG - F^2 &= (1 + 4x^2)(1 + 4y^2) - (4xy)^2 = 1 + 4x^2 + 4y^2 + 16x^2y^2 - 16x^2y^2 = 1 + 4r^2 \\ LN - M^2 &= \left(\frac{2}{\sqrt{1+4r^2}} \right) \left(\frac{2}{\sqrt{1+4r^2}} \right) - 0 = \frac{4}{1+4r^2} \end{aligned}$$

The Gaussian curvature is

$$K = \frac{LN - M^2}{EG - F^2} = \frac{4/(1+4r^2)}{1+4r^2} = \frac{4}{(1+4r^2)^2}$$

The mean curvature is

$$H = \frac{EN + GL}{2(EG - F^2)} = \frac{(1+4x^2)\frac{2}{\sqrt{1+4r^2}} + (1+4y^2)\frac{2}{\sqrt{1+4r^2}}}{2(1+4r^2)} = \frac{2(2+4r^2)}{2(1+4r^2)^{3/2}} = \frac{2+4r^2}{(1+4r^2)^{3/2}}$$

Both curvatures are highest at the origin ($r = 0$) and decrease as one moves away from the origin. ■

The second fundamental form encodes the extrinsic curvature of a surface. Its coefficients L, M, N are obtained from the derivatives of the unit normal. The shape operator, it gives the normal curvature, principal curvatures, and the Gaussian and mean curvatures. The Theorema Egregium reveals that K is an intrinsic invariant, a deep insight that paved the way for modern Riemannian geometry.

We know that any plane drawn through a point P on a surface cuts it in a curve which is called the section of the surface by that plane. If, however, the plane is such that it contains the normal to the surface at P , then this section is called the normal section, and all other sections will be termed as oblique sections. Observe that the principal normal for the normal section is parallel to the surface normal. There are infinitely many normal sections at each point of the surface.

At its core, normal curvature measures how a regular surface S bends at a point P in a specific tangent direction. It can be visualized as the curvature of the curve formed by slicing the surface with a plane containing both the direction vector and the surface normal $\vec{N}$.

Definition 2.5.9 (Normal Curvature). Let S be an oriented surface, $P \in S$, and $\vec{t}$ be a unit tangent vector in the tangent plane $T_P(S)$. The **normal curvature** of S at P in the direction $\vec{t}$, denoted $k_n(\vec{t})$, is defined as the second fundamental form evaluated on $\vec{t}$:

$$k_n(\vec{t}) = \langle L_P(\vec{t}), \vec{t} \rangle$$

where L_P is the shape operator at P . This value is equivalent to the projection of the acceleration of any curve $\vec{\alpha}(s)$ with tangent $\vec{t}$ onto the surface normal:

$$k_n(\vec{t}) = \langle \vec{\alpha}''(s), \vec{N} \rangle$$

Remark 2.5.10. A fundamental property is that k_n depends only on the direction $\vec{t}$ at the point P , not on the specific curve passing through P with that tangent.

Theorem 2.5.11. Prove that the normal curvature of a surface $\vec{r}(u, v)$ at any point P is

$$\kappa_n = \frac{II_P}{I_P} = \frac{Ldu^2 + 2Mdudv + Ndv^2}{Edu^2 + 2Fdudv + Gdv^2}$$

Proof. The proof begins with the definition of normal curvature and expresses it in terms of the two fundamental forms.

The normal curvature k_n at a point P is defined for a direction specified by a *unit* tangent vector $\vec{t} \in T_P(S)$, and is given by the action of the shape operator L_P :

$$k_n(\vec{t}) = \langle L_P(\vec{t}), \vec{t} \rangle$$

A general tangent direction can be represented by an arbitrary non-zero tangent vector $\vec{v}$, which corresponds to an infinitesimal displacement $\vec{v} = \vec{r}_1 du + \vec{r}_2 dv$. To use the definition, we normalize this vector to get the corresponding unit tangent vector $\vec{t} = \vec{v}/|\vec{v}|$.

Substituting this into the definition of k_n , and using the linearity of the shape operator and the bilinearity of the inner product, we have

$$k_n = \langle L_P \left(\frac{\vec{v}}{|\vec{v}|} \right), \frac{\vec{v}}{|\vec{v}|} \rangle = \frac{1}{|\vec{v}|^2} \langle L_P(\vec{v}), \vec{v} \rangle$$

We now identify the numerator and denominator of this expression.

The numerator, $\langle L_P(\vec{v}), \vec{v} \rangle$, is by definition the second fundamental form II_P evaluated on the vector $\vec{v}$. Expanding this gives

$$\begin{aligned} \langle L_P(\vec{r}_1 du + \vec{r}_2 dv), \vec{r}_1 du + \vec{r}_2 dv \rangle &= \langle L_P(\vec{r}_1), \vec{r}_1 \rangle du^2 + 2\langle L_P(\vec{r}_1), \vec{r}_2 \rangle du dv + \langle L_P(\vec{r}_2), \vec{r}_2 \rangle dv^2 \\ &= L du^2 + 2M du dv + N dv^2 = II_P \end{aligned}$$

The denominator, $|\vec{v}|^2 = \langle \vec{v}, \vec{v} \rangle$, is the first fundamental form I_P evaluated on $\vec{v}$:

$$\begin{aligned} \langle \vec{r}_1 du + \vec{r}_2 dv, \vec{r}_1 du + \vec{r}_2 dv \rangle &= \langle \vec{r}_1, \vec{r}_1 \rangle du^2 + 2\langle \vec{r}_1, \vec{r}_2 \rangle du dv + \langle \vec{r}_2, \vec{r}_2 \rangle dv^2 \\ &= E du^2 + 2F du dv + G dv^2 = I_P \end{aligned}$$

Substituting these expressions back into our formula for k_n yields the desired result:

$$k_n = \frac{\langle L_P(\vec{v}), \vec{v} \rangle}{|\vec{v}|^2} = \frac{II_P}{I_P} = \frac{L du^2 + 2M du dv + N dv^2}{E du^2 + 2F du dv + G dv^2}$$

This proves the result. ■

2nd Proof. This proof begins with the geometric definition of normal curvature and derives the formula by analyzing the acceleration of a curve on the surface.

Let $\vec{\alpha}(s)$ be a curve on the surface S parametrized by arc length s , passing through the point P . The curve can be expressed in terms of the surface parametrization as $\vec{\alpha}(s) = \vec{r}(u(s), v(s))$. The normal curvature at P is defined as

$$k_n = \langle \vec{\alpha}''(s), \vec{N} \rangle$$

where $\vec{N}$ is the unit normal to the surface at P . Our goal is to express $\vec{\alpha}''(s)$ in terms of the parametrization $\vec{r}$.

First compute the tangent vector $\vec{\alpha}'(s)$ using the chain rule. We use the notation $u' = \frac{du}{ds}$ and $v' = \frac{dv}{ds}$.

$$\vec{\alpha}'(s) = \frac{\partial \vec{r}}{\partial u} \frac{du}{ds} + \frac{\partial \vec{r}}{\partial v} \frac{dv}{ds} = \vec{r}_1 u' + \vec{r}_2 v'$$

Next differentiate $\vec{\alpha}'(s)$ with respect to s to find the acceleration vector $\vec{\alpha}''(s)$. This requires both the product rule and the chain rule, as the basis vectors $\vec{r}_1$ and $\vec{r}_2$ change with position $(u(s), v(s))$:

$$\begin{aligned} \vec{\alpha}''(s) &= \frac{d}{ds}(\vec{r}_1 u' + \vec{r}_2 v') \\ &= \left(\frac{d\vec{r}_1}{ds} \right) u' + \vec{r}_1 u'' + \left(\frac{d\vec{r}_2}{ds} \right) v' + \vec{r}_2 v'' \end{aligned}$$

Applying the chain rule to the derivatives of the basis vectors:

$$\begin{aligned} \frac{d\vec{r}_1}{ds} &= \frac{\partial \vec{r}_1}{\partial u} \frac{du}{ds} + \frac{\partial \vec{r}_1}{\partial v} \frac{dv}{ds} = \vec{r}_{11} u' + \vec{r}_{12} v' \\ \frac{d\vec{r}_2}{ds} &= \frac{\partial \vec{r}_2}{\partial u} \frac{du}{ds} + \frac{\partial \vec{r}_2}{\partial v} \frac{dv}{ds} = \vec{r}_{21} u' + \vec{r}_{22} v' \end{aligned}$$

Substituting these back into the expression for $\vec{\alpha}''(s)$ and using the equality of mixed partials ($\vec{r}_{12} = \vec{r}_{21}$), we obtain

$$\vec{\alpha}''(s) = \vec{r}_{11}(u')^2 + 2\vec{r}_{12}u'v' + \vec{r}_{22}(v')^2 + \vec{r}_1 u'' + \vec{r}_2 v''$$

Now we take the inner product of $\vec{\alpha}''(s)$ with the surface normal $\vec{N}$ to find k_n :

$$k_n = \langle \vec{r}_{11}(u')^2 + 2\vec{r}_{12}u'v' + \vec{r}_{22}(v')^2 + \vec{r}_1 u'' + \vec{r}_2 v'', \vec{N} \rangle$$

By definition, the tangent vectors $\vec{r}_1$ and $\vec{r}_2$ are orthogonal to the surface normal $\vec{N}$. Therefore, $\langle \vec{r}_1, \vec{N} \rangle = 0$ and $\langle \vec{r}_2, \vec{N} \rangle = 0$. The last two terms of the expression vanish, leaving:

$$k_n = \langle \vec{r}_{11}, \vec{N} \rangle (u')^2 + 2\langle \vec{r}_{12}, \vec{N} \rangle u'v' + \langle \vec{r}_{22}, \vec{N} \rangle (v')^2$$

We recognize the inner products as the coefficients of the second fundamental form: $L = \langle \vec{r}_{11}, \vec{N} \rangle$, $M = \langle \vec{r}_{12}, \vec{N} \rangle$, and $N = \langle \vec{r}_{22}, \vec{N} \rangle$. This simplifies the expression to:

$$k_n = L(u')^2 + 2Mu'v' + N(v')^2$$

This formula for k_n is in terms of derivatives with respect to arc length. The theorem, however, is stated in terms of the differentials du and dv , which represent a direction independent of any specific curve's parametrization. To bridge this, we recall that for an arc length parametrization, $|\vec{\alpha}'(s)|^2 = 1$. This implies:

$$1 = \langle \vec{r}_1 u' + \vec{r}_2 v', \vec{r}_1 u' + \vec{r}_2 v' \rangle = E(u')^2 + 2Fu'v' + G(v')^2$$

We can therefore write k_n as a ratio where the denominator is unity:

$$k_n = \frac{L(u')^2 + 2Mu'v' + N(v')^2}{E(u')^2 + 2Fu'v' + G(v')^2}$$

Finally, observe that the direction of the curve is given by the ratio $v' : u' = \frac{dv}{ds} : \frac{du}{ds}$, which is the same as the ratio $dv : du$. We can multiply the numerator and denominator by the scalar $(ds)^2$ without changing the value of the fraction. This converts the derivatives into differentials:

$$\begin{aligned} k_n &= \frac{\left(L\left(\frac{du}{ds}\right)^2 + 2M\frac{du}{ds}\frac{dv}{ds} + N\left(\frac{dv}{ds}\right)^2\right)(ds)^2}{\left(E\left(\frac{du}{ds}\right)^2 + 2F\frac{du}{ds}\frac{dv}{ds} + G\left(\frac{dv}{ds}\right)^2\right)(ds)^2} \\ &= \frac{L du^2 + 2M du dv + N dv^2}{E du^2 + 2F du dv + G dv^2} \end{aligned}$$

This proves the result. ■

Geometric Interpretation

This formula provides a powerful way to understand normal curvature.

- The denominator, $I_P = ds^2$, is the squared infinitesimal distance one travels along the surface in a given direction.
- The numerator, II_P , measures the component of the acceleration vector normal to the surface. It quantifies how much the surface "pulls away" from the tangent plane in that direction.

The normal curvature is therefore the ratio of how much the surface bends away from its tangent plane to the squared distance traveled. This ratio depends only on the direction of travel (i.e., the ratio $dv : du$), not on the speed.

Definition 2.5.12 (Normal curvature). For a unit tangent vector $\mathbf{v} \in T_p S$, the **normal curvature** in the direction $\mathbf{v}$ is $\kappa_n(\mathbf{v}) = II_P(\mathbf{v}, \mathbf{v}) = \langle S_p(\mathbf{v}), \mathbf{v} \rangle$.

Theorem 2.5.13 (Euler's Theorem). Let k_1 and k_2 be the principal curvatures at a point P on a surface, with corresponding orthonormal principal directions $\vec{e}_1$ and $\vec{e}_2$. The normal curvature k_n in the direction of a unit tangent vector $\vec{t}$ that makes an angle θ with $\vec{e}_1$ is given by $k_n(\vec{t}) = k_1 \cos^2 \theta + k_2 \sin^2 \theta$.

Proof. The shape operator L_P is self-adjoint, so there exists an orthonormal basis $\{\vec{e}_1, \vec{e}_2\}$ of the tangent plane $T_P(S)$ consisting of eigenvectors of L_P . Let k_1 and k_2 be the corresponding eigenvalues (the principal curvatures), so $L_P(\vec{e}_1) = k_1 \vec{e}_1$ and $L_P(\vec{e}_2) = k_2 \vec{e}_2$.

A unit tangent vector $\vec{t}$ making an angle θ with $\vec{e}_1$ can be expressed in this basis as:

$$\vec{t} = (\cos \theta) \vec{e}_1 + (\sin \theta) \vec{e}_2$$

The normal curvature in the direction of $\vec{t}$ is given by $k_n(\vec{t}) = \langle L_P(\vec{t}), \vec{t} \rangle$. We compute this directly:

$$\begin{aligned} k_n(\vec{t}) &= \langle L_P(\vec{t}), \vec{t} \rangle = \langle L_P((\cos \theta) \vec{e}_1 + (\sin \theta) \vec{e}_2), (\cos \theta) \vec{e}_1 + (\sin \theta) \vec{e}_2 \rangle \\ &= \langle (\cos \theta) L_P(\vec{e}_1) + (\sin \theta) L_P(\vec{e}_2), (\cos \theta) \vec{e}_1 + (\sin \theta) \vec{e}_2 \rangle \quad (\text{Linearity of } L_P) \\ &= \langle (k_1 \cos \theta) \vec{e}_1 + (k_2 \sin \theta) \vec{e}_2, (\cos \theta) \vec{e}_1 + (\sin \theta) \vec{e}_2 \rangle \quad (\text{Eigenvector property}) \end{aligned}$$

Expanding the inner product and using the orthonormality of the basis ($\langle \vec{e}_i, \vec{e}_j \rangle = \delta_{ij}$), the cross terms vanish:

$$\begin{aligned} k_n(\vec{t}) &= (k_1 \cos^2 \theta) \langle \vec{e}_1, \vec{e}_1 \rangle + (k_2 \sin^2 \theta) \langle \vec{e}_2, \vec{e}_2 \rangle \\ &= k_1 \cos^2 \theta + k_2 \sin^2 \theta \end{aligned}$$

This establishes the theorem. ■

Theorem 2.5.14 (Meusnier's Theorem). *Let κ_n denote the normal curvature of a surface at a point in a given direction, and let κ be the curvature of an arbitrary curve on the surface passing through that point with the same tangent. Then:*

$$\kappa_n = \kappa \cos \theta$$

where θ is the angle between the principal normal of the curve and the surface normal.

Proof. Let $\alpha(s)$ be a unit-speed curve on the surface, $\mathbf{N}$ the surface normal, and $\mathbf{N}_c$ the principal normal of the curve. The curvature vector is $\mathbf{k} = \alpha''(s) = \kappa \mathbf{N}_c$.

By definition, the normal curvature is the projection of the curve's curvature vector onto the surface normal:

$$\kappa_n = \mathbf{k} \cdot \mathbf{N}$$

Substituting $\mathbf{k} = \kappa \mathbf{N}_c$:

$$\kappa_n = (\kappa \mathbf{N}_c) \cdot \mathbf{N} = \kappa (\mathbf{N}_c \cdot \mathbf{N})$$

Since $\mathbf{N}_c$ and $\mathbf{N}$ are both unit vectors, their dot product is $\cos \theta$, where θ is the angle between them. Thus

$$\kappa_n = \kappa \cos \theta$$

■

2.5.1 Geometric Interpretation

The elegance of Meusnier's theorem lies in its geometric interpretation.

- The normal curvature k_n is fixed for a given direction $\vec{t}$ at point P .
- The curvature k of a curve $\vec{\alpha}$ through P with tangent $\vec{t}$ can vary depending on how the curve twists on the surface.
- The theorem $k = k_n / \cos \theta$ implies that the curvature k is minimized when $\cos \theta$ is maximized. This occurs when $\theta = 0$, meaning the principal normal of the curve $\vec{n}_\alpha$ is aligned with the surface normal $\vec{N}$.

Example 2.5.15. If a surface has a property that $L = M = N = 0$ at any point P , then the surface is a plane.

Solution. The coefficients L, M, N are given by $L = \mathbf{n} \cdot \mathbf{r}_{uu}$, $M = \mathbf{n} \cdot \mathbf{r}_{uv}$, and $N = \mathbf{n} \cdot \mathbf{r}_{vv}$. If $L = M = N = 0$, then the normal vector $\mathbf{n}$ is orthogonal to all second-order partial derivatives of the surface patch $\mathbf{r}$.

Since $\mathbf{n}$ is also orthogonal to the tangent vectors $\mathbf{r}_u$ and $\mathbf{r}_v$, the derivatives of the normal vector satisfy $\mathbf{n}_u \cdot \mathbf{r}_u = -\mathbf{n} \cdot \mathbf{r}_{uu} = 0$, $\mathbf{n}_u \cdot \mathbf{r}_v = -\mathbf{n} \cdot \mathbf{r}_{uv} = 0$, and similarly for $\mathbf{n}_v$. Thus, $\mathbf{n}_u = \mathbf{n}_v = 0$, which implies $\mathbf{n}$ is a constant vector. Consequently, the surface is a plane. ■

2.6 Principal Curvatures and Euler's Theorem

Since S_p is self-adjoint, it has two real eigenvalues κ_1, κ_2 (the **principal curvatures**) with orthogonal eigenvectors (the **principal directions**). The **Gaussian curvature** and **mean curvature** are

$$K = \kappa_1 \kappa_2 = \frac{LN - M^2}{EG - F^2}, \quad H = \frac{\kappa_1 + \kappa_2}{2} = \frac{EN - 2FM + GL}{2(EG - F^2)}.$$

Theorem 2.6.1. *Let P be a point on a regular surface S . The maximum and minimum values of the normal curvature $k_n(\vec{v})$ as $\vec{v}$ ranges over all unit tangent vectors in $S_P(S)$ are exactly the principal curvatures k_1 and k_2 .*

Proof. Let $k_1 \geq k_2$ be the principal curvatures with corresponding orthonormal principal directions $\vec{e}_1$ and $\vec{e}_2$. Any unit tangent vector $\vec{v}$ can be written in this basis as:

$$\vec{v} = (\cos \theta)\vec{e}_1 + (\sin \theta)\vec{e}_2$$

From Euler's Theorem, the normal curvature in this direction is $k_n(\theta) = k_1 \cos^2 \theta + k_2 \sin^2 \theta$. Using the identity $\sin^2 \theta = 1 - \cos^2 \theta$, we rewrite this as:

$$k_n(\theta) = k_1 \cos^2 \theta + k_2(1 - \cos^2 \theta) = (k_1 - k_2) \cos^2 \theta + k_2$$

Since $0 \leq \cos^2 \theta \leq 1$, we can find the bounds for k_n :

Maximum: The expression is maximized when $\cos^2 \theta$ is as large as possible, i.e., $\cos^2 \theta = 1$ (which means $\theta = 0$).

$$k_{max} = (k_1 - k_2)(1) + k_2 = k_1$$

Minimum: The expression is minimized when $\cos^2 \theta$ is as small as possible, i.e., $\cos^2 \theta = 0$ (which means $\theta = \pi/2$).

$$k_{min} = (k_1 - k_2)(0) + k_2 = k_2$$

Thus, the normal curvature k_n is constrained by $k_2 \leq k_n \leq k_1$, and these bounds are exactly the principal curvatures. ■

Remark 2.6.2. This result confirms that the principal directions are the "axes of curvature" for the surface. When $\vec{v}$ is aligned with a principal direction, the normal curvature is extremized. For any other direction, Euler's Theorem tells us that k_n is a weighted average of these two extreme values, always lying between them:

$$\min(k_1, k_2) \leq k_n(\vec{v}) \leq \max(k_1, k_2)$$

2.7 Lines of Curvature

Definition 2.7.1. A **line of curvature** on a surface S is a regular curve $\vec{\alpha}(s)$ such that its tangent vector $\vec{\alpha}'(s)$ at every point is a principal direction of the surface.

In other words, at every point along the curve, the tangent vector to the curve must be an eigenvector of the shape operator L_P . This implies that the surface normal $\vec{N}$ changes in the direction of the curve's tangent, keeping the curve "aligned" with the principal directions of bending.

Differential Equation for Lines of Curvature

Theorem 2.7.2. A curve $\vec{r}(u(t), v(t))$ is a line of curvature if and only if its tangent vector $\vec{v} = \frac{\partial u}{\partial t} \vec{r}_1 + \frac{\partial v}{\partial t} \vec{r}_2$ satisfies $(EM - FL)du^2 + (EN - GL)du dv + (FN - GM)dv^2 = 0$. This can be put in the matrix form as

$$\det \begin{pmatrix} dv^2 & -du dv & du^2 \\ E & F & G \\ L & M & N \end{pmatrix} = 0 \quad (2.6)$$

Proof. A vector $\vec{v}$ is a principal direction if and only if it is an eigenvector of the shape operator, i.e., if and only if $\vec{v} \times L_P(\vec{v}) = 0$. That is $\vec{v}$ and $L_P(\vec{v})$ are parallel meaning that $\det(\vec{v}, L_P(\vec{v})) = 0$ in the basis $\{\vec{r}_1, \vec{r}_2\}$.

Let $A = [a_{ij}] = M_1^{-1}M_2$ be the matrix of L_P . $L_P(\vec{v}) = A\vec{v}$ can be expressed in the basis $\{\vec{r}_1, \vec{r}_2\}$ as $L_P(\vec{v}) = (a_{11}du + a_{12}dv)\vec{r}_1 + (a_{21}du + a_{22}dv)\vec{r}_2$. The condition $\det(\vec{v}, L_P(\vec{v})) = 0$ becomes

$$\det \begin{pmatrix} du & a_{11}du + a_{12}dv \\ dv & a_{21}du + a_{22}dv \end{pmatrix} = a_{21}du^2 + (a_{22} - a_{11})du dv - a_{12}dv^2 = 0$$

Using the matrix relation $M_1 A = M_2$, where $M_1 = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$ and $M_2 = \begin{pmatrix} L & M \\ M & N \end{pmatrix}$, we have

$$\begin{aligned} a_{21} &= \frac{1}{\det M_1} (EM - FL), & a_{12} &= \frac{1}{\det M_1} (FN - GM) \\ a_{22} - a_{11} &= \frac{1}{\det M_1} (EN - GL) \end{aligned}$$

Substituting these into the determinant equation and multiplying by $\det M_1 = EG - F^2$, we immediately obtain:

$$(EM - FL)du^2 + (EN - GL)du dv + (FN - GM)dv^2 = 0$$

This proves the result. ■

Theorem 2.7.3. *The principal curvatures k at a point P are the roots of the quadratic equation:*

$$k^2(EG - F^2) - k(EN - 2FM + GL) + (LN - M^2) = 0$$

Proof. We seek the extrema of the second fundamental form $II = L du^2 + 2M du dv + N dv^2$ subject to the constraint $I = E du^2 + 2F du dv + G dv^2 = 1$. Using the method of Lagrange multipliers, we define the Lagrangian:

$$\mathcal{L}(du, dv, k) = (L du^2 + 2M du dv + N dv^2) - k(E du^2 + 2F du dv + G dv^2 - 1)$$

Taking partial derivatives with respect to du and dv and setting them to zero gives

$$\frac{\partial \mathcal{L}}{\partial(du)} = 2(L du + M dv) - 2k(E du + F dv) = 0$$

$$\frac{\partial \mathcal{L}}{\partial(dv)} = 2(M du + N dv) - 2k(F du + G dv) = 0$$

Rearranging these yields the linear system:

$$\begin{cases} (L - kE)du + (M - kF)dv = 0 \\ (M - kF)du + (N - kG)dv = 0 \end{cases}$$

This system of equations can be rewritten as:

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix} \begin{pmatrix} du \\ dv \end{pmatrix} = k \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} du \\ dv \end{pmatrix}$$

For a non-trivial solution $(du, dv) \neq (0, 0)$ to exist, the determinant of the system must vanish:

$$\det \begin{pmatrix} L - kE & M - kF \\ M - kF & N - kG \end{pmatrix} = 0$$

Expanding this determinant:

$$(L - kE)(N - kG) - (M - kF)^2 = 0$$

$$(LN - kLG - kEN + k^2EG) - (M^2 - 2kMF + k^2F^2) = 0$$

Collecting the terms in powers of k :

$$k^2(EG - F^2) - k(EN - 2FM + GL) + (LN - M^2) = 0$$

This is the characteristic equation for the principal curvatures k_1 and k_2 . ■

Note: The Gaussian curvature is the product of principal curvatures

$$k_1 k_2 = \frac{LN - M^2}{EG - F^2}$$

and the Mean curvature is

$$\frac{1}{2}(k_1 + k_2) = \frac{EN - 2FM + GL}{EG - F^2}.$$

Theorem 2.7.4. *The parametric curves are lines of curvature if and only if $F = 0$ and $M = 0$.*

Proof. Let $\{\vec{r}_1, \vec{r}_2\}$ be the basis of the tangent plane $T_P(S)$. The parametric curves are lines of curvature if and only if $\vec{r}_1$ and $\vec{r}_2$ are eigenvectors of the shape operator L_P . This is equivalent to requiring the matrix of L_P relative to this basis to be diagonal.

The shape operator L_P is related to the fundamental forms by the matrix identity $b = gA$, where $g = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$, $b = \begin{pmatrix} L & M \\ M & N \end{pmatrix}$, and $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ is the matrix of L_P . Thus

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix} = \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

Sufficiency ($\Leftarrow$): If $F = 0$ and $M = 0$, the identity becomes

$$\begin{pmatrix} L & 0 \\ 0 & N \end{pmatrix} = \begin{pmatrix} E & 0 \\ 0 & G \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

This yields $L = Ea_{11}$, $N = Ga_{22}$, and $0 = Ea_{12} = Ga_{21}$. Since $E, G \neq 0$, it follows $a_{12} = a_{21} = 0$. The matrix A is diagonal, so $\vec{r}_1, \vec{r}_2$ are eigenvectors, and the parametric curves are lines of curvature.

Necessity ($\Rightarrow$): If the parametric curves are lines of curvature, A must be diagonal, i.e., $a_{12} = a_{21} = 0$. Expanding $b = gA$:

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix} = \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} k_1 & 0 \\ 0 & k_2 \end{pmatrix} = \begin{pmatrix} Ek_1 & Fk_2 \\ Fk_1 & Gk_2 \end{pmatrix}$$

Equating the off-diagonal entries: $M = Fk_2$ and $M = Fk_1$. This implies $F(k_1 - k_2) = 0$. For a non-umbilic point ($k_1 \neq k_2$), we must have $F = 0$, which subsequently forces $M = 0$. ■

Remark 2.7.5. When a surface is parametrized by lines of curvature, the first and second fundamental forms are both diagonal. Specifically, the orthogonality of the principal directions implies $F = 0$, and the alignment of these directions with the coordinate grid implies $M = 0$. Thus, for such a parametrization, the fundamental forms simplify to:

$$I = E du^2 + G dv^2 \quad \text{and} \quad II = L du^2 + N dv^2$$

2.7.1 Rodrigues' Formula

The most important theorem concerning lines of curvature relates the derivative of the normal vector directly to the tangent vector.

Theorem 2.7.6 (Rodrigues' Formula). *Prove that a curve is a line of curvature if and only if*

$$k d\vec{r} + d\vec{N} = 0.$$

Proof. The system for the principal curvature k and direction (du, dv) is given by

$$(L - kE)du + (M - kF)dv = 0 \tag{2.7}$$

$$(M - kF)du + (N - kG)dv = 0 \tag{2.8}$$

We wish to show this is equivalent to $d\vec{N} + k d\vec{r} = 0$.

Recall the Weingarten equations, which express the derivatives of the unit normal $\vec{N}$ in terms of the tangent basis $\{\vec{r}_1, \vec{r}_2\}$:

$$\begin{aligned}\vec{N}_1 &= \frac{FM - EN}{EG - F^2} \vec{r}_1 + \frac{FL - EM}{EG - F^2} \vec{r}_2 \\ \vec{N}_2 &= \frac{FN - GM}{EG - F^2} \vec{r}_1 + \frac{FM - EN}{EG - F^2} \vec{r}_2\end{aligned}$$

Using the definitions $L = -\langle \vec{N}_1, \vec{r}_1 \rangle$, $M = -\langle \vec{N}_1, \vec{r}_2 \rangle = -\langle \vec{N}_2, \vec{r}_1 \rangle$, and $N = -\langle \vec{N}_2, \vec{r}_2 \rangle$, we recognize that the system of equations (1) and (2) can be rewritten as:

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix} \begin{pmatrix} du \\ dv \end{pmatrix} = k \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} du \\ dv \end{pmatrix}$$

Expanding this, the first row is $Ldu + Mdv = k(Edu + Fdv)$ and the second is $Mdu + Ndv = k(Fdu + Gdv)$. Observe that $d\vec{r} = \vec{r}_1 du + \vec{r}_2 dv$. The quantities $(Edu + Fdv)$ and $(Fdu + Gdv)$ are exactly the inner products $\langle d\vec{r}, \vec{r}_1 \rangle$ and $\langle d\vec{r}, \vec{r}_2 \rangle$. Similarly, $Ldu + Mdv$ and $Mdu + Ndv$ are exactly $-\langle d\vec{N}, \vec{r}_1 \rangle$ and $-\langle d\vec{N}, \vec{r}_2 \rangle$.

Substituting these into the expanded system:

$$\begin{aligned}-\langle d\vec{N}, \vec{r}_1 \rangle &= k \langle d\vec{r}, \vec{r}_1 \rangle \implies \langle d\vec{N} + kd\vec{r}, \vec{r}_1 \rangle = 0 \\ -\langle d\vec{N}, \vec{r}_2 \rangle &= k \langle d\vec{r}, \vec{r}_2 \rangle \implies \langle d\vec{N} + kd\vec{r}, \vec{r}_2 \rangle = 0\end{aligned}$$

Since the vector $\vec{X} = d\vec{N} + kd\vec{r}$ is orthogonal to both basis vectors $\vec{r}_1$ and $\vec{r}_2$, it must be orthogonal to the entire tangent plane. However, $d\vec{N}$ and $d\vec{r}$ are tangent vectors by definition; thus their sum $\vec{X}$ is also a tangent vector. The only vector that is both tangent to the surface and orthogonal to the tangent plane is the zero vector:

$$d\vec{N} + kd\vec{r} = 0$$

This is Rodrigues' formula. ■

2.7.2 Joachimsthal's Theorem

Lines of curvature have a special relationship with the angle of intersection between surfaces.

Definition 2.7.7 (Lines of Curvature). A regular curve $\vec{\alpha}(s)$ on a surface S is called a **line of curvature** if the tangent vector $\vec{\alpha}'(s)$ is a principal direction at every point along the curve.

Theorem 2.7.8 (Joachimsthal). Let S_1 and S_2 be two regular surfaces intersecting along a regular curve C .

1. If C is a line of curvature on both S_1 and S_2 , then the angle of intersection θ between the surfaces is constant along C .
2. If the surfaces intersect at a constant angle θ and C is a line of curvature on S_1 , then C is also a line of curvature on S_2 .

Proof. Let $\vec{\alpha}(s)$ be the intersection curve parametrized by arc length s , with unit tangent $\vec{t} = \vec{\alpha}'(s)$. Let $\vec{N}_1$ and $\vec{N}_2$ be the unit normal fields of S_1 and S_2 along C . The angle θ between the surfaces is defined by $\cos \theta = \langle \vec{N}_1, \vec{N}_2 \rangle$.

Differentiating with respect to s :

$$-\sin \theta \frac{d\theta}{ds} = \frac{d}{ds} \langle \vec{N}_1, \vec{N}_2 \rangle = \left\langle \frac{d\vec{N}_1}{ds}, \vec{N}_2 \right\rangle + \left\langle \vec{N}_1, \frac{d\vec{N}_2}{ds} \right\rangle$$

By Rodrigues' formula, which states that a curve is a line of curvature if and only if the derivative of the normal vector along the curve is proportional to the tangent vector: $\frac{d\vec{N}_i}{ds} = -k_i \vec{t}$, where k_i is the principal curvature of S_i in the direction of $\vec{t}$. Substituting this into the derivative equation:

$$-\sin \theta \frac{d\theta}{ds} = \langle -k_1 \vec{t}, \vec{N}_2 \rangle + \langle \vec{N}_1, -k_2 \vec{t} \rangle = -k_1 \langle \vec{t}, \vec{N}_2 \rangle - k_2 \langle \vec{N}_1, \vec{t} \rangle$$

Since $\vec{t}$ is tangent to the curve C which lies on both surfaces, $\vec{t}$ must be perpendicular to both normals: $\langle \vec{t}, \vec{N}_1 \rangle = 0$ and $\langle \vec{t}, \vec{N}_2 \rangle = 0$. Consequently:

$$-\sin \theta \frac{d\theta}{ds} = -k_1(0) - k_2(0) = 0$$

Necessity: If C is a line of curvature on both surfaces, the derivatives are well-defined by Rodrigues' formula, and the expression above vanishes. Assuming the surfaces are not tangent ($\sin \theta \neq 0$), we must have $\frac{d\theta}{ds} = 0$, implying θ is constant.

Sufficiency: Conversely, suppose θ is constant ($\frac{d\theta}{ds} = 0$) and C is a line of curvature on S_1 . Then $d\vec{N}_1/ds = -k_1\vec{t}$. The derivative equation becomes

$$0 = \langle -k_1\vec{t}, \vec{N}_2 \rangle + \langle \vec{N}_1, \frac{d\vec{N}_2}{ds} \rangle$$

Since $\langle \vec{t}, \vec{N}_2 \rangle = 0$, we are left with $\langle \vec{N}_1, \frac{d\vec{N}_2}{ds} \rangle = 0$. This implies the vector $d\vec{N}_2/ds$ is orthogonal to $\vec{N}_1$. Since $d\vec{N}_2/ds$ is also orthogonal to $\vec{N}_2$ (because $\vec{N}_2$ is a unit vector), $d\vec{N}_2/ds$ must be parallel to the cross product $\vec{N}_1 \times \vec{N}_2$. However, $\vec{N}_1 \times \vec{N}_2$ is parallel to the tangent $\vec{t}$. Thus, $d\vec{N}_2/ds$ is proportional to $\vec{t}$, satisfying Rodrigues' formula for S_2 . Therefore, C is a line of curvature on S_2 . ■

Theorem 2.7.9 (Dupin). *The sum of the normal curvatures of a surface in any two orthogonal directions is invariant and equal to the sum of the principal curvatures.*

Proof. Let k_1 and k_2 be the principal curvatures at a point P . By Euler's Theorem, the normal curvature k_n in a direction making an angle θ with the principal direction $\vec{e}_1$ is $k_n(\theta) = k_1 \cos^2 \theta + k_2 \sin^2 \theta$. Let k'_n be the normal curvature in the orthogonal direction $\theta + \pi/2$. Using the trigonometric identities $\cos(\theta + \pi/2) = -\sin \theta$ and $\sin(\theta + \pi/2) = \cos \theta$:

$$k'_n = k_1 \cos^2(\theta + \pi/2) + k_2 \sin^2(\theta + \pi/2) = k_1 \sin^2 \theta + k_2 \cos^2 \theta$$

Adding these two expressions:

$$k_n + k'_n = k_1(\cos^2 \theta + \sin^2 \theta) + k_2(\sin^2 \theta + \cos^2 \theta) = k_1 + k_2$$

Since $k_1 + k_2 = 2H$ (where H is the mean curvature), the sum of normal curvatures in any two orthogonal directions is equal to $2H$, which is constant at P . ■

2.8 Gauss's Theorema Egregium

A landmark result: the Gaussian curvature K depends only on the first fundamental form (i.e., it is intrinsic).

Theorem 2.8.1 (Theorema Egregium). *The Gaussian curvature K can be expressed solely in terms of E, F, G and their derivatives. Consequently, K is invariant under local isometries.*

The explicit formula (Brioschi) is

$$K = \frac{1}{(EG - F^2)^2} \begin{vmatrix} -\frac{1}{2}E_{vv} + F_{uv} - \frac{1}{2}G_{uu} & \frac{1}{2}E_u & F_u - \frac{1}{2}E_v \\ F_v - \frac{1}{2}G_u & E & F \\ \frac{1}{2}G_v & F & G \end{vmatrix} - \frac{1}{(EG - F^2)^2} \begin{vmatrix} 0 & \frac{1}{2}E_v & \frac{1}{2}G_u \\ \frac{1}{2}E_v & E & F \\ \frac{1}{2}G_u & F & G \end{vmatrix}.$$

Remark 2.8.2. Geometrically, a line of curvature is a curve whose tangent vector always aligns with one of the two directions of extremal normal curvature. Consequently, a surface is covered by two families of lines of curvature (corresponding to the two principal curvatures k_1 and k_2), which intersect each other orthogonally at every point where $k_1 \neq k_2$.

2.8.1 Isometry between Surfaces

An isometry is a map that preserves the intrinsic geometry of surfaces—specifically, the metric properties such as lengths of curves, angles between curves, and areas.

Definition 2.8.3. Let S_1 and S_2 be two regular surfaces. A differentiable map $f : S_1 \rightarrow S_2$ is called an **isometry** if f is a diffeomorphism (i.e., f is differentiable, bijective, and f^{-1} is differentiable) and if the differential map $df_p : T_p S_1 \rightarrow T_{f(p)} S_2$ preserves the inner product for all $p \in S_1$. That is,

$$\langle u, v \rangle_p = \langle df_p(u), df_p(v) \rangle_{f(p)} \quad \forall u, v \in T_p S_1. \quad (2.9)$$

Definition 2.8.4. A differentiable map $f : S_1 \rightarrow S_2$ is a **local isometry** at $p \in S_1$ if there exists a neighborhood U of p such that $f|_U : U \rightarrow f(U)$ is an isometry. If f is a local isometry at every point $p \in S_1$, then f is called a local isometry.

Example 2.8.5 (Local Isometry: Plane to Cylinder). Let S_1 be the xy -plane and S_2 be the cylinder defined by the map $f : S_1 \rightarrow S_2$ where $f(x, y, 0) = (\cos x, \sin x, y)$. We verify that f is a local isometry by calculating the differential df_p at a point $p \in S_1$.

Solution. Let $\{e_1, e_2\}$ be the natural basis for $T_p S_1$, where $e_1 = (1, 0, 0)$ and $e_2 = (0, 1, 0)$. To compute the differential $df_p(v)$, we use a curve $\alpha(t) = p + tv$ such that $\alpha(0) = p$ and $\alpha'(0) = v$. Then $df_p(v) = \left. \frac{d}{dt} \right|_{t=0} f(\alpha(t))$.

As $\alpha(t) = p + te_1 = (x + t, y, 0)$. WE have

$$df_p(e_1) = \left. \frac{d}{dt} \right|_{t=0} f(x + t, y, 0) = \left. \frac{d}{dt} \right|_{t=0} (\cos(x + t), \sin(x + t), y) = (-\sin x, \cos x, 0)$$

and

$$df_p(e_2) = \left. \frac{d}{dt} \right|_{t=0} f(x, y + t, 0) = \left. \frac{d}{dt} \right|_{t=0} (\cos x, \sin x, y + t) = (0, 0, 1)$$

Therefore,

$$\langle df_p(e_1), df_p(e_2) \rangle = (-\sin x, \cos x, 0) \cdot (0, 0, 1) = 0$$

Since $\langle e_1, e_2 \rangle = 0$ and the squared norms are preserved ($\|df_p(e_1)\|^2 = \sin^2 x + \cos^2 x = 1 = \|e_1\|^2$ and $\|df_p(e_2)\|^2 = 1 = \|e_2\|^2$), the map f acts as a local isometry between the plane and the cylinder. ■

Note:

- The first fundamental form of both plane and the cylinder is $du^2 + dv^2$, confirming that distances are preserved.
- A local isometry does not need to be injective globally, but it must be a local diffeomorphism. Observe that an isometry is always a local isometry, but a local isometry is not necessarily an isometry (it may fail to be a global bijection).

The central result in the study of isometries is **Gauss's Theorema Egregium**, which states that the Gaussian curvature K is invariant under local isometry.

Theorem 2.8.6. Let $X : U \rightarrow S$ and $\bar{X} : U \rightarrow \bar{S}$ be two parameterized surfaces defined on the same open parameter domain $U \subset \mathbb{R}^2$. Suppose that their first fundamental forms are equal on U ; that is, $E = \bar{E}$, $F = \bar{F}$, and $G = \bar{G}$. Then there exists a local isometry between S and $\bar{S}$.

Proof. To prove the theorem, we will construct a mapping between the two surfaces and show that its differential preserves the lengths of all tangent vectors.

Since X is a regular parametrization, $X^{-1} : X(U) \rightarrow U$ is a well-defined smooth map. We define a mapping $\varphi : X(U) \rightarrow \bar{S}$ by composing X^{-1} with $\bar{X}$:

$$\varphi = \bar{X} \circ X^{-1}$$

Intuitively, φ takes a point p on S with coordinates (u, v) , maps it back to $(u, v) \in U$, and then maps it forward to the corresponding point on $\bar{S}$ with the exact same coordinates.

To show φ is a local isometry, we must look at how its differential $d\varphi_p$ acts on a tangent vector. Let $p \in X(U)$ and let $w \in T_p S$ be a tangent vector.

Every tangent vector can be realized as the velocity of some curve. Let $\alpha(t) = (u(t), v(t))$ be a curve in U such that $X(\alpha(0)) = p$. The velocity vector of the surface curve $X(\alpha(t))$ at $t = 0$ gives us w :

$$w = \left. \frac{d}{dt} \right|_{t=0} X(u(t), v(t)) = X_u u' + X_v v'$$

By the definition of the differential, $d\varphi_p(w)$ is the velocity vector of the *image* curve under φ . The image curve is

$$\varphi(X(\alpha(t))) = \bar{X}(X^{-1}(X(\alpha(t)))) = \bar{X}(\alpha(t)) = \bar{X}(u(t), v(t))$$

Taking the derivative of this image curve at $t = 0$ yields

$$d\varphi_p(w) = \left. \frac{d}{dt} \right|_{t=0} \bar{X}(u(t), v(t)) = \bar{X}_u u' + \bar{X}_v v'$$

The First Fundamental Form I measures the squared length of tangent vectors. For our original vector w on S , the squared length is $I_p(w) = \langle w, w \rangle_p = E(u')^2 + 2F u' v' + G(v')^2$. For the mapped vector $d\varphi_p(w)$ on $\bar{S}$, the squared length is measured using the coefficients of $\bar{S}$:

$$I_{\varphi(p)}(d\varphi_p(w)) = \langle d\varphi_p(w), d\varphi_p(w) \rangle_{\varphi(p)} = \bar{E}(u')^2 + 2\bar{F} u' v' + \bar{G}(v')^2$$

By our initial hypothesis, the metric coefficients are identical: $E = \bar{E}$, $F = \bar{F}$, and $G = \bar{G}$. Substituting these into the equation for the mapped vector yields $I_{\varphi(p)}(d\varphi_p(w)) = E(u')^2 + 2F u' v' + G(v')^2 = I_p(w)$. Therefore, we have

$$\langle d\varphi_p(w), d\varphi_p(w) \rangle_{\varphi(p)} = \langle w, w \rangle_p$$

Since the differential $d\varphi_p$ preserves the squared norm (length) of all tangent vectors, it completely preserves the inner product. Thus, φ is a local isometry. ■

The above theorem can also be restated and proved as follows.

Theorem 2.8.7. Let $\phi : U \subset \mathbb{R}^2 \rightarrow S_1$ and $\psi : U \subset \mathbb{R}^2 \rightarrow S_2$ be two parametrizations of surfaces S_1 and S_2 such that their first fundamental forms are identical in the coordinates (u, v) :

$$E_1 = E_2, \quad F_1 = F_2, \quad G_1 = G_2.$$

Then the map $f = \psi \circ \phi^{-1} : \phi(U) \rightarrow \psi(U)$ is a local isometry.

Proof. Let $p = \phi(u_0, v_0) \in S_1$ and let $\mathbf{w}_1, \mathbf{w}_2 \in T_p S_1$. We express these tangent vectors in the coordinate basis $\{\phi_u, \phi_v\}$ as $\mathbf{w}_1 = a_1 \phi_u + b_1 \phi_v$ and $\mathbf{w}_2 = a_2 \phi_u + b_2 \phi_v$. By definition of the First Fundamental Form I_1 :

$$I_1(\mathbf{w}_1, \mathbf{w}_2) = a_1 a_2 E_1 + (a_1 b_2 + a_2 b_1) F_1 + b_1 b_2 G_1.$$

The differential map df_p of the composition $f = \psi \circ \phi^{-1}$ acts on the basis vectors as $df_p(\phi_u) = \psi_u$ and $df_p(\phi_v) = \psi_v$. Thus:

$$df_p(\mathbf{w}_i) = a_i \psi_u + b_i \psi_v.$$

Evaluating the inner product in S_2 using the First Fundamental Form I_2 :

$$I_2(df_p(\mathbf{w}_1), df_p(\mathbf{w}_2)) = a_1 a_2 E_2 + (a_1 b_2 + a_2 b_1) F_2 + b_1 b_2 G_2.$$

Given that $E_1 = E_2$, $F_1 = F_2$, and $G_1 = G_2$, it follows that $I_1(\mathbf{w}_1, \mathbf{w}_2) = I_2(df_p(\mathbf{w}_1), df_p(\mathbf{w}_2))$. Since df_p preserves the inner product, f is a local isometry. ■

Example 2.8.8. Show that the Catenoid and the Helicoid are locally isometric.

Solution. The first fundamental form of both the surfaces is $ds^2 = a^2 \cosh^2 v (du^2 + dv^2)$ so, they are locally isometric. ■

Chapter 3

Curvature and Special Surfaces

This chapter develops the local geometry of surfaces through normal curvature, principal curvatures, and the Gaussian and mean curvatures. It then studies the variation of the surface normal through the shape operator, Weingarten equations, and the geometry of lines of curvature. Special attention is given to the Dupin indicatrix, asymptotic and conjugate directions, ruled surfaces, minimal surfaces, and isothermal coordinates. The chapter concludes with intrinsic aspects of surface geometry, including isometries, local isometries, and the harmonic characterization of minimal surfaces.

3.1 Principal, Mean, and Gaussian Curvatures

Let $\mathbf{r} = \mathbf{r}(u, v)$ be a regular surface with unit normal $\mathbf{N} = (\mathbf{r}_u \times \mathbf{r}_v) / \|\mathbf{r}_u \times \mathbf{r}_v\|$. The first and second fundamental forms are

$$I = E du^2 + 2F du dv + G dv^2, \quad II = L du^2 + 2M du dv + N dv^2,$$

where $E = \mathbf{r}_u \cdot \mathbf{r}_u$, $F = \mathbf{r}_u \cdot \mathbf{r}_v$, $G = \mathbf{r}_v \cdot \mathbf{r}_v$, and $L = \mathbf{r}_{uu} \cdot \mathbf{N}$, $M = \mathbf{r}_{uv} \cdot \mathbf{N}$, $N = \mathbf{r}_{vv} \cdot \mathbf{N}$. We write $H_0^2 = EG - F^2 > 0$; the notation H_0 is used to distinguish this quantity from the mean curvature H .

Definition 3.1.1 (Normal Curvature). Let a curve on the surface have tangent direction (du, dv) at a point P . The **normal curvature** in this direction is

$$k_n = \frac{L du^2 + 2M du dv + N dv^2}{E du^2 + 2F du dv + G dv^2}.$$

If the direction is parameterized by arc length s , put $l = du/ds$ and $m = dv/ds$. Then $El^2 + 2Flm + Gm^2 = 1$, and consequently $k_n = Ll^2 + 2Mlm + Nm^2$.

Definition 3.1.2 (Principal Directions and Principal Curvatures). At a point P of a regular surface, the tangent directions in which the normal curvature assumes its maximum and minimum values are called the **principal directions**. The corresponding normal curvatures are called the **principal curvatures** and are denoted by k_1 and k_2 .

Theorem 3.1.3 (Equation for the Principal Curvatures). *The principal curvatures are the roots of*

$$(EG - F^2)k^2 - (EN + GL - 2FM)k + (LN - M^2) = 0.$$

Proof. Let $l = du/ds$ and $m = dv/ds$. We seek the stationary values of

$$k_n = Ll^2 + 2Mlm + Nm^2$$

subject to

$$El^2 + 2Flm + Gm^2 = 1.$$

Using a Lagrange multiplier λ , consider

$$\Phi = Ll^2 + 2Mlm + Nm^2 - \lambda(El^2 + 2Flm + Gm^2 - 1).$$

The equations $\partial\Phi/\partial l = 0$ and $\partial\Phi/\partial m = 0$ give

$$(L - \lambda E)l + (M - \lambda F)m = 0$$

and

$$(M - \lambda F)l + (N - \lambda G)m = 0.$$

Multiplying the first equation by l , the second by m , and adding, we obtain $\lambda = k_n$. Hence a principal curvature k satisfies

$$(L - kE)l + (M - kF)m = 0, \quad (M - kF)l + (N - kG)m = 0.$$

A non-zero solution (l, m) exists if and only if

$$\begin{vmatrix} L - kE & M - kF \\ M - kF & N - kG \end{vmatrix} = 0.$$

Expanding the determinant gives $(EG - F^2)k^2 - (EN + GL - 2FM)k + (LN - M^2) = 0$. ■

Definition 3.1.4 (Mean Curvature). The **mean curvature** of a surface is $H = \frac{k_1 + k_2}{2}$. Thus

$$H = \frac{EN + GL - 2FM}{2(EG - F^2)}.$$

Definition 3.1.5 (Gaussian Curvature). The **Gaussian curvature** is the product of the principal curvatures:

$$K = k_1 k_2.$$

Consequently,

$$K = \frac{LN - M^2}{EG - F^2}.$$

Proposition 3.1.6. The principal curvatures may be recovered from the mean and Gaussian curvatures by

$$k_1 = H + \sqrt{H^2 - K}, \quad k_2 = H - \sqrt{H^2 - K}.$$

Moreover,

$$H^2 - K = \frac{(k_1 - k_2)^2}{4} \geq 0,$$

and a point is umbilic if and only if $H^2 = K$.

Proof. Since $k_1 + k_2 = 2H$ and $k_1 k_2 = K$, the principal curvatures are the roots of $x^2 - 2Hx + K = 0$. Hence the quadratic formula gives the stated expressions. Also,

$$H^2 - K = \frac{(k_1 + k_2)^2}{4} - k_1 k_2 = \frac{(k_1 - k_2)^2}{4}.$$

Thus $H^2 = K$ precisely when $k_1 = k_2$, that is, precisely at an umbilic. ■

Remark 3.1.7. The Gaussian curvature is independent of the choice of regular parameters. Indeed, under a change of parameters, both $LN - M^2$ and $EG - F^2$ acquire the same factor, namely the square of the Jacobian.

Proposition 3.1.8 (Classification by Gaussian Curvature). At a non-singular point P :

1. $K(P) > 0$ if and only if P is elliptic;
2. $K(P) < 0$ if and only if P is hyperbolic;
3. $K(P) = 0$ if and only if at least one principal curvature vanishes.

Proof. Since $EG - F^2 > 0$, the sign of K is the sign of $LN - M^2$. Equivalently, $K = k_1 k_2$. Hence $K > 0$ when the principal curvatures have the same sign, $K < 0$ when they have opposite signs, and $K = 0$ when at least one of them vanishes. ■

Theorem 3.1.9 (Equation of the Principal Directions). *The principal directions satisfy*

$$(EM - FL) du^2 + (EN - GL) du dv + (FN - GM) dv^2 = 0.$$

At a non-umbilic point, the two principal directions are orthogonal.

Proof. For a principal curvature k ,

$$(L - kE)du + (M - kF)dv = 0,$$

$$(M - kF)du + (N - kG)dv = 0.$$

Eliminating k gives $(EM - FL) du^2 + (EN - GL) du dv + (FN - GM) dv^2 = 0$. The standard orthogonality condition for the two directions of $P du^2 + Q du dv + R dv^2 = 0$ reduces here to an identity, so the two principal directions are perpendicular whenever they are distinct. ■

Definition 3.1.10 (Umbilic). A point P is called an **umbilic** if the normal curvature is the same in every tangent direction. Equivalently,

$$\frac{L}{E} = \frac{M}{F} = \frac{N}{G}$$

whenever these ratios are meaningful, or, invariantly, the second fundamental form is a scalar multiple of the first fundamental form at P .

Example 3.1.11 (Sphere). Show that every point of a sphere of radius a is an umbilic and determine its Gaussian and mean curvatures.

Solution. Use the parameterization

$$\mathbf{r}(u, v) = (a \sin u \cos v, a \sin u \sin v, a \cos u).$$

A direct calculation gives $E = a^2$, $F = 0$, $G = a^2 \sin^2 u$. With a consistent choice of orientation,

$$L = -a, \quad M = 0, \quad N = -a \sin^2 u.$$

Hence

$$\frac{L}{E} = \frac{N}{G} = -\frac{1}{a},$$

so every point is umbilic. Thus

$$k_1 = k_2 = -\frac{1}{a}, \quad H = -\frac{1}{a}, \quad K = \frac{1}{a^2}.$$

Changing the orientation reverses the signs of k_1, k_2 and H , but leaves K unchanged. ■

Theorem 3.1.12 (Totally Umbilic Surfaces). *Let S be a connected regular surface for which every point is umbilic. Then S is contained in either a plane or a sphere.*

Proof. Since every point is umbilic, the shape operator has the form $S = \lambda I$ for some smooth function λ . Hence $d\mathbf{N} = -\lambda d\mathbf{r}$, or equivalently $\mathbf{N}_u = -\lambda \mathbf{r}_u$ and $\mathbf{N}_v = -\lambda \mathbf{r}_v$. Equality of mixed derivatives gives

$$\mathbf{N}_{uv} - \mathbf{N}_{vu} = -\lambda_v \mathbf{r}_u + \lambda_u \mathbf{r}_v = 0.$$

Since $\mathbf{r}_u$ and $\mathbf{r}_v$ are linearly independent, $\lambda_u = \lambda_v = 0$; thus λ is constant on the connected surface.

If $\lambda = 0$, then $d\mathbf{N} = 0$, so $\mathbf{N}$ is constant and $\mathbf{r} \cdot \mathbf{N}$ is constant. Hence S lies in a plane. If $\lambda \neq 0$, then $d(\mathbf{r} + \lambda^{-1} \mathbf{N}) = 0$, so $\mathbf{r} + \lambda^{-1} \mathbf{N} = \mathbf{c}$ for a fixed vector $\mathbf{c}$. Therefore $\|\mathbf{r} - \mathbf{c}\| = 1/|\lambda|$, and S lies on a sphere. ■

Example 3.1.13 (Circular Cylinder). Find the principal, mean, and Gaussian curvatures of the circular cylinder $\mathbf{r}(u, v) = (a \cos u, a \sin u, v)$, where $a > 0$.

Solution. We have $\mathbf{r}_u = (-a \sin u, a \cos u, 0)$ and $\mathbf{r}_v = (0, 0, 1)$, so $E = a^2$, $F = 0$, and $G = 1$. Choosing the outward unit normal $\mathbf{N} = (\cos u, \sin u, 0)$ gives $L = -a$, $M = 0$, and $N = 0$. Since the coordinate curves are principal, $k_1 = L/E = -1/a$ and $k_2 = N/G = 0$. Hence $H = -1/(2a)$ and $K = 0$. ■

Example 3.1.14 (Hyperbolic Paraboloid at the Origin). Find the principal curvatures of the surface $z = xy$ at the origin.

Solution. Take $\mathbf{r}(u, v) = (u, v, uv)$. At $(0, 0)$, $E = G = 1$ and $F = 0$. With the upward unit normal, $L = N = 0$ and $M = 1$. Therefore the principal curvature equation becomes $k^2 - 1 = 0$. Hence $k_1 = 1$, $k_2 = -1$, so $H = 0$ and $K = -1$. Thus the origin is a hyperbolic point. ■

Example 3.1.15 (Torus). (See Figure 34 in the Atlas.) For the torus $\mathbf{r}(u, v) = ((R + a \cos u) \cos v, (R + a \cos u) \sin v, a \sin u)$, where $R > a > 0$, determine the Gaussian curvature.

Solution. The first fundamental coefficients are $E = a^2$, $F = 0$, and $G = (R + a \cos u)^2$. With the outward normal $\mathbf{N} = (\cos u \cos v, \cos u \sin v, \sin u)$, one obtains $L = -a$, $M = 0$, and $N = -(R + a \cos u) \cos u$. Therefore

$$K = \frac{LN}{EG} = \frac{\cos u}{a(R + a \cos u)}.$$

Thus $K > 0$ on the outer part of the torus, $K < 0$ on the inner part, and $K = 0$ along the two circles corresponding to $\cos u = 0$. ■

Example 3.1.16 (Umbilic Criterion). Suppose a point of a surface has $H = 3$ and $K = 9$. Show that the point is umbilic and determine the principal curvatures.

Solution. Since $H^2 - K = 9 - 9 = 0$, the two principal curvatures are equal. Indeed, $k_1 = k_2 = H = 3$. Hence the point is umbilic. ■

3.2 Hilbert's Theorem and Its Applications

Remark 3.2.1. The name “Hilbert’s theorem” is used for more than one classical result in surface theory. In this unit we use the compact-surface form below. It should not be confused with Hilbert’s deeper non-immersion theorem, which states that no complete regular surface of constant negative Gaussian curvature can be immersed in $\mathbb{R}^3$.

Lemma 3.2.2 (Supporting Sphere Lemma). *Let S be a compact regular surface and let O lie in the bounded region enclosed by S . If $P_0 \in S$ maximizes the distance from O , then the sphere centered at O through P_0 is tangent to S at P_0 , and S lies locally on one side of that sphere.*

Proof. The function $f(P) = \|P - O\|^2$ attains its maximum at P_0 . For every $X \in T_{P_0}S$, the directional derivative of f vanishes, so $(P_0 - O) \cdot X = 0$. Hence $P_0 - O$ is normal to S at P_0 . The maximality of the distance implies that nearby points of S lie inside or on the sphere. ■

Theorem 3.2.3 (Hilbert’s Theorem). *Let $S \subset \mathbb{R}^3$ be a compact regular surface without boundary. Then there exists at least one point $P \in S$ at which the Gaussian curvature is positive.*

Proof. Since S is compact and has no boundary, it encloses a bounded region. Choose a point O in the enclosed region and consider the function $f(P) = \|P - O\|^2$ on S . By compactness, f attains its maximum at some point $P_0 \in S$.

Let $R = \|P_0 - O\|$. The sphere of radius R centered at O contains the surface and is tangent to S at P_0 . Thus the tangent plane of the sphere at P_0 agrees with the tangent plane of S at P_0 . With the normal chosen toward O , every normal section of S at P_0 bends at least as much toward the interior as the corresponding normal section of the supporting sphere. Consequently both principal curvatures at P_0 have the same non-zero sign. Therefore

$$K(P_0) = k_1(P_0)k_2(P_0) > 0.$$

Corollary 3.2.4. *There is no compact regular minimal surface without boundary in $\mathbb{R}^3$.* ■

Proof. If S were compact and minimal, then $H = 0$, so $k_1 = -k_2$ at every point. Hence

$$K = k_1 k_2 = -k_1^2 \leq 0.$$

Hilbert's theorem gives a point at which $K > 0$, a contradiction. ■

Corollary 3.2.5. *No compact regular surface without boundary can have $K \leq 0$ everywhere.*

Proof. This follows immediately from Hilbert's theorem. ■

3.3 The Shape Operator and the Weingarten Equations

Definition 3.3.1 (Shape Operator). The **shape operator** or **Weingarten map** at P is the linear map

$$S_P : T_P S \rightarrow T_P S, \quad S_P(X) = -d\mathbf{N}_P(X).$$

Its eigenvalues are the principal curvatures and its eigenvectors are the principal directions.

Proposition 3.3.2. The shape operator is self-adjoint:

$$\langle S(X), Y \rangle = \langle X, S(Y) \rangle$$

for all $X, Y \in T_P S$.

Proof. It is enough to verify the identity on the coordinate basis $\{\mathbf{r}_u, \mathbf{r}_v\}$. Differentiating $\mathbf{N} \cdot \mathbf{r}_u = 0$ and $\mathbf{N} \cdot \mathbf{r}_v = 0$ gives

$$-\mathbf{N}_u \cdot \mathbf{r}_v = \mathbf{N} \cdot \mathbf{r}_{uv} = -\mathbf{N}_v \cdot \mathbf{r}_u.$$

Hence $\langle S(\mathbf{r}_u), \mathbf{r}_v \rangle = \langle \mathbf{r}_u, S(\mathbf{r}_v) \rangle$, and bilinearity completes the proof. ■

Remark 3.3.3. The Gaussian and mean curvatures are $K = \det S$, $H = \frac{1}{2} \operatorname{tr} S$.

The derivatives of the unit normal are tangent to the surface. Indeed, $\mathbf{N} \cdot \mathbf{N} = 1$ implies $\mathbf{N} \cdot \mathbf{N}_u = \mathbf{N} \cdot \mathbf{N}_v = 0$. Hence $\mathbf{N}_u$ and $\mathbf{N}_v$ may be expressed as linear combinations of $\mathbf{r}_u$ and $\mathbf{r}_v$.

Theorem 3.3.4 (Weingarten Equations). *Let*

$$H_0^2 = EG - F^2.$$

Then

$$\mathbf{N}_u = -\frac{1}{H_0^2} [(GL - FM)\mathbf{r}_u + (EM - FL)\mathbf{r}_v],$$

and

$$\mathbf{N}_v = -\frac{1}{H_0^2} [(GM - FN)\mathbf{r}_u + (EN - FM)\mathbf{r}_v].$$

Equivalently,

$$\begin{aligned} \mathbf{N}_u &= \frac{FM - GL}{H_0^2} \mathbf{r}_u + \frac{FL - EM}{H_0^2} \mathbf{r}_v, \\ \mathbf{N}_v &= \frac{FN - GM}{H_0^2} \mathbf{r}_u + \frac{FM - EN}{H_0^2} \mathbf{r}_v. \end{aligned}$$

Proof. Write

$$\mathbf{N}_u = a\mathbf{r}_u + b\mathbf{r}_v.$$

Differentiating $\mathbf{N} \cdot \mathbf{r}_u = 0$ and $\mathbf{N} \cdot \mathbf{r}_v = 0$ with respect to u gives

$$\mathbf{N}_u \cdot \mathbf{r}_u = -L, \quad \mathbf{N}_u \cdot \mathbf{r}_v = -M.$$

Therefore

$$Ea + Fb = -L, \quad Fa + Gb = -M.$$

Solving,

$$a = \frac{FM - GL}{EG - F^2}, \quad b = \frac{FL - EM}{EG - F^2}.$$

This gives the formula for $\mathbf{N}_u$.

Similarly, if

$$\mathbf{N}_v = c\mathbf{r}_u + d\mathbf{r}_v,$$

then

$$Ec + Fd = -M, \quad Fc + Gd = -N.$$

Solving,

$$c = \frac{FN - GM}{EG - F^2}, \quad d = \frac{FM - EN}{EG - F^2},$$

which gives the formula for $\mathbf{N}_v$. ■

Corollary 3.3.5. *If the coordinate curves are lines of curvature, then $F = M = 0$, and*

$$\mathbf{N}_u = -\frac{L}{E}\mathbf{r}_u, \quad \mathbf{N}_v = -\frac{N}{G}\mathbf{r}_v.$$

Thus

$$\mathbf{N}_u = -k_1\mathbf{r}_u, \quad \mathbf{N}_v = -k_2\mathbf{r}_v.$$

Example 3.3.6 (Weingarten Equations for a Sphere). Verify the Weingarten equations for the sphere $\mathbf{r}(u, v) = (a \sin u \cos v, a \sin u \sin v, a \cos u)$.

Solution. For the outward orientation, $\mathbf{N} = \mathbf{r}/a$. Hence $\mathbf{N}_u = \mathbf{r}_u/a$ and $\mathbf{N}_v = \mathbf{r}_v/a$. With this orientation the principal curvatures are both $-1/a$ under the convention $S = -d\mathbf{N}$, so the Weingarten equations read $\mathbf{N}_u = -k_1\mathbf{r}_u = \mathbf{r}_u/a$ and $\mathbf{N}_v = -k_2\mathbf{r}_v = \mathbf{r}_v/a$, as required. ■

Example 3.3.7 (Shape Operator of a Cylinder). Determine the matrix of the shape operator of the cylinder $\mathbf{r}(u, v) = (a \cos u, a \sin u, v)$ relative to the orthogonal basis $\{\mathbf{r}_u, \mathbf{r}_v\}$.

Solution. With the outward normal $\mathbf{N} = (\cos u, \sin u, 0)$, $\mathbf{N}_u = \mathbf{r}_u/a$ and $\mathbf{N}_v = 0$. Since $S = -d\mathbf{N}$, we have $S(\mathbf{r}_u) = -(1/a)\mathbf{r}_u$ and $S(\mathbf{r}_v) = 0$. Thus the matrix of S in this basis is

$$\begin{pmatrix} -1/a & 0 \\ 0 & 0 \end{pmatrix}.$$

Its eigenvalues are the principal curvatures $-1/a$ and 0 . ■

3.4 Gaussian Curvature as an Area Ratio

Definition 3.4.1 (Gauss Map). The **Gauss map** of an oriented surface is the map

$$\mathcal{G} : S \longrightarrow \mathbb{S}^2, \quad \mathcal{G}(P) = \mathbf{N}(P).$$

Theorem 3.4.2. *The derivatives of the Gauss map satisfy*

$$\mathbf{N}_u \times \mathbf{N}_v = K(\mathbf{r}_u \times \mathbf{r}_v).$$

Consequently, if dA is the area element on S and $dA_{\mathbb{S}^2}$ is the area element of the spherical image, then

$$dA_{\mathbb{S}^2} = |K| dA.$$

With oriented areas,

$$K = \frac{dA_{\mathbb{S}^2}}{dA}.$$

Proof. Using the Weingarten equations, write

$$\mathbf{N}_u = a\mathbf{r}_u + b\mathbf{r}_v, \quad \mathbf{N}_v = c\mathbf{r}_u + d\mathbf{r}_v.$$

Then

$$\mathbf{N}_u \times \mathbf{N}_v = (ad - bc)(\mathbf{r}_u \times \mathbf{r}_v).$$

The determinant $ad - bc$ is the determinant of the shape operator and equals $\frac{LN - M^2}{EG - F^2} = K$. Hence

$$\mathbf{N}_u \times \mathbf{N}_v = K(\mathbf{r}_u \times \mathbf{r}_v).$$

Taking norms gives

$$\|\mathbf{N}_u \times \mathbf{N}_v\| = |K| \|\mathbf{r}_u \times \mathbf{r}_v\|.$$

Multiplying by $du dv$ yields $dA_{\mathbb{S}^2} = |K| dA$. ■

Theorem 3.4.3 (Gaussian Curvature in Orthogonal Coordinates). *If $F = 0$, then*

$$K = -\frac{1}{2\sqrt{EG}} \left[\frac{\partial}{\partial u} \left(\frac{G_u}{\sqrt{EG}} \right) + \frac{\partial}{\partial v} \left(\frac{E_v}{\sqrt{EG}} \right) \right].$$

Proof. For an orthogonal metric $ds^2 = E du^2 + G dv^2$, Liouville's formula expresses the geodesic curvature of the coordinate curves in terms of E and G . Applying Green's theorem to a small coordinate rectangle and using the Gauss–Bonnet formula gives the stated expression. Thus, in orthogonal coordinates, K is determined entirely by the first fundamental form. ■

Corollary 3.4.4 (Geodesic Polar Coordinates). *If*

$$ds^2 = du^2 + g(u, v)^2 dv^2,$$

then

$$K = -\frac{g_{uu}}{g},$$

or equivalently $g_{uu} + Kg = 0$.

Proof. Substitute $E = 1$ and $G = g^2$ in the preceding formula. ■

Theorem 3.4.5 (Gaussian Curvature from Geodesic Triangles). *(See Figure 55 in the Atlas.) Let a geodesic triangle with area A and angles α, β, γ shrink to a point P . Then*

$$K(P) = \lim_{A \rightarrow 0} \frac{\alpha + \beta + \gamma - \pi}{A}.$$

Proof. For a geodesic triangle, the geodesic curvature of each side vanishes. The Gauss–Bonnet formula therefore gives

$$\iint_R K dA = \alpha + \beta + \gamma - \pi.$$

Dividing by the area of R and allowing the triangle to shrink to P gives the stated limit by continuity of K . ■

Example 3.4.6 (Sphere). For a sphere of radius a , a spherical triangle satisfies $A = a^2(\alpha + \beta + \gamma - \pi)$. Hence

$$K = \frac{\alpha + \beta + \gamma - \pi}{A} = \frac{1}{a^2}.$$

Theorem 3.4.7 (Curvature from Small Geodesic Circles). *Let $C(r)$ be the circumference of a geodesic circle of radius r centered at P , and let $A(r)$ be the area of the corresponding geodesic disc. Then*

$$K(P) = \lim_{r \rightarrow 0} \frac{3(2\pi r - C(r))}{\pi r^3},$$

and

$$K(P) = \lim_{r \rightarrow 0} \frac{12(\pi r^2 - A(r))}{\pi r^4}.$$

Proof. In geodesic polar coordinates centered at P ,

$$ds^2 = du^2 + g(u, v)^2 dv^2, \quad g_{uu} + Kg = 0,$$

with $g(0, v) = 0$ and $g_u(0, v) = 1$. If $K(P) = K_0$, Taylor expansion gives $g(u, v) = u - \frac{K_0}{6}u^3 + O(u^4)$. Thus

$$C(r) = 2\pi r - \frac{\pi K_0}{3}r^3 + O(r^4),$$

and

$$A(r) = \pi r^2 - \frac{\pi K_0}{12}r^4 + O(r^5).$$

Solving for K_0 and taking $r \rightarrow 0$ gives the two formulas. ■

Example 3.4.8 (Gauss Map of a Cylinder). Describe the Gauss map of the circular cylinder and explain why its Gaussian curvature vanishes.

Solution. For $\mathbf{r}(u, v) = (a \cos u, a \sin u, v)$, the outward unit normal is $\mathbf{N}(u, v) = (\cos u, \sin u, 0)$. Its image lies on the equator of the unit sphere and is independent of v . Thus the differential of the Gauss map has rank one and sends every infinitesimal surface patch to zero spherical area. Hence its area ratio is zero, so $K = 0$. ■

Example 3.4.9 (Geodesic Circle on a Sphere). (See Figure 45 in the Atlas.) Use the small geodesic-circle formula to recover the Gaussian curvature of a sphere of radius a .

Solution. A geodesic circle of radius r on a sphere of radius a has circumference $C(r) = 2\pi a \sin(r/a)$. Using $\sin(r/a) = r/a - r^3/(6a^3) + O(r^5)$, we obtain $C(r) = 2\pi r - \pi r^3/(3a^2) + O(r^5)$. Therefore

$$\lim_{r \rightarrow 0} \frac{3(2\pi r - C(r))}{\pi r^3} = \frac{1}{a^2},$$

which is the Gaussian curvature of the sphere. ■

3.5 Lines of Curvature

Definition 3.5.1 (Line of Curvature). A curve on a surface is called a **line of curvature** if its tangent at every point is a principal direction.

The differential equation of the two families of lines of curvature is $(EM - FL) du^2 + (EN - GL) du dv + (FN - GM) dv^2 = 0$.

Theorem 3.5.2. *The coordinate curves are lines of curvature if and only if*

$$F = 0, \quad M = 0$$

away from umbilic points.

Proof. If the coordinate curves are lines of curvature, they are principal directions and hence are orthogonal, so $F = 0$. The equation of the lines of curvature must then agree with $du dv = 0$. Therefore its du^2 and dv^2 coefficients vanish:

$$EM - FL = 0, \quad FN - GM = 0.$$

Since $F = 0$ and $E, G > 0$, we obtain $M = 0$.

Conversely, if $F = M = 0$, the equation of lines of curvature becomes $(EN - GL) du dv = 0$. Away from an umbilic, $EN - GL \neq 0$, so $du dv = 0$. Hence the coordinate curves are lines of curvature. ■

Theorem 3.5.3 (Rodrigues' Formula). *A curve on a surface is a line of curvature if and only if*

$$d\mathbf{N} + k_n d\mathbf{r} = 0$$

along the curve, where k_n is its normal curvature. Equivalently, for an arc length parameter s ,

$$\frac{d\mathbf{N}}{ds} = -k_n \frac{d\mathbf{r}}{ds}.$$

Proof. Suppose first that the curve is a line of curvature with principal curvature k . Its tangent direction (du, dv) satisfies

$$(L - kE)du + (M - kF)dv = 0,$$

$$(M - kF)du + (N - kG)dv = 0.$$

Using

$$d\mathbf{r} = \mathbf{r}_u du + \mathbf{r}_v dv, \quad d\mathbf{N} = \mathbf{N}_u du + \mathbf{N}_v dv,$$

together with

$$L = -\mathbf{N}_u \cdot \mathbf{r}_u, \quad M = -\mathbf{N}_u \cdot \mathbf{r}_v = -\mathbf{N}_v \cdot \mathbf{r}_u, \quad N = -\mathbf{N}_v \cdot \mathbf{r}_v,$$

the two principal-direction equations imply

$$(d\mathbf{N} + k d\mathbf{r}) \cdot \mathbf{r}_u = 0, \quad (d\mathbf{N} + k d\mathbf{r}) \cdot \mathbf{r}_v = 0.$$

Both $d\mathbf{N}$ and $d\mathbf{r}$ are tangent to the surface, so their sum is tangential. A tangential vector perpendicular to both $\mathbf{r}_u$ and $\mathbf{r}_v$ must vanish. Hence

$$d\mathbf{N} + k d\mathbf{r} = 0.$$

Conversely, assume

$$d\mathbf{N} + k d\mathbf{r} = 0.$$

Taking scalar products with $\mathbf{r}_u$ and $\mathbf{r}_v$ recovers

$$(L - kE)du + (M - kF)dv = 0,$$

$$(M - kF)du + (N - kG)dv = 0.$$

Thus the tangent direction is principal. Moreover, taking the scalar product with $d\mathbf{r}$ gives

$$k = \frac{II}{I} = k_n.$$

Therefore the curve is a line of curvature. ■

Theorem 3.5.4 (Euler's Formula). *Let a tangent direction make an angle θ with a principal direction corresponding to k_1 . Then its normal curvature is $k_n = k_1 \cos^2 \theta + k_2 \sin^2 \theta$.*

Proof. Choose the principal directions as orthogonal coordinate directions at the point. Then $F = M = 0$. If (l, m) is a unit tangent direction, then

$$El^2 + Gm^2 = 1, \quad k_n = Ll^2 + Nm^2.$$

Because the direction makes angle θ with the first principal direction,

$$l = \frac{\cos \theta}{\sqrt{E}}, \quad m = \frac{\sin \theta}{\sqrt{G}}.$$

Hence

$$k_n = \frac{L}{E} \cos^2 \theta + \frac{N}{G} \sin^2 \theta = k_1 \cos^2 \theta + k_2 \sin^2 \theta. ■$$

Theorem 3.5.5 (Dupin's Theorem). *At a point of a surface, the sum of the normal curvatures in any two mutually orthogonal tangent directions is constant and is equal to*

$$k_1 + k_2 = 2H.$$

Proof. Let the first direction make an angle θ with a principal direction. The orthogonal direction then makes an angle $\theta + \pi/2$. By Euler's formula,

$$k_n(\theta) = k_1 \cos^2 \theta + k_2 \sin^2 \theta,$$

and

$$k_n\left(\theta + \frac{\pi}{2}\right) = k_1 \sin^2 \theta + k_2 \cos^2 \theta.$$

Adding,

$$k_n(\theta) + k_n\left(\theta + \frac{\pi}{2}\right) = k_1 + k_2 = 2H.$$

■

Example 3.5.6 (Surface of Revolution). (See Figure 36 in the Atlas.) Show that the meridians and parallels of

$$\mathbf{r}(u, v) = (u \cos v, u \sin v, f(u))$$

are lines of curvature.

Solution. We have

$$\mathbf{r}_u = (\cos v, \sin v, f'(u)), \quad \mathbf{r}_v = (-u \sin v, u \cos v, 0).$$

Therefore

$$F = \mathbf{r}_u \cdot \mathbf{r}_v = 0.$$

Also,

$$\mathbf{r}_{uv} = (-\sin v, \cos v, 0),$$

while $\mathbf{r}_u \times \mathbf{r}_v$ is orthogonal to $\mathbf{r}_{uv}$. Hence $M = 0$. Therefore the coordinate curves $u = \text{constant}$ and $v = \text{constant}$ are lines of curvature. ■

Example 3.5.7 (Lines of Curvature on a Cylinder). Find the lines of curvature on the circular cylinder $\mathbf{r}(u, v) = (a \cos u, a \sin u, v)$.

Solution. For the cylinder, $F = M = 0$. Hence the coordinate curves are lines of curvature. Thus $v = \text{constant}$ gives the circular sections and $u = \text{constant}$ gives the generators. The corresponding principal curvatures are $-1/a$ and 0 , respectively. ■

Example 3.5.8 (Euler's Formula on a Cylinder). Find the normal curvature of a cylinder of radius a in a direction making an angle θ with a circular principal direction.

Solution. The principal curvatures are $k_1 = -1/a$ along the circles and $k_2 = 0$ along the generators. Euler's formula gives $k_n = -(1/a) \cos^2 \theta$. In particular, the normal curvature vanishes along the generators and has maximum absolute value along the circular sections. ■

3.6 Dupin Indicatrix, Conjugate Directions, and Asymptotic Lines

(See Figure 35 in the Atlas.)

Definition 3.6.1 (Dupin Indicatrix). Let P be a point of a surface and choose rectangular coordinates in the tangent plane along the principal directions. The section of the surface by a plane parallel and sufficiently close to the tangent plane is, to second order, represented by a conic called the **Dupin indicatrix** at P .

Theorem 3.6.2 (Equation of the Dupin Indicatrix). If k_1 and k_2 are the principal curvatures at P , then, to second order, the section by the plane $z = h$ has equation

$$k_1 x^2 + k_2 y^2 = 2h.$$

Equivalently, if $R_1 = 1/k_1$ and $R_2 = 1/k_2$ are the principal radii, then $x^2/R_1 + y^2/R_2 = 2h$.

Proof. Choose principal coordinates at P . Then $F = M = 0$, and the coordinate tangent vectors are orthogonal. If x and y denote arc length coordinates along the principal directions, the first fundamental form at P is $dx^2 + dy^2$ and the second fundamental form is $k_1 dx^2 + k_2 dy^2$. The normal displacement of a nearby point from the tangent plane is, to second order, $2h = k_1 x^2 + k_2 y^2$. This is the required equation. ■

Corollary 3.6.3. *The Dupin indicatrix is an ellipse at an elliptic point and a hyperbola at a hyperbolic point. At a parabolic point it degenerates into a pair of parallel lines in the quadratic approximation. At a non-planar minimal point it is a rectangular hyperbola.*

Proof. The type of the conic is determined by the sign of $K = k_1 k_2$. If $H = 0$, then $k_2 = -k_1$, so the equation becomes $k_1(x^2 - y^2) = 2h$, which is a rectangular hyperbola. ■

Definition 3.6.4 (Conjugate Directions). Two tangent directions represented by (l_1, m_1) and (l_2, m_2) are called **conjugate** if

$$Ll_1 l_2 + M(l_1 m_2 + m_1 l_2) + N m_1 m_2 = 0.$$

Equivalently, if X and Y are tangent vectors, they are conjugate when $II(X, Y) = 0$.

Proposition 3.6.5. The two principal directions are conjugate. More generally, the coordinate curves are conjugate if and only if $M = 0$.

Proof. In principal coordinates the mixed coefficient of the second fundamental form vanishes, so $II(\mathbf{r}_u, \mathbf{r}_v) = M = 0$. The same identity shows that the coordinate directions are conjugate exactly when $M = 0$. ■

Definition 3.6.6 (Asymptotic Direction and Asymptotic Line). A tangent direction is called **asymptotic** if its normal curvature is zero. Thus its direction ratios satisfy

$$L du^2 + 2M du dv + N dv^2 = 0.$$

A curve whose tangent direction is asymptotic at every point is called an **asymptotic line**.

Proposition 3.6.7. At an elliptic point there are no real asymptotic directions; at a hyperbolic point there are two distinct asymptotic directions; and at a parabolic point there is one repeated asymptotic direction.

Proof. The asymptotic directions are the zeros of the quadratic form II . Its discriminant is proportional to $M^2 - LN$, whose sign is the opposite of the sign of $K = (LN - M^2)/(EG - F^2)$. The three cases follow immediately. ■

Theorem 3.6.8. *At a hyperbolic point, the principal directions bisect the angles between the two asymptotic directions.*

Proof. Choose the principal directions as an orthonormal basis of the tangent plane. If an asymptotic direction makes angle θ with the first principal direction, Euler's formula gives $k_1 \cos^2 \theta + k_2 \sin^2 \theta = 0$. The second solution has the opposite angle with the same principal axis. Hence the principal axes are the angle bisectors of the two asymptotic directions. ■

Theorem 3.6.9 (Beltrami–Enneper). *At a point of an asymptotic line on a surface with $K < 0$, the torsion τ of the asymptotic line satisfies*

$$\tau^2 = -K, \quad \text{hence} \quad \tau = \pm \sqrt{-K}.$$

Proof. Let the asymptotic direction make angle θ with the principal direction corresponding to k_1 . Since $k_n = 0$, Euler's formula gives $k_1 \cos^2 \theta + k_2 \sin^2 \theta = 0$. The geodesic torsion in this direction is $\tau_g = (k_2 - k_1) \sin \theta \cos \theta$. Using the preceding relation,

$$\tau_g^2 = (k_2 - k_1)^2 \sin^2 \theta \cos^2 \theta = -k_1 k_2 = -K.$$

For an asymptotic line the principal normal of the curve is tangent to the surface, so the curve binormal agrees with the surface normal up to sign. Therefore $|\tau| = |\tau_g|$, and the result follows. ■

Example 3.6.10 (Dupin Indicatrix of a Sphere). Determine the Dupin indicatrix at a point of a sphere of radius a .

Solution. At every point of the sphere the principal curvatures are equal in magnitude, $k_1 = k_2 = \pm 1/a$, according to orientation. Hence the Dupin indicatrix is a circle. This agrees with the fact that every tangent direction at an umbilic is principal. ■

Example 3.6.11 (Asymptotic Directions on $z = xy$). Find the asymptotic directions of the surface $z = xy$ at the origin.

Solution. For $\mathbf{r}(u, v) = (u, v, uv)$, at the origin we have $L = N = 0$ and $M = 1$. The asymptotic equation is therefore $2 du dv = 0$. Hence the two asymptotic directions are $du = 0$ and $dv = 0$, corresponding to the coordinate axes in the tangent plane. ■

Example 3.6.12 (Conjugate Directions on a Sphere). Show that meridian and parallel directions on a sphere are conjugate.

Solution. For the usual spherical coordinates, $M = 0$. The coordinate directions $(du, 0)$ and $(0, dv)$ are conjugate precisely when $L du \cdot 0 + M(du dv) + N 0 \cdot dv = 0$, which holds because $M = 0$. Hence meridian and parallel directions are conjugate. ■

3.7 Joachimsthal's Theorem

Theorem 3.7.1 (Joachimsthal). *Let two regular surfaces S_1 and S_2 intersect along a regular curve C . If the surfaces meet at a constant angle along C , then C is a line of curvature on S_1 if and only if it is a line of curvature on S_2 .*

Proof. Let $\mathbf{N}_1$ and $\mathbf{N}_2$ be the unit normals to S_1 and S_2 along C , and let $\mathbf{t}$ be the unit tangent to C . Since the angle θ between the surfaces is constant,

$$\mathbf{N}_1 \cdot \mathbf{N}_2 = \cos \theta.$$

Differentiating with respect to arc length s gives

$$\frac{d\mathbf{N}_1}{ds} \cdot \mathbf{N}_2 + \mathbf{N}_1 \cdot \frac{d\mathbf{N}_2}{ds} = 0.$$

Assume that C is a line of curvature on S_1 . By Rodrigues' formula,

$$\frac{d\mathbf{N}_1}{ds} = -k_1 \mathbf{t}.$$

Since $\mathbf{t} \cdot \mathbf{N}_2 = 0$, it follows that

$$\frac{d\mathbf{N}_1}{ds} \cdot \mathbf{N}_2 = 0.$$

Hence

$$\mathbf{N}_1 \cdot \frac{d\mathbf{N}_2}{ds} = 0.$$

Also, because $\mathbf{N}_2$ is a unit vector,

$$\mathbf{N}_2 \cdot \frac{d\mathbf{N}_2}{ds} = 0.$$

Thus $d\mathbf{N}_2/ds$ is perpendicular to both $\mathbf{N}_1$ and $\mathbf{N}_2$. The tangent $\mathbf{t}$ is also perpendicular to both normals, so

$$\frac{d\mathbf{N}_2}{ds} = -k_2 \mathbf{t}$$

for some scalar function k_2 . By Rodrigues' formula, C is a line of curvature on S_2 .

The converse follows by interchanging S_1 and S_2 . ■

Corollary 3.7.2. *If the curve of intersection of two surfaces is a line of curvature on both surfaces, then the angle of intersection is constant.*

Proof. If C is a line of curvature on both surfaces, then

$$\frac{d\mathbf{N}_1}{ds} = -k_1\mathbf{t}, \quad \frac{d\mathbf{N}_2}{ds} = -k_2\mathbf{t}.$$

Therefore

$$\frac{d}{ds}(\mathbf{N}_1 \cdot \mathbf{N}_2) = -k_1\mathbf{t} \cdot \mathbf{N}_2 - k_2\mathbf{N}_1 \cdot \mathbf{t} = 0.$$

Thus $\mathbf{N}_1 \cdot \mathbf{N}_2$ is constant, so the angle between the surfaces is constant. ■

Example 3.7.3 (Sphere and Plane). (See Figure 37 in the Atlas.) A plane cuts a sphere in a circle. Show that the circle is a line of curvature of the sphere.

Solution. A plane has constant unit normal, while the angle between that normal and the radial normal of the sphere is constant along the circle of intersection. Every curve on a plane is a line of curvature because its shape operator is zero. By Joachimsthal's theorem, the intersection circle is therefore a line of curvature on the sphere. This is also consistent with the fact that every direction on a sphere is principal. ■

3.8 Ruled Surfaces

Definition 3.8.1 (Ruled Surface). A surface generated by a one-parameter family of straight lines is called a **ruled surface**. The generating lines are called **rulings** or **generators**.

Definition 3.8.2 (Directrix). A curve on a ruled surface which meets each generator once is called a **directrix** or **base curve**.

Definition 3.8.3 (Developable and Skew Ruled Surfaces). A ruled surface is called **developable** if its tangent plane is constant along each generator. A ruled surface which is not developable is called a **skew ruled surface** or a **scroll**.

A ruled surface may be written in the form

$$\mathbf{R}(u, v) = \mathbf{r}(u) + v\mathbf{g}(u),$$

where $\mathbf{r}(u)$ is a directrix and $\mathbf{g}(u)$ is a unit vector in the direction of the generator.

Theorem 3.8.4 (Gaussian Curvature of a Ruled Surface). *For*

$$\mathbf{R}(u, v) = \mathbf{r}(u) + v\mathbf{g}(u), \quad \|\mathbf{g}(u)\| = 1,$$

the Gaussian curvature satisfies

$$K = -\frac{[\mathbf{r}', \mathbf{g}, \mathbf{g}']^2}{\|\mathbf{R}_u \times \mathbf{R}_v\|^4} \leq 0,$$

where $[\mathbf{a}, \mathbf{b}, \mathbf{c}]$ denotes the scalar triple product.

Proof. Differentiating,

$$\mathbf{R}_u = \mathbf{r}' + v\mathbf{g}', \quad \mathbf{R}_v = \mathbf{g}.$$

Since $\|\mathbf{g}\| = 1$,

$$\mathbf{g} \cdot \mathbf{g}' = 0.$$

Also,

$$\mathbf{R}_{uv} = \mathbf{g}', \quad \mathbf{R}_{vv} = 0.$$

Hence the second fundamental coefficient corresponding to dv^2 is $N = 0$. If $H_0 = \|\mathbf{R}_u \times \mathbf{R}_v\|$, then

$$H_0 M = [\mathbf{R}_{uv}, \mathbf{R}_u, \mathbf{R}_v] = [\mathbf{g}', \mathbf{r}' + v\mathbf{g}', \mathbf{g}] = [\mathbf{g}', \mathbf{r}', \mathbf{g}].$$

Therefore

$$M^2 = \frac{[\mathbf{r}', \mathbf{g}, \mathbf{g}']^2}{H_0^2}.$$

Since

$$K = \frac{LN - M^2}{H_0^2}$$

and $N = 0$, we obtain

$$K = -\frac{[\mathbf{r}', \mathbf{g}, \mathbf{g}']^2}{H_0^4} \leq 0.$$

■

Corollary 3.8.5. *A ruled surface is developable if and only if*

$$[\mathbf{r}', \mathbf{g}, \mathbf{g}'] = 0.$$

Proof. A ruled surface is developable precisely when $K = 0$. The preceding formula shows that this occurs exactly when the scalar triple product vanishes. ■

Theorem 3.8.6 (Equivalent Characterizations of a Developable Ruled Surface). *For a regular ruled surface $\mathbf{R}(u, v) = \mathbf{r}(u) + v\mathbf{g}(u)$, the following conditions are equivalent:*

1. *the surface is developable;*
2. *the tangent plane is constant along each generator;*
3. $[\mathbf{r}', \mathbf{g}, \mathbf{g}'] = 0$;
4. $K = 0$ *at every regular point.*

Proof. The surface normal is parallel to $(\mathbf{r}' + v\mathbf{g}') \times \mathbf{g}$. Its direction is independent of v precisely when $\mathbf{r}', \mathbf{g}, \mathbf{g}'$ are linearly dependent, which is equivalent to the vanishing of their scalar triple product. By the formula for the Gaussian curvature of a ruled surface, the same condition is equivalent to $K = 0$. These are precisely the conditions for the tangent plane to remain fixed along a generator. ■

Definition 3.8.7 (Parameter of Distribution). For a ruled surface

$$\mathbf{R}(u, v) = \mathbf{r}(u) + v\mathbf{g}(u),$$

the function

$$p(u) = \frac{[\mathbf{r}'(u), \mathbf{g}(u), \mathbf{g}'(u)]}{\|\mathbf{g}'(u)\|^2},$$

whenever $\mathbf{g}'(u) \neq 0$, is called the **parameter of distribution**.

Remark 3.8.8. The parameter of distribution is constant along each generator and is independent of the chosen directrix. For a developable ruled surface, $p(u) = 0$.

Proposition 3.8.9. The generators of a ruled surface form one family of asymptotic lines.

Proof. The equation of the asymptotic directions is $L du^2 + 2M du dv + N dv^2 = 0$. For a ruled surface $N = 0$, so

$$du(L du + 2M dv) = 0.$$

One family is therefore $du = 0$, that is, $u = \text{constant}$. These are precisely the generators. ■

Example 3.8.10 (Right Helicoid). Consider

$$\mathbf{r}(u, v) = (u \cos v, u \sin v, av).$$

Show that it is a ruled minimal surface and find its Gaussian curvature.

Solution. For fixed v , the curve $u \mapsto \mathbf{r}(u, v)$ is a straight line, so the surface is ruled. We have

$$\mathbf{r}_u = (\cos v, \sin v, 0), \quad \mathbf{r}_v = (-u \sin v, u \cos v, a).$$

Thus

$$E = 1, \quad F = 0, \quad G = u^2 + a^2.$$

Also,

$$\mathbf{r}_{uu} = 0, \quad \mathbf{r}_{uv} = (-\sin v, \cos v, 0), \quad \mathbf{r}_{vv} = (-u \cos v, -u \sin v, 0).$$

A unit normal is

$$\mathbf{N} = \frac{(a \sin v, -a \cos v, u)}{\sqrt{u^2 + a^2}}.$$

Hence

$$L = 0, \quad M = -\frac{a}{\sqrt{u^2 + a^2}}, \quad N = 0.$$

Therefore

$$H = \frac{EN + GL - 2FM}{2(EG - F^2)} = 0,$$

so the helicoid is minimal, and

$$K = \frac{LN - M^2}{EG - F^2} = -\frac{a^2}{(u^2 + a^2)^2}.$$

■

Example 3.8.11 (Hyperbolic Paraboloid as a Ruled Surface). (See Figure 38 in the Atlas.) Show that the surface $z = xy$ is doubly ruled.

Solution. For any constant c , the set $x = c$ gives the straight line (c, y, cy) as y varies. Thus one family of rulings is obtained by fixing x . Similarly, fixing $y = c$ gives the straight line (x, c, cx) , producing a second family. Hence the hyperbolic paraboloid is doubly ruled. ■

Example 3.8.12 (Developable Cylinder). Verify the developability criterion for the cylinder $\mathbf{R}(u, v) = (a \cos u, a \sin u, 0) + v(0, 0, 1)$.

Solution. Here $\mathbf{r}'(u) = (-a \sin u, a \cos u, 0)$, $\mathbf{g} = (0, 0, 1)$, and $\mathbf{g}' = 0$. Therefore $[\mathbf{r}', \mathbf{g}, \mathbf{g}'] = 0$. Hence the cylinder is developable, in agreement with $K = 0$. ■

3.9 Minimal Surfaces

Definition 3.9.1 (Minimal Surface). A regular surface is called a **minimal surface** if its mean curvature vanishes identically:

$$H = 0.$$

Equivalently,

$$EN + GL - 2FM = 0.$$

Remark 3.9.2. If the coordinate curves are orthogonal, $F = 0$, and the minimal-surface condition reduces to

$$EN + GL = 0.$$

If the coordinate curves are also lines of curvature, then

$$k_1 + k_2 = 0.$$

Thus the principal curvatures of a minimal surface are equal in magnitude and opposite in sign.

Proposition 3.9.3. At every non-planar point of a minimal surface,

$$K \leq 0.$$

More precisely,

$$K = -k_1^2 = -k_2^2.$$

Proof. Since $H = 0$, we have $k_2 = -k_1$. Hence $K = k_1 k_2 = -k_1^2 \leq 0$. ■

Example 3.9.4 (A Scherk-Type Minimal Surface). (See Figure 39 in the Atlas.) Show that the surface

$$e^z \cos x = \cos y$$

is minimal wherever it is regular.

Solution. Write

$$z = \log(\cos y) - \log(\cos x)$$

and use x, y as parameters:

$$\mathbf{r}(x, y) = (x, y, \log(\cos y) - \log(\cos x)).$$

Then

$$\mathbf{r}_x = (1, 0, \tan x), \quad \mathbf{r}_y = (0, 1, -\tan y),$$

so

$$E = \sec^2 x, \quad F = -\tan x \tan y, \quad G = \sec^2 y.$$

Further,

$$\mathbf{r}_{xx} = (0, 0, \sec^2 x), \quad \mathbf{r}_{xy} = 0, \quad \mathbf{r}_{yy} = (0, 0, -\sec^2 y).$$

If $H_0 = \|\mathbf{r}_x \times \mathbf{r}_y\|$, then

$$L = \frac{\sec^2 x}{H_0}, \quad M = 0, \quad N = -\frac{\sec^2 y}{H_0}.$$

Therefore

$$EN + GL - 2FM = \frac{-\sec^2 x \sec^2 y + \sec^2 y \sec^2 x}{H_0} = 0.$$

Hence the surface is minimal. ■

Theorem 3.9.5. The two asymptotic directions at a point are orthogonal if and only if the surface is minimal at that point.

Proof. The asymptotic directions satisfy

$$L du^2 + 2M du dv + N dv^2 = 0.$$

For a double family

$$P du^2 + 2Q du dv + R dv^2 = 0,$$

the two directions are orthogonal precisely when

$$ER - 2FQ + GP = 0.$$

Taking $P = L$, $Q = M$, and $R = N$, the condition becomes $EN - 2FM + GL = 0$, which is exactly the condition $H = 0$. ■

Theorem 3.9.6 (Minimal Surfaces of Revolution). Apart from the plane, the catenoid is the only minimal surface of revolution.

Proof. Let

$$\mathbf{r}(u, v) = (u \cos v, u \sin v, f(u)).$$

Then

$$E = 1 + f'^2, \quad F = 0, \quad G = u^2,$$

and, for a suitable orientation,

$$L = \frac{f''}{\sqrt{1 + f'^2}}, \quad M = 0, \quad N = \frac{uf'}{\sqrt{1 + f'^2}}.$$

The condition $H = 0$ becomes $uf'' + f'(1 + f'^2) = 0$. If $f' \equiv 0$, the surface is a plane. Otherwise, putting $p = f'$ gives $\frac{p'}{p(1+p^2)} = -\frac{1}{u}$. Integration yields $\frac{p}{\sqrt{1+p^2}} = \frac{a}{u}$, so

$$f'(u) = \frac{a}{\sqrt{u^2 - a^2}}.$$

A second integration gives $f(u) = a \log(u + \sqrt{u^2 - a^2}) + c$. This is a catenary; its surface of revolution is a catenoid. Hence, apart from the plane, the catenoid is the only minimal surface of revolution. ■

Example 3.9.7 (Catenoid). (See Figure 40 in the Atlas.) The standard catenoid

$$\mathbf{r}(u, v) = (a \cosh u \cos v, a \cosh u \sin v, au)$$

has $E = G = a^2 \cosh^2 u$ and $F = 0$. Thus (u, v) are isothermal coordinates. A direct computation gives $H = 0$, so the catenoid is minimal.

Theorem 3.9.8 (First Variation of Area). *If a smooth surface bounded by a fixed closed curve has least area among all sufficiently small normal variations with the same boundary, then it is minimal.*

Proof. Let

$$\mathbf{R}(u, v) = \mathbf{r}(u, v) + \varepsilon \phi(u, v) \mathbf{N}(u, v),$$

where ϕ vanishes on the boundary. Denote the area of the varied surface by $A(\varepsilon)$. To first order in ε , the first fundamental coefficients change as

$$E_\varepsilon = E - 2\varepsilon \phi L + O(\varepsilon^2),$$

$$F_\varepsilon = F - 2\varepsilon \phi M + O(\varepsilon^2),$$

$$G_\varepsilon = G - 2\varepsilon \phi N + O(\varepsilon^2).$$

Thus

$$E_\varepsilon G_\varepsilon - F_\varepsilon^2 = (EG - F^2) - 2\varepsilon \phi (EN + GL - 2FM) + O(\varepsilon^2).$$

Since

$$EN + GL - 2FM = 2H(EG - F^2),$$

we obtain

$$dA_\varepsilon = (1 - 2\varepsilon \phi H) dA + O(\varepsilon^2).$$

Therefore

$$A'(0) = -2 \iint_S H \phi dA.$$

If the original surface minimizes area, then $A'(0) = 0$ for every admissible ϕ . Hence

$$\iint_S H \phi dA = 0$$

for every such ϕ , which implies $H = 0$ throughout the surface. Thus the surface is minimal. ■

Example 3.9.9 (Enneper Surface). (See Figure 41 in the Atlas.) Show that

$$\mathbf{r}(u, v) = \left(u - \frac{u^3}{3} + uv^2, v - \frac{v^3}{3} + u^2v, u^2 - v^2 \right)$$

is a minimal surface.

Solution. Differentiation gives

$$\mathbf{r}_u = (1 - u^2 + v^2, 2uv, 2u), \quad \mathbf{r}_v = (2uv, 1 - v^2 + u^2, -2v).$$

A direct calculation shows that $\mathbf{r}_u \cdot \mathbf{r}_v = 0$ and $\|\mathbf{r}_u\|^2 = \|\mathbf{r}_v\|^2 = (1 + u^2 + v^2)^2$. Hence (u, v) are isothermal coordinates. Moreover, each coordinate function is harmonic: $\mathbf{r}_{uu} + \mathbf{r}_{vv} = 0$. By the harmonic characterization of minimal surfaces, the Enneper surface is minimal. ■

Example 3.9.10 (No Non-Planar Minimal Surface with Positive Gaussian Curvature). Show that a minimal surface cannot have positive Gaussian curvature at a regular point.

Solution. For a minimal surface, $k_1 + k_2 = 0$, so $k_2 = -k_1$. Therefore $K = k_1 k_2 = -k_1^2 \leq 0$. Thus positive Gaussian curvature is impossible at a regular point of a minimal surface. ■

3.10 Isothermal Coordinates

Definition 3.10.1 (Isothermal Coordinates). A coordinate system (u, v) on a surface is called **isothermal** or **isothermic** if the first fundamental form has the form

$$ds^2 = \lambda^2(u, v)(du^2 + dv^2), \quad \lambda > 0.$$

Equivalently,

$$E = G = \lambda^2, \quad F = 0.$$

Remark 3.10.2. Isothermal coordinates are orthogonal and have the same scale factor in both coordinate directions.

Theorem 3.10.3 (Local Existence of Isothermal Coordinates). *Every point of a regular surface has a neighbourhood on which isothermal coordinates may be introduced.*

Proof. Locally, one may first introduce complex isotropic parameters in which the metric takes the form

$$ds^2 = \Lambda(u, v) du dv.$$

Set

$$u = U + iV, \quad v = U - iV.$$

Then

$$du dv = dU^2 + dV^2,$$

and therefore

$$ds^2 = \Lambda(U, V)(dU^2 + dV^2).$$

Thus (U, V) are local isothermal coordinates. ■

Theorem 3.10.4. *From one isothermal coordinate system, infinitely many others may be obtained locally through analytic changes of the complex variable $u + iv$.*

Proof. Suppose (u, v) and (U, V) are both isothermal systems:

$$ds^2 = \lambda^2(du^2 + dv^2) = \mu^2(dU^2 + dV^2).$$

Writing $U = U(u, v)$ and $V = V(u, v)$ and comparing the coefficients of du^2 , $du dv$, and dv^2 gives

$$U_u^2 + V_u^2 = U_v^2 + V_v^2, \quad U_u U_v + V_u V_v = 0.$$

These equations imply, up to orientation,

$$U_u = V_v, \quad U_v = -V_u,$$

which are the Cauchy–Riemann equations. Hence $U + iV$ is locally an analytic function of $u + iv$. Conversely, an analytic change of variable preserves the conformal form of the metric. ■

Example 3.10.5 (Isothermal Coordinates on a Surface of Revolution). Let

$$\mathbf{r}(u, v) = (g(u) \cos v, g(u) \sin v, f(u)).$$

Find an isothermal coordinate system.

Solution. The first fundamental form is

$$ds^2 = (g'(u)^2 + f'(u)^2) du^2 + g(u)^2 dv^2.$$

Define

$$U = \int \frac{\sqrt{g'(u)^2 + f'(u)^2}}{g(u)} du, \quad V = v.$$

Then

$$dU^2 = \frac{g'(u)^2 + f'(u)^2}{g(u)^2} du^2,$$

and hence

$$ds^2 = g(u)^2 (dU^2 + dV^2).$$

Thus (U, V) is an isothermal system. ■

Example 3.10.6 (Isothermal Coordinates on the Sphere). (See Figure 46 in the Atlas.) Find an isothermal coordinate for the unit sphere using longitude v and colatitude u .

Solution. The metric is $ds^2 = du^2 + \sin^2 u dv^2$. Define $U = \log \tan(u/2)$ and $V = v$. Since $dU = \csc u du$, we have $du^2 = \sin^2 u dU^2$. Hence

$$ds^2 = \sin^2 u (dU^2 + dV^2),$$

so (U, V) is an isothermal system. This is the coordinate form underlying the Mercator projection. ■

3.11 Minimal Surfaces and Harmonic Coordinate Functions

Theorem 3.11.1. *Let*

$$\mathbf{r}(u, v) = (x(u, v), y(u, v), z(u, v))$$

be a regular surface expressed in isothermal coordinates, so that $ds^2 = \lambda^2(du^2 + dv^2)$. Then

$$\mathbf{r}_{uu} + \mathbf{r}_{vv} = 2\lambda^2 H \mathbf{N}.$$

Consequently, the surface is minimal if and only if its coordinate functions x, y, z are harmonic:

$$x_{uu} + x_{vv} = 0, \quad y_{uu} + y_{vv} = 0, \quad z_{uu} + z_{vv} = 0.$$

Proof. For isothermal coordinates,

$$E = G = \lambda^2, \quad F = 0.$$

The Gauss formulas decompose $\mathbf{r}_{uu}$ and $\mathbf{r}_{vv}$ into tangential and normal parts. For a conformal metric $\lambda^2(du^2 + dv^2)$, the tangential parts cancel in the sum $\mathbf{r}_{uu} + \mathbf{r}_{vv}$. Its normal part is

$$(L + N)\mathbf{N}.$$

Hence

$$\mathbf{r}_{uu} + \mathbf{r}_{vv} = (L + N)\mathbf{N}.$$

But

$$H = \frac{EN + GL}{2EG} = \frac{\lambda^2(N + L)}{2\lambda^4} = \frac{L + N}{2\lambda^2},$$

so

$$\mathbf{r}_{uu} + \mathbf{r}_{vv} = 2\lambda^2 H \mathbf{N}.$$

If the surface is minimal, $H = 0$, and therefore $\mathbf{r}_{uu} + \mathbf{r}_{vv} = 0$. Since

$$\mathbf{r} = (x, y, z),$$

this is equivalent to

$$x_{uu} + x_{vv} = 0, \quad y_{uu} + y_{vv} = 0, \quad z_{uu} + z_{vv} = 0.$$

Conversely, if the coordinate functions are harmonic, then $\mathbf{r}_{uu} + \mathbf{r}_{vv} = 0$, and the displayed identity gives $2\lambda^2 H \mathbf{N} = 0$. Since $\lambda > 0$ and $\|\mathbf{N}\| = 1$, we obtain $H = 0$. Hence the surface is minimal. ■

Corollary 3.11.2. *In isothermal coordinates, a minimal surface is locally represented by three harmonic functions satisfying the conformality conditions*

$$\mathbf{r}_u \cdot \mathbf{r}_v = 0, \quad \|\mathbf{r}_u\|^2 = \|\mathbf{r}_v\|^2.$$

Corollary 3.11.3 (No Compact Minimal Surface without Boundary). *There is no compact regular minimal surface without boundary in $\mathbb{R}^3$.*

Proof. In local isothermal coordinates, the coordinate functions of a minimal surface are harmonic. If the surface were compact without boundary, each ambient coordinate function restricted to the surface would attain a maximum and a minimum. The maximum principle forces each coordinate function to be constant, which is incompatible with regularity. Hence no such surface exists. ■

Example 3.11.4 (Harmonic Coordinate Test). Let $\mathbf{r}(u, v) = (u, v, u^2 - v^2)$. Determine whether the parameterization is isothermal and whether the surface is minimal by the harmonic-coordinate criterion.

Solution. The coordinate functions u , v , and $u^2 - v^2$ are harmonic. However, $\mathbf{r}_u = (1, 0, 2u)$ and $\mathbf{r}_v = (0, 1, -2v)$ satisfy $\mathbf{r}_u \cdot \mathbf{r}_v = -4uv$, which is not identically zero. Thus the given parameters are not isothermal, so harmonicity of the coordinate functions in these parameters does not by itself imply minimality. Indeed the surface is not minimal in general. This example emphasizes that the harmonic criterion requires isothermal coordinates. ■

3.12 Isometries and Local Isometries

Definition 3.12.1 (Isometry). Let S and $\tilde{S}$ be regular surfaces. A differentiable homeomorphism

$$\Phi : S \longrightarrow \tilde{S}$$

is called an **isometry** if it preserves the length of every tangent vector:

$$\|d\Phi_P(X)\| = \|X\|$$

for every $P \in S$ and every $X \in T_P S$.

Definition 3.12.2 (Local Isometry). A smooth map

$$\Phi : S \longrightarrow \tilde{S}$$

is called a **local isometry** if every point $P \in S$ has a neighbourhood U such that

$$\Phi|_U : U \longrightarrow \Phi(U)$$

is an isometry.

Theorem 3.12.3 (Characterization of an Isometry by the First Fundamental Form). *A smooth local diffeomorphism*

$$\Phi : S \longrightarrow \tilde{S}$$

is a local isometry if and only if it preserves the first fundamental form:

$$\tilde{I}(d\Phi_P X, d\Phi_P Y) = I(X, Y)$$

for all $X, Y \in T_P S$.

Proof. If Φ is a local isometry, then it preserves lengths. Using the polarization identity,

$$2I(X, Y) = \|X + Y\|^2 - \|X\|^2 - \|Y\|^2,$$

we see that it also preserves inner products. Hence $\tilde{I}(d\Phi X, d\Phi Y) = I(X, Y)$.

Conversely, if the first fundamental form is preserved, then

$$\|d\Phi_P X\|^2 = \tilde{I}(d\Phi_P X, d\Phi_P X) = I(X, X) = \|X\|^2.$$

Thus the differential preserves lengths of tangent vectors. Consequently the length of every curve is preserved locally, and Φ is a local isometry. ■

Corollary 3.12.4 (Coordinate Criterion). *Suppose corresponding points on two surfaces are represented by the same local parameters (u, v) :*

$$\mathbf{r} = \mathbf{r}(u, v), \quad \tilde{\mathbf{r}} = \tilde{\mathbf{r}}(u, v).$$

Then the correspondence is a local isometry if and only if

$$E = \tilde{E}, \quad F = \tilde{F}, \quad G = \tilde{G}.$$

Proof. The equality of the first fundamental forms

$$E du^2 + 2F du dv + G dv^2 = \tilde{E} du^2 + 2\tilde{F} du dv + \tilde{G} dv^2$$

for all du, dv is equivalent to equality of the three coefficients. ■

Proposition 3.12.5. A local isometry preserves lengths of curves, angles, areas, and Gaussian curvature.

Proof. A local isometry preserves the first fundamental form. Hence it preserves lengths and inner products of tangent vectors, and therefore angles. Since

$$dA = \sqrt{EG - F^2} du dv,$$

it preserves area. Gaussian curvature is intrinsic and is determined by the first fundamental form, so it is also preserved. ■

Theorem 3.12.6. *A local isometry carries geodesics to geodesics and preserves geodesic curvature.*

Proof. A local isometry preserves the first fundamental form and therefore the Levi-Civita connection determined by that metric. Hence the equation $\nabla_T T = 0$ is preserved, so geodesics map to geodesics. Since geodesic curvature is the norm of the tangential component of $\nabla_T T$, it is also preserved. ■

Example 3.12.7 (Plane and Cylinder). (See Figure 42 in the Atlas.) The plane

$$\mathbf{r}(u, v) = (u, v, 0)$$

has metric

$$ds^2 = du^2 + dv^2.$$

The circular cylinder

$$\tilde{\mathbf{r}}(u, v) = \left(a \cos \frac{u}{a}, a \sin \frac{u}{a}, v \right)$$

also has

$$d\tilde{s}^2 = du^2 + dv^2.$$

Hence the plane and cylinder are locally isometric. The correspondence is not globally one-to-one when the whole plane is used, because the coordinate u is periodic on the cylinder.

Example 3.12.8 (Catenoid and Right Helicoid). (See Figure 43 in the Atlas.) Show that a catenoid and a right helicoid are locally isometric.

Solution. Consider the catenoid

$$\mathbf{r}(u, v) = (a \cosh u \cos v, a \cosh u \sin v, au).$$

Its first fundamental form is

$$ds^2 = a^2 \cosh^2 u (du^2 + dv^2).$$

Consider the helicoid

$$\tilde{\mathbf{r}}(\rho, v) = (\rho \cos v, \rho \sin v, av).$$

Its metric is

$$d\tilde{s}^2 = d\rho^2 + (\rho^2 + a^2) dv^2.$$

Set

$$\rho = a \sinh u.$$

Then

$$d\rho = a \cosh u du, \quad \rho^2 + a^2 = a^2 \cosh^2 u.$$

Therefore

$$d\tilde{s}^2 = a^2 \cosh^2 u du^2 + a^2 \cosh^2 u dv^2 = ds^2.$$

Hence the two surfaces are locally isometric. ■

Theorem 3.12.9 (Gauss's Theorema Egregium). *The Gaussian curvature of a surface is determined entirely by its first fundamental form. Consequently, Gaussian curvature is preserved by every local isometry.*

Proof. In local coordinates, the Christoffel symbols are determined by E, F, G and their first derivatives. The Gauss equation expresses the curvature of the induced metric, and hence K , in terms of these Christoffel symbols and their first derivatives. Thus K depends only on the coefficients of the first fundamental form and their derivatives. Since a local isometry preserves the first fundamental form, it preserves K . ■

Theorem 3.12.10 (Minding's Theorem). *Any two surfaces having the same constant Gaussian curvature are locally isometric.*

Proof. Choose geodesic coordinates in a neighbourhood of a point so that the metric has the form

$$ds^2 = du^2 + g(u, v)^2 dv^2.$$

For such coordinates the Gaussian curvature satisfies $g_{uu} + Kg = 0$. Along the initial geodesic one may impose the standard initial conditions

$$g(0, v) = 1, \quad g_u(0, v) = 0.$$

If two surfaces have the same constant curvature K_0 , the function g satisfies the same ordinary differential equation in u with the same initial conditions on both surfaces. Uniqueness therefore gives the same local metric.

For $K_0 = 0$ one obtains $g = 1$, giving the Euclidean metric. For $K_0 = 1/a^2$ one obtains

$$g = \cos \frac{u}{a},$$

which is locally the metric of a sphere of radius a . For $K_0 = -1/a^2$ one obtains

$$g = \cosh \frac{u}{a},$$

which gives the standard local metric of a surface of constant negative curvature. Hence surfaces with the same constant Gaussian curvature have locally identical first fundamental forms and are therefore locally isometric. ■

Example 3.12.11 (A Local Isometry from the Plane to a Cone). (See Figure 44 in the Atlas.) Consider the cone $\mathbf{r}(u, v) = (u \cos(v/\sin \alpha), u \sin(v/\sin \alpha), u \cot \alpha)$, where $0 < \alpha < \pi/2$. Show that it is locally isometric to a planar sector.

Solution. Differentiation gives $E = \csc^2 \alpha$, $F = 0$, and $G = u^2/\sin^2 \alpha$. Put $\rho = u/\sin \alpha$. Then $d\rho = du/\sin \alpha$, and the metric becomes $ds^2 = d\rho^2 + \rho^2 dv^2$, which is the polar metric of the plane. Therefore the cone is locally isometric to a planar sector. ■

3.13 Summary of the Main Formulae

For quick reference, the principal formulae of this unit are

$$\begin{aligned} k_n &= \frac{L du^2 + 2M du dv + N dv^2}{E du^2 + 2F du dv + G dv^2}, \\ (EG - F^2)k^2 - (EN + GL - 2FM)k + (LN - M^2) &= 0, \\ H &= \frac{EN + GL - 2FM}{2(EG - F^2)}, \quad K = \frac{LN - M^2}{EG - F^2}, \\ (EM - FL) du^2 + (EN - GL) du dv + (FN - GM) dv^2 &= 0, \\ d\mathbf{N} + k_n d\mathbf{r} &= 0 \quad \text{along a line of curvature,} \\ k_n &= k_1 \cos^2 \theta + k_2 \sin^2 \theta, \quad H^2 - K = \frac{(k_1 - k_2)^2}{4}, \\ \mathbf{N}_u \times \mathbf{N}_v &= K(\mathbf{r}_u \times \mathbf{r}_v), \\ dA_{\mathbb{S}^2} &= |K| dA, \\ H = 0 &\iff EN + GL - 2FM = 0, \end{aligned}$$

and, in isothermal coordinates,

$$\mathbf{r}_{uu} + \mathbf{r}_{vv} = 2\lambda^2 H \mathbf{N}.$$

Thus, in isothermal coordinates,

$$H = 0 \iff \Delta x = \Delta y = \Delta z = 0.$$

3.14 Exercises

Exercise 3.14.1. • Find the principal curvatures, mean curvature, and Gaussian curvature of the paraboloid $\mathbf{r}(u, v) = (u, v, u^2 + v^2)$ at the origin.

- Determine the principal curvatures of the circular cylinder $x^2 + y^2 = a^2$ directly from the equation for the principal curvatures.
- For the saddle surface $z = x^2 - y^2$, classify the origin and determine its asymptotic directions.

- Show that every point of a plane is an umbilic and determine its mean and Gaussian curvatures.
- Verify the Weingarten equations for the circular cylinder and identify the principal directions from them.
- Find the Gauss map of the torus $\mathbf{r}(u, v) = ((R + a \cos u) \cos v, (R + a \cos u) \sin v, a \sin u)$ and determine where the Gauss map is singular.
- Determine the lines of curvature on the sphere using Rodrigues' formula.
- Prove directly from Euler's formula that the normal curvatures in two perpendicular directions have constant sum.
- Find the Dupin indicatrix at the origin of the surface $z = x^2 + 2y^2$ and classify the point.
- Find the Dupin indicatrix at the origin of the surface $z = x^2 - y^2$ and determine its asymptotes.
- Show that the right helicoid $\mathbf{r}(u, v) = (u \cos v, u \sin v, av)$ is a ruled surface and determine both families of asymptotic lines.
- Verify that the cone $\mathbf{R}(u, v) = (u \cos v, u \sin v, cu)$ is developable away from its vertex.
- Show directly that the catenoid $\mathbf{r}(u, v) = (a \cosh u \cos v, a \cosh u \sin v, au)$ has zero mean curvature.
- Determine whether the graph $z = x^2 - y^2$ is minimal. Justify the conclusion from the mean-curvature equation for a graph.
- Find an isothermal coordinate system for the circular cylinder and write its metric in conformal form.
- Show that the map from the plane with polar coordinates (r, θ) to the cone obtained by identifying an angular sector is a local isometry, and explain why it cannot be a global isometry of the whole plane onto the cone.

Chapter 4

Intrinsic Geometry and Geodesics

This chapter explores the intrinsic and extrinsic geometry of surfaces through Christoffel symbols, the celebrated Theorema Egregium, and the fundamental Gauss and Mainardi-Codazzi compatibility equations.

We develop the mathematical framework of geodesics, defining their intrinsic curvature, establishing their governing differential equations, and examining their trajectories on spheres and pseudospheres. Furthermore, we state the fundamental theorem for regular surfaces, which ensures the unique determination of a surface up to rigid motions from its fundamental forms.

Finally, we introduce the Gauss–Bonnet theorem, exploring its profound implications for geodesic triangles and demonstrating how local curvature dictates global topological properties.

4.1 Christoffel Symbols and Fundamental Equations of Surfaces

Let $S \subset \mathbb{R}^3$ be a smooth regular surface and let $\mathbf{r} : U \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a local parametrization of S with coordinates (u^1, u^2) . The tangent space $T_p S$ at $p = \mathbf{r}(u, v)$ is spanned by the coordinate tangent vectors:

$$\mathbf{r}_1 = \frac{\partial \mathbf{r}}{\partial u}, \quad \mathbf{r}_2 = \frac{\partial \mathbf{r}}{\partial v}.$$

The first fundamental magnitudes are given by

$$E = \mathbf{r}_1 \cdot \mathbf{r}_1 = r_1^2, \quad F = \mathbf{r}_1 \cdot \mathbf{r}_2 = r_2 \cdot \mathbf{r}_1, \quad G = \mathbf{r}_2 \cdot \mathbf{r}_2 = r_2^2$$

If $\vec{N}$ denotes the unit normal vector to the surface, then the second fundamental magnitudes are given by

$$L = r_{11} \cdot \vec{N}, \quad M = r_{12} \cdot \vec{N}, \quad N = r_{22} \cdot \vec{N}$$

Definition 4.1.1. For $i, j, k = 1, 2$, the notations

$$\Gamma_{ijk} = \frac{1}{2} \{ (\mathbf{r}_i \cdot \mathbf{r}_j)_k + (\mathbf{r}_i \cdot \mathbf{r}_k)_j - (\mathbf{r}_j \cdot \mathbf{r}_k)_i \}$$

where $(\)_k, (\)_j, (\)_i$ denote differentiation with respect to $\mathbf{r}_k, \mathbf{r}_j$ and $\mathbf{r}_i$ respectively, are called the Christoffel symbols.

Using the definition, the above expression simplifies to $\Gamma_{ijk} = \mathbf{r}_i \cdot \mathbf{r}_{jk}$ as shown below.

$$\begin{aligned} \Gamma_{ijk} &= \mathbf{r}_i \cdot \mathbf{r}_{jk} \\ &= \frac{1}{2} \{ (r_{ik} \cdot r_j) + r_i \cdot r_{jk} + r_{ij} \cdot r_k + r_i \cdot r_{jk} - r_j \cdot r_{ki} - r_k \cdot r_{ij} \} \\ &= \frac{1}{2} \{ 2r_i \cdot r_{jk} \} = r_i \cdot r_{jk}. \end{aligned}$$

For particular values of $i, j, k = 1, 2$, we have

$$\begin{aligned}\Gamma_{111} &= \mathbf{r}_1 \cdot \mathbf{r}_{11} = \frac{1}{2}E_1 & \Gamma_{211} &= \mathbf{r}_2 \cdot \mathbf{r}_{11} = F_1 - \frac{1}{2}E_2 \\ \Gamma_{112} &= \mathbf{r}_1 \cdot \mathbf{r}_{12} = \frac{1}{2}E_2 & \Gamma_{212} &= \mathbf{r}_2 \cdot \mathbf{r}_{12} = \frac{1}{2}G_1 \\ \Gamma_{121} &= \mathbf{r}_1 \cdot \mathbf{r}_{21} = \frac{1}{2}E_2 & \Gamma_{221} &= \mathbf{r}_2 \cdot \mathbf{r}_{21} = \frac{1}{2}G_1 \\ \Gamma_{122} &= \mathbf{r}_1 \cdot \mathbf{r}_{22} = F_2 - \frac{1}{2}G_1 & \Gamma_{222} &= \mathbf{r}_2 \cdot \mathbf{r}_{22} = \frac{1}{2}G_2\end{aligned}$$

Also we note from the above

$$\Gamma_{112} = \Gamma_{121} \quad \text{and} \quad \Gamma_{221} = \Gamma_{212}$$

The Christoffel symbols of the second kind denoted by

Γ_{jk}^i , $i = 1, 2$, $j, k = 1, 2$ are defined as

$$\Gamma_{jk}^1 = \mathbf{H}^{-2}(G\Gamma_{1jk} - F\Gamma_{2jk}), \quad \Gamma_{jk}^2 = \mathbf{H}^{-2}(E\Gamma_{2jk} - F\Gamma_{1jk})$$

For particular values i, j, k , we have the following

$$\begin{aligned}\Gamma_{11}^1 &= \frac{1}{\mathbf{H}^2}(G\Gamma_{111} - F\Gamma_{211}) = \frac{1}{2\mathbf{H}^2}(GE_1 - 2FF_1 + FE_2) \\ \Gamma_{12}^1 &= \frac{1}{\mathbf{H}^2}(G\Gamma_{112} - F\Gamma_{212}) = \frac{1}{2\mathbf{H}^2}(GE_2 - FG_1) \\ \Gamma_{22}^1 &= \frac{1}{\mathbf{H}^2}(G\Gamma_{122} - F\Gamma_{222}) = \frac{1}{2\mathbf{H}^2}(2GF_2 - GG_1 - FG_2) \\ \Gamma_{11}^2 &= \frac{1}{\mathbf{H}^2}(E\Gamma_{211} - F\Gamma_{111}) = \frac{1}{2\mathbf{H}^2}(2EF_1 - EE_2 - FE_1) \\ \Gamma_{12}^2 &= \frac{1}{\mathbf{H}^2}(E\Gamma_{212} - F\Gamma_{112}) = \frac{1}{2\mathbf{H}^2}(EG_1 - FE_2) \\ \Gamma_{22}^2 &= \frac{1}{\mathbf{H}^2}(E\Gamma_{222} - F\Gamma_{122}) = \frac{1}{2\mathbf{H}^2}(EG_2 - 2FF_2 + FG_1)\end{aligned}$$

Theorem 4.1.2. *If $\mathbf{r}_1$, $\mathbf{r}_2$ and $\mathbf{N}$ are three linearly independent vectors at a point P on the surface $\mathbf{r} = \mathbf{r}(u, v)$, then*

$$\begin{aligned}\mathbf{r}_{11} &= \Gamma_{11}^1\mathbf{r}_1 + \Gamma_{11}^2\mathbf{r}_2 + L\mathbf{N} \\ \mathbf{r}_{12} &= \Gamma_{12}^1\mathbf{r}_1 + \Gamma_{12}^2\mathbf{r}_2 + M\mathbf{N} \\ \mathbf{r}_{22} &= \Gamma_{22}^1\mathbf{r}_1 + \Gamma_{22}^2\mathbf{r}_2 + N\mathbf{N}\end{aligned}$$

Proof. At a point P on surface, let us introduce a coordinate system consisting of three linearly independent vectors $\mathbf{r}_1$, $\mathbf{r}_2$ and $\mathbf{N}$ of which $\mathbf{r}_1$ and $\mathbf{r}_2$ lie in the tangent plane at P and $\mathbf{N}$ is the unit normal to the surface. With respect to this coordinate system, let us express $\mathbf{r}_{11}$, $\mathbf{r}_{12}$ and $\mathbf{r}_{22}$ at P as linear combinations of these basic vectors.

Hence we can represent

$$\mathbf{r}_{11} = \alpha_1\mathbf{r}_1 + \alpha_2\mathbf{r}_2 + \alpha_3\mathbf{N} \quad (1)$$

$$\mathbf{r}_{12} = \beta_1\mathbf{r}_1 + \beta_2\mathbf{r}_2 + \beta_3\mathbf{N} \quad (2)$$

$$\mathbf{r}_{22} = \gamma_1\mathbf{r}_1 + \gamma_2\mathbf{r}_2 + \gamma_3\mathbf{N} \quad (3)$$

where we determine $\alpha_i, \beta_i, \gamma_i, i = 1, 2, 3$.

Taking dot product with $\mathbf{N}$ on both sides of the above equations and using $\mathbf{r}_1 \cdot \mathbf{N} = \mathbf{r}_2 \cdot \mathbf{N} = 0$, we obtain

$$\mathbf{N} \cdot \mathbf{r}_{11} = \alpha_3 \mathbf{N} \cdot \mathbf{N}, \quad \mathbf{N} \cdot \mathbf{r}_{12} = \beta_3 \mathbf{N} \cdot \mathbf{N}, \quad \mathbf{N} \cdot \mathbf{r}_{22} = \gamma_3 \mathbf{N} \cdot \mathbf{N}$$

Since $\mathbf{N} \cdot \mathbf{r}_{11} = L$, $\mathbf{N} \cdot \mathbf{r}_{12} = M$, $\mathbf{N} \cdot \mathbf{r}_{22} = N$ and $\mathbf{N} \cdot \mathbf{N} = 1$, we obtain

$$\alpha_3 = L, \quad \beta_3 = M \text{ and } \gamma_3 = N \quad (4)$$

Having determined α_3, β_3 and γ_3 , we find $\alpha_1, \beta_1, \gamma_1$, and $\alpha_2, \beta_2, \gamma_2$ as follows. To determine these, we need the derivatives of the fundamental coefficients E, F and G .

$$E_1 = \frac{\partial E}{\partial u} = \frac{\partial}{\partial u}(\mathbf{r}_1^2) = 2\mathbf{r}_1 \cdot \mathbf{r}_{11}, \quad \mathbf{r}_1 \cdot \mathbf{r}_{11} = \frac{1}{2}E_1 \quad (5)$$

$$E_2 = \frac{\partial E}{\partial v} = \frac{\partial}{\partial v}(\mathbf{r}_1^2) = 2\mathbf{r}_1 \cdot \mathbf{r}_{12}, \quad \mathbf{r}_1 \cdot \mathbf{r}_{12} = \frac{1}{2}E_2 \quad (6)$$

$$G_1 = \frac{\partial G}{\partial u} = \frac{\partial}{\partial u}(\mathbf{r}_2^2) = 2\mathbf{r}_2 \cdot \mathbf{r}_{21}, \quad \mathbf{r}_2 \cdot \mathbf{r}_{21} = \frac{1}{2}G_1 \quad (7)$$

$$G_2 = \frac{\partial G}{\partial v} = \frac{\partial}{\partial v}(\mathbf{r}_2^2) = 2\mathbf{r}_2 \cdot \mathbf{r}_{22}, \quad \mathbf{r}_2 \cdot \mathbf{r}_{22} = \frac{1}{2}G_2 \quad (8)$$

$$F_1 = \frac{\partial}{\partial u}(\mathbf{r}_1 \cdot \mathbf{r}_2) = \mathbf{r}_2 \cdot \mathbf{r}_{11} + \mathbf{r}_1 \cdot \mathbf{r}_{12}, \text{ so, } \mathbf{r}_2 \cdot \mathbf{r}_{11} = F_1 - \mathbf{r}_1 \cdot \mathbf{r}_{12} = F_1 - \frac{1}{2}E_2 \quad (9)$$

and

$$F_2 = \frac{\partial}{\partial v}(\mathbf{r}_1 \cdot \mathbf{r}_2) = \mathbf{r}_2 \cdot \mathbf{r}_{12} + \mathbf{r}_1 \cdot \mathbf{r}_{22}, \text{ so, } \mathbf{r}_1 \cdot \mathbf{r}_{22} = F_2 - \mathbf{r}_2 \cdot \mathbf{r}_{12} = F_2 - \frac{1}{2}G_1 \quad (10)$$

Now, let us take the dot product of (1) with $\mathbf{r}_1$ and $\mathbf{r}_2$ respectively. Then we have

$$\mathbf{r}_1 \cdot \mathbf{r}_{11} = \alpha_1 \mathbf{r}_1 \cdot \mathbf{r}_1 + \alpha_2 \mathbf{r}_2 \cdot \mathbf{r}_1$$

$$\mathbf{r}_2 \cdot \mathbf{r}_{11} = \alpha_1 \mathbf{r}_1 \cdot \mathbf{r}_2 + \alpha_2 \mathbf{r}_2 \cdot \mathbf{r}_2$$

Since $\mathbf{r}_1 \cdot \mathbf{r}_1 = E$, $\mathbf{r}_2 \cdot \mathbf{r}_2 = G$, using (5) and (9), we obtain

$$\alpha_1 E + F \alpha_2 - \frac{1}{2}E_1 = 0$$

$$\alpha_1 F + G \alpha_2 - \left(F_1 - \frac{1}{2}E_2\right) = 0$$

Solving the above two equations,

$$\frac{2\alpha_1}{GE_1 - 2FF_1 + FE_2} = \frac{2\alpha_2}{2EF_1 - EE_2 - E_1F} = \frac{1}{EG - F^2}$$

Thus we obtain

$$\alpha_1 = \frac{GE_1 - 2FF_1 + FE_2}{2(EG - F^2)} = \Gamma_{11}^1$$

$$\alpha_2 = \frac{2EF_1 - EE_2 - E_1F}{2(EG - F^2)} = \Gamma_{11}^2$$

Hence the equation (1) becomes

$$\mathbf{r}_{11} = \Gamma_{11}^1 \mathbf{r}_1 + \Gamma_{11}^2 \mathbf{r}_2 + L\mathbf{N}$$

Taking dot product with $\mathbf{r}_1$ and $\mathbf{r}_2$ of (2),

$$\mathbf{r}_1 \cdot \mathbf{r}_{12} = \beta_1 \mathbf{r}_1 \cdot \mathbf{r}_1 + \beta_2 \mathbf{r}_2 \cdot \mathbf{r}_1$$

$$\mathbf{r}_2 \cdot \mathbf{r}_{12} = \beta_1 \mathbf{r}_1 \cdot \mathbf{r}_2 + \beta_2 \mathbf{r}_2 \cdot \mathbf{r}_2$$

Using the fundamental coefficients and (6) and (7),

$$\beta_1 E + \beta_2 F - \frac{1}{2} E_2 = 0$$

$$\beta_1 F + \beta_2 G - \frac{1}{2} G_1 = 0.$$

Solving the above two equations, we obtain

$$\frac{2\beta_1}{GE_2 - FG_1} = \frac{2\beta_2}{EG_1 - FE_2} = \frac{1}{EG - F^2}$$

Hence

$$\beta_1 = \frac{GE_2 - FG_1}{2(EG - F^2)} = \Gamma_{12}^1$$

$$\beta_2 = \frac{EG_1 - FE_2}{2(EG - F^2)} = \Gamma_{12}^2$$

Let us take dot product of (3) with $\mathbf{r}_1$ and $\mathbf{r}_2$ then

$$\mathbf{r}_1 \cdot \mathbf{r}_{22} = \gamma_1 \mathbf{r}_1 \cdot \mathbf{r}_1 + \gamma_2 \mathbf{r}_1 \cdot \mathbf{r}_2$$

$$\mathbf{r}_2 \cdot \mathbf{r}_{22} = \gamma_1 \mathbf{r}_1 \cdot \mathbf{r}_2 + \gamma_2 \mathbf{r}_2 \cdot \mathbf{r}_2$$

So the simultaneous equations in γ_1 and γ_2 become

$$E\gamma_1 + F\gamma_2 - \left(F_2 - \frac{1}{2}G_1\right) = 0$$

$$F\gamma_1 + G\gamma_2 - \frac{1}{2}G_2 = 0$$

Solving these two equations, we obtain

$$\gamma_1 = \frac{2GF_2 - GG_1 - FG_2}{2(EG - F^2)} = \Gamma_{22}^1$$

$$\gamma_2 = \frac{EG_2 - 2FF_2 + FG_1}{2(EG - F^2)} = \Gamma_{22}^2$$

Thus the equations (1), (2) and (3) become

$$\mathbf{r}_{11} = \Gamma_{11}^1 \mathbf{r}_1 + \Gamma_{11}^2 \mathbf{r}_2 + L\mathbf{N}$$

$$\mathbf{r}_{12} = \Gamma_{12}^1 \mathbf{r}_1 + \Gamma_{12}^2 \mathbf{r}_2 + M\mathbf{N}$$

$$\mathbf{r}_{22} = \Gamma_{22}^1 \mathbf{r}_1 + \Gamma_{22}^2 \mathbf{r}_2 + N\mathbf{N}$$

The above three equations are known as Gauss equations. $\mathbf{r}_{11}$, $\mathbf{r}_{12}$ and $\mathbf{r}_{22}$ are called Gauss coefficients. Thus the Gauss equations give Gauss coefficients in terms of E , F and G and their partial derivatives and L , M , N . These equations are also known as the partial differential equations of the surface theory. ■

Corollary 4.1.3. *If the parametric curves are orthogonal, then the Gauss equations are*

$$\mathbf{r}_{11} = \frac{E_1}{2E} \mathbf{r}_1 - \frac{E_2}{2G} \mathbf{r}_2 + L\mathbf{N}$$

$$\mathbf{r}_{12} = \frac{E_2}{2E} \mathbf{r}_1 + \frac{G_1}{2G} \mathbf{r}_2 + M\mathbf{N}$$

$$\mathbf{r}_{22} = -\frac{G_1}{2E} \mathbf{r}_1 + \frac{G_2}{2G} \mathbf{r}_2 + N\mathbf{N}$$

Proof. Since the parametric curves are orthogonal, $F = 0$ and $\mathbf{H}^2 = EG$. Using these conditions, the Christoffel symbol coefficients in the Gauss equations became

$$\begin{aligned}\Gamma_{11}^1 &= \frac{GE_1}{2EG} = \frac{E_1}{2E} & \Gamma_{11}^2 &= \frac{-EE_2}{2EG} = \frac{-E_2}{2G} \\ \Gamma_{12}^1 &= \frac{GE_2}{2EG} = \frac{E_2}{2E} & \Gamma_{12}^2 &= \frac{EG_1}{2EG} = \frac{G_1}{2G} \\ \Gamma_{22}^1 &= \frac{-GG_1}{2EG} = \frac{-G_1}{2E} & \Gamma_{22}^2 &= \frac{EG_2}{2EG} = \frac{G_2}{2G}\end{aligned}$$

Using these coefficients, the Gauss equations take the given form. ■

Corollary 4.1.4. *If the lines of curvature are parametric curves on the surface, then the Gauss equations are*

$$\begin{aligned}\mathbf{r}_{11} &= \frac{E_1}{2E}\mathbf{r}_1 - \frac{E_2}{2G}\mathbf{r}_2 + L\mathbf{N} \\ \mathbf{r}_{12} &= \frac{E_2}{2E}\mathbf{r}_1 + \frac{G_1}{2G}\mathbf{r}_2 \\ \mathbf{r}_{22} &= \frac{-G_1}{2E}\mathbf{r}_1 + \frac{G_2}{2G}\mathbf{r}_2 + N\mathbf{N}\end{aligned}$$

Proof. When the lines of curvature are chosen as parametric curves, then $F = 0$, $M = 0$ and $\mathbf{H}^2 = EG$. Using these conditions, we can rewrite the Gauss equations in the above form. ■

Theorem 4.1.5 (Mainardi-Codazzi Equations). *The Mainardi-Codazzi equations are*

$$\begin{aligned}\frac{\partial L}{\partial v} - \frac{\partial M}{\partial u} &= L\Gamma_{12}^1 - N\Gamma_{11}^2 + M(\Gamma_{12}^2 - \Gamma_{11}^1), \\ \frac{\partial M}{\partial v} - \frac{\partial N}{\partial u} &= L\Gamma_{22}^1 + M(\Gamma_{22}^2 - \Gamma_{12}^1) - N\Gamma_{12}^2.\end{aligned}$$

Proof. We have have the compatibility conditions

$$\begin{aligned}\frac{\partial}{\partial v} \left(\frac{\partial^2 r}{\partial u^2} \right) &= \frac{\partial}{\partial u} \left(\frac{\partial^2 r}{\partial v \partial u} \right), \text{ or } (r_{11})_2 = (r_{12})_1 \\ \frac{\partial}{\partial u} \left(\frac{\partial^2 r}{\partial v^2} \right) &= \frac{\partial}{\partial v} \left(\frac{\partial^2 r}{\partial v \partial u} \right), \text{ or } (r_{12})_2 = (r_{22})_1\end{aligned}$$

Substituting for $\mathbf{r}_{11}$, $\mathbf{r}_{12}$ and $\mathbf{r}_{22}$ from the Gauss equations into the compatibility conditions:

$$\frac{\partial}{\partial v} [\Gamma_{11}^1 \mathbf{r}_1 + \Gamma_{11}^2 \mathbf{r}_2 + L\mathbf{N}] = \frac{\partial}{\partial u} [\Gamma_{12}^1 \mathbf{r}_1 + \Gamma_{12}^2 \mathbf{r}_2 + M\mathbf{N}], \quad (1)$$

$$\frac{\partial}{\partial v} [\Gamma_{12}^1 \mathbf{r}_1 + \Gamma_{12}^2 \mathbf{r}_2 + M\mathbf{N}] = \frac{\partial}{\partial u} [\Gamma_{22}^1 \mathbf{r}_1 + \Gamma_{22}^2 \mathbf{r}_2 + N\mathbf{N}]. \quad (2)$$

Using Gauss equations, we obtain each term of the first equation above:

$$\begin{aligned}\frac{\partial}{\partial v} [\Gamma_{11}^1 \mathbf{r}_1] &= \frac{\partial}{\partial v} (\Gamma_{11}^1) \mathbf{r}_1 + \Gamma_{11}^1 \mathbf{r}_{12} \\ &= \frac{\partial}{\partial v} (\Gamma_{11}^1) \mathbf{r}_1 + \Gamma_{11}^1 (\Gamma_{12}^1 \mathbf{r}_1 + \Gamma_{12}^2 \mathbf{r}_2 + M\mathbf{N}), \\ \frac{\partial}{\partial v} [\Gamma_{11}^2 \mathbf{r}_2] &= \frac{\partial}{\partial v} (\Gamma_{11}^2) \mathbf{r}_2 + \Gamma_{11}^2 \mathbf{r}_{22} \\ &= \frac{\partial}{\partial v} (\Gamma_{11}^2) \mathbf{r}_2 + \Gamma_{11}^2 (\Gamma_{22}^1 \mathbf{r}_1 + \Gamma_{22}^2 \mathbf{r}_2 + N\mathbf{N}).\end{aligned}$$

$$\begin{aligned}
\frac{\partial}{\partial u}(\Gamma_{12}^1 \mathbf{r}_1) &= \frac{\partial}{\partial u}(\Gamma_{12}^1) \mathbf{r}_1 + \Gamma_{12}^1 \mathbf{r}_{11} \\
&= \frac{\partial}{\partial u}(\Gamma_{12}^1) \mathbf{r}_1 + \Gamma_{12}^1 [\Gamma_{11}^1 \mathbf{r}_1 + \Gamma_{11}^2 \mathbf{r}_2 + L\mathbf{N}], \\
\frac{\partial}{\partial u}[\Gamma_{12}^2 \mathbf{r}_2] &= \frac{\partial}{\partial u}(\Gamma_{12}^2) \mathbf{r}_2 + \Gamma_{12}^2 \mathbf{r}_{21} \\
&= \frac{\partial}{\partial u}(\Gamma_{12}^2) \mathbf{r}_2 + \Gamma_{12}^2 [\Gamma_{12}^1 \mathbf{r}_1 + \Gamma_{12}^2 \mathbf{r}_2 + M\mathbf{N}].
\end{aligned}$$

Using Weingarten equations,

$$\begin{aligned}
\frac{\partial}{\partial v}[L\mathbf{N}] &= \frac{\partial L}{\partial v} \mathbf{N} + L \frac{\partial \mathbf{N}}{\partial v} = \frac{\partial L}{\partial v} \mathbf{N} + LN_2 \\
&= \frac{\partial L}{\partial v} \mathbf{N} + \frac{L}{H^2} [(FN - GM)\mathbf{r}_1 + (FM - EN)\mathbf{r}_2], \\
\frac{\partial}{\partial u}(MN) &= \frac{\partial M}{\partial u} \mathbf{N} + M \frac{\partial \mathbf{N}}{\partial u} = \frac{\partial M}{\partial u} \mathbf{N} + MN_1 \\
&= \frac{\partial M}{\partial u} \mathbf{N} + \frac{M}{H^2} [(FM - GL)\mathbf{r}_1 + (FL - EM)\mathbf{r}_2].
\end{aligned}$$

Now gathering the coefficients of $\mathbf{r}_1$, $\mathbf{r}_2$ and $\mathbf{N}$, we have the left-hand side of the equation (1) as:

$$\begin{aligned}
\left[\frac{\partial}{\partial v}(\Gamma_{11}^1) + \Gamma_{11}^1 \Gamma_{12}^1 + \Gamma_{11}^2 \Gamma_{22}^1 + \frac{L(FN - GM)}{EG - F^2} \right] \mathbf{r}_1 &+ \left[\frac{\partial}{\partial v}(\Gamma_{11}^2) + \Gamma_{11}^1 \Gamma_{12}^2 + \Gamma_{11}^2 \Gamma_{22}^2 + \frac{L(FM - EN)}{EG - F^2} \right] \mathbf{r}_2 \\
&+ \left[\Gamma_{11}^1 M + \Gamma_{11}^2 N + \frac{\partial L}{\partial v} \right] \mathbf{N}
\end{aligned} \quad (3)$$

and gathering the coefficients of $\mathbf{r}_1$, $\mathbf{r}_2$ and $\mathbf{N}$, the right-hand side of the equation (1) gives

$$\begin{aligned}
\left[\frac{\partial}{\partial u}(\Gamma_{12}^1) + \Gamma_{12}^1 \Gamma_{11}^1 + \Gamma_{12}^2 \Gamma_{12}^1 + \frac{M}{H^2}(FM - GL) \right] \mathbf{r}_1 &+ \left[\frac{\partial}{\partial u}(\Gamma_{12}^2) + \Gamma_{12}^1 \Gamma_{12}^2 + \Gamma_{12}^2 \Gamma_{22}^2 + \frac{M}{H^2}(FL - EM) \right] \mathbf{r}_2 \\
&+ \left[\Gamma_{12}^1 L + \Gamma_{12}^2 M + \frac{\partial M}{\partial u} \right] \mathbf{N}.
\end{aligned} \quad (4)$$

The coefficient of $\mathbf{N}$ gives

$$\frac{\partial L}{\partial v} - \frac{\partial M}{\partial u} = L\Gamma_{12}^1 - N\Gamma_{11}^2 + M(\Gamma_{12}^2 - \Gamma_{11}^1).$$

which is the first Mainardi-Codazzi equation. The other Mainardi-Codazzi equation can similarly be obtained from equation 3.2. ■

Theorem 4.1.6 (Theorema Egregium).

Proof. Since the expressions (3) and (4) represent both sides of the equation (1), we equate the coefficients of $\mathbf{r}_1$, $\mathbf{r}_2$ and $\mathbf{N}$ in both parts.

Equating the coefficient of $\mathbf{r}_1$, we obtain

$$\frac{\partial}{\partial v}(\Gamma_{11}^1) + \Gamma_{11}^1 \Gamma_{12}^1 + \Gamma_{11}^2 \Gamma_{22}^1 + \frac{L(FN - GM)}{EG - F^2} = \frac{\partial}{\partial u}(\Gamma_{12}^1) + \Gamma_{12}^1 \Gamma_{11}^1 + \Gamma_{12}^2 \Gamma_{12}^1 + \frac{M(FM - GL)}{EG - F^2}.$$

Simplifying the above equation, we have

$$F \frac{(LN - M^2)}{EG - F^2} = \frac{\partial}{\partial u}(\Gamma_{12}^1) - \frac{\partial}{\partial v}(\Gamma_{11}^1) + \Gamma_{12}^2 \Gamma_{12}^1 - \Gamma_{11}^2 \Gamma_{22}^1. \quad (1)$$

Equating the coefficient of $\mathbf{r}_2$, we have

$$\frac{\partial}{\partial v}(\Gamma_{11}^2) + \Gamma_{11}^1 \Gamma_{12}^2 + \Gamma_{11}^2 \Gamma_{22}^2 + \frac{L(FM - EN)}{EG - F^2} = \frac{\partial}{\partial u}(\Gamma_{12}^2) + \Gamma_{12}^1 \Gamma_{11}^2 + \Gamma_{12}^2 \Gamma_{12}^2 + \frac{M(FL - EM)}{EG - F^2}.$$

Simplifying the above equation, we obtain

$$-E \frac{(LN - M^2)}{EG - F^2} = \frac{\partial}{\partial u}(\Gamma_{12}^2) - \frac{\partial}{\partial v}(\Gamma_{11}^2) + \Gamma_{12}^1 \Gamma_{11}^2 - \Gamma_{11}^1 \Gamma_{12}^2 + \Gamma_{12}^2 \Gamma_{12}^2 - \Gamma_{11}^2 \Gamma_{22}^2. \quad (2)$$

Using $\frac{LN - M^2}{EG - F^2} = k_n$ in equations (1) and (2), we obtain

$$\begin{aligned} k_n &= \frac{1}{F} \frac{\partial}{\partial u}(\Gamma_{12}^1) - \frac{\partial}{\partial v}(\Gamma_{11}^1) + \Gamma_{12}^2 \Gamma_{12}^1 - \Gamma_{11}^2 \Gamma_{22}^1 \\ k_n &= \frac{-1}{E} \frac{\partial}{\partial u}(\Gamma_{12}^2) - \frac{\partial}{\partial v}(\Gamma_{11}^2) + \Gamma_{12}^1 \Gamma_{11}^2 - \Gamma_{11}^1 \Gamma_{12}^2 + \Gamma_{12}^2 \Gamma_{12}^2 - \Gamma_{11}^2 \Gamma_{22}^2 \end{aligned}$$

These equations, obtained from the coefficients of $\mathbf{r}_1$ and $\mathbf{r}_2$ give the Gaussian curvature in terms of E, F and G with their partial derivatives of first and second order. A property depending only on E, F, G and their partial derivatives of first and second order is known as an intrinsically invariant property. Thus the Gaussian curvature is intrinsically invariant. Gauss considered this as a most excellent theorem. In the words of Gauss, “it is Theorema egregium.” ■

4.2 Vector Fields

Definition 4.2.1 (Vector Field). A **vector field** on a surface S (or more generally on a manifold) is a rule that assigns to each point $p \in S$ a tangent vector $X(p)$ that lies in the tangent plane $T_p S$. In other words, it is a function $X : S \rightarrow TS$ such that $X(p) \in T_p S$ for every p , where TS denotes the tangent bundle (the collection of all tangent planes).

Example 4.2.2 (Constant Vector Field on a Plane). Let $S = \mathbb{R}^2$ be the plane. At each point (x, y) we assign the same tangent vector $(1, 0)$. This vector field is written as $X(x, y) = (1, 0)$. Every arrow points horizontally to the right, and all have length 1.

Example 4.2.3 (Radial Vector Field on a Sphere). Consider the unit sphere $S^2 = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$. At a point $p = (x, y, z)$, the tangent plane consists of vectors orthogonal to the position vector p . Define

$$X(x, y, z) = (-y, x, 0).$$

This vector is tangent to the sphere because $(-y, x, 0) \cdot (x, y, z) = -yx + xy + 0 = 0$. The vectors point in the direction of increasing longitude (like wind blowing around the equator). This is a non-zero vector field on the sphere (though it vanishes at the poles? Actually at $(0, 0, \pm 1)$ we obtain $(0, 0, 0)$ – so it is zero at the poles; still a valid vector field).

Example 4.2.4 (Gradient Vector Field). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a smooth function, e.g., $f(x, y) = x^2 + y^2$. Its gradient $\nabla f = (2x, 2y)$ is a vector field on the plane. At each point, the gradient points in the direction of steepest ascent and is tangent to the plane (since the plane is the whole $\mathbb{R}^2$).

Definition 4.2.5 (Parallel Surface). Let S be a regular surface in $\mathbb{R}^3$ with a unit normal vector field $\mathbf{N}(p)$. For a fixed constant $d \in \mathbb{R}$ (with $|d|$ small enough so that the new surface is regular), the **parallel surface** S_d is defined by $S_d = \{p + d\mathbf{N}(p) \mid p \in S\}$. In other words, from each point p of the original surface, we move a distance $|d|$ along the normal direction (outward if $d > 0$, inward if $d < 0$) to obtain a new point. The set of all such points forms the parallel surface.

Example 4.2.6 (Parallel Surfaces of a Sphere). (See Figure 50 in the Atlas.) Take S to be the unit sphere: $S = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$. The outward unit normal at a point p is simply $\mathbf{N}(p) = p$ (since for a sphere centred at the origin, the normal points radially outward). For a given d (with $d > -1$ to avoid collapse), the parallel surface is $S_d = \{p + dp \mid \|p\| = 1\} = \{(1+d)p \mid \|p\| = 1\}$. Thus S_d is a sphere of radius $|1+d|$. If $d > 0$, we obtain a larger sphere; if $-1 < d < 0$, we obtain a smaller sphere (inside the original). For $d = -1$, the sphere shrinks to a point.

Example 4.2.7 (Parallel Surfaces of a Cylinder). Let S be a right circular cylinder of radius R : $S = \{(x, y, z) \mid x^2 + y^2 = R^2, z \in \mathbb{R}\}$. The outward unit normal at (x, y, z) is $\mathbf{N}(x, y, z) = \frac{1}{R}(x, y, 0)$. For a constant d , the parallel surface consists of points

$$(x, y, z) + d \cdot \frac{1}{R}(x, y, 0) = \left(x \left(1 + \frac{d}{R} \right), y \left(1 + \frac{d}{R} \right), z \right),$$

with $x^2 + y^2 = R^2$. This is again a cylinder, but with radius $R + d$ (since $\|(x(1 + d/R), y(1 + d/R))\| = |R + d|$). Hence S_d is a cylinder of radius $|R + d|$. If $d = -R$, the cylinder collapses to the z -axis.

Example 4.2.8 (Parallel Surfaces of a Plane). Let S be a plane, e.g., $S = \{(x, y, 0) \mid x, y \in \mathbb{R}\}$. The unit normal is constant: $\mathbf{N} = (0, 0, 1)$ (up to sign). For any d , the parallel surface is

$$S_d = \{(x, y, 0) + d(0, 0, 1) \mid x, y \in \mathbb{R}\} = \{(x, y, d) \mid x, y \in \mathbb{R}\}.$$

So S_d is just another plane parallel to the original one, at height d . All parallel surfaces are again planes.

Definition 4.2.9. Let S be a regular surface, $U \subset S$ an open set, and $W : U \rightarrow TS$ a differentiable vector field. For a point $p \in U$ and a tangent vector $y \in T_p S$, choose a smooth curve $\alpha : (-\varepsilon, \varepsilon) \rightarrow U$ such that $\alpha(0) = p$ and $\alpha'(0) = y$. Let $W(t) = W(\alpha(t))$ denote the restriction of W to α . Then the **covariant derivative** of W at p with respect to y is the orthogonal projection of $\frac{dW}{dt}\big|_{t=0}$ onto the tangent plane $T_p S$. It is denoted by

$$\frac{DW}{dt}\bigg|_{t=0}, \quad (D_y W)(p), \quad \text{or} \quad (\nabla_y W)(p).$$

Definition 4.2.10 (Parallel Vector Field Along a Curve). Let $\gamma : I \rightarrow S$ be a smooth curve. A vector field $W(t)$ along γ (i.e., $W(t) \in T_{\gamma(t)} S$ for each t) is called **parallel** if its covariant derivative along γ vanishes:

$$\frac{DW}{dt} = 0 \quad \text{for all } t.$$

Example 4.2.11 (Parallel Vector Field on a Great Circle of the Sphere). (See Figure 49 in the Atlas.) Take a great circle (e.g., the equator) on the sphere. If you start with a tangent vector pointing north at a point on the equator, parallel transporting it along the equator will rotate it so that after a quarter circle it points east. This shows that a non-zero parallel vector field can exist *along a curve* even though no global parallel field exists on the sphere.

Example 4.2.12. Discuss the notion of covariant derivative for a plane.

Solution. Consider a plane passing through the point $p_0 = (x_0, y_0, z_0)$ and spanned by the orthonormal vectors

$$\mathbf{w}_1 = (a_1, a_2, a_3), \quad \mathbf{w}_2 = (b_1, b_2, b_3).$$

A parametrization of the plane is

$$\mathbf{X}(u, v) = p_0 + u \mathbf{w}_1 + v \mathbf{w}_2, \quad (u, v) \in \mathbb{R}^2.$$

Then

$$\mathbf{X}_u = \mathbf{w}_1, \quad \mathbf{X}_v = \mathbf{w}_2,$$

so the coefficients of the first fundamental form are

$$E = \mathbf{X}_u \cdot \mathbf{X}_u = 1, \quad F = \mathbf{X}_u \cdot \mathbf{X}_v = 0, \quad G = \mathbf{X}_v \cdot \mathbf{X}_v = 1.$$

For a surface with metric coefficients E, F, G , the Christoffel symbols are given by

$$\begin{aligned}\Gamma_{11}^1 &= \frac{2EF_u - EE_v - FE_u}{2(EG - F^2)}, \\ \Gamma_{12}^1 &= \frac{GE_v - FG_u}{2(EG - F^2)}, \\ \Gamma_{22}^1 &= \frac{2GF_v - GG_u - FG_v}{2(EG - F^2)}, \\ \Gamma_{11}^2 &= \frac{2EF_u - EE_v - FE_u}{2(EG - F^2)} \quad (\text{similar}),\end{aligned}$$

and cyclic permutations. In the case of the plane we have $E = G = 1$, $F = 0$, and all partial derivatives of E, F, G vanish. Substituting these values yields $\Gamma_{ij}^k = 0$ for all $i, j, k \in \{1, 2\}$.

Now let $\alpha(t) = \mathbf{X}(u(t), v(t))$ be a curve on the plane and

$$w(t) = a(t) \mathbf{X}_u + b(t) \mathbf{X}_v$$

a vector field along α . The covariant derivative of w along α is (from the general formula)

$$\frac{Dw}{dt} = (a' + \Gamma_{11}^1 au' + \Gamma_{12}^1 av' + \Gamma_{12}^1 bu' + \Gamma_{22}^1 bv') \mathbf{X}_u + (b' + \Gamma_{11}^2 au' + \Gamma_{12}^2 av' + \Gamma_{12}^2 bu' + \Gamma_{22}^2 bv') \mathbf{X}_v.$$

Since all $\Gamma_{ij}^k = 0$, this simplifies to

$$\frac{Dw}{dt} = a' \mathbf{X}_u + b' \mathbf{X}_v.$$

But the ordinary derivative of $w(t)$ in $\mathbb{R}^3$ is

$$\frac{dw}{dt} = a' \mathbf{X}_u + b' \mathbf{X}_v + a \frac{d}{dt}(\mathbf{X}_u) + b \frac{d}{dt}(\mathbf{X}_v).$$

On the plane, $\mathbf{X}_u = \mathbf{w}_1$ and $\mathbf{X}_v = \mathbf{w}_2$ are constant vectors, so their derivatives vanish. Hence $\frac{dw}{dt} = a' \mathbf{X}_u + b' \mathbf{X}_v$ as well. Thus

$$\frac{Dw}{dt} = \frac{dw}{dt}.$$

Therefore, for a plane, the covariant derivative coincides with the ordinary derivative in $\mathbb{R}^3$. This shows that the covariant derivative is a natural generalization of the usual directional derivative of vector fields on Euclidean space to curved surfaces. ■

Definition 4.2.13. Let S be a regular surface. A smooth curve $\alpha : I \rightarrow S$ is called a **geodesic** if its tangent vector field is parallel along the curve:

$$\frac{D\alpha'(t)}{dt} = 0 \quad \text{for all } t \in I.$$

Equivalently, geodesics have zero intrinsic acceleration; they are the “straightest possible” curves on the surface.

Example 4.2.14. Examples of Geodesics:

1. **Straight lines in the Euclidean plane.** For $S = \mathbb{R}^2$ with the standard flat metric, the geodesic equations reduce to $\ddot{x} = 0$, $\ddot{y} = 0$. Hence the geodesics are straight lines $\gamma(t) = (a_1 t + b_1, a_2 t + b_2)$.
2. **Great circles on a sphere.** On the unit sphere S^2 with the induced metric, the geodesics are the great circles (intersections of the sphere with planes through the centre). For example, the equator $\gamma(\theta) = (\cos \theta, \sin \theta, 0)$ is a geodesic. Any arc of a great circle gives a locally distance-minimizing path.

Theorem 4.2.15 (Differential Equations of Geodesics). *Let S be a regular surface with a local parametrization $\mathbf{x}(u, v)$, and let $\alpha : I \rightarrow S$ be a parametrized curve on S given locally by $\alpha(t) = \mathbf{x}(u(t), v(t))$. Then, α is a geodesic if and only if the coordinate functions $u(t)$ and $v(t)$ satisfy the system of second-order differential equations:*

$$u'' + \Gamma_{11}^1(u')^2 + 2\Gamma_{12}^1 u'v' + \Gamma_{22}^1(v')^2 = 0 \quad (4.1)$$

$$v'' + \Gamma_{11}^2(u')^2 + 2\Gamma_{12}^2 u'v' + \Gamma_{22}^2(v')^2 = 0 \quad (4.2)$$

where Γ_{ij}^k are the Christoffel symbols of the second kind associated with the parametrization $\mathbf{x}$.

Proof. Let $W(t) = \alpha'(t)$ be the tangent vector field along the curve α . By the chain rule, we can express W in terms of the coordinate basis $\{\mathbf{x}_u, \mathbf{x}_v\}$ as:

$$W = \mathbf{x}_u u' + \mathbf{x}_v v' \quad (4.3)$$

To evaluate the geodesic condition, we compute the ordinary derivative of W in $\mathbb{R}^3$ with respect to t . Using the product and chain rules, we have

$$\begin{aligned} \frac{dW}{dt} &= \frac{d}{dt}(\mathbf{x}_u) u' + \mathbf{x}_u u'' + \frac{d}{dt}(\mathbf{x}_v) v' + \mathbf{x}_v v'' \\ &= (\mathbf{x}_{uu}u' + \mathbf{x}_{uv}v') u' + \mathbf{x}_u u'' + (\mathbf{x}_{vu}u' + \mathbf{x}_{vv}v') v' + \mathbf{x}_v v'' \end{aligned}$$

Assuming the mixed partial derivatives of the parametrization are symmetric ($\mathbf{x}_{uv} = \mathbf{x}_{vu}$) and grouping the terms, we obtain

$$\frac{dW}{dt} = u'(\mathbf{x}_{uu}u' + \mathbf{x}_{uv}v') + v'(\mathbf{x}_{uv}u' + \mathbf{x}_{vv}v') + u''\mathbf{x}_u + v''\mathbf{x}_v \quad (4.4)$$

Next, we substitute the Gauss equations for the second partial derivatives of the parametrization $\mathbf{x}$:

$$\begin{aligned} \mathbf{x}_{uu} &= \Gamma_{11}^1 \mathbf{x}_u + \Gamma_{11}^2 \mathbf{x}_v + L\hat{N} \\ \mathbf{x}_{uv} &= \Gamma_{12}^1 \mathbf{x}_u + \Gamma_{12}^2 \mathbf{x}_v + M\hat{N} \\ \mathbf{x}_{vv} &= \Gamma_{22}^1 \mathbf{x}_u + \Gamma_{22}^2 \mathbf{x}_v + N\hat{N} \end{aligned}$$

where $\hat{N}$ is the unit normal vector to the surface, and L, M, N are the coefficients of the second fundamental form. Substituting these expressions into $\frac{dW}{dt}$ yields

$$\begin{aligned} \frac{dW}{dt} &= u' \left[\left(\Gamma_{11}^1 \mathbf{x}_u + \Gamma_{11}^2 \mathbf{x}_v + L\hat{N} \right) u' + \left(\Gamma_{12}^1 \mathbf{x}_u + \Gamma_{12}^2 \mathbf{x}_v + M\hat{N} \right) v' \right] \\ &\quad + v' \left[\left(\Gamma_{12}^1 \mathbf{x}_u + \Gamma_{12}^2 \mathbf{x}_v + M\hat{N} \right) u' + \left(\Gamma_{22}^1 \mathbf{x}_u + \Gamma_{22}^2 \mathbf{x}_v + N\hat{N} \right) v' \right] \\ &\quad + u''\mathbf{x}_u + v''\mathbf{x}_v \end{aligned}$$

Collecting the terms with respect to $\mathbf{x}_u$, $\mathbf{x}_v$, and $\hat{N}$, and using the symmetry $\Gamma_{12}^k = \Gamma_{21}^k$, we expand and simplify:

$$\begin{aligned} \frac{dW}{dt} &= (u'' + \Gamma_{11}^1(u')^2 + \Gamma_{12}^1 u'v' + \Gamma_{12}^1 v'u' + \Gamma_{22}^1(v')^2) \mathbf{x}_u \\ &\quad + (v'' + \Gamma_{11}^2(u')^2 + \Gamma_{12}^2 u'v' + \Gamma_{12}^2 v'u' + \Gamma_{22}^2(v')^2) \mathbf{x}_v \\ &\quad + (L(u')^2 + 2Mu'v' + N(v')^2) \hat{N} \end{aligned}$$

By definition, a curve α is a geodesic if and only if its covariant acceleration $\frac{DW}{dt}$ along the curve is identically zero:

$$\frac{DW}{dt} = 0 \quad (4.5)$$

Since the covariant derivative $\frac{DW}{dt}$ is the orthogonal projection of $\frac{dW}{dt}$ onto the tangent plane $T_p S$, we discard the component normal to the surface (the term containing the normal vector N):

$$\frac{DW}{dt} = (u'' + \Gamma_{11}^1(u')^2 + 2\Gamma_{12}^1 u'v' + \Gamma_{22}^1(v')^2) \mathbf{x}_u + (v'' + \Gamma_{11}^2(u')^2 + 2\Gamma_{12}^2 u'v' + \Gamma_{22}^2(v')^2) \mathbf{x}_v \quad (4.6)$$

Setting $\frac{DW}{dt} = 0$ yields

$$(u'' + \Gamma_{11}^1(u')^2 + 2\Gamma_{12}^1 u'v' + \Gamma_{22}^1(v')^2) \mathbf{x}_u + (v'' + \Gamma_{11}^2(u')^2 + 2\Gamma_{12}^2 u'v' + \Gamma_{22}^2(v')^2) \mathbf{x}_v = 0$$

At any point on S , the coordinate vectors $\mathbf{x}_u$ and $\mathbf{x}_v$ are linearly independent. Therefore, this linear combination is equal to zero if and only if the coefficients of both basis vectors vanish independently. This yields the desired system of differential equations:

$$\begin{aligned} u'' + \Gamma_{11}^1(u')^2 + 2\Gamma_{12}^1 u'v' + \Gamma_{22}^1(v')^2 &= 0 \\ v'' + \Gamma_{11}^2(u')^2 + 2\Gamma_{12}^2 u'v' + \Gamma_{22}^2(v')^2 &= 0 \end{aligned}$$

■

Remark 4.2.16. From the system of geodesic equations, we observe that the equations depend exclusively on the Christoffel symbols. Since properties depending on Christoffel symbols are intrinsic to the metric and are preserved under isometries, it follows that an isometry between two surfaces maps the geodesics of one surface to the geodesics of the other surface.

Example 4.2.17. Find the geodesics on a surface of revolution. (See Figure 52 in the Atlas.)

Solution. See Figure 4.1. **Parametrization.** A surface of revolution about the z -axis is given by

$$\mathbf{X}(u, v) = (f(v) \cos u, f(v) \sin u, g(v)), \quad f > 0, \quad f'^2 + g'^2 \neq 0.$$

Here u is the rotation angle, v the profile parameter.

Metric and Christoffel symbols.

$$E = f^2, \quad F = 0, \quad G = f'^2 + g'^2.$$

The non-zero Christoffel symbols (diagonal metric) are

$$\Gamma_{12}^1 = \frac{f'}{f}, \quad \Gamma_{11}^2 = -\frac{f f'}{f'^2 + g'^2}, \quad \Gamma_{22}^2 = \frac{f' f'' + g' g''}{f'^2 + g'^2}, \quad \text{others } 0.$$

Geodesic equations (arc length s).

$$\ddot{u} + 2\frac{f'}{f} \dot{u}\dot{v} = 0, \quad \ddot{v} - \frac{f f'}{f'^2 + g'^2} \dot{u}^2 + \frac{f' f'' + g' g''}{f'^2 + g'^2} \dot{v}^2 = 0. \quad (1)$$

$$\text{The first equation gives } \frac{d}{ds}(f^2 \dot{u}) = 0, \text{ hence } f(v)^2 \dot{u}(s) = C \quad (\text{constant}). \quad (2)$$

Meridians ($u = \text{const}$, $\dot{u} = 0$). Then (2) holds trivially. From the arc-length condition $G\dot{v}^2 = 1$ we obtain $\dot{v}^2 = 1/(f'^2 + g'^2)$. Differentiating yields the second equation of (1). Hence every meridian is a geodesic.

Parallels ($v = \text{const}$, $\dot{v} = 0$). Equation (2) gives $\ddot{u} = 0$, so $\dot{u} = \text{const}$. The second equation of (1) becomes $-\frac{f f'}{f'^2 + g'^2} \dot{u}^2 = 0$. Thus a parallel $v = v_0$ is a geodesic iff $f'(v_0) = 0$ (the profile has a horizontal tangent). Such a parallel is a geodesic circle.

Remark. Other geodesics (spirals) exist when both $\dot{u}, \dot{v} \neq 0$; they are found by integrating (1) with (2) and $|\gamma'| = 1$. ■

Example 4.2.18. Discuss the notion of covariant derivative for a plane.

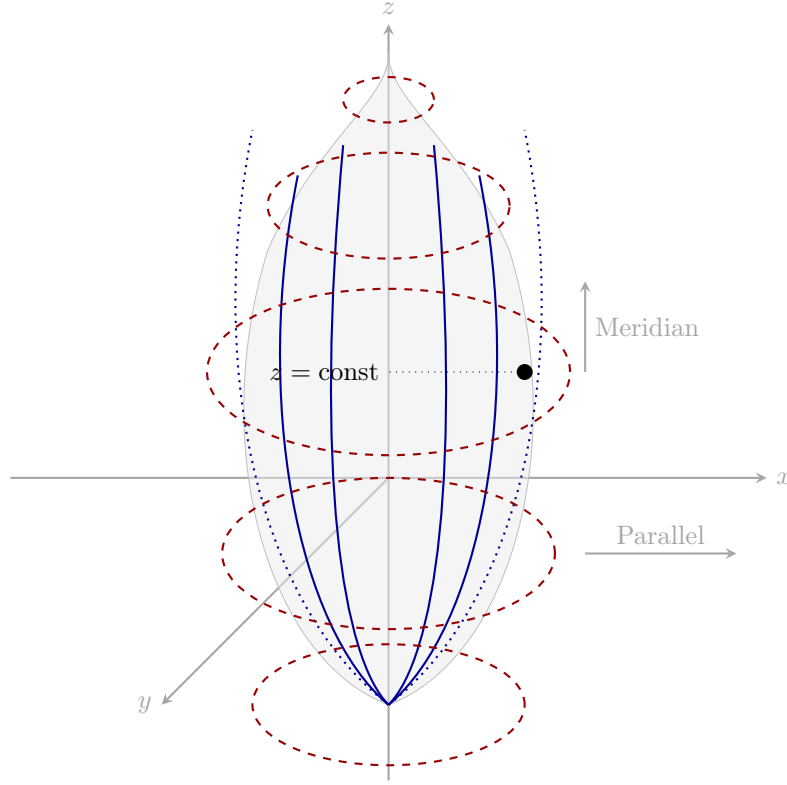


Figure 4.1: Meridians and Parallels of Surface of Revolution

Solution. A plane passing through $p_0 = (x_0, y_0, z_0)$ and containing the orthonormal vectors $w_1 = (a_1, a_2, a_3)$ and $w_2 = (b_1, b_2, b_3)$ is given by $\mathbf{x}(u, v) = p_0 + uw_1 + vw_2$, $(u, v) \in \mathbb{R}^2$. Then:

$$\mathbf{x}_u = w_1, \quad \mathbf{x}_v = w_2 \implies E = G = 1, \quad F = 0$$

Also we know the formulas for the Christoffel symbols of the second kind:

$$\begin{aligned} \Gamma_{11}^1 &= \frac{GE_u - 2FF_u + FE_v}{2(EG - F^2)}, & \Gamma_{12}^1 &= \frac{GE_v - FG_u}{2(EG - F^2)}, & \Gamma_{22}^1 &= \frac{2GF_v - GG_u - FG_v}{2(EG - F^2)} \\ \Gamma_{11}^2 &= \frac{2EF_u - EE_v - FE_u}{2(EG - F^2)}, & \Gamma_{12}^2 &= \frac{EG_u - FE_v}{2(EG - F^2)}, & \Gamma_{22}^2 &= \frac{EG_v - 2FF_v + FG_u}{2(EG - F^2)} \end{aligned}$$

Since E, F , and G are constants ($E = 1, G = 1, F = 0$), all of their partial derivatives with respect to u and v are zero. Substitution gives $\Gamma_{ij}^k = 0 \quad \forall i, j, k \in \{1, 2\}$. Therefore, from the general equation of the covariant derivative,

$$\frac{DW}{dt} = (a' + \Gamma_{11}^1(u')^2 + 2\Gamma_{12}^1u'v' + \Gamma_{22}^1(v')^2) \mathbf{x}_u + (b' + \Gamma_{11}^2(u')^2 + 2\Gamma_{12}^2u'v' + \Gamma_{22}^2(v')^2) \mathbf{x}_v.$$

We obtain

$$\frac{DW}{dt} = a' \mathbf{x}_u + b' \mathbf{x}_v$$

This shows that the covariant derivative agrees with the usual derivative in the plane; therefore, the covariant derivative is the generalization of the usual derivative of a vector in a plane. ■

Theorem 4.2.19. *A curve on a surface $\mathbf{r} = \mathbf{r}(u, v)$ is a geodesic if and only if at every point its principal normal is parallel to the surface normal.*

Proof. To prove the theorem, we will establish that the principal normal $\mathbf{r}'' = \kappa \mathbf{n}$ of the curve is orthogonal to the tangential directions $\mathbf{r}_1$ and $\mathbf{r}_2$ at P so that the principal normal of the curve coincides with the surface normal. Thus, we show that $\mathbf{r}'' \cdot \mathbf{r}_1 = 0$ and $\mathbf{r}'' \cdot \mathbf{r}_2 = 0$.

We derive these conditions from the geodesic equations

$$U(t) = \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{u}} \right) - \frac{\partial T}{\partial u} = 0 \text{ and } V(t) = \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{v}} \right) - \frac{\partial T}{\partial v} = 0$$

Let us consider

$$\dot{\mathbf{r}} = \frac{d\mathbf{r}}{dt} = \mathbf{r}_1 \dot{u} + \mathbf{r}_2 \dot{v}, \quad \text{so that, } \frac{\partial \dot{\mathbf{r}}}{\partial \dot{u}} = \mathbf{r}_1 \quad \text{and} \quad \frac{\partial \dot{\mathbf{r}}}{\partial \dot{v}} = \mathbf{r}_2 \quad (1)$$

Hence

$$\dot{\mathbf{r}} \cdot \dot{\mathbf{r}} = (\mathbf{r}_1 \dot{u} + \mathbf{r}_2 \dot{v})^2 = E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2$$

so that we can take

$$T = \frac{1}{2} \dot{\mathbf{r}}^2 \quad (2)$$

Differentiating (2) partially, we obtain

$$\begin{aligned} \frac{\partial T}{\partial u} &= \dot{\mathbf{r}} \cdot \frac{\partial \dot{\mathbf{r}}}{\partial u} = \dot{\mathbf{r}} \cdot [\mathbf{r}_{11} \dot{u} + \mathbf{r}_{21} \dot{v}], & \frac{\partial T}{\partial v} &= \dot{\mathbf{r}} \cdot \frac{\partial \dot{\mathbf{r}}}{\partial v} = \dot{\mathbf{r}} \cdot [\mathbf{r}_{12} \dot{u} + \mathbf{r}_{22} \dot{v}] \\ \frac{\partial T}{\partial \dot{u}} &= \dot{\mathbf{r}} \cdot \frac{\partial \dot{\mathbf{r}}}{\partial \dot{u}} = \dot{\mathbf{r}} \cdot \mathbf{r}_1, & \frac{\partial T}{\partial \dot{v}} &= \dot{\mathbf{r}} \cdot \frac{\partial \dot{\mathbf{r}}}{\partial \dot{v}} = \dot{\mathbf{r}} \cdot \mathbf{r}_2 \end{aligned}$$

Therefore,

$$\begin{aligned} U(t) &= \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{u}} \right) - \frac{\partial T}{\partial u} \\ &= \frac{d}{dt} (\dot{\mathbf{r}} \cdot \mathbf{r}_1) - \dot{\mathbf{r}} \cdot (\mathbf{r}_{11} \dot{u} + \mathbf{r}_{21} \dot{v}) \\ &= \ddot{\mathbf{r}} \cdot \mathbf{r}_1 + \dot{\mathbf{r}} \cdot \frac{d}{dt} \mathbf{r}_1 - \dot{\mathbf{r}} \cdot (\mathbf{r}_{11} \dot{u} + \mathbf{r}_{21} \dot{v}) \\ &= \ddot{\mathbf{r}} \cdot \mathbf{r}_1 + \dot{\mathbf{r}} \cdot (\mathbf{r}_{11} \dot{u} + \mathbf{r}_{12} \dot{v}) - \dot{\mathbf{r}} \cdot (\mathbf{r}_{11} \dot{u} + \mathbf{r}_{21} \dot{v}) \\ &= \ddot{\mathbf{r}} \cdot \mathbf{r}_2, \text{ because } \mathbf{r}_{12} = \mathbf{r}_{21} \end{aligned}$$

Further,

$$\begin{aligned} V(t) &= \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{v}} \right) - \frac{\partial T}{\partial v} \\ &= \frac{d}{dt} (\dot{\mathbf{r}} \cdot \mathbf{r}_2) - \dot{\mathbf{r}} \cdot (\mathbf{r}_{12} \dot{u} + \mathbf{r}_{22} \dot{v}) \\ &= \ddot{\mathbf{r}} \cdot \mathbf{r}_2 + \dot{\mathbf{r}} \cdot \frac{d}{dt} \mathbf{r}_2 - \dot{\mathbf{r}} \cdot (\mathbf{r}_{12} \dot{u} + \mathbf{r}_{22} \dot{v}) \\ &= \ddot{\mathbf{r}} \cdot \mathbf{r}_2 + \dot{\mathbf{r}} \cdot (\mathbf{r}_{12} \dot{u} + \mathbf{r}_{22} \dot{v}) - \dot{\mathbf{r}} \cdot (\mathbf{r}_{12} \dot{u} + \mathbf{r}_{22} \dot{v}) \\ &= \ddot{\mathbf{r}} \cdot \mathbf{r}_2 \end{aligned}$$

Hence the geodesic equations $U(t) = 0$, $V(t) = 0$ imply

$$\ddot{\mathbf{r}} \cdot \mathbf{r}_1 = 0 \quad \text{and} \quad \ddot{\mathbf{r}} \cdot \mathbf{r}_2 = 0$$

which shows that $\ddot{\mathbf{r}}$ is perpendicular to the vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ lying in the tangent plane at P . Hence $\ddot{\mathbf{r}}$ is along the surface normal at P .

But

$$\ddot{\mathbf{r}} = \kappa \mathbf{n}$$

so that $\ddot{\mathbf{r}}$ is along the principal normal of the curve at P . Hence, the principal normal at every point of the geodesic is normal to the surface at P . Since all the above steps are reversible, the converse is also true. This completes the proof of the theorem. ■

Note: Instead of taking the parameter t , we can use as well the parameter s . In that case we have to replace t by s in the above working so that we have

$$U(s) = \mathbf{r}'' \cdot \mathbf{r}_1 \quad \text{and} \quad V(s) = \mathbf{r}'' \cdot \mathbf{r}_2$$

Theorem 4.2.20 (Clairaut's Theorem). *Let α be a geodesic on a surface of revolution S , and ρ be the distance of a point of S from the axis of rotation, and let ψ be the angle between α' and the meridian of S . Then $\rho \sin \psi$ is constant along α .*

Conversely, if $\rho \sin \psi$ is constant along some curve α on the surface, and if no part of α is part of some parallel of S , then α is a geodesic.

Proof. Suppose S is parametrized by,

$$\mathbf{x}(u, v) = (f(v) \cos u, f(v) \sin u, g(v))$$

then obviously $\rho = f(v)$. Here:

$$\mathbf{x}_u = (-f \sin u, f \cos u, 0)$$

$$\mathbf{x}_v = (f' \cos u, f' \sin u, g') \implies \|\mathbf{x}_u\| = f = \rho, \quad \|\mathbf{x}_v\| = \sqrt{f'^2 + g'^2} = c$$

Since $E = f^2$, $G = f'^2 + g'^2$, and $F = 0$, the coordinate system has an orthogonal parametrization. Thus, the normalized vectors

$$\left\{ \frac{\mathbf{x}_u}{\|\mathbf{x}_u\|}, \frac{\mathbf{x}_v}{\|\mathbf{x}_v\|} \right\}$$

form an orthonormal basis for $T_p S$ for all $p \in S$.

Thus, we can decompose the unit velocity vector α' in this basis as:

$$\begin{aligned} \alpha' &= \cos \psi \frac{\mathbf{x}_v}{\|\mathbf{x}_v\|} + \sin \psi \frac{\mathbf{x}_u}{\|\mathbf{x}_u\|} \\ &= \cos \psi \frac{\mathbf{x}_v}{c} + \sin \psi \rho^{-1} \mathbf{x}_u \\ \implies \alpha' \times \mathbf{x}_v &= \sin \psi \rho^{-1} (\mathbf{x}_u \times \mathbf{x}_v) \end{aligned} \tag{C_1}$$

Also, since we can consider the curve parameterized by arc length $\alpha(s) = \mathbf{x}(u(s), v(s))$:

$$\begin{aligned} \alpha' &= \mathbf{x}_u u' + \mathbf{x}_v v' \\ \implies \alpha' \times \mathbf{x}_v &= (\mathbf{x}_u \times \mathbf{x}_v) u' \end{aligned} \tag{C_2}$$

Combining equations (C_1) and (C_2) , we have

$$\begin{aligned} (\mathbf{x}_u \times \mathbf{x}_v) u' &= \sin \psi \rho^{-1} (\mathbf{x}_u \times \mathbf{x}_v) \\ \text{i.e. } u' \rho &= \sin \psi \implies \rho^2 u' = \rho \sin \psi \end{aligned}$$

Since $\rho = f$, we obtain

$$\boxed{f^2 u' = \rho \sin \psi}$$

From the first equation of geodesic equations in the surface of revolution case:

$$\frac{d}{ds}(f^2 u') = 0 \implies f^2 u' = \text{constant}$$

This implies that $\rho \sin \psi$ is constant, say Ω , along the geodesic.

Conversely, if $\rho \sin \psi$ is constant along a unit-speed curve α on S , the above argument shows that the first equation $\frac{d}{ds}(f^2 u') = 0$ is satisfied. Now we need to check the second equation, i.e.,

$$\frac{f' f'' + g' g''}{f'^2 + g'^2} (v')^2 - \frac{f f'}{f'^2 + g'^2} u'^2 + v'' = 0 \tag{C_3}$$

is satisfied too.

Since $u' = \frac{\sin \psi}{\rho} = \frac{\Omega}{\rho^2}$, the first fundamental form for the curve α is

$$\begin{aligned} 1 &= |\alpha'|^2 = ds^2 = Eu'^2 + 2Fu'v' + Gv'^2 \\ &\Rightarrow 1 = f^2 u'^2 + (f'^2 + g'^2) v'^2 \\ &\Rightarrow 1 = f^2 \frac{\Omega^2}{f^4} + (f'^2 + g'^2) v'^2 \\ &\Rightarrow 1 = \frac{\Omega^2}{f^2} + (f'^2 + g'^2) v'^2 \end{aligned}$$

Differentiating this relation with respect to the arc length s :

$$0 = \Omega^2 \left(-\frac{2f'v'}{f^3} \right) + (2f'f''v' + 2g'g''v') v'^2 + (f'^2 + g'^2) 2v'v'' = 0$$

Dividing by $2v'$:

$$\Rightarrow -\frac{ff'}{f^4} \Omega^2 + (f'f'' + g'g'') v'^2 + (f'^2 + g'^2) v'' = 0$$

Dividing by $(f'^2 + g'^2)$:

$$\Rightarrow v'' + \frac{f'f'' + g'g''}{f'^2 + g'^2} v'^2 - \frac{ff'}{f'^2 + g'^2} u'^2 = 0 \quad \left(\text{since } \frac{\Omega^2}{f^4} = u'^2 \right)$$

which is exactly equation (C_3) . ■

Note: With the help of Clairaut's theorem and the exercise before it, we can completely classify the geodesics on a revolution surface.

Example 4.2.21. Find the geodesics on a pseudosphere. (See Figure 53 in the Atlas.)

Solution. The pseudosphere is a surface of constant negative curvature ($K = -1$). It can be obtained by revolving the tractrix around its asymptote (the z -axis). A convenient parametrization is

$$\mathbf{x}(u, v) = (\text{sech } v \cos u, \text{sech } v \sin u, v - \tanh v),$$

where $u \in (0, 2\pi)$ is the rotation angle and $v \in [0, \infty)$. The first fundamental form is

$$E = \langle \mathbf{x}_u, \mathbf{x}_u \rangle = \text{sech}^2 v, \quad F = 0, \quad G = \langle \mathbf{x}_v, \mathbf{x}_v \rangle = \tanh^2 v.$$

Thus the line element becomes $ds^2 = \text{sech}^2 v du^2 + \tanh^2 v dv^2$.

For a surface of revolution with $F = 0$, the geodesic equations admit a first integral (Clairaut's theorem). From the u -geodesic equation we obtain

$$\frac{d}{ds}(E\dot{u}) = 0 \quad \Rightarrow \quad E\dot{u} = \text{sech}^2 v \frac{du}{ds} = \text{constant} \equiv h.$$

This constant h is the conserved angular momentum. The arc-length condition $E\dot{u}^2 + G\dot{v}^2 = 1$ gives

$$\text{sech}^2 v \left(\frac{du}{ds} \right)^2 + \tanh^2 v \left(\frac{dv}{ds} \right)^2 = 1.$$

Eliminating $du/ds = h \cosh^2 v$ (since $\text{sech}^2 v = 1/\cosh^2 v$), we obtain

$$\frac{h^2}{\cosh^2 v} + \tanh^2 v \left(\frac{dv}{ds} \right)^2 = 1 \quad \Rightarrow \quad \left(\frac{dv}{ds} \right)^2 = \frac{1 - h^2 \text{sech}^2 v}{\tanh^2 v}.$$

Special geodesics:

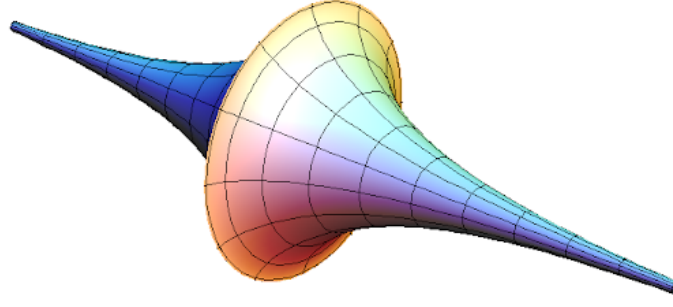


Figure 4.2: Pseudosphere - surface of revolution of a tractrix with constant negative Gaussian curvature $K = -1/a^2$.

- **Meridians:** $h = 0$. Then $du/ds = 0$, so $u = \text{constant}$. From the arc-length condition $dv/ds = \coth v$. Integrating yields a parametrization of the tractrix. These curves are geodesics.
- **Parallels:** $dv/ds = 0$. This requires $\tanh^2 v \neq 0$ and $1 - h^2 \operatorname{sech}^2 v = 0$, i.e. $h = \cosh v$. Since h is constant, this can only hold for a single value of v . Hence a parallel (circle of latitude) is a geodesic iff $v = \operatorname{arcosh} h$, which occurs only for one specific circle (the geodesic parallel). On the pseudosphere this is the circle at $v = \operatorname{arcosh} 1 = 0$? Actually $h = \cosh v_0$ gives a constant. For a given h , there is a unique v_0 such that the parallel is a geodesic. On the pseudosphere such a geodesic parallel does not exist because $\cosh v \geq 1$ and the pseudosphere has a boundary at $v = 0$ (a cusp). Indeed, the only possible geodesic parallel would be at $v = 0$ where the surface degenerates; thus there are no non-trivial geodesic parallels on the regular part.
- **General geodesics:** For $0 < |h| < 1$, the expression $(dv/ds)^2$ becomes positive for all v and the geodesic winds around the pseudosphere, approaching the boundary $v = 0$ (the cusp) asymptotically. In fact, one can integrate to obtain explicit formulas in terms of hyperbolic functions. The general solution describes curves that spiral towards the equator or the cusp depending on h .

Thus the geodesics on the pseudosphere consist of:

1. All meridians (the generating curves).
2. A one-parameter family of non-meridional geodesics that are not closed and that approach the cusp.

No closed geodesics other than the meridians exist on the pseudosphere. This behaviour is characteristic of surfaces with constant negative curvature. ■

4.3 Geodesic Curvatures

If $\mathbf{r} = \mathbf{r}(s)$ is the position vector of any point P on a curve on a surface, then the curvature vector $\mathbf{r}'' = \mathbf{t}' = \kappa \mathbf{n}$ where κ is the curvature and $\mathbf{n}$ is the principal normal at P to the curve. As we have already noted, at a point P on the surface, there exist three linearly independent non-coplanar vectors $\mathbf{N}$, $\mathbf{r}_1$ and $\mathbf{r}_2$ where $\mathbf{r}_1$ and $\mathbf{r}_2$ are tangential to the surface and $\mathbf{N}$ is the surface normal. Hence $\mathbf{r}''$ at P can be expressed as a linear combination of the vectors $\mathbf{N}$, $\mathbf{r}_1$ and $\mathbf{r}_2$ so that we can write

$$\mathbf{r}'' = \kappa_n \mathbf{N} + \lambda \mathbf{r}_1 + \mu \mathbf{r}_2$$

where κ_n is the normal component of $\mathbf{r}''$.

Definition 4.3.1. The vector $\lambda \mathbf{r}_1 + \mu \mathbf{r}_2$ with components (λ, μ) is tangential to the surface. The vector with components (λ, μ) of the tangential vector $\lambda \mathbf{r}_1 + \mu \mathbf{r}_2$ to the surface is called the geodesic curvature vector at P . It is denoted by K_g .

Theorem 4.3.2. A curve on a surface is a geodesic if and only if the geodesic curvature vector is zero.

Proof. Let $\mathbf{r} = \mathbf{r}(s)$ be any curve on the surface with the principal normal $\mathbf{n}$ and surface normal $\mathbf{N}$ at P . If (λ, μ) is the geodesic curvature vector at P , then:

$$\mathbf{r}'' = \kappa_n \mathbf{N} + \lambda \mathbf{r}_1 + \mu \mathbf{r}_2 \quad \text{or} \quad \kappa \mathbf{n} = \kappa_n \mathbf{N} + \lambda \mathbf{r}_1 + \mu \mathbf{r}_2 \quad (1)$$

Let the curve be a geodesic. Then, by the normal property of a geodesic:

$$\mathbf{n} = \mathbf{N} \quad (2)$$

Using (2) in (1), we obtain

$$\kappa \mathbf{N} = \kappa_n \mathbf{N} + \lambda \mathbf{r}_1 + \mu \mathbf{r}_2$$

Equating the coefficients of $\mathbf{r}_1$ and $\mathbf{r}_2$ on both sides, we obtain $\lambda = \mu = 0$, so that $K_g = 0$.

Conversely, let $K_g = 0$. This implies $\lambda = 0$, $\mu = 0$. Hence, from (1), we obtain $\kappa \mathbf{n} = \kappa_n \mathbf{N}$. Thus, the principal normal to the curve is parallel to the surface normal. Therefore, the curve is a geodesic by the normal property. ■

Lemma 4.3.3. *Any normal section of a surface is a geodesic.*

Proof. We know a normal section is the intersection curve α of S with a plane P such that P is perpendicular ($\perp$) to the surface at each point of α , or P contains both the surface normal (N) and the principal normal (n) of α at each point of α .

That is, $\alpha'' \parallel n \parallel N$ at each point of α . This implies, α is a geodesic. ■

Great circle: A curve (circle) obtained as a normal section of a plane on S^2 passing through the centre of S^2 . (See Figure 48 in the Atlas.)

Theorem 4.3.4. *The necessary and sufficient condition for a curve on a sphere to be a geodesic is that the curve is a great circle (sometimes also referred to as a meridian). (See Figure 51 in the Atlas.)*

Proof. Suppose α is a geodesic on $S^2(c, r)$ [a sphere with centre c and radius r]. Also, let n be the principal normal of α and N be the normal vector field of $S^2(c, r)$. Since α is a geodesic, we have $n \parallel N$. Also we know that the normal to S^2 passes through the centre c at every point of S^2 . This implies that the principal normal n at every point of the curve passes through the centre c , so that we can write:

$$\vec{\alpha} - c = \lambda n$$

where $\vec{\alpha}$ is the position vector of any point on α , and λ is a non-zero differentiable function. Therefore,

$$\vec{\alpha} = c + \lambda n$$

Differentiating with respect to the arc length s , we obtain

$$\begin{aligned} t &= \lambda' n + \lambda n' \\ t &= \lambda' n + \lambda(-\kappa t + \tau b) \\ &= -\lambda \kappa t + \lambda' n + \lambda \tau b \end{aligned}$$

Comparing the coefficients of the binormal vector b on both sides, we obtain $\lambda \tau = 0 \implies \tau = 0$. That is, α is a planar curve such that the normal at every point of α passes through the centre c of the sphere. This implies that α is the section of the sphere obtained by a plane passing through its centre, which therefore is a great circle of the sphere.

Conversely, suppose $\alpha \subset S^2$ is a great circle. Then, by the definition of a great circle and the lemma above, α is always a geodesic. ■

Example 4.3.5. Find all the geodesics on a Cylinder. (See Figure 54 in the Atlas.)

Solution. Let $S \subset \mathbb{R}^3$ be a right circular cylinder parameterized by $\mathbf{x}(u, v) = (\cos u, \sin u, v)$. Differentiating with respect to the coordinates yields the tangent vectors $\mathbf{x}_u = (-\sin u, \cos u, 0)$ and $\mathbf{x}_v = (0, 0, 1)$, from which we compute the coefficients of the first fundamental form:

$$E = \langle \mathbf{x}_u, \mathbf{x}_u \rangle = 1, \quad F = \langle \mathbf{x}_u, \mathbf{x}_v \rangle = 0, \quad G = \langle \mathbf{x}_v, \mathbf{x}_v \rangle = 1$$

Because these coefficients are constant, all of their partial derivatives vanish. This implies that all Christoffel symbols of the second kind are zero ($\Gamma_{ij}^k = 0$). Consequently, the geodesic equations for a curve $\gamma(s) = \mathbf{x}(u(s), v(s))$ parameterized by arc length s ,

$$u'' + \sum_{i,j} \Gamma_{ij}^1 u'_i u'_j = 0 \quad \text{and} \quad v'' + \sum_{i,j} \Gamma_{ij}^2 u'_i u'_j = 0$$

simplify directly to $u''(s) = 0$ and $v''(s) = 0$. Integrating these equations twice yields the linear coordinate relations:

$$u(s) = as + b \quad \text{and} \quad v(s) = cs + d$$

where a, b, c , and d are constants of integration.

Substituting these back into the parametrization of the cylinder gives the general vector equation of any geodesic curve on the surface:

$$\gamma(s) = (\cos(as + b), \sin(as + b), cs + d)$$

The constants a and c classify these curves into three distinct geometric families:

1. **Vertical straight lines** (generators of the cylinder) when $a = 0$,
2. **Horizontal circles** (parallel cross-sections) when $c = 0$, and
3. **Circular helices** winding around the cylinder when both $a \neq 0$ and $c \neq 0$.

This classification is intrinsically explained by the flat geometry of the cylinder. Since the metric coefficients are $E = G = 1$ and $F = 0$, the surface is locally isometric to the Euclidean plane. Under this isometry, the straight lines of the plane (which are its geodesics) map directly to these three families of curves on the cylinder. ■

Example 4.3.6. Find all geodesics on a right circular cylinder of radius R .

Solution. Let the cylinder be given by the parametrization

$$\mathbf{x}(\theta, z) = (R \cos \theta, R \sin \theta, z), \quad 0 \leq \theta < 2\pi, \quad z \in \mathbb{R},$$

where $R > 0$ is the radius. The first fundamental form is

$$E = \|\mathbf{x}_\theta\|^2 = R^2, \quad F = \mathbf{x}_\theta \cdot \mathbf{x}_z = 0, \quad G = \|\mathbf{x}_z\|^2 = 1,$$

so the line element is $ds^2 = R^2 d\theta^2 + dz^2$. The Christoffel symbols for this metric are all zero (since the metric coefficients are constant). Consequently the geodesic equations become

$$\frac{d^2\theta}{ds^2} = 0, \quad \frac{d^2z}{ds^2} = 0.$$

Hence

$$\theta(s) = as + \theta_0, \quad z(s) = cs + z_0,$$

with constants a, c satisfying the unit-speed condition

$$R^2 \dot{\theta}^2 + \dot{z}^2 = R^2 a^2 + c^2 = 1.$$

The explicit curves in $\mathbb{R}^3$ are therefore

$$\gamma(s) = (R \cos(as + \theta_0), R \sin(as + \theta_0), cs + z_0).$$

We distinguish three cases:

- **Meridians (vertical lines):** $a = 0$, $c = \pm 1$. Then θ is constant and the curve is a straight line parallel to the axis:

$$\gamma(s) = (R \cos \theta_0, R \sin \theta_0, \pm s + z_0).$$

- **Parallels (horizontal circles):** $c = 0$, $a = \pm 1/R$. Then z is constant and the curve is a circle:

$$\gamma(s) = (R \cos(\pm s/R + \theta_0), R \sin(\pm s/R + \theta_0), z_0).$$

- **Helices:** both $a \neq 0$ and $c \neq 0$. These are curves of constant pitch:

$$\gamma(s) = (R \cos(as + \theta_0), R \sin(as + \theta_0), cs + z_0), \quad a^2 + \frac{c^2}{R^2} = \frac{1}{R^2}.$$

When the cylinder is cut along a generator and flattened into a plane, these helices become ordinary straight lines (slanted lines) in the (θ, z) -plane.

Thus all geodesics on a cylinder are exactly the meridians, the parallels, and the helices (which include the two previous families as limiting cases when $c = 0$ or $a = 0$). ■

2nd Solution. We have already shown that the map:

$$f(x, y, 0) = (\cos x, \sin x, y)$$

acts as a local isometry between a plane and a cylinder. So, f (geodesic of plane) will be a geodesic on the cylinder.

We know that in the case of a plane, the geodesics are all straight lines, which one can classify as:

- the x -axis,
- the y -axis, or
- an arbitrary line: $y = mx + c$.

Applying the isometry map f to each class of lines:

$$\begin{aligned} f(x\text{-axis}) &= f(x, 0, 0) = (\cos x, \sin x, 0) \\ &= \text{a circle (normal section)} \end{aligned}$$

$$\begin{aligned} f(y\text{-axis}) &= f(0, y, 0) = (1, 0, y) \\ &= \text{generator of the cylinder (straight line)} \end{aligned}$$

$$\begin{aligned} f(\text{arbitrary line}) &= f(x, mx + c, 0) = (\cos x, \sin x, mx + c) \\ &= \text{a helix on the cylinder} \end{aligned}$$

Thus, the geodesics on a cylinder are: **circles, straight lines, and helices.** ■

4.4 The Gauss–Bonnet Theorem

(See Figure 56 in the Atlas.)

The Gauss–Bonnet theorem is one of the deepest results in differential geometry. It connects the local geometric quantity (Gaussian curvature) with a global topological invariant (the Euler characteristic). The theorem has several versions, each revealing a different aspect of this interplay.

4.4.1 Geometric and Topological Preliminaries

Definition 4.4.1 (Gaussian curvature). For a smooth surface S , the Gaussian curvature $K(p)$ at a point p is the product of the principal curvatures: $K = \kappa_1 \kappa_2$. It measures the intrinsic curvature of the surface.

Definition 4.4.2 (Euler characteristic). The Euler characteristic $\chi(S)$ of a compact surface S is a topological invariant. For a sphere $\chi = 2$, for a torus $\chi = 0$, for a surface of genus g (orientable) $\chi = 2 - 2g$, for a projective plane $\chi = 1$, etc.

4.4.2 Global Gauss–Bonnet Theorem (Closed Surfaces)

Theorem 4.4.3 (Gauss–Bonnet for closed surfaces). *Let S be a compact, orientable, two-dimensional Riemannian manifold without boundary. Then*

$$\int_S K dA = 2\pi \chi(S),$$

where K is the Gaussian curvature, dA the area element, and $\chi(S)$ the Euler characteristic of S .

Explanation: The total Gaussian curvature of a closed surface is an integer multiple of 2π determined solely by the topology of the surface – it is independent of the metric. For example:

- A sphere (any smooth metric) has total curvature 4π .
- A torus (genus 1) has total curvature 0.
- A surface of genus 2 has total curvature -4π .

If the surface is non-orientable, the same formula holds with the Euler characteristic defined appropriately; for the projective plane $\chi = 1$ gives total curvature 2π .

4.4.3 Gauss–Bonnet Theorem for Surfaces with Boundary

Theorem 4.4.4 (Gauss–Bonnet with boundary). *Let S be a compact, orientable Riemannian surface with smooth boundary ∂S . Then*

$$\int_S K dA + \int_{\partial S} \kappa_g ds = 2\pi \chi(S),$$

where κ_g is the geodesic curvature of the boundary curve (taken with respect to the inward or outward orientation, consistently), and ds is the arc length element.

Explanation: The boundary contributes its total geodesic curvature. If the boundary is piecewise smooth (e.g., a polygon), the geodesic curvature is replaced by the sum of the exterior angles at the vertices.

4.4.4 Local (Geodesic Polygon) Version

Theorem 4.4.5 (Gauss–Bonnet for a geodesic polygon). *Let R be a simply connected region in a surface whose boundary consists of a finite number of geodesic arcs meeting at vertices with exterior angles $\theta_1, \dots, \theta_n$ (the turning angles). Then*

$$\int_R K dA + \sum_{i=1}^n \theta_i = 2\pi.$$

If the boundary arcs are not geodesic, one integrates the geodesic curvature along each arc:

$$\int_R K dA + \sum_{i=1}^n \int_{\gamma_i} \kappa_g ds + \sum_{i=1}^n \theta_i = 2\pi.$$

Explanation: This is the original formulation used by Gauss. For a triangle on a sphere with angles α, β, γ and constant curvature $K = 1$, we have $\int K dA = \text{area}$, and the exterior angles are $\pi - \alpha$, etc. The formula gives

$$\text{area} + (\pi - \alpha) + (\pi - \beta) + (\pi - \gamma) = 2\pi \implies \text{area} = \alpha + \beta + \gamma - \pi,$$

which is the well-known spherical excess formula.

4.4.5 Combinatorial Gauss–Bonnet (Discrete Version)

For a polyhedral surface (a triangulated surface with flat faces), the curvature is concentrated at vertices. The discrete Gauss–Bonnet theorem states:

$$\sum_{\text{vertices } v} (2\pi - \sum_{\text{faces around } v} \text{angle at } v) = 2\pi \chi(S).$$

This is a fundamental tool in discrete differential geometry and computer graphics.

4.4.6 Remarks and Consequences

- The Gauss–Bonnet theorem shows that the total curvature is a topological invariant, which is remarkable because curvature is a local geometric quantity.
- It implies that a metric of positive curvature can exist only on surfaces with positive Euler characteristic (sphere, projective plane); a metric of zero curvature only on tori or Klein bottles; negative curvature only on surfaces of genus at least 2.
- The theorem also holds for non-orientable surfaces if the Euler characteristic is defined appropriately (using a triangulation).

Atlas of Figures

Standard Curves and Geometrical Results

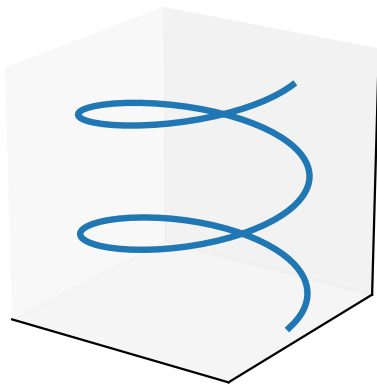


Figure 1. Circular helix: $\alpha(t) = (a \cos t, a \sin t, ct)$.

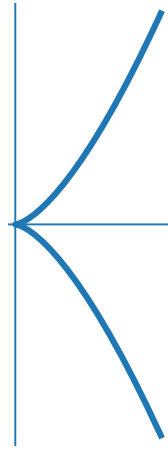


Figure 2. Semicubical parabola: $\alpha(t) = (t^2, t^3)$.

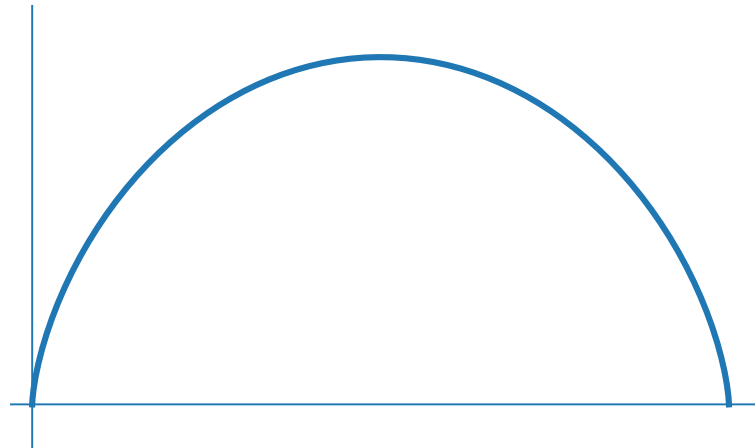


Figure 3. Cycloid arch: $\alpha(t) = (R(t - \sin t), R(1 - \cos t))$.

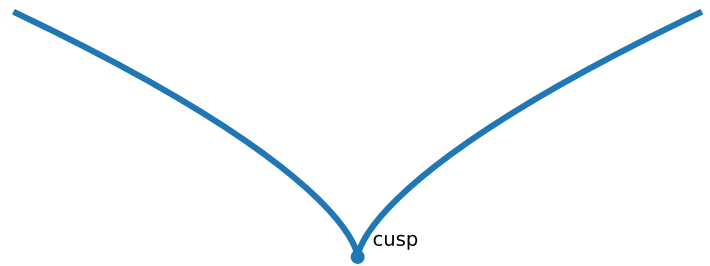


Figure 4. Cusp: $\alpha(t) = (t^3, t^2)$.

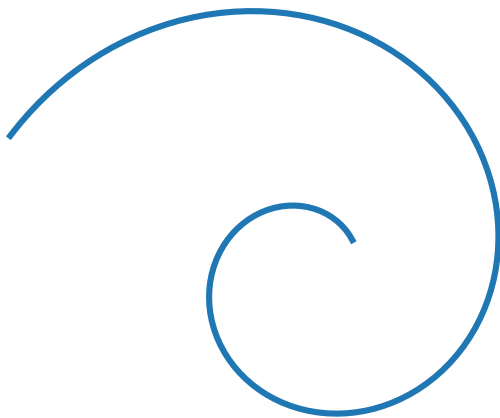


Figure 5. Logarithmic spiral: $\alpha(t) = (e^t \cos t, e^t \sin t)$.

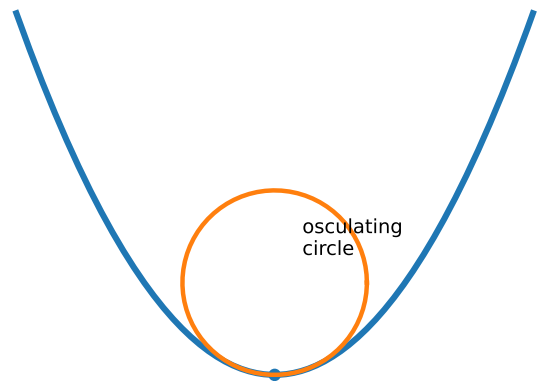


Figure 6. Parabola and its osculating circle at the vertex.

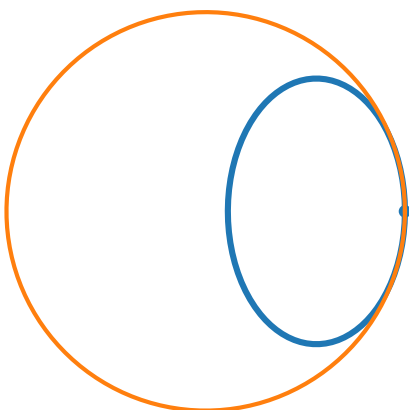


Figure 7. Ellipse and its osculating circle at $(2, 0)$.

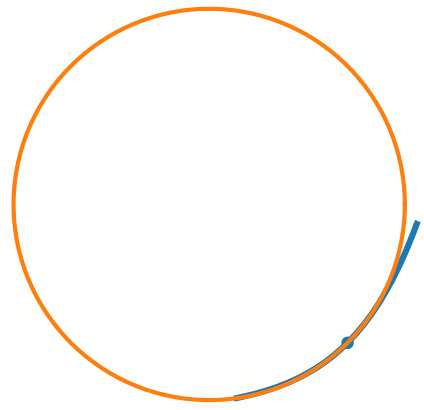


Figure 8. Curve $y = e^x$ and its osculating circle at $(0, 1)$.

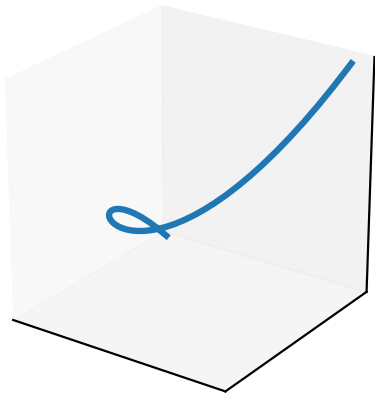


Figure 9. Twisted cubic: $\alpha(t) = (t, t^2, t^3)$.

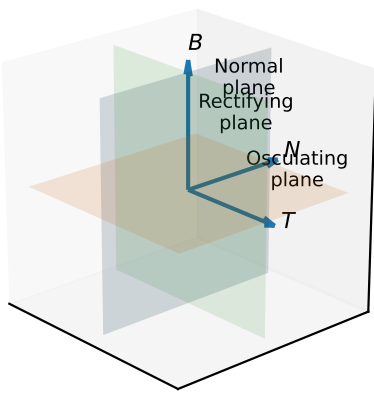


Figure 10. Frenet planes: normal, osculating, and rectifying planes.

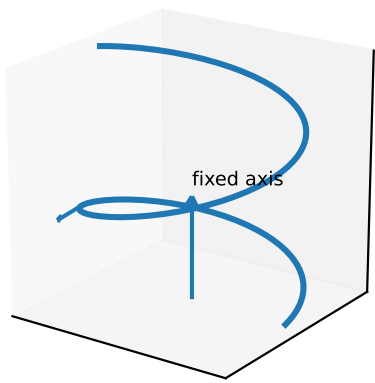


Figure 11. General helix: tangent making a constant angle with a fixed axis.

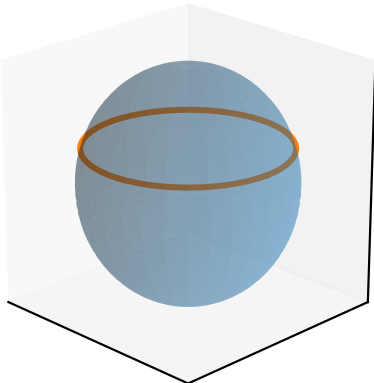


Figure 12. Spherical curve.

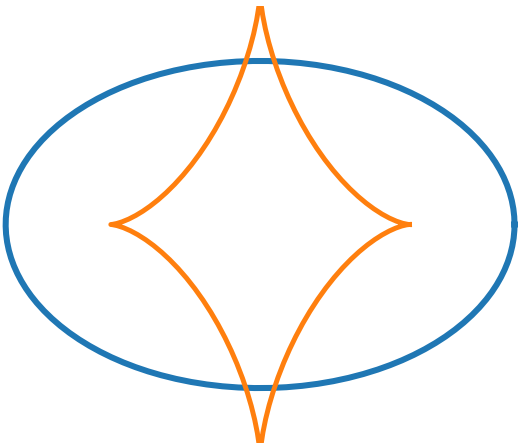


Figure 13. Ellipse and evolute.

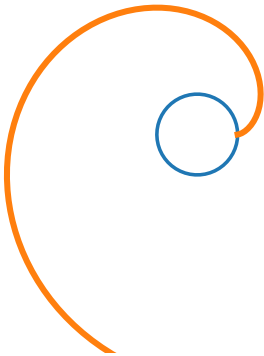


Figure 14. Circle and involute.

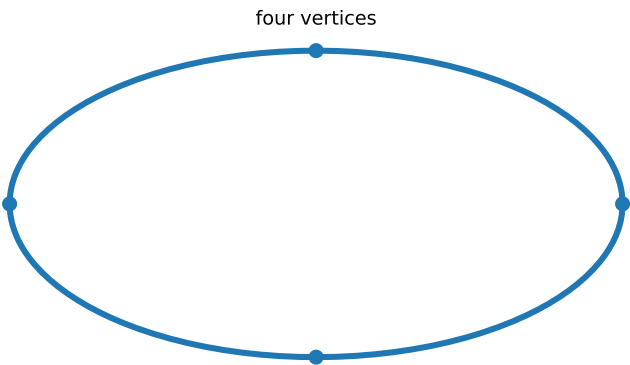


Figure 15. Four-vertex theorem illustrated on an ellipse.

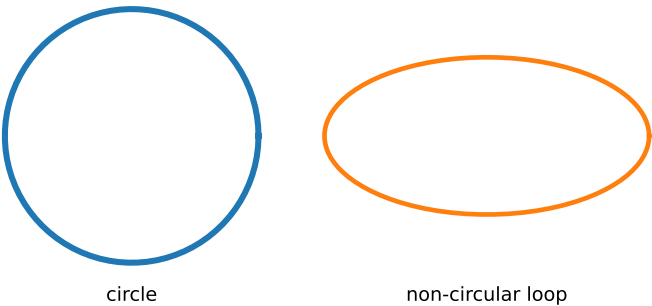


Figure 16. Isoperimetric principle: circle versus a non-circular loop.

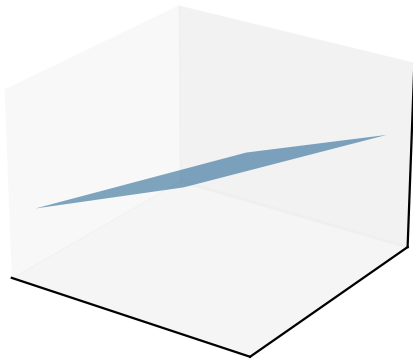


Figure 17. Plane in $\mathbb{R}^3$.

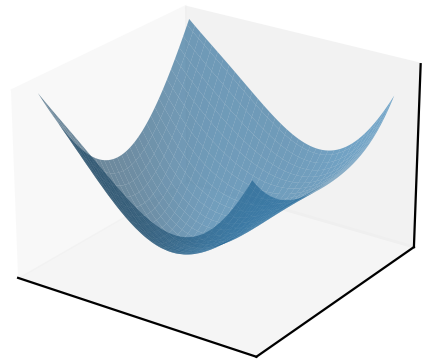


Figure 18. Graph of a smooth function $z = f(x, y)$.

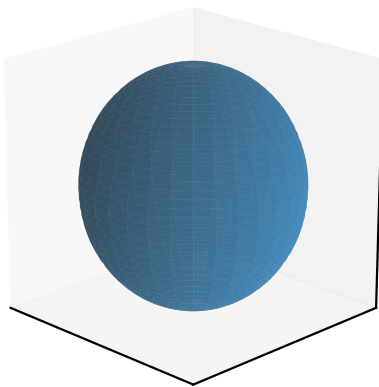


Figure 19. Sphere.

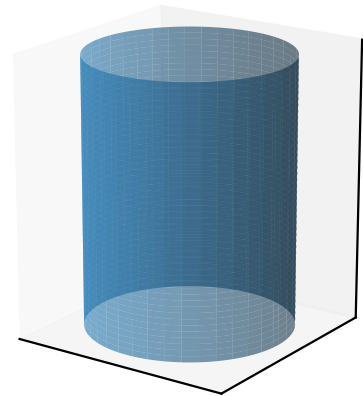


Figure 20. Right circular cylinder.

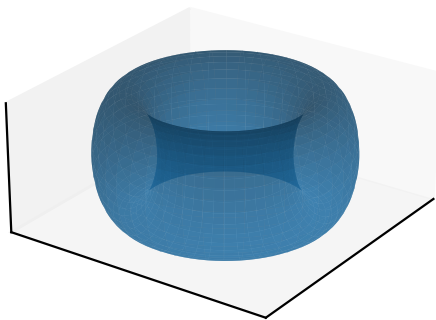


Figure 21. Torus.

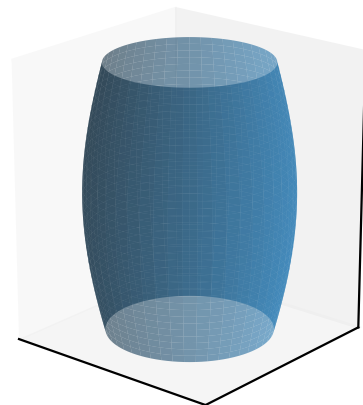


Figure 22. Surface of revolution.

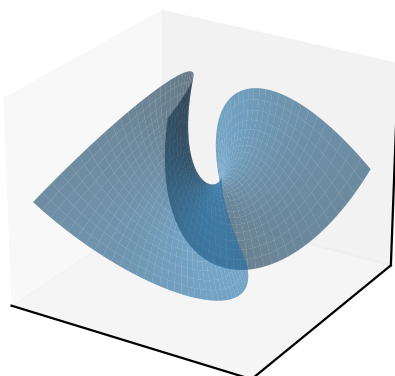


Figure 23. Enneper surface.

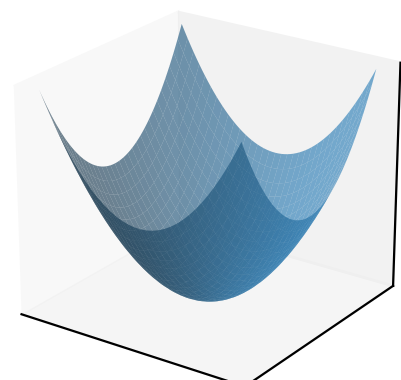


Figure 24. Elliptic paraboloid: $z = x^2 + y^2$.

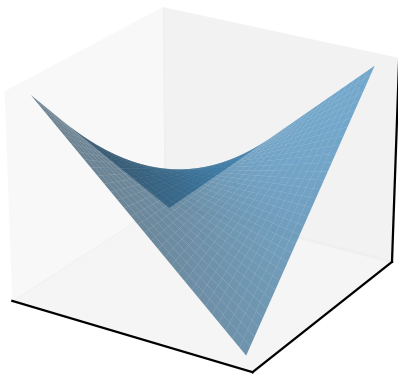


Figure 25. Hyperbolic paraboloid: $z = xy$.

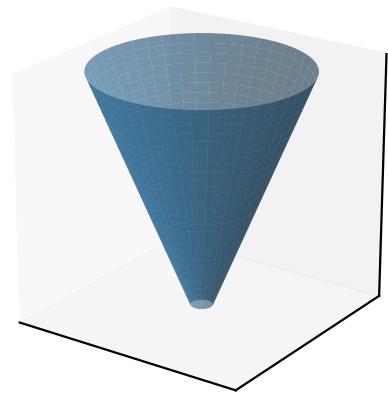


Figure 26. Right circular cone.

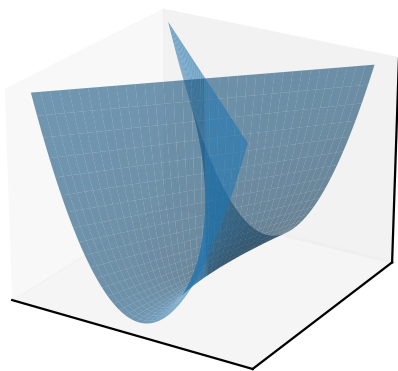


Figure 27. Whitney umbrella: $X(u, v) = (uv, u, v^2)$.

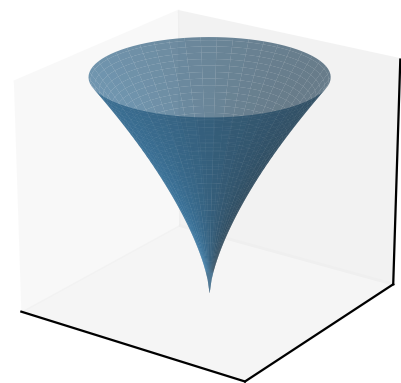


Figure 28. Cusp surface: $z = (x^2 + y^2)^{1/3}$.

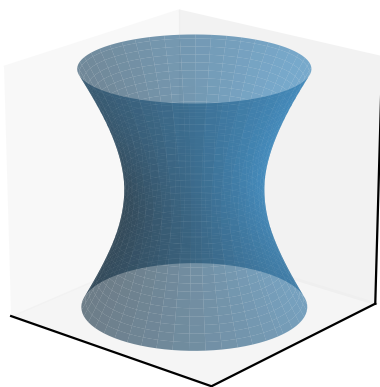


Figure 29. One-sheeted hyperboloid: $x^2 + y^2 - z^2 = 1$.

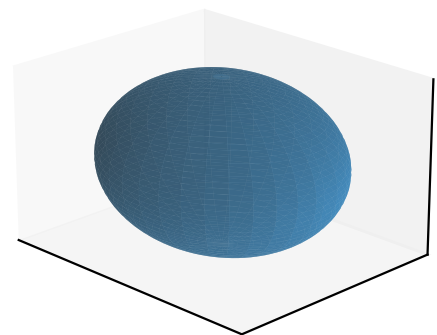


Figure 30. Ellipsoid.

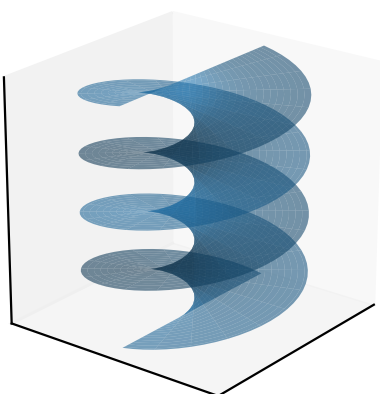


Figure 31. Helicoid.

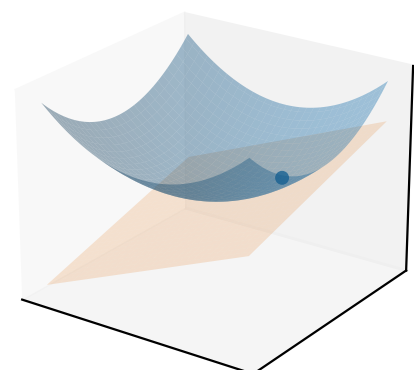


Figure 32. Tangent plane to a regular surface.

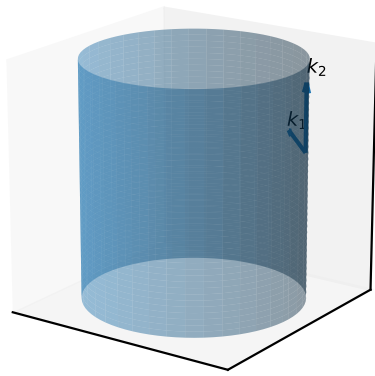


Figure 33. Principal directions on a cylinder.

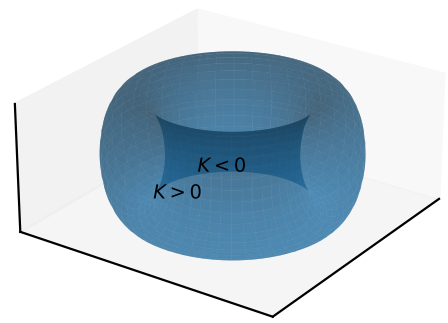


Figure 34. Torus: outer region $K > 0$ and inner region $K < 0$.

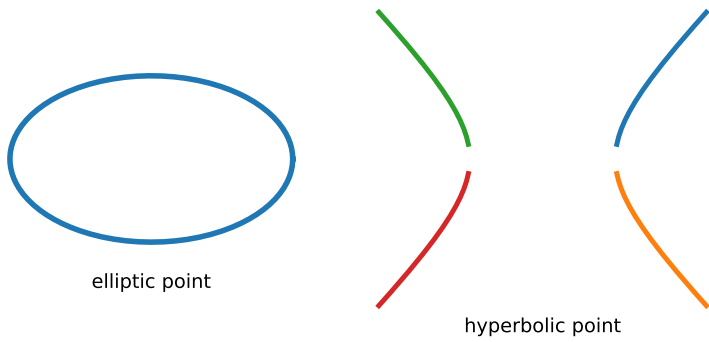


Figure 35. Dupin indicatrix: ellipse and hyperbola.

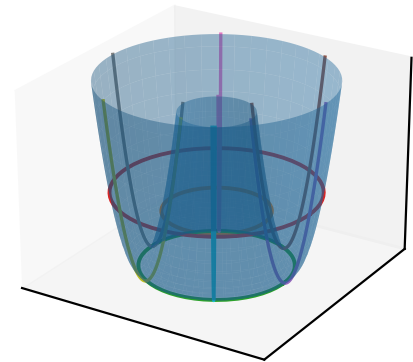


Figure 36. Lines of curvature on a surface of revolution.

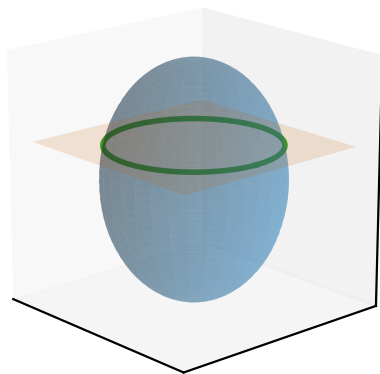


Figure 37. Joachimsthal configuration: a plane cutting a sphere in a circle.

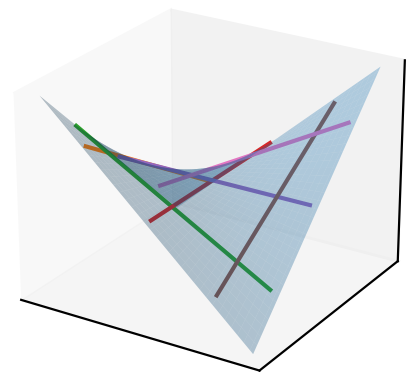


Figure 38. Doubly ruled hyperbolic paraboloid.

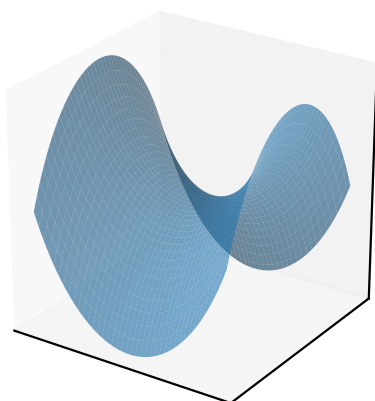


Figure 39. Scherk-type minimal surface.

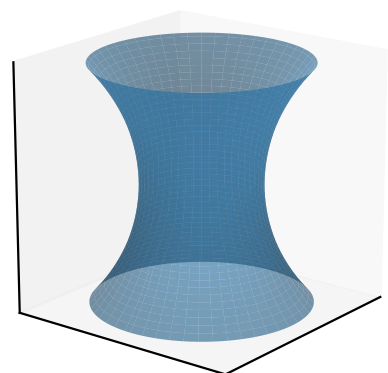


Figure 40. Catenoid.

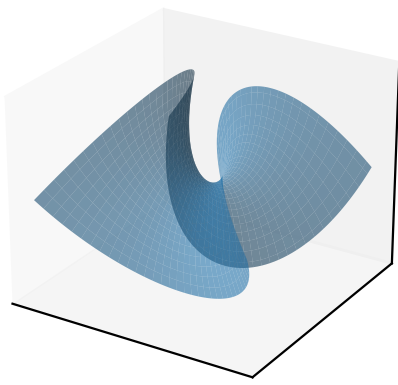


Figure 41. Enneper minimal surface.

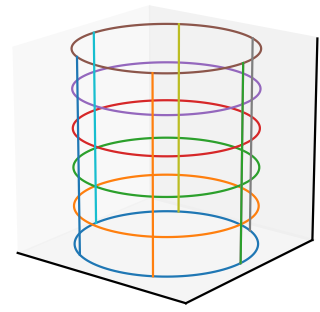
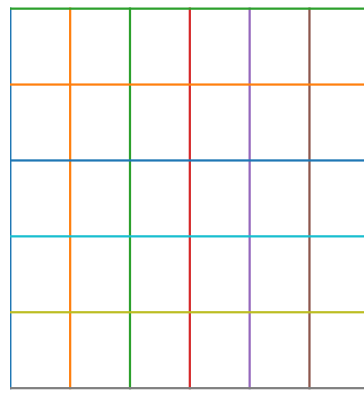


Figure 42. Local isometry: plane strip and cylinder.

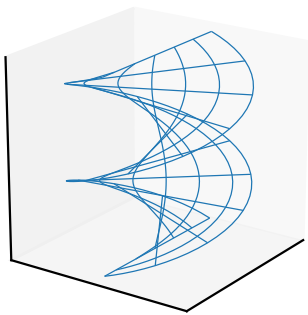
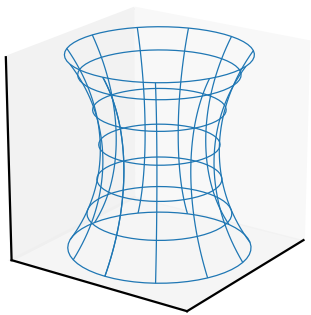


Figure 43. Local isometry: catenoid and helicoid.

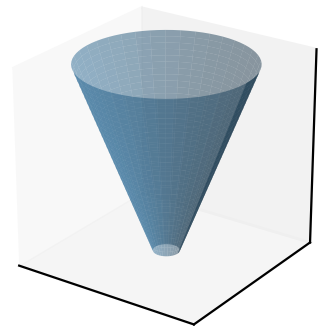
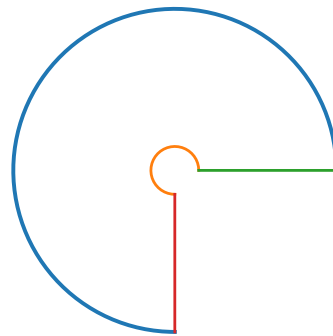


Figure 44. Local isometry: planar sector and cone.

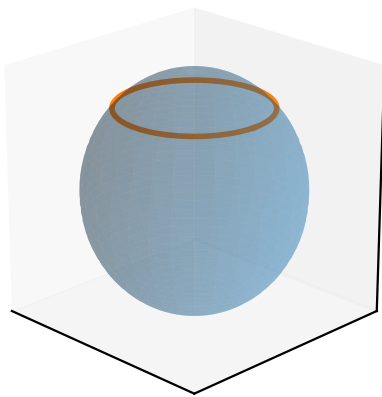


Figure 45. Geodesic circle on a sphere.

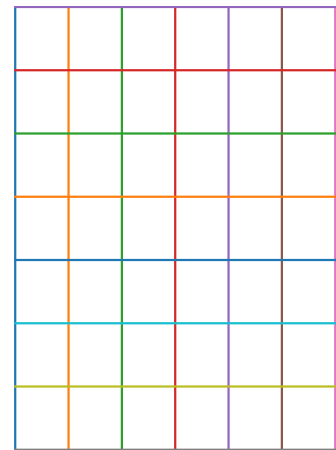
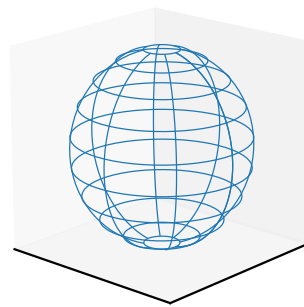


Figure 46. Isothermal coordinates on the sphere.

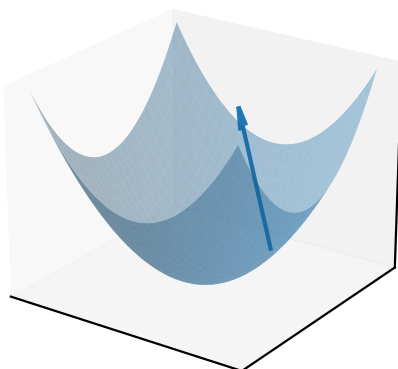


Figure 47. Surface normal / Gauss-map direction at a point.

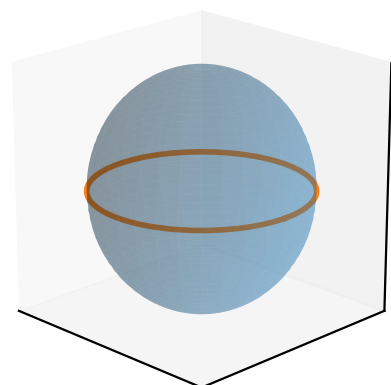


Figure 48. Great circle on the sphere.

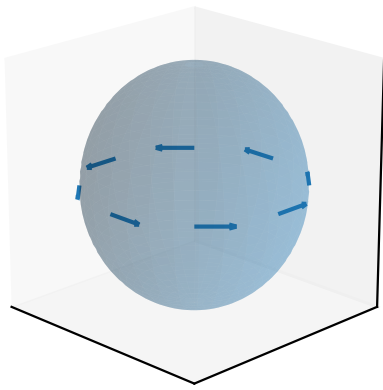


Figure 49. Tangent vector field along the equator of a sphere.

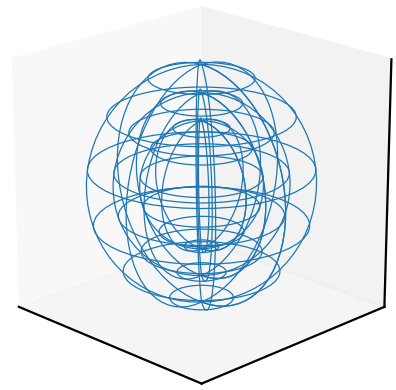


Figure 50. Parallel surfaces of a sphere.

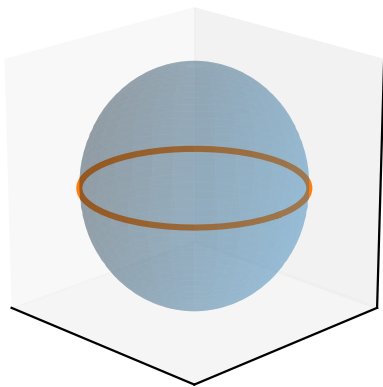


Figure 51. Great-circle geodesic on a sphere.

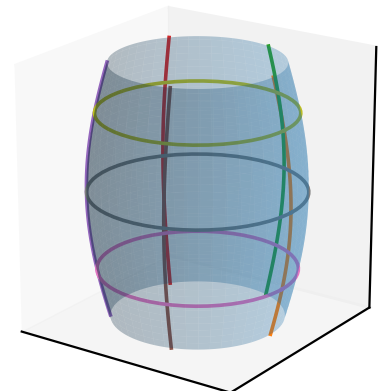


Figure 52. Meridians and parallels of a surface of revolution.

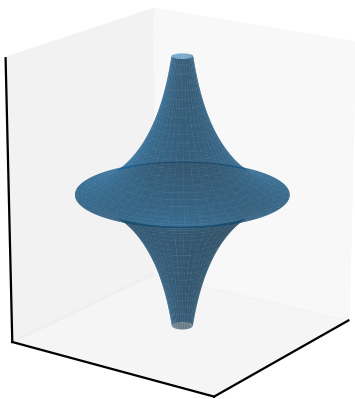


Figure 53. Pseudosphere: surface of revolution of a tractrix.

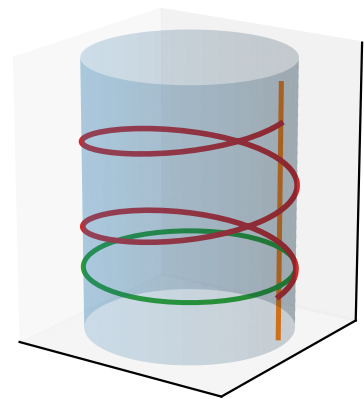


Figure 54. Geodesics on a cylinder: generator, parallel, and helix.

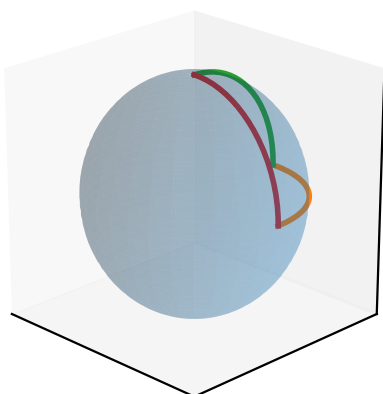


Figure 55. Geodesic triangle on the sphere.

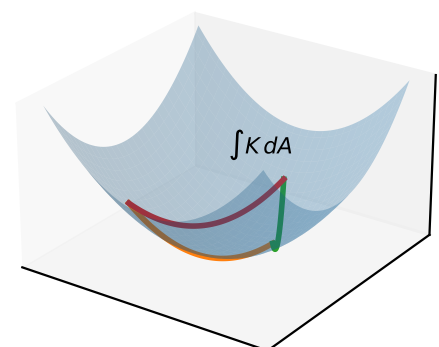


Figure 56. Gauss--Bonnet region on a curved surface.